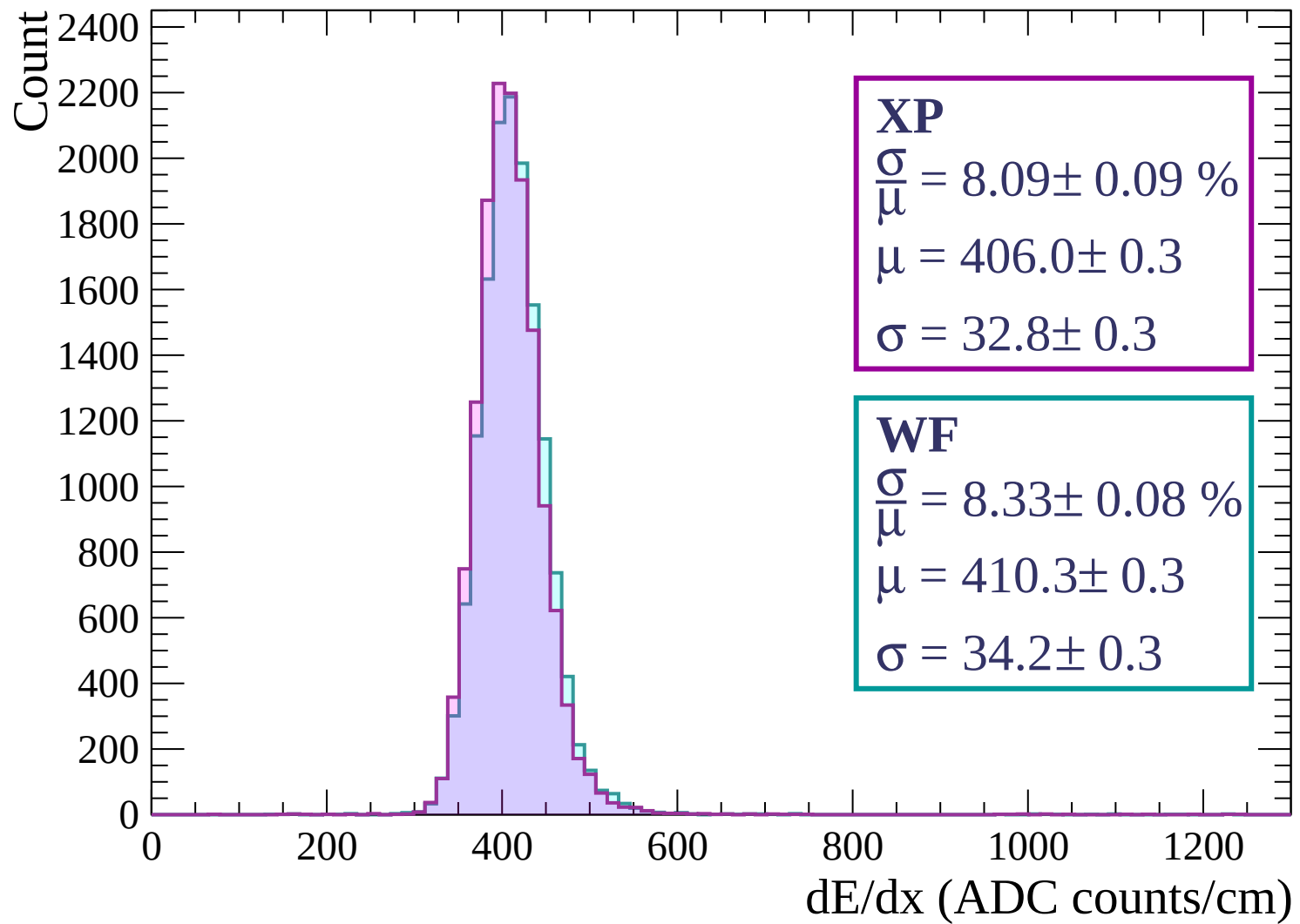
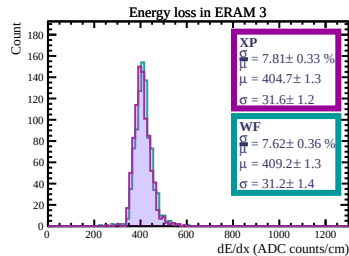
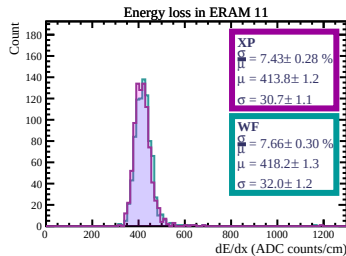
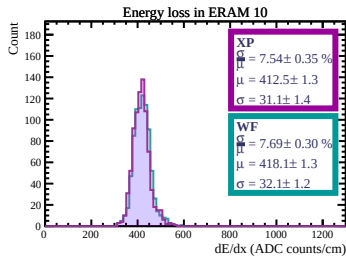
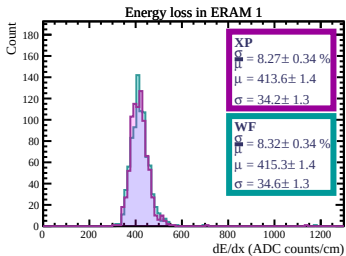
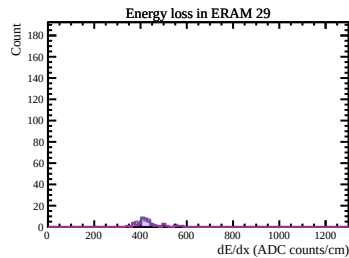
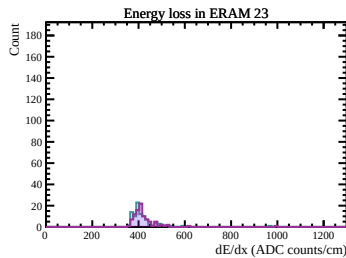
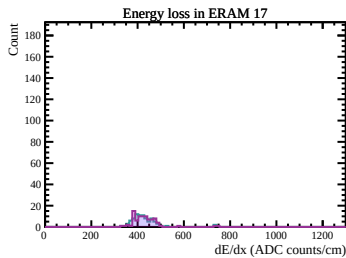
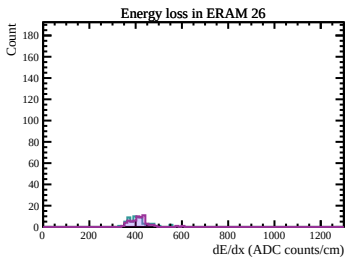
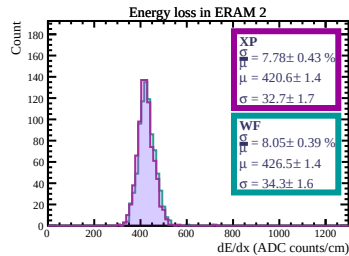
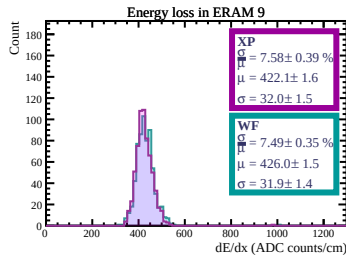
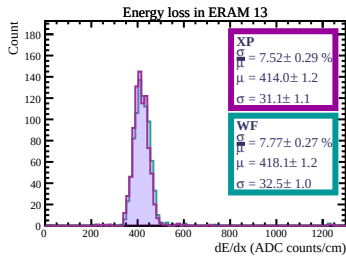
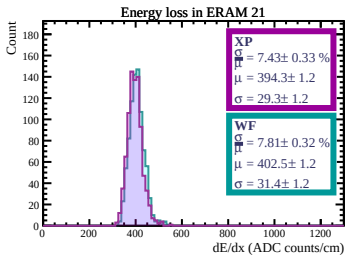
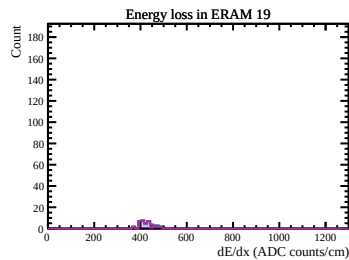
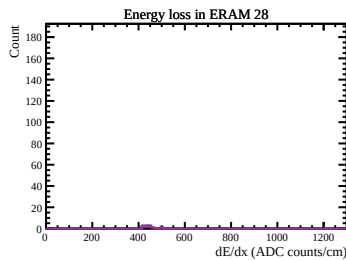
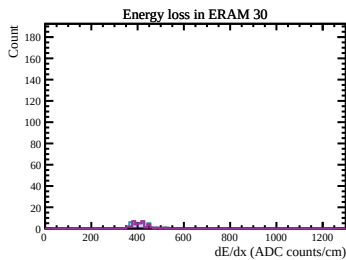
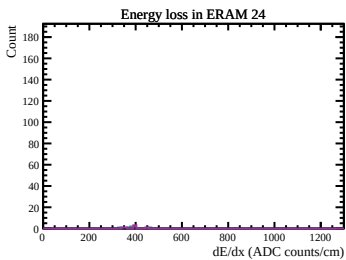
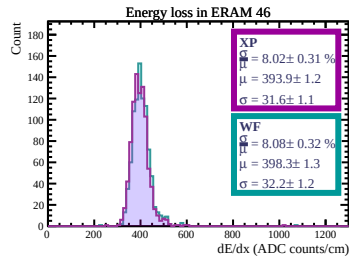
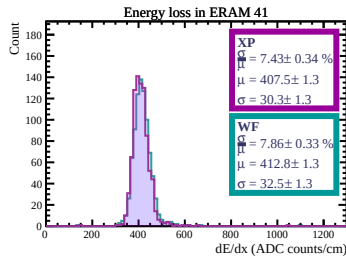
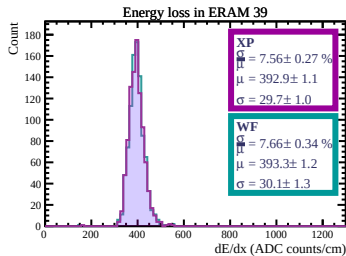
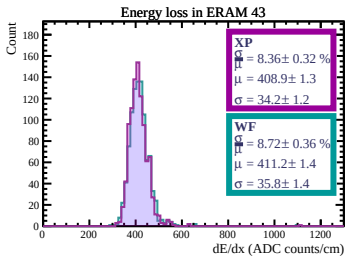
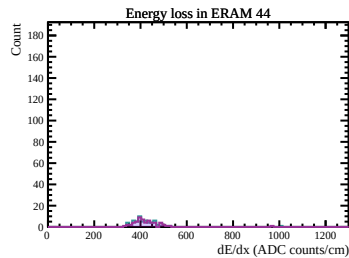
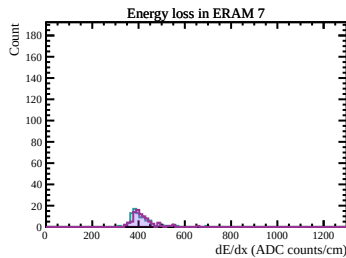
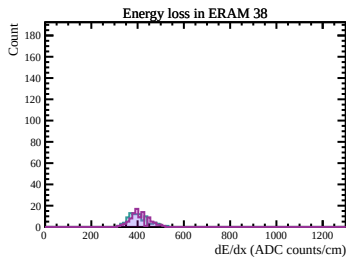
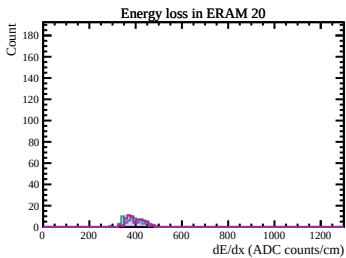
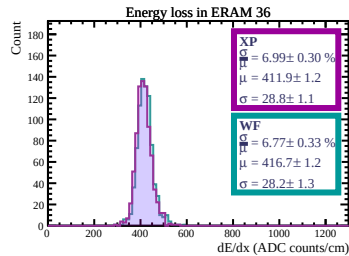
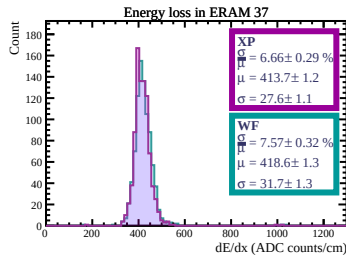
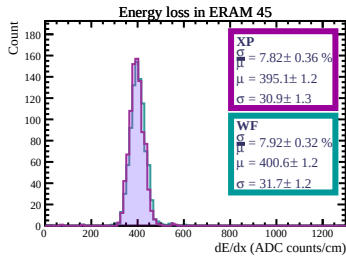
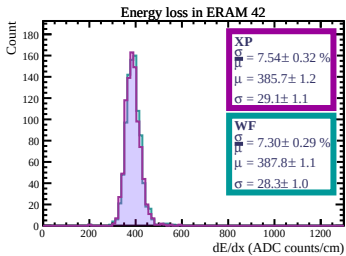
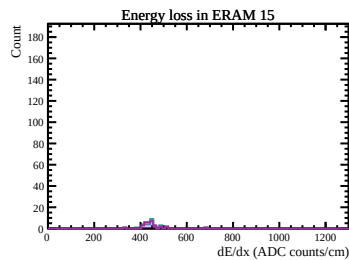
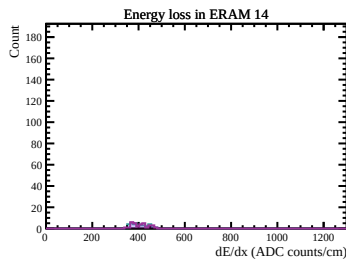
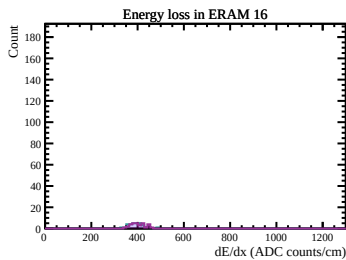
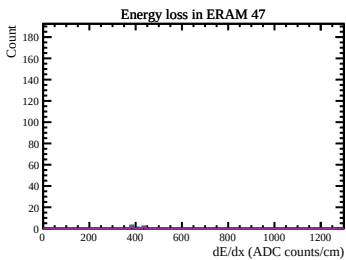
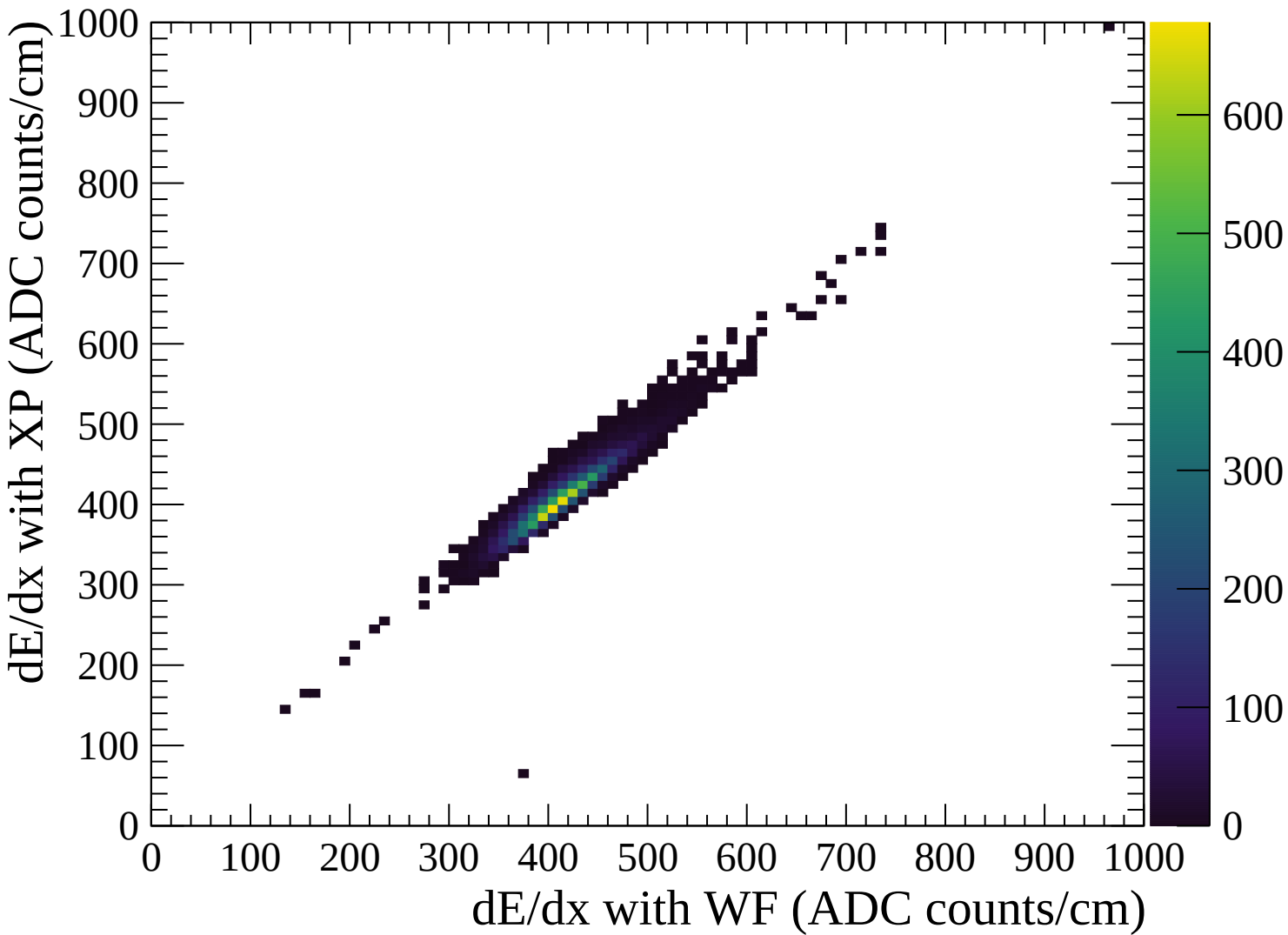
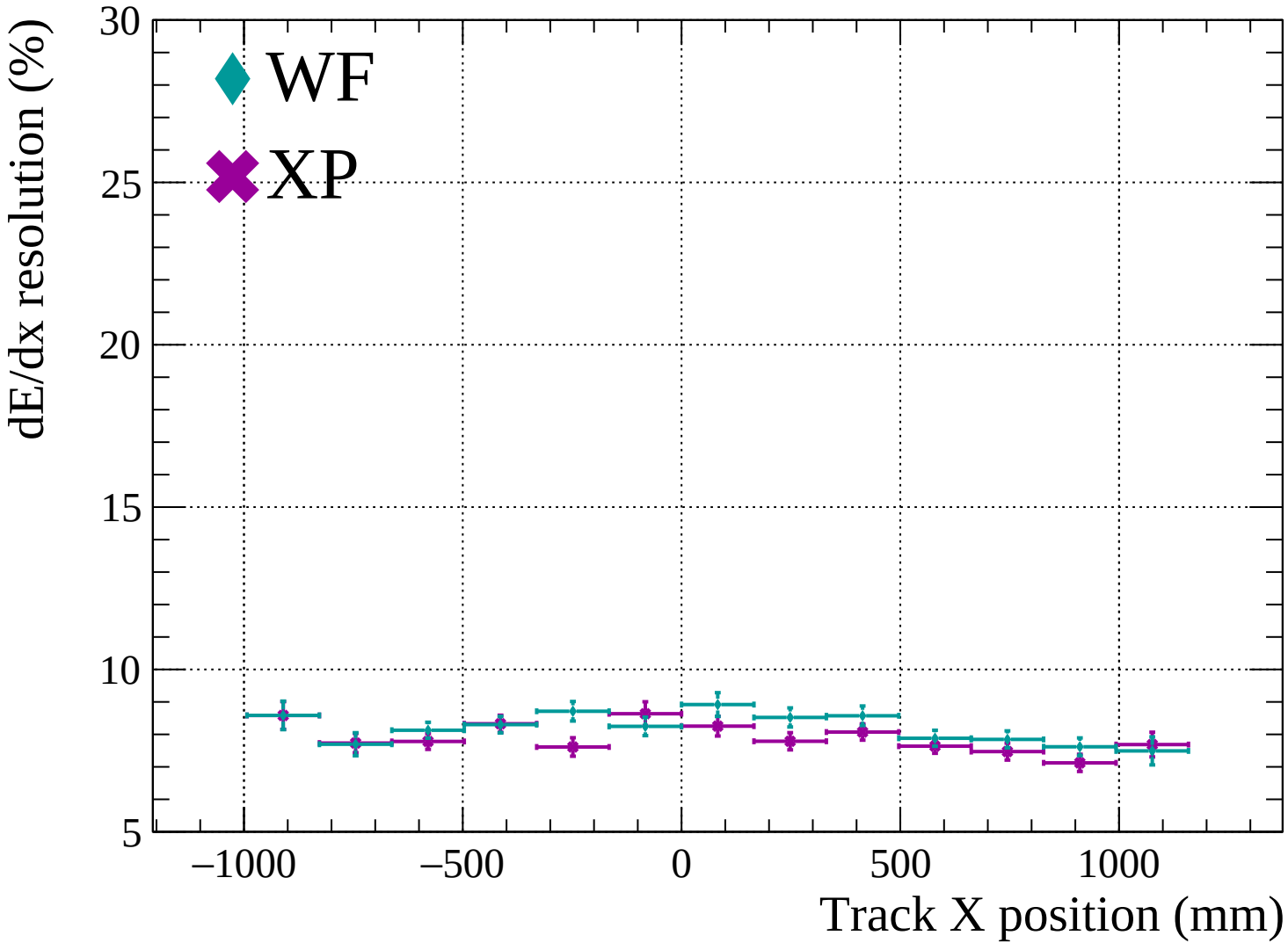


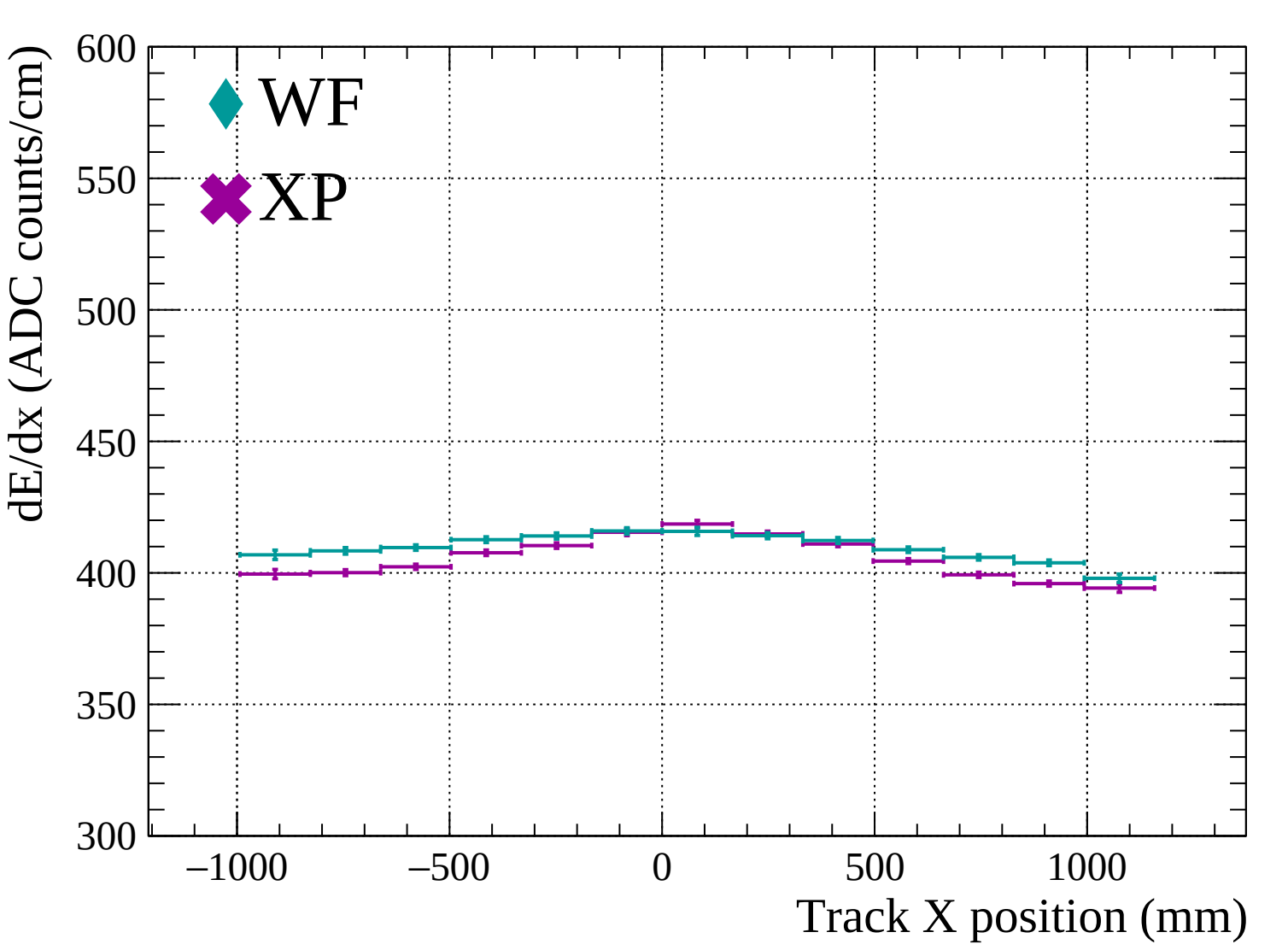


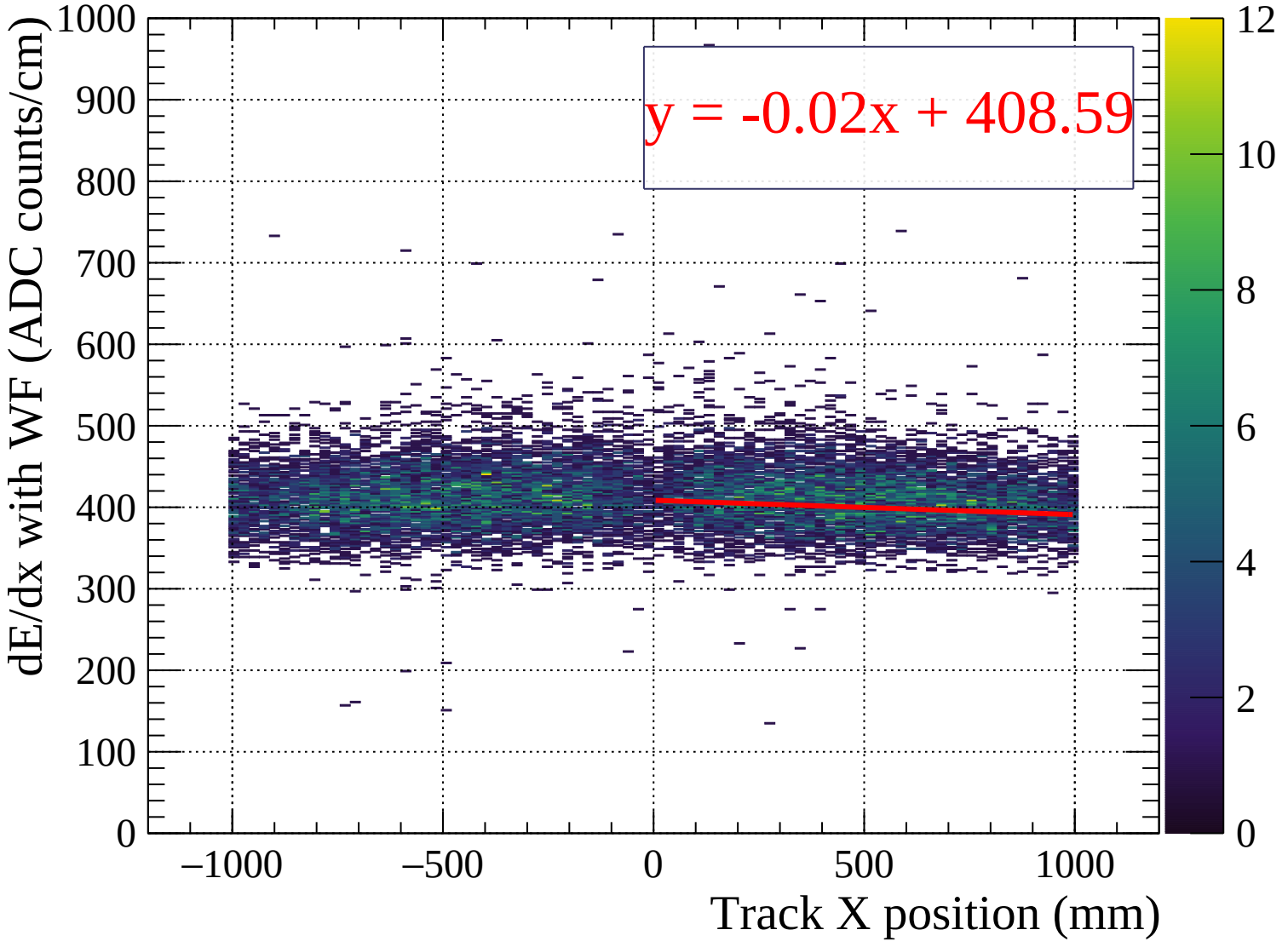


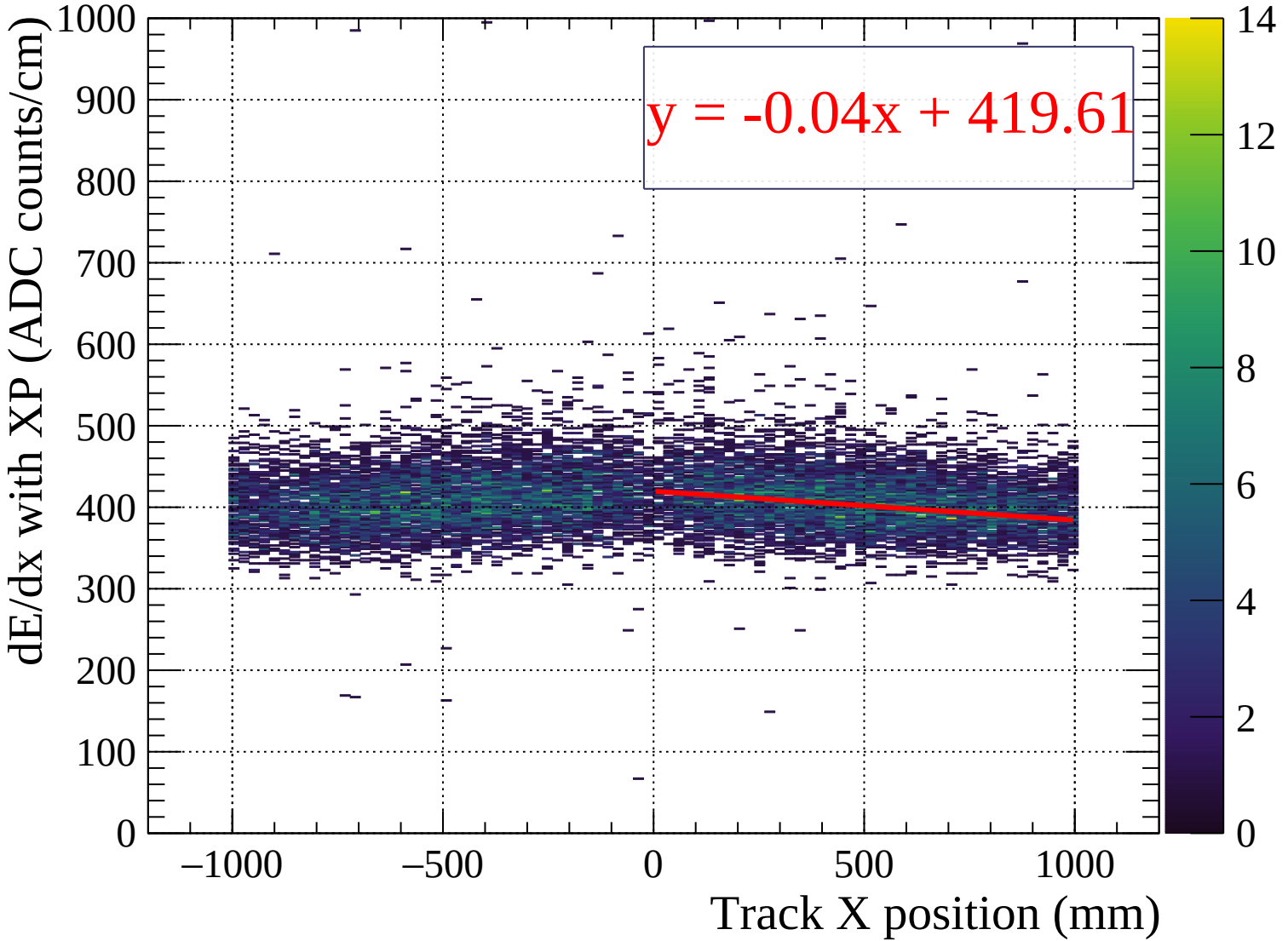


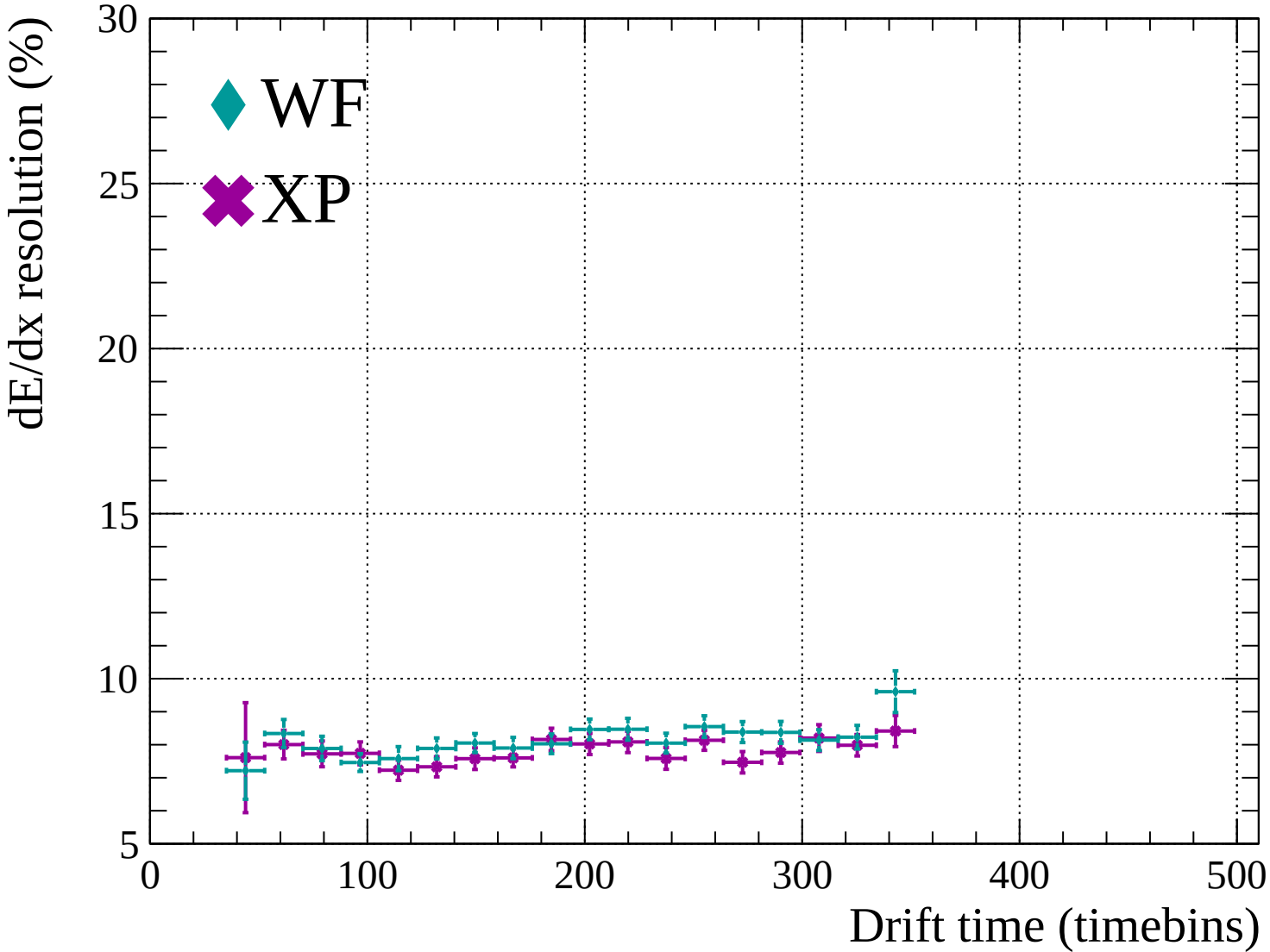


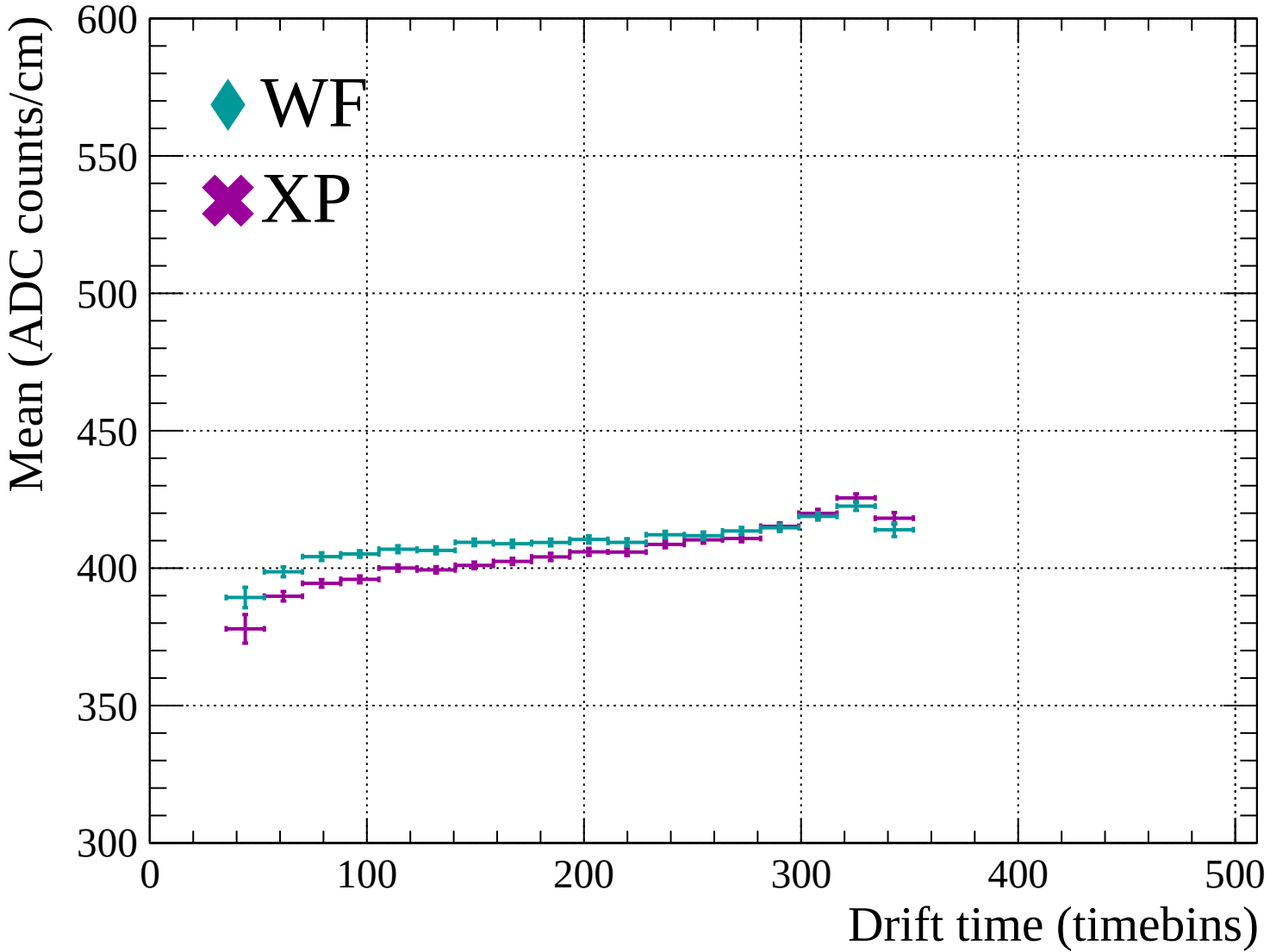


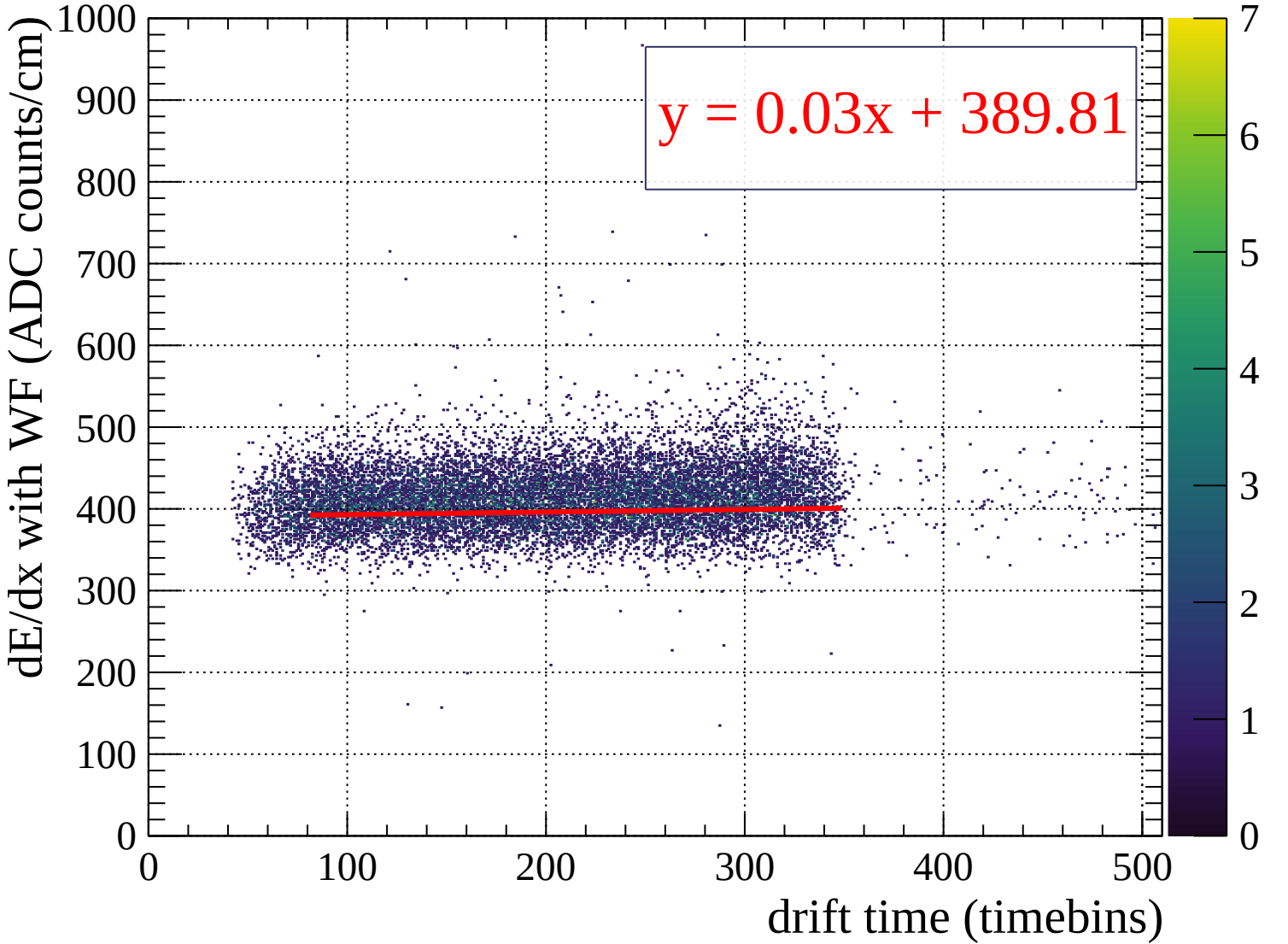


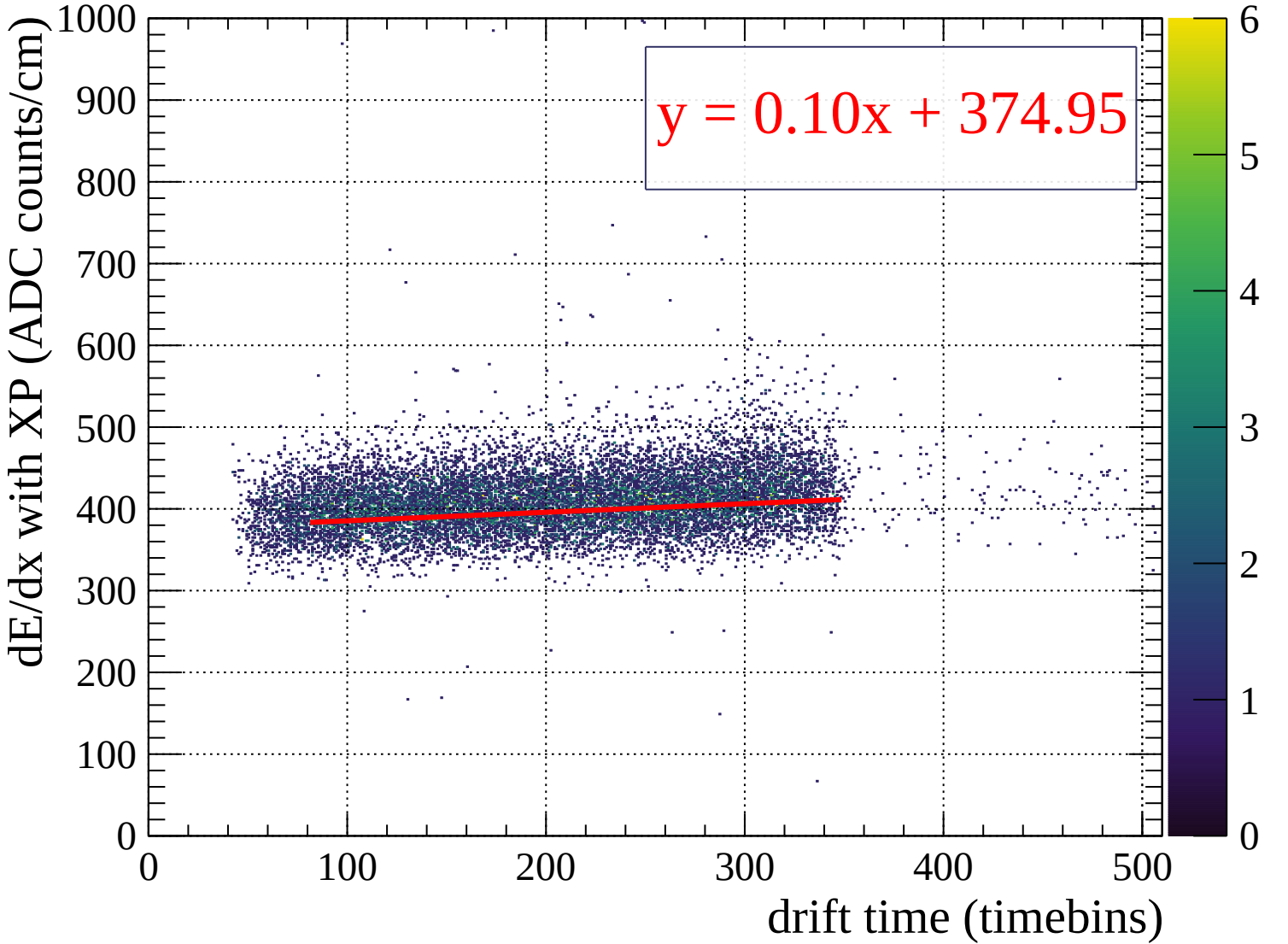


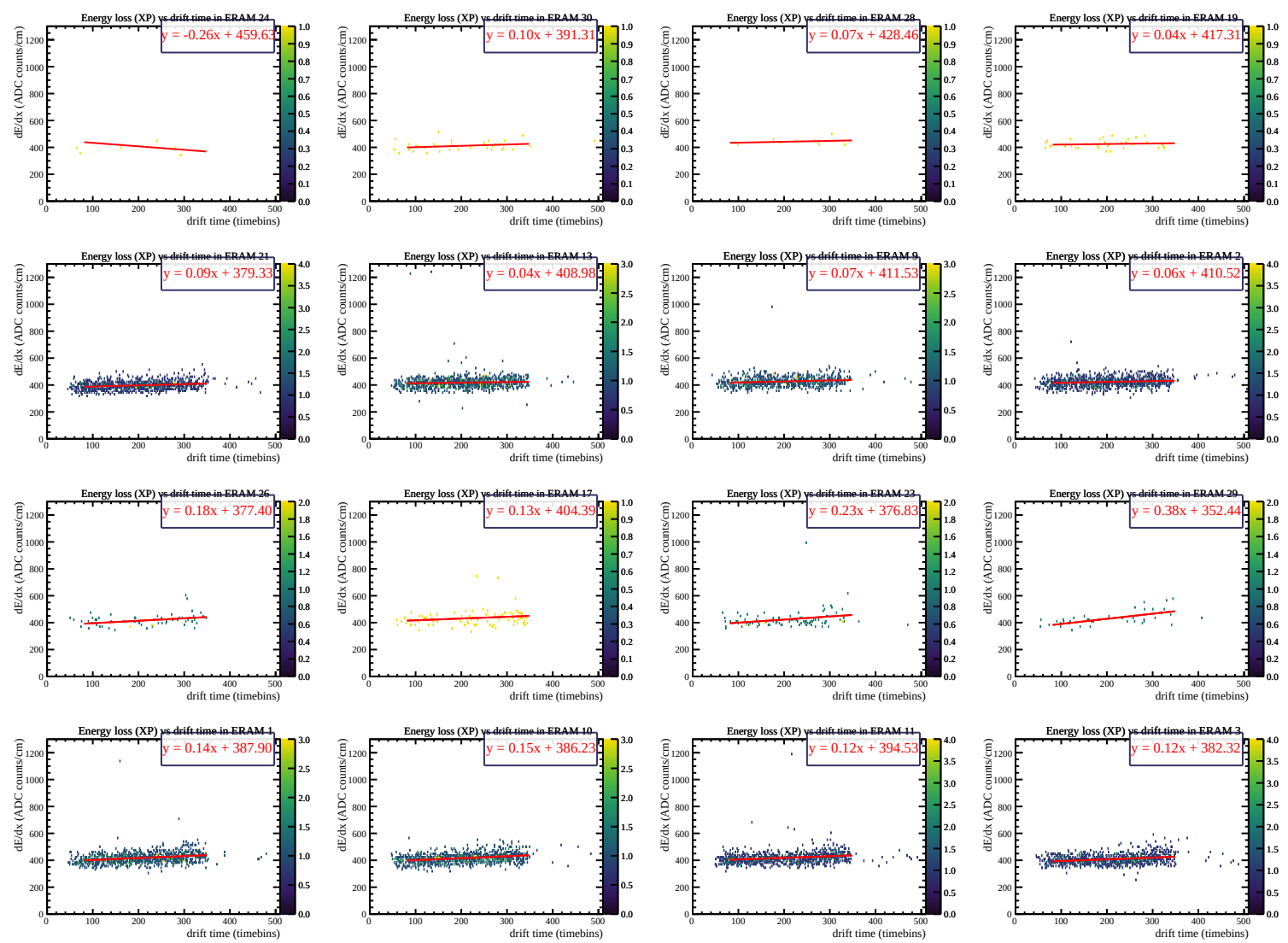


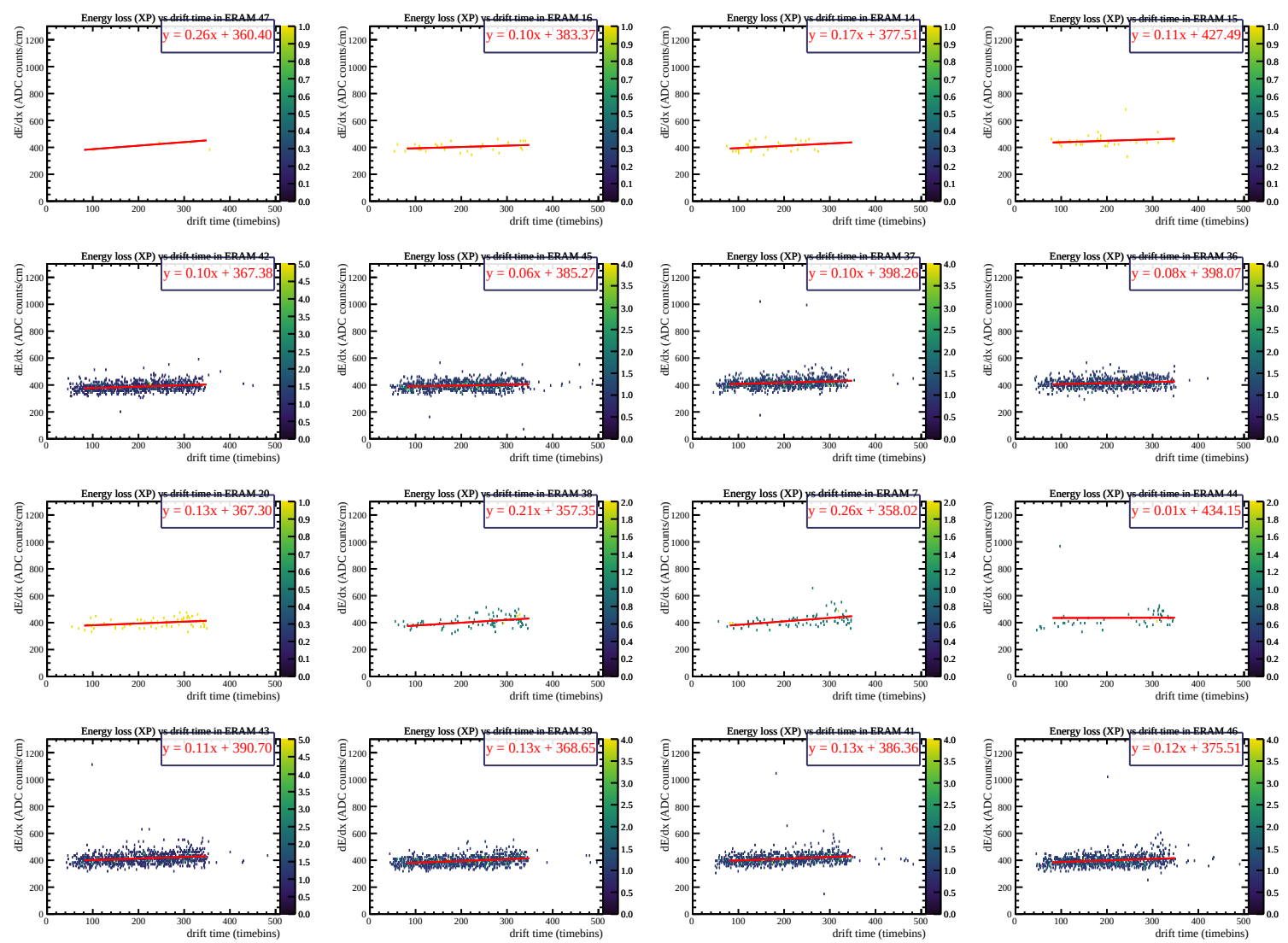


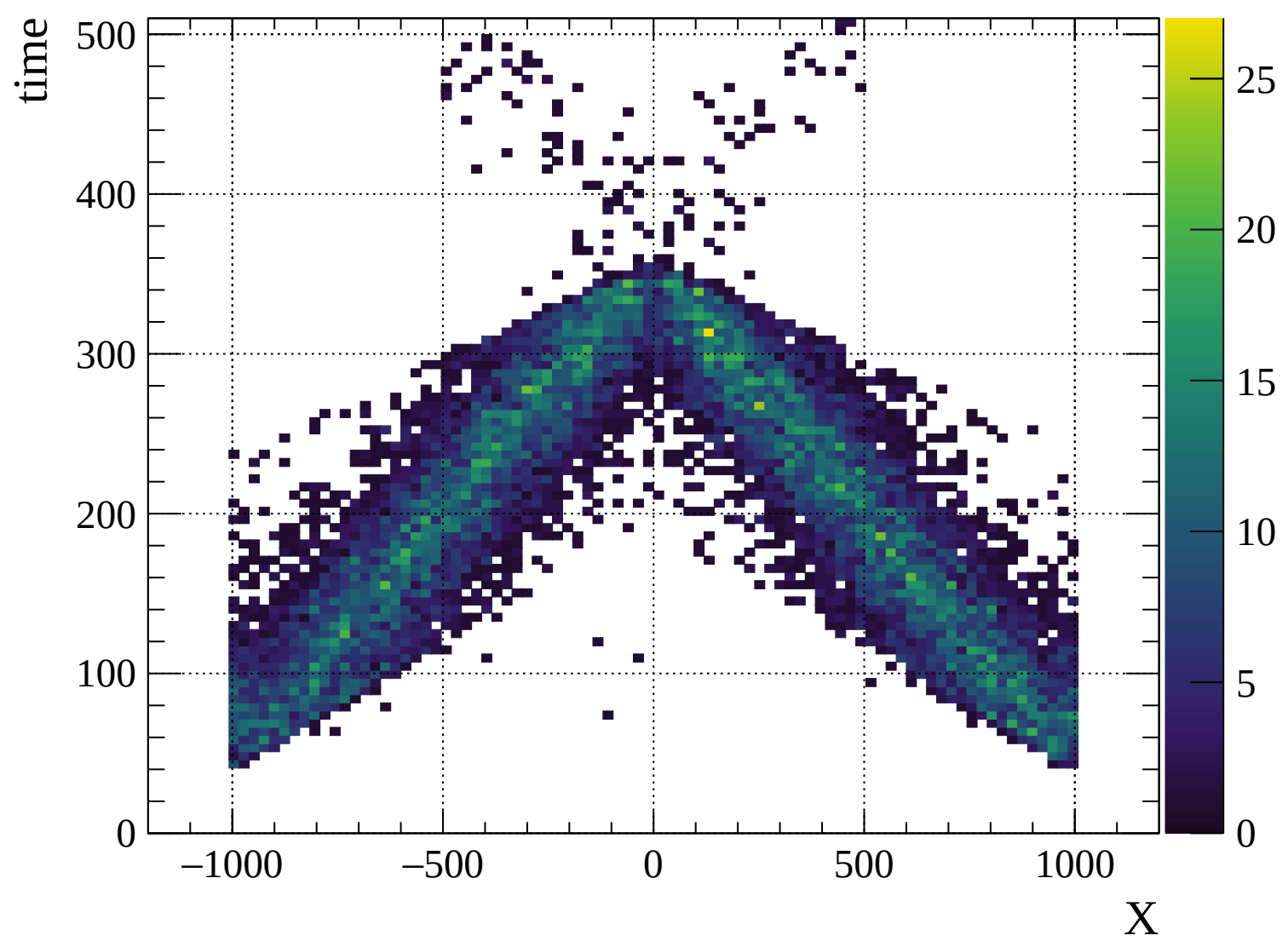


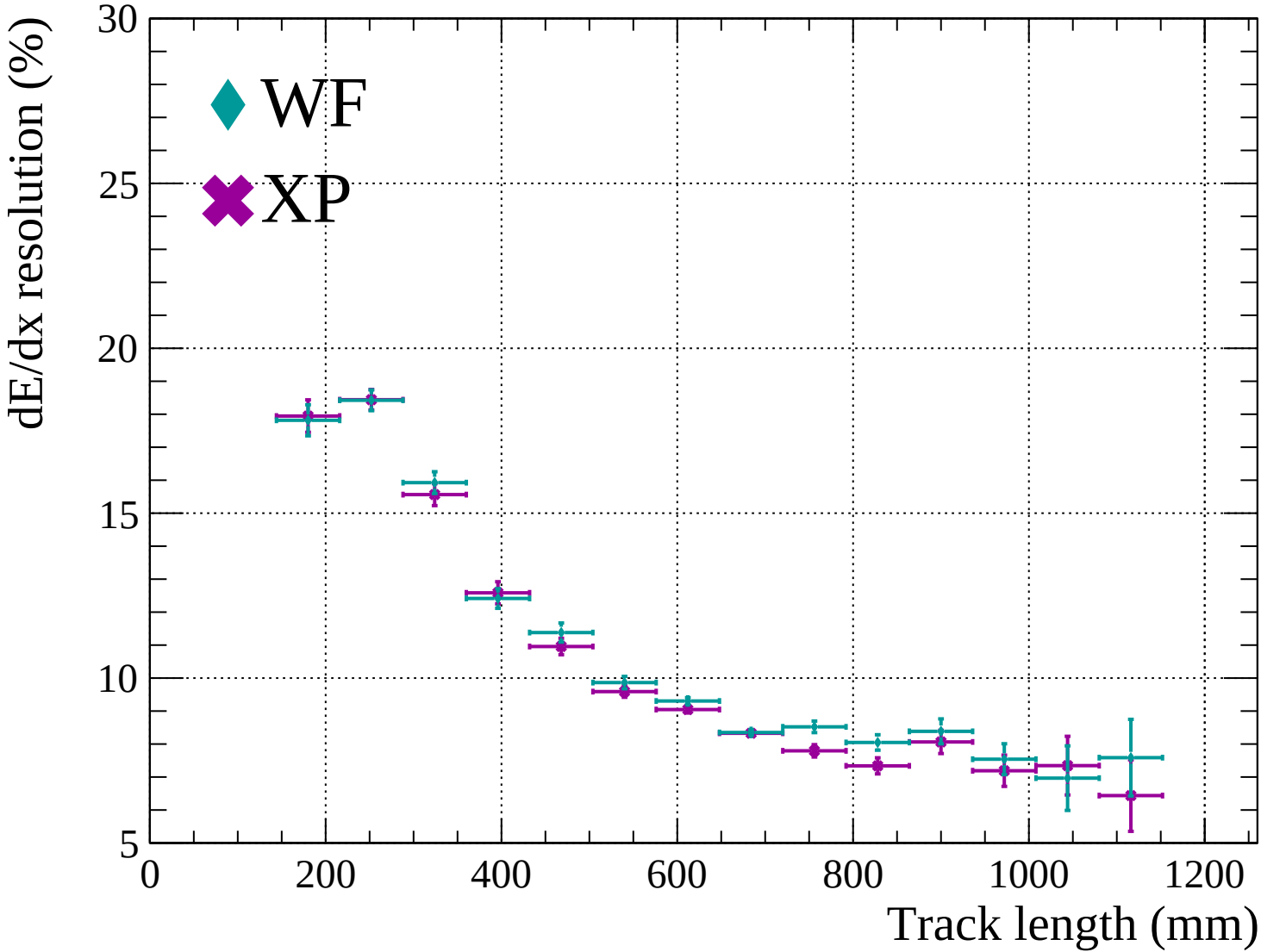


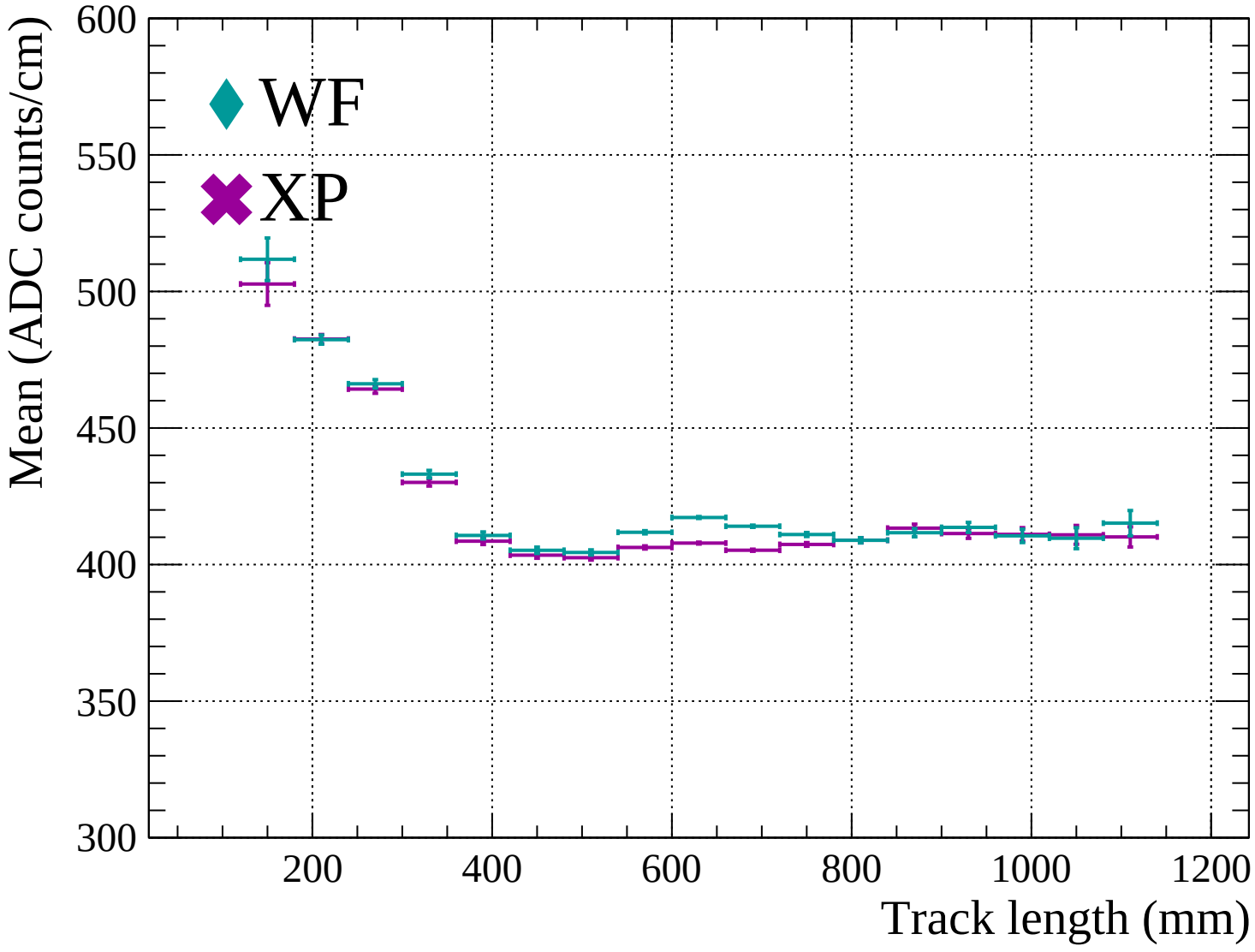


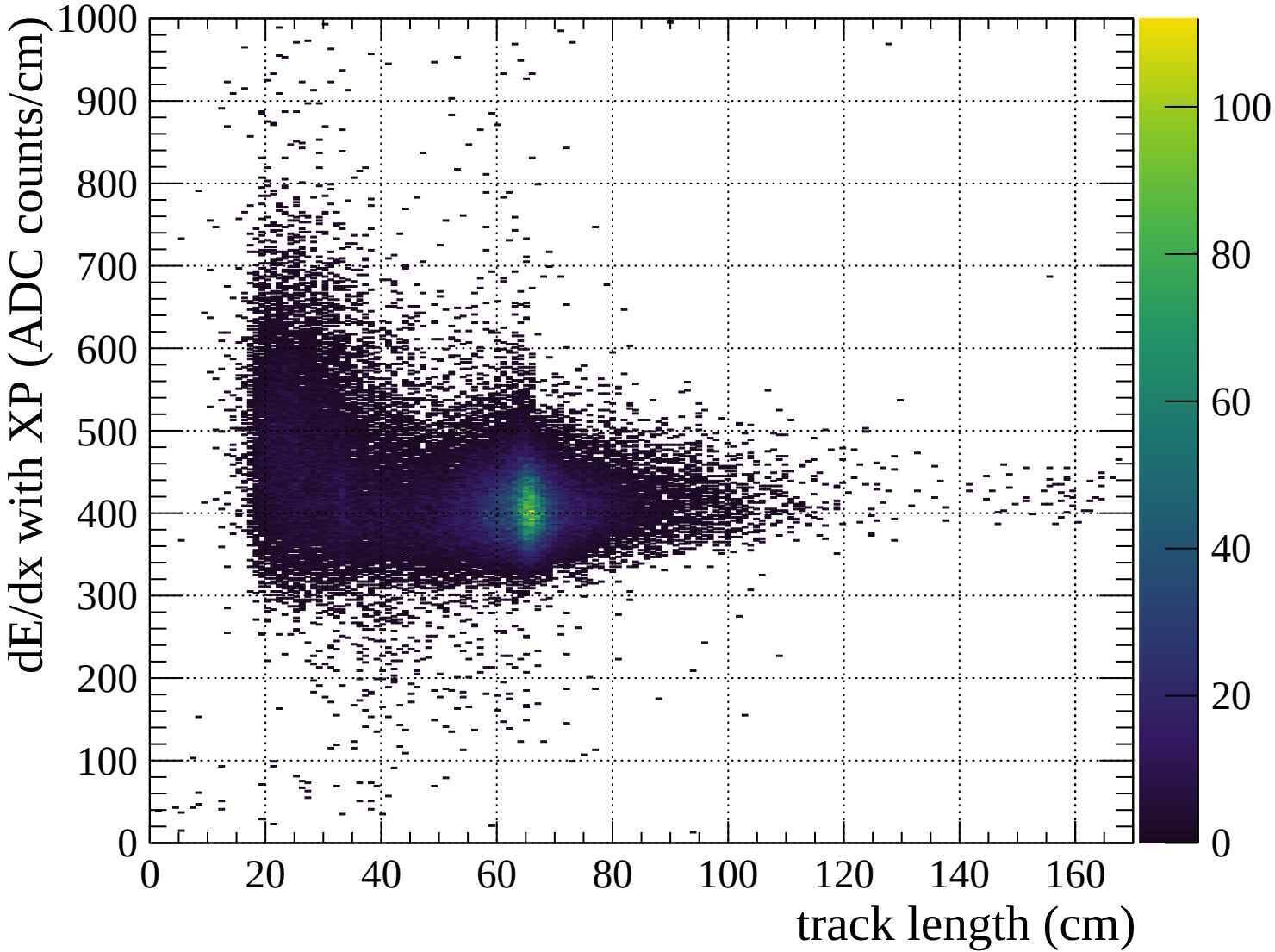


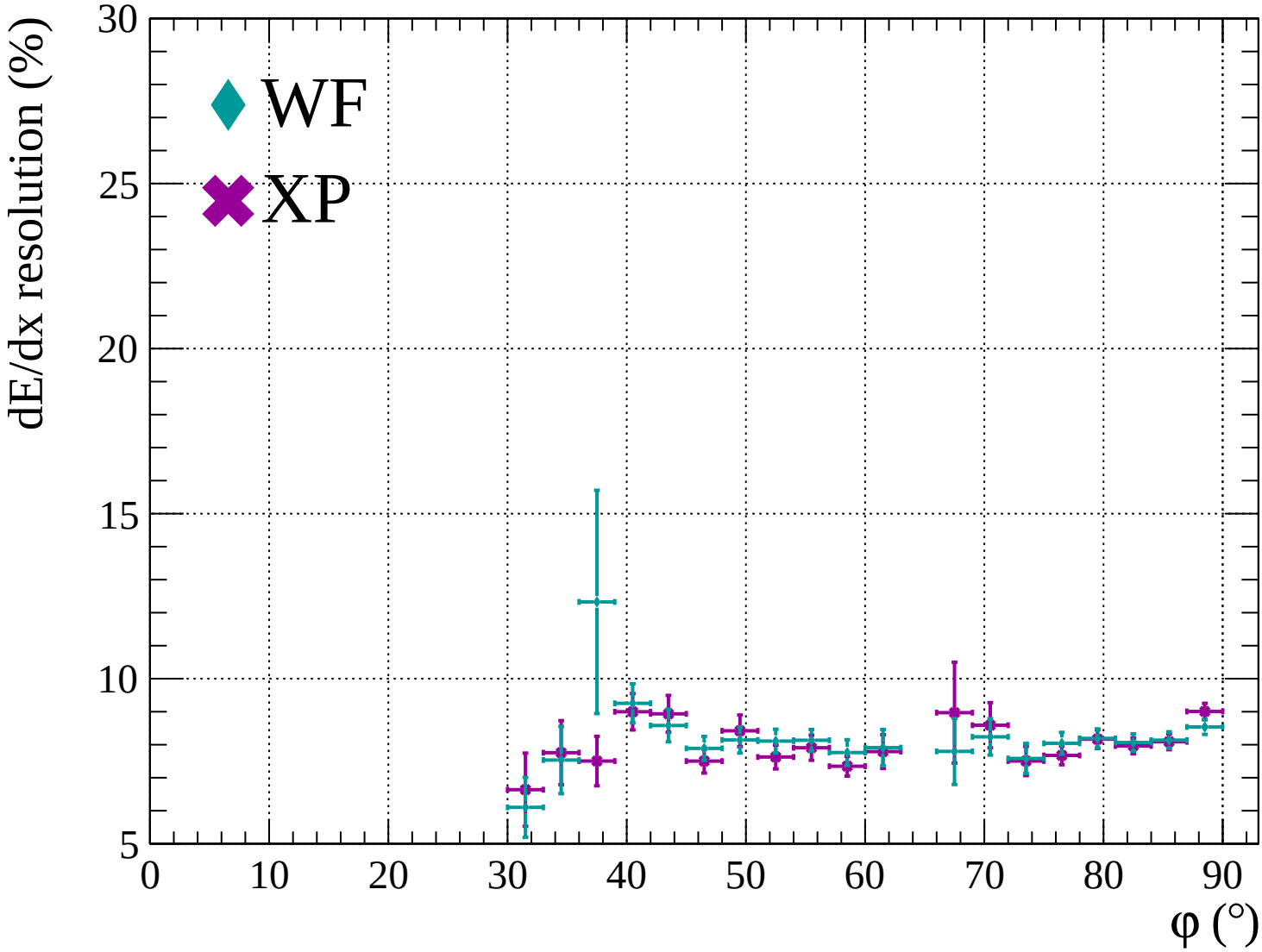


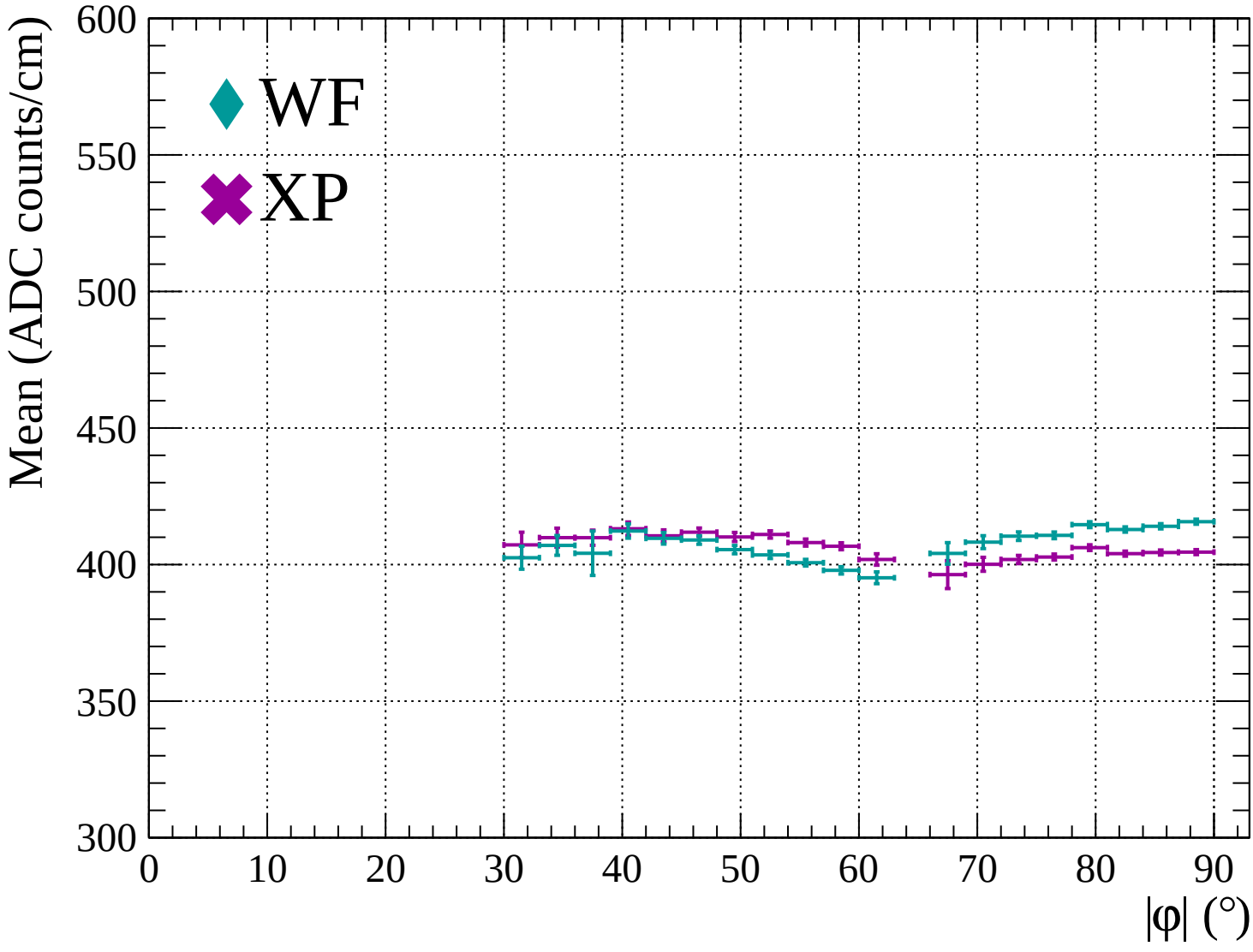


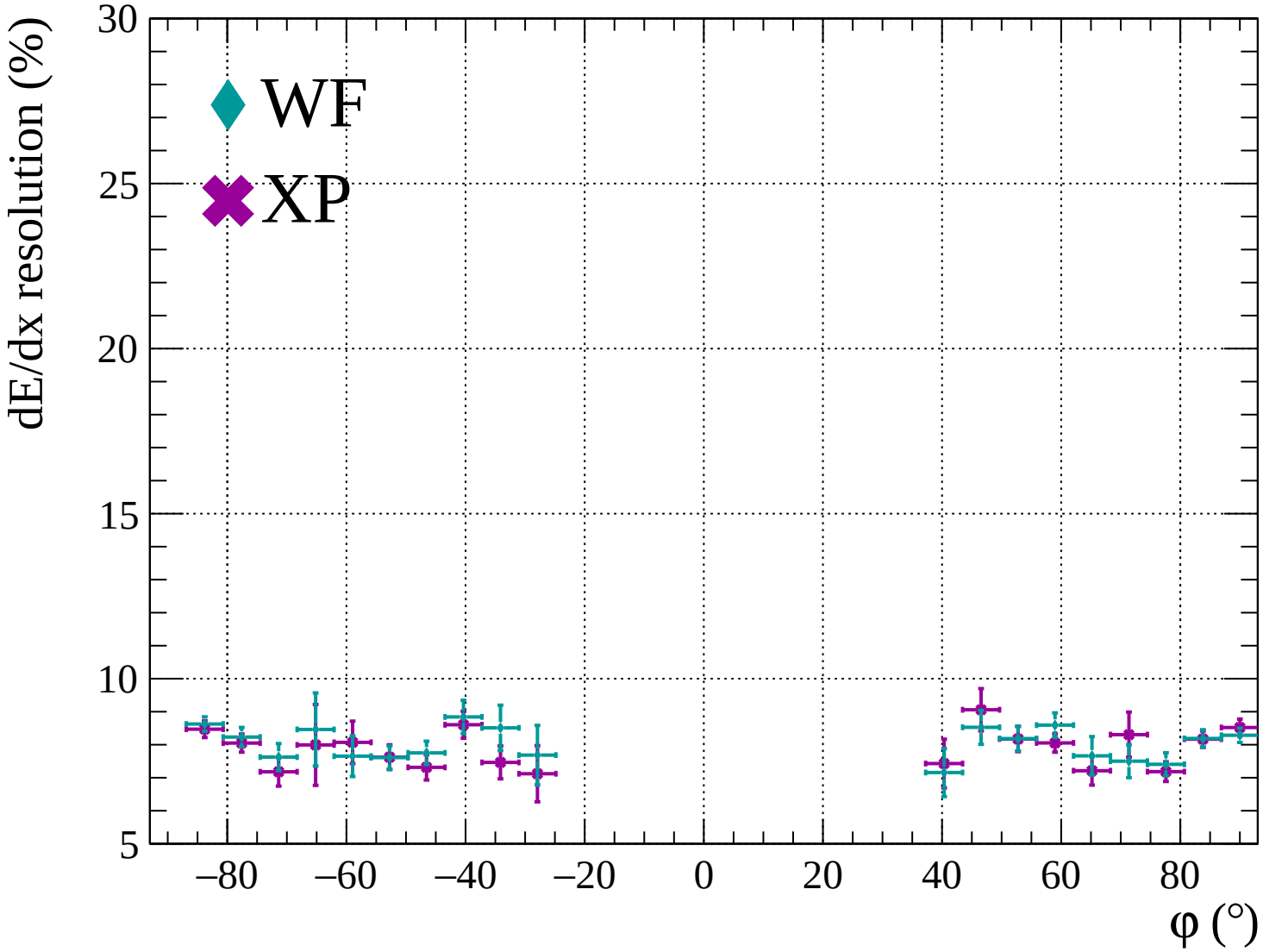


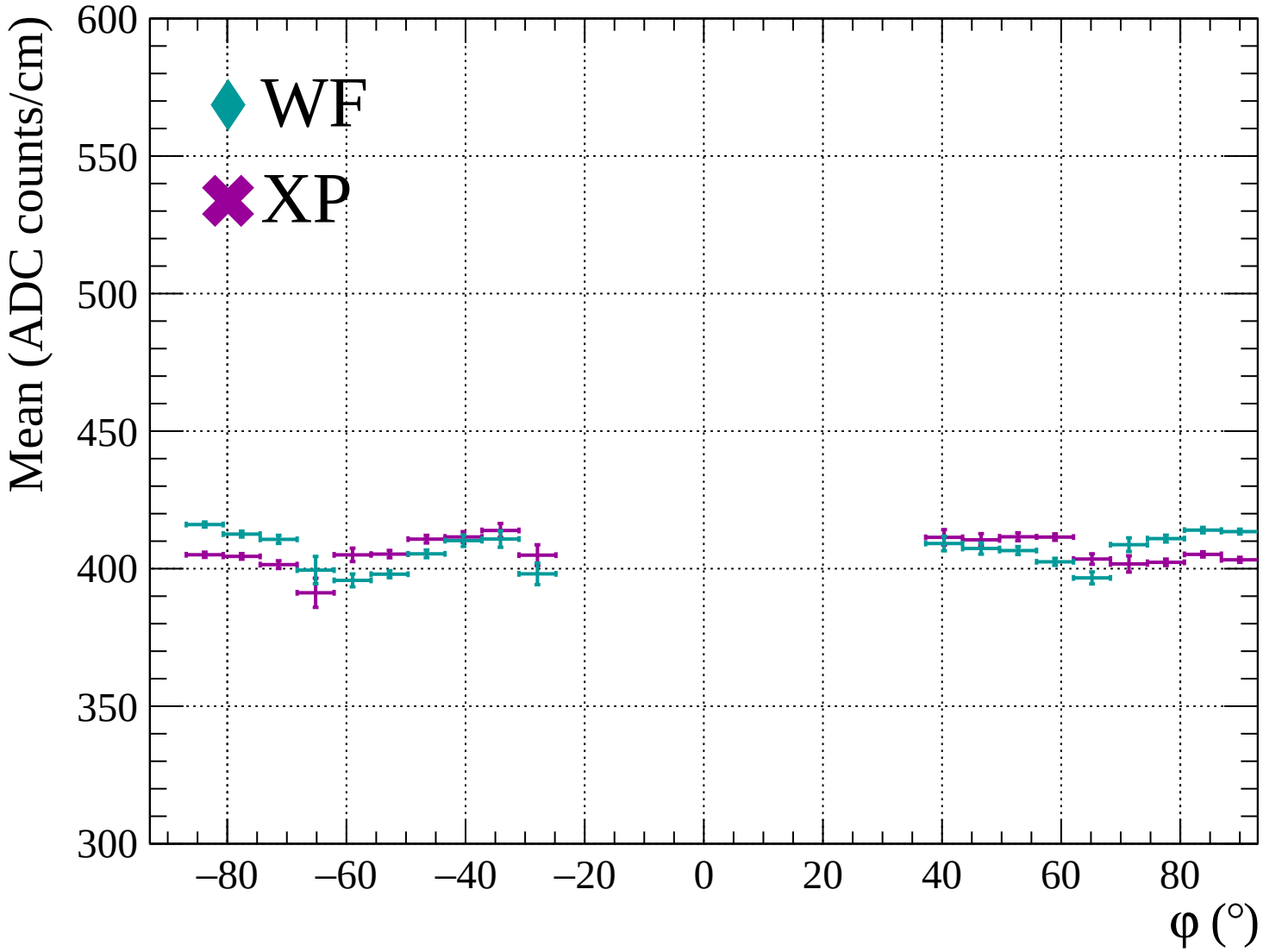


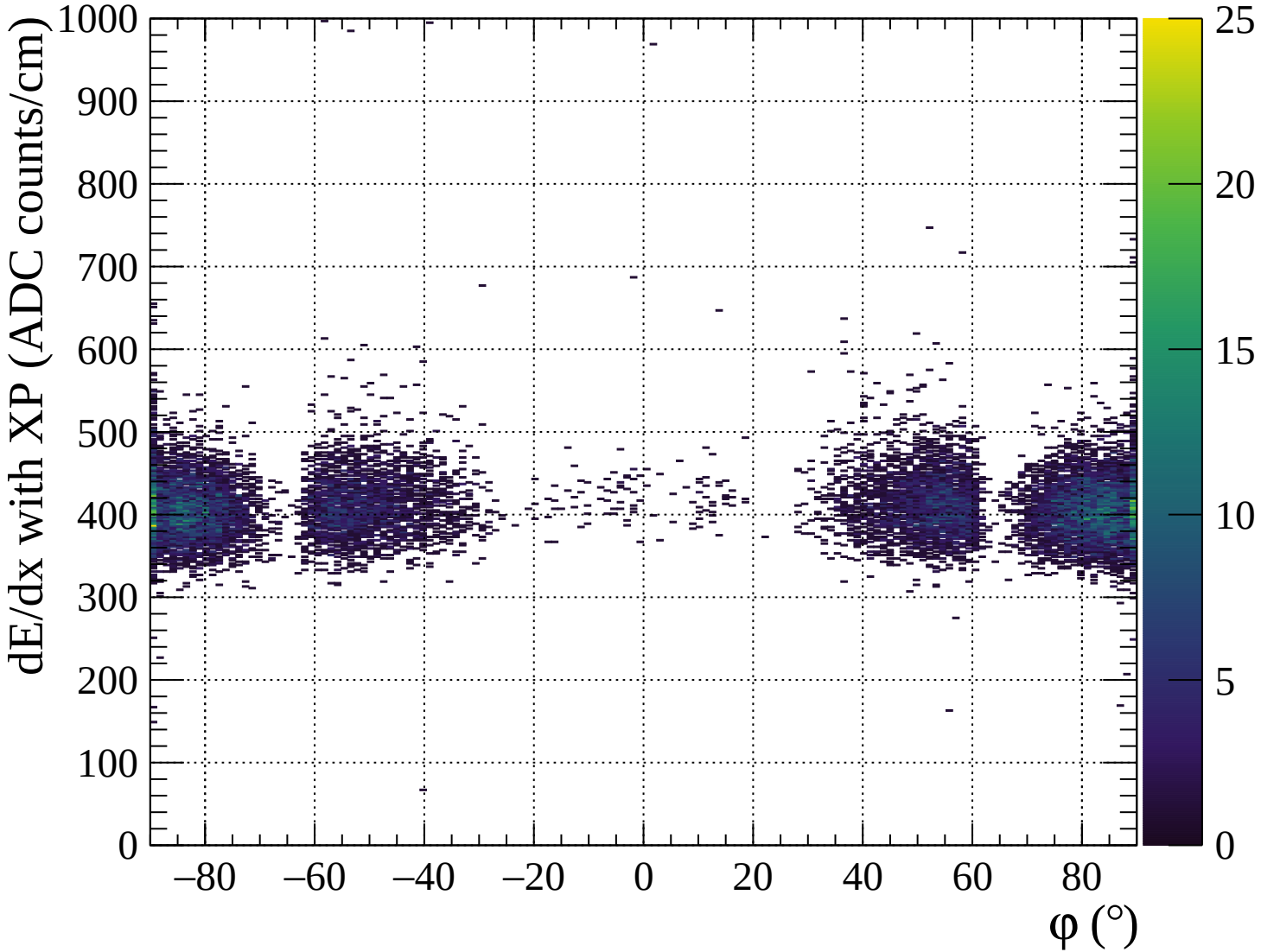


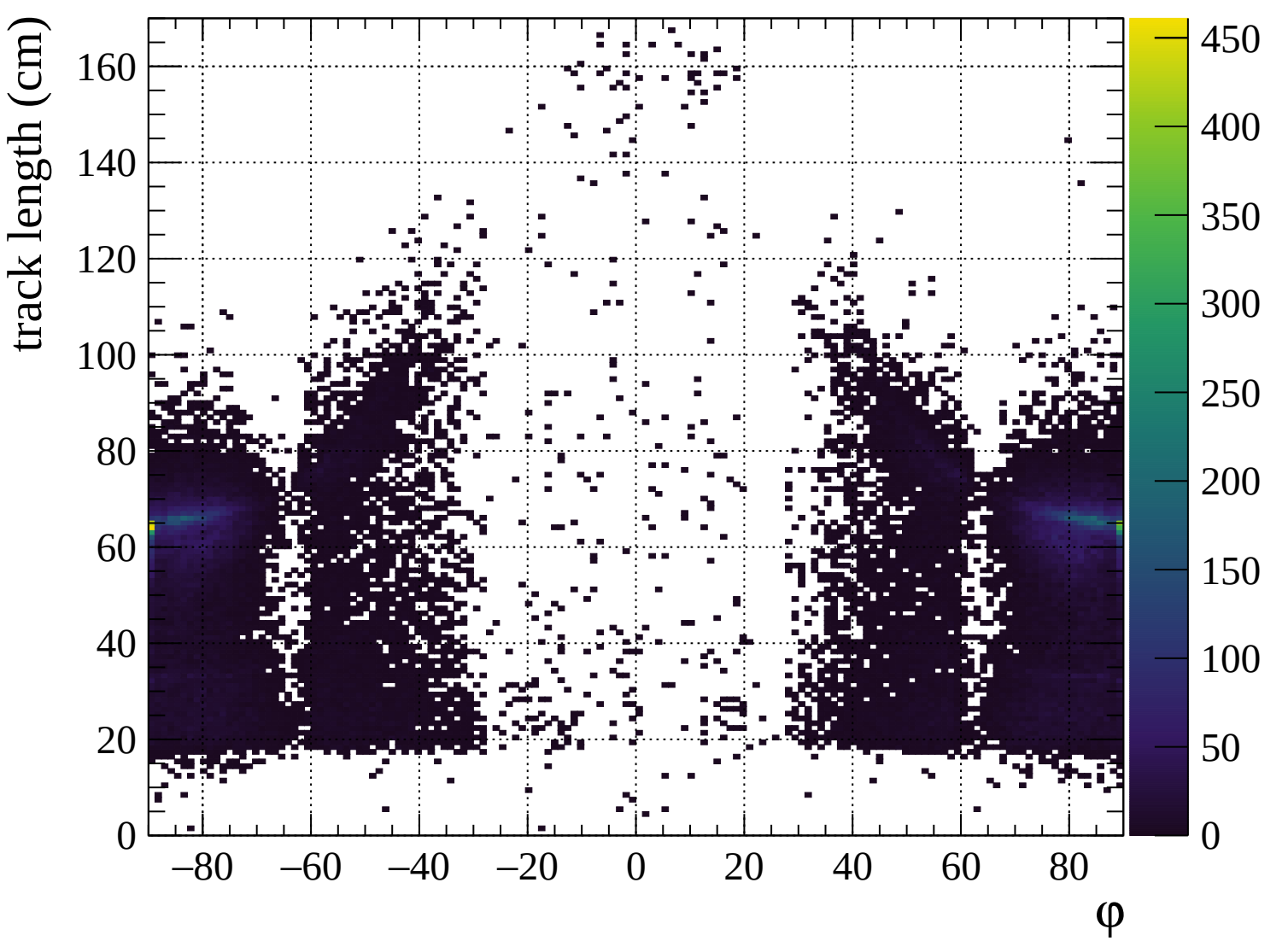


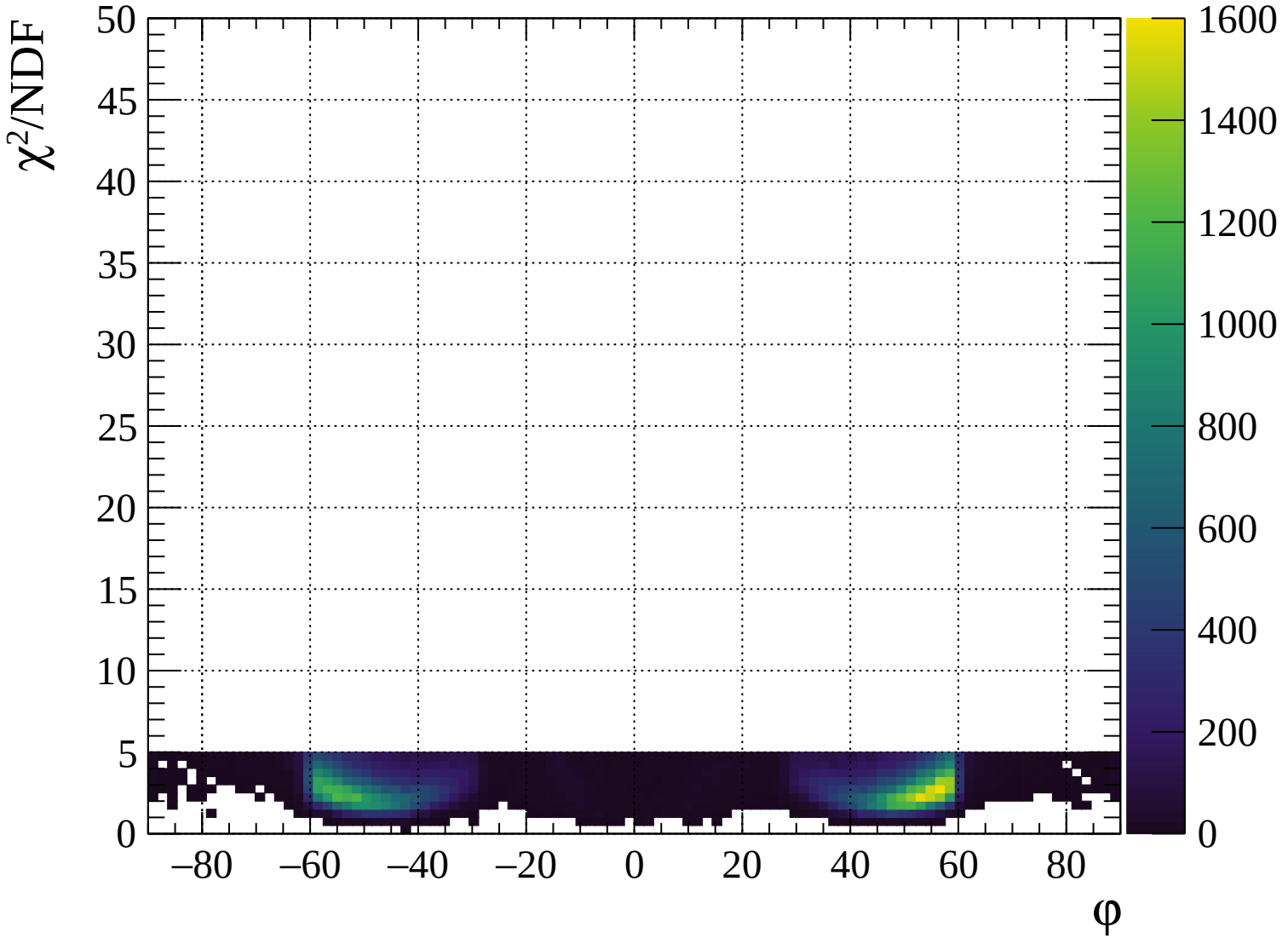


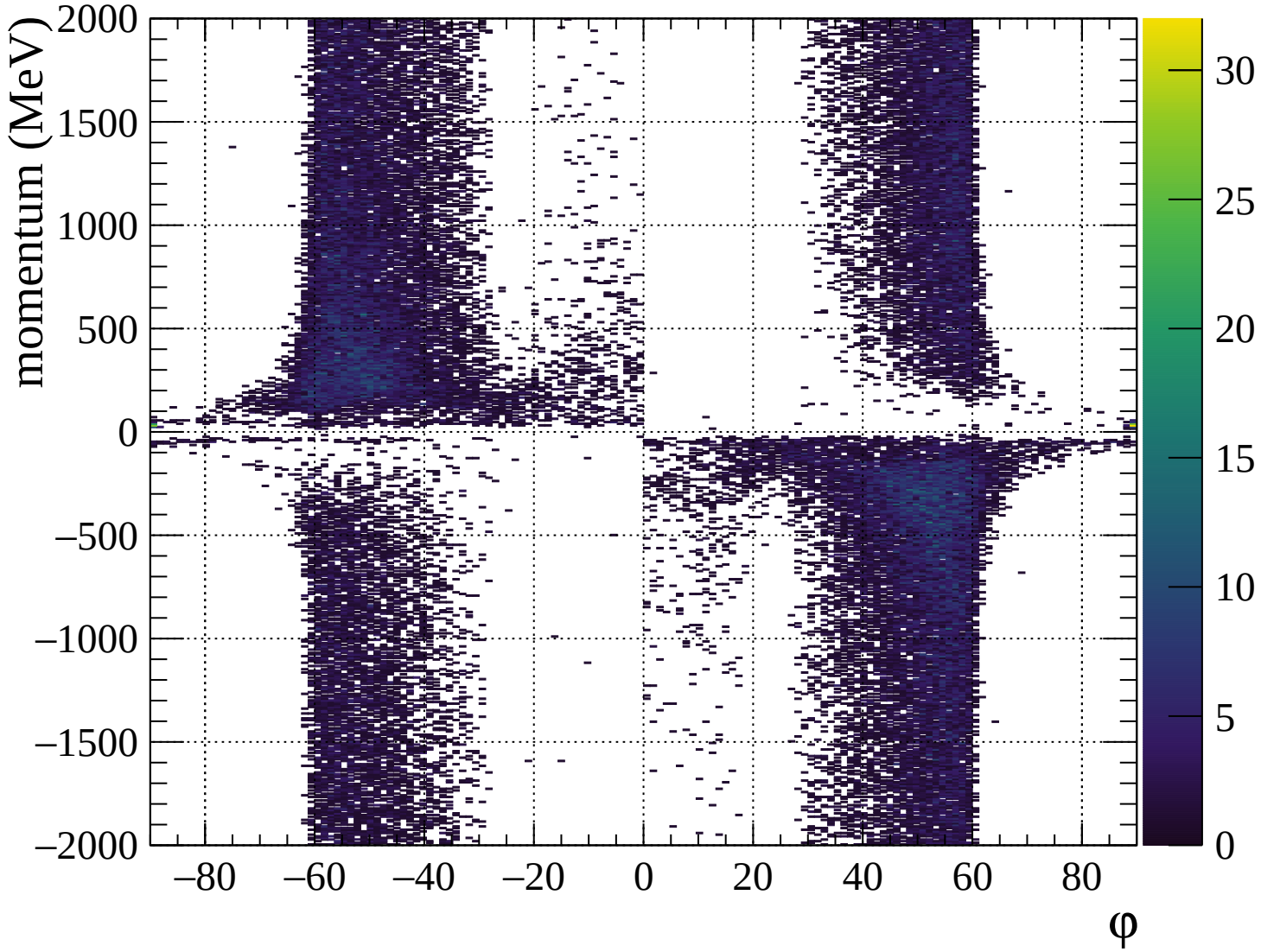


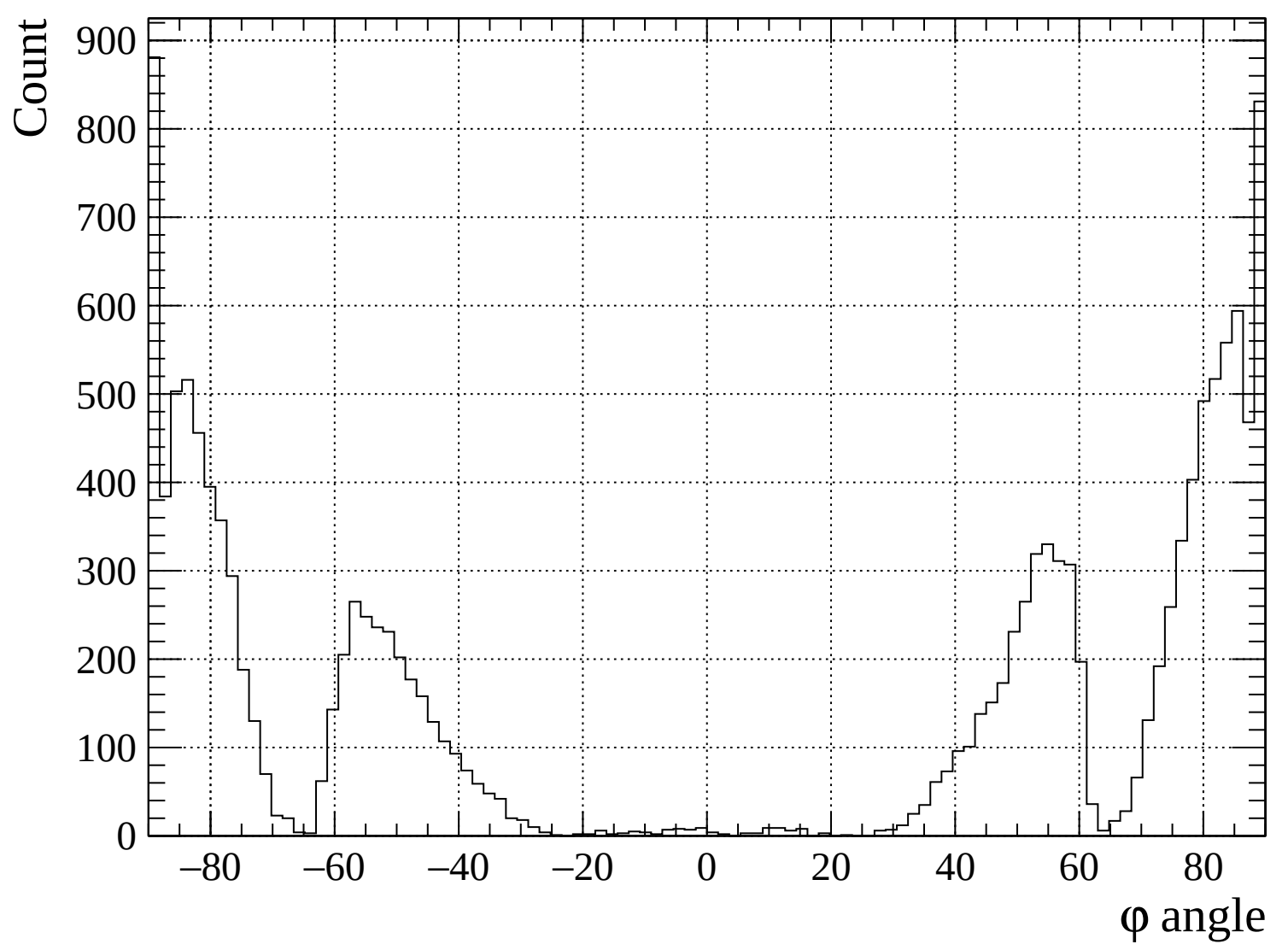


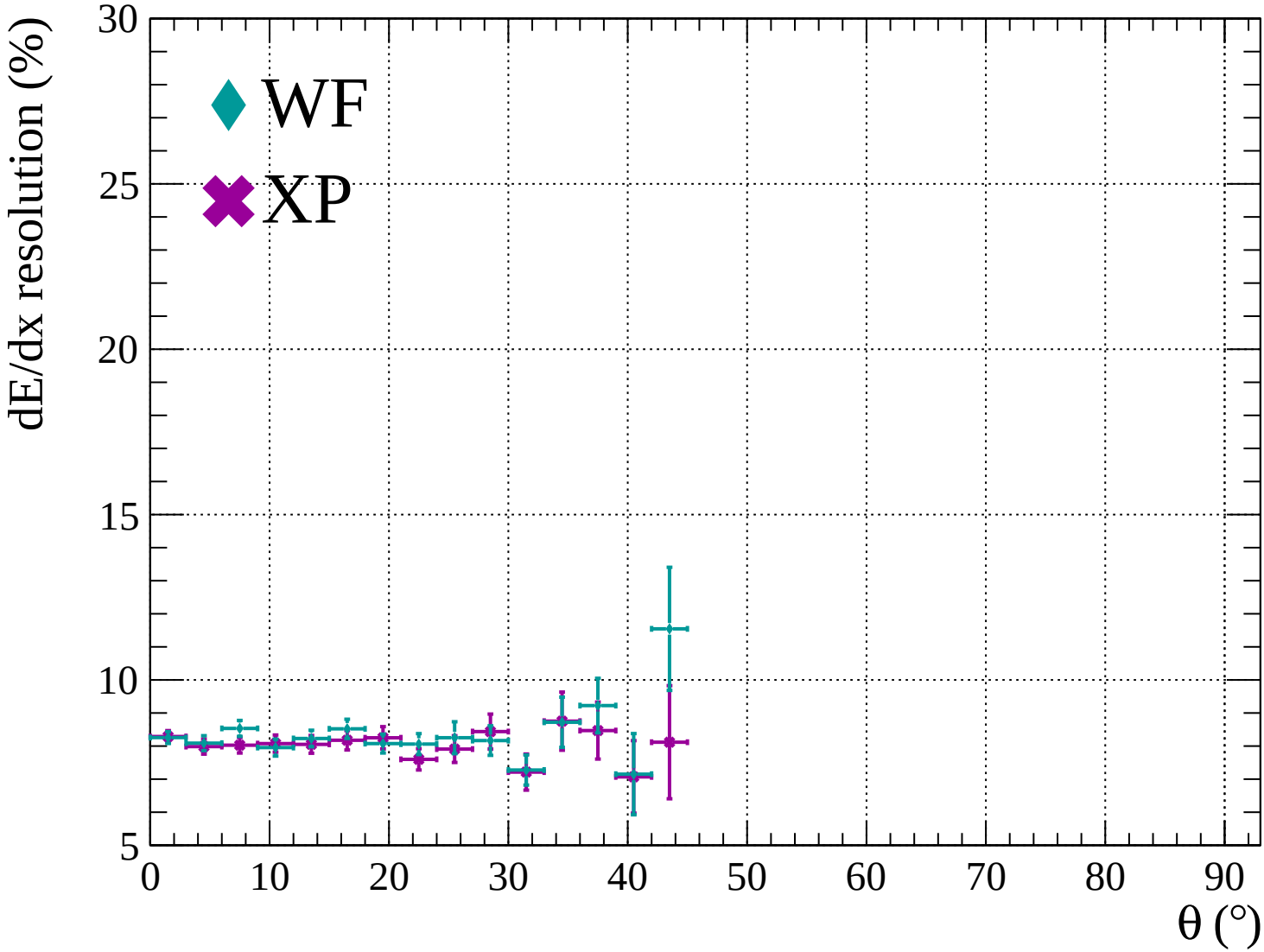


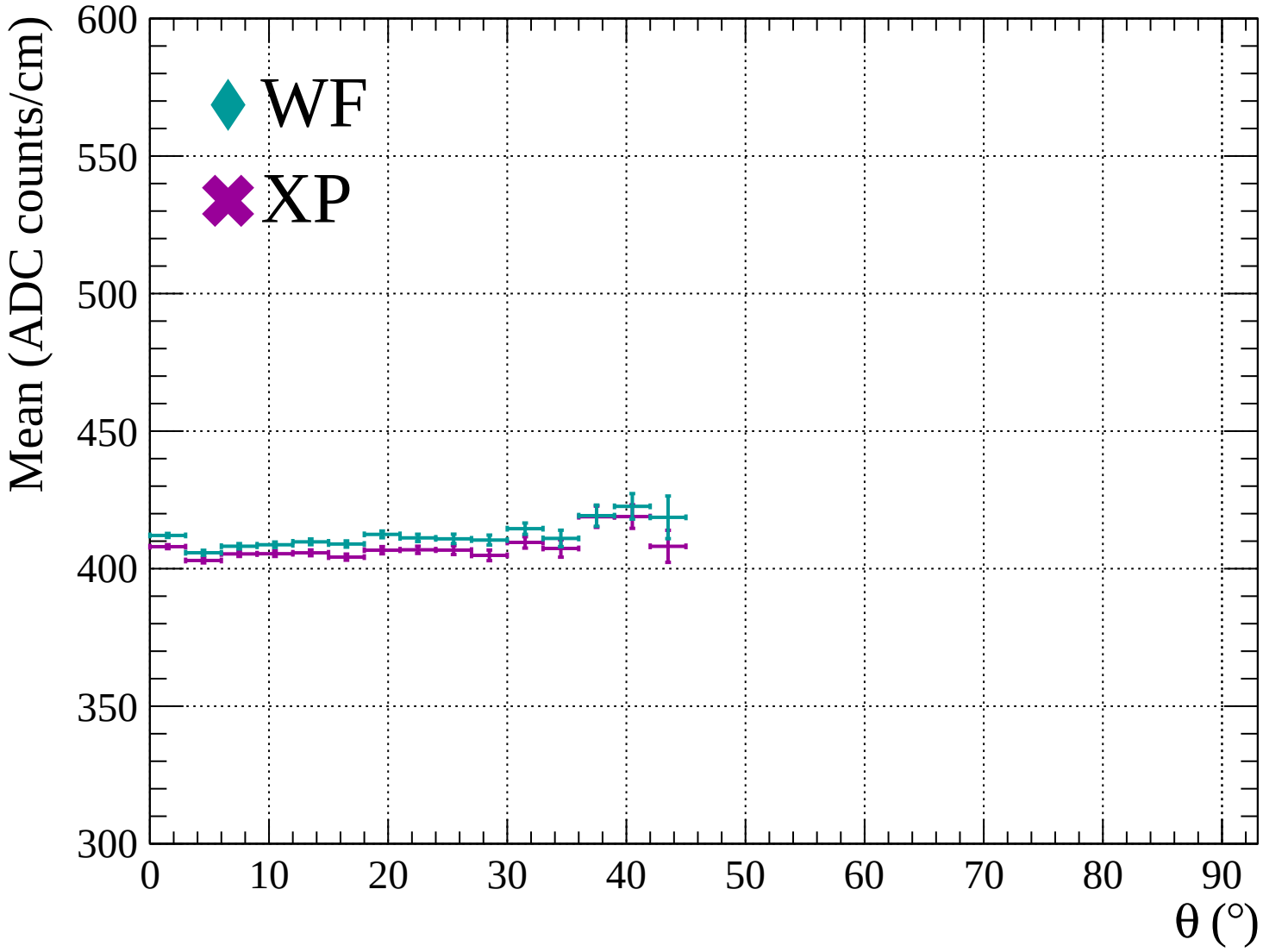


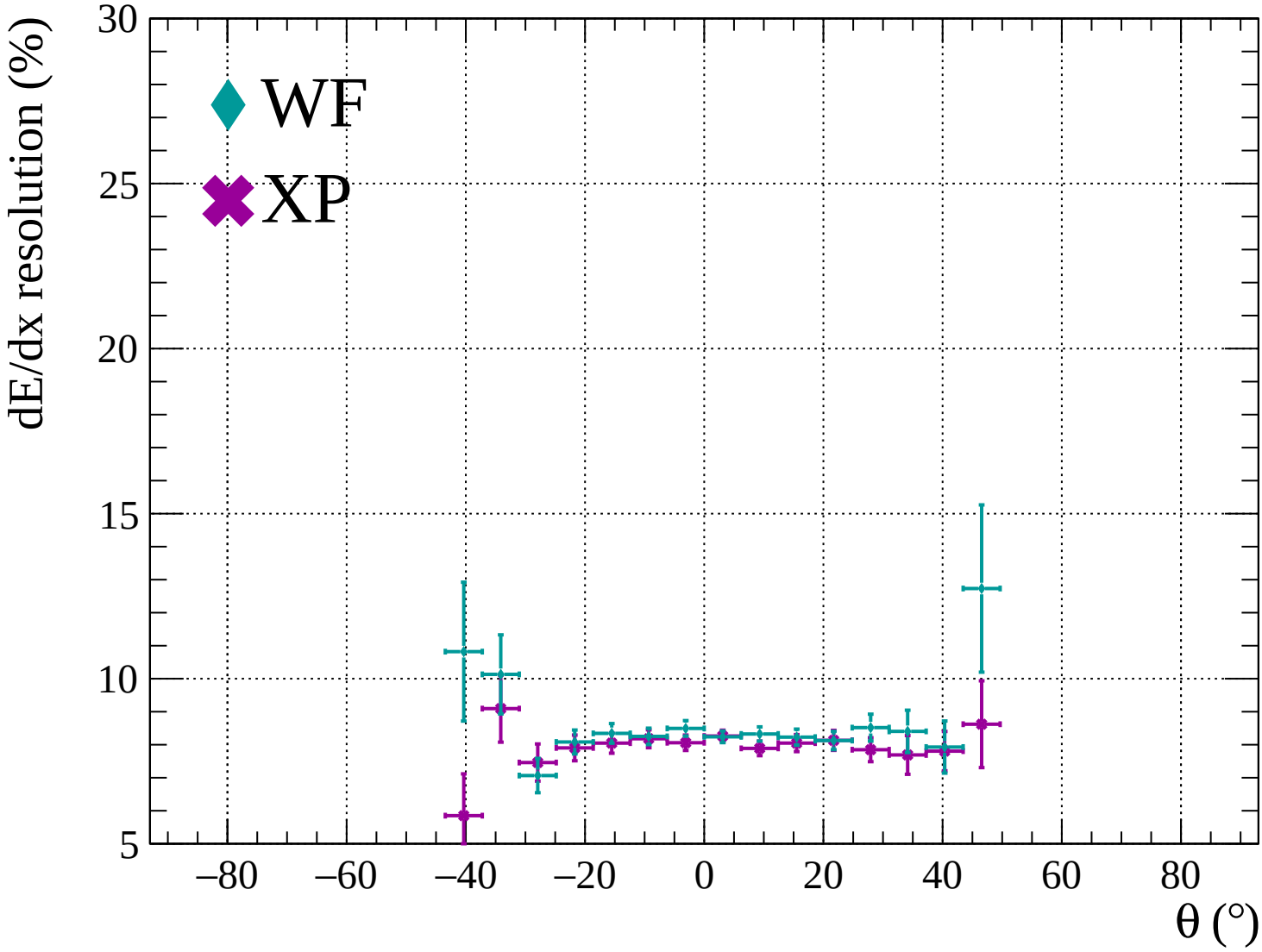


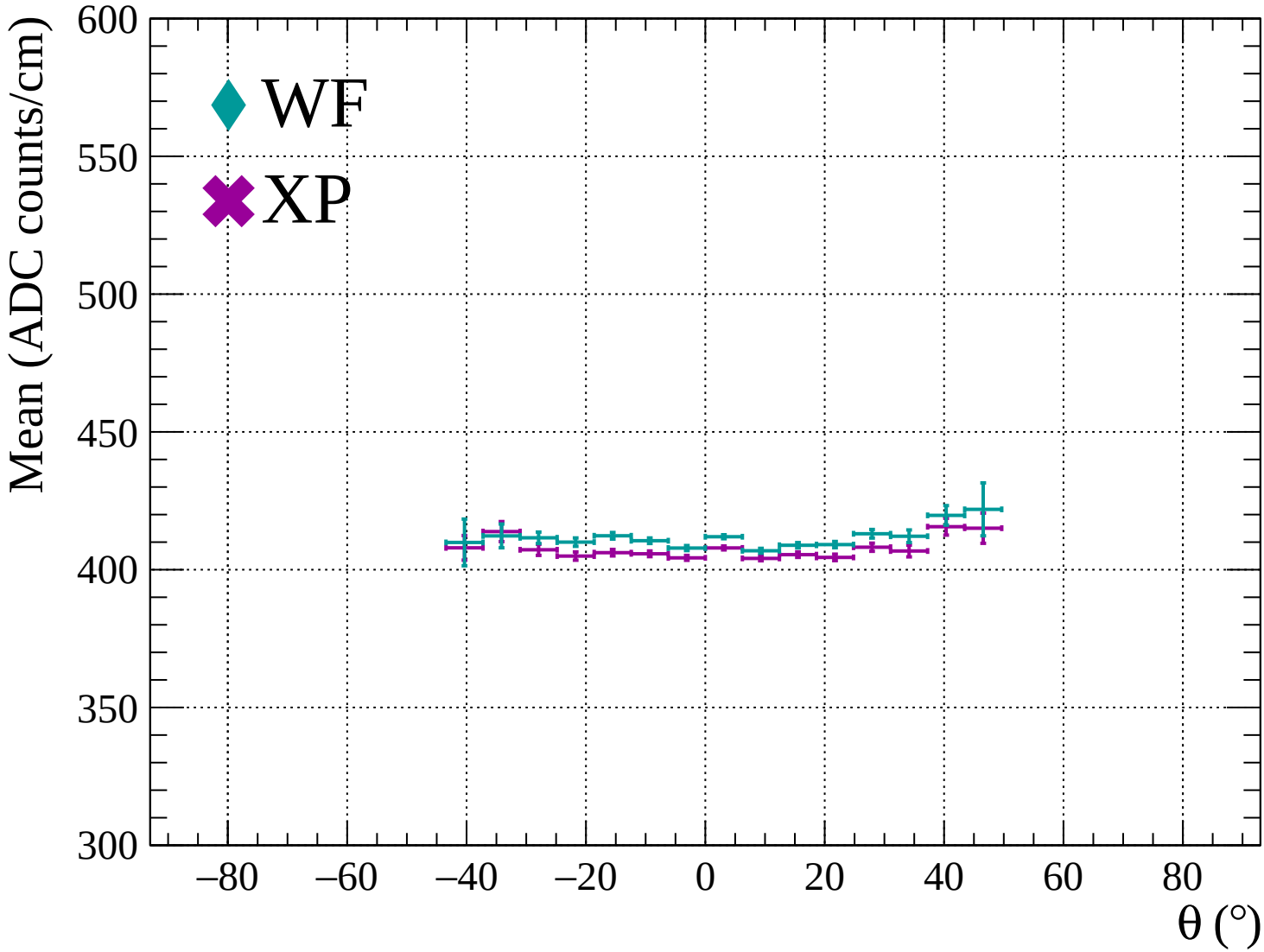


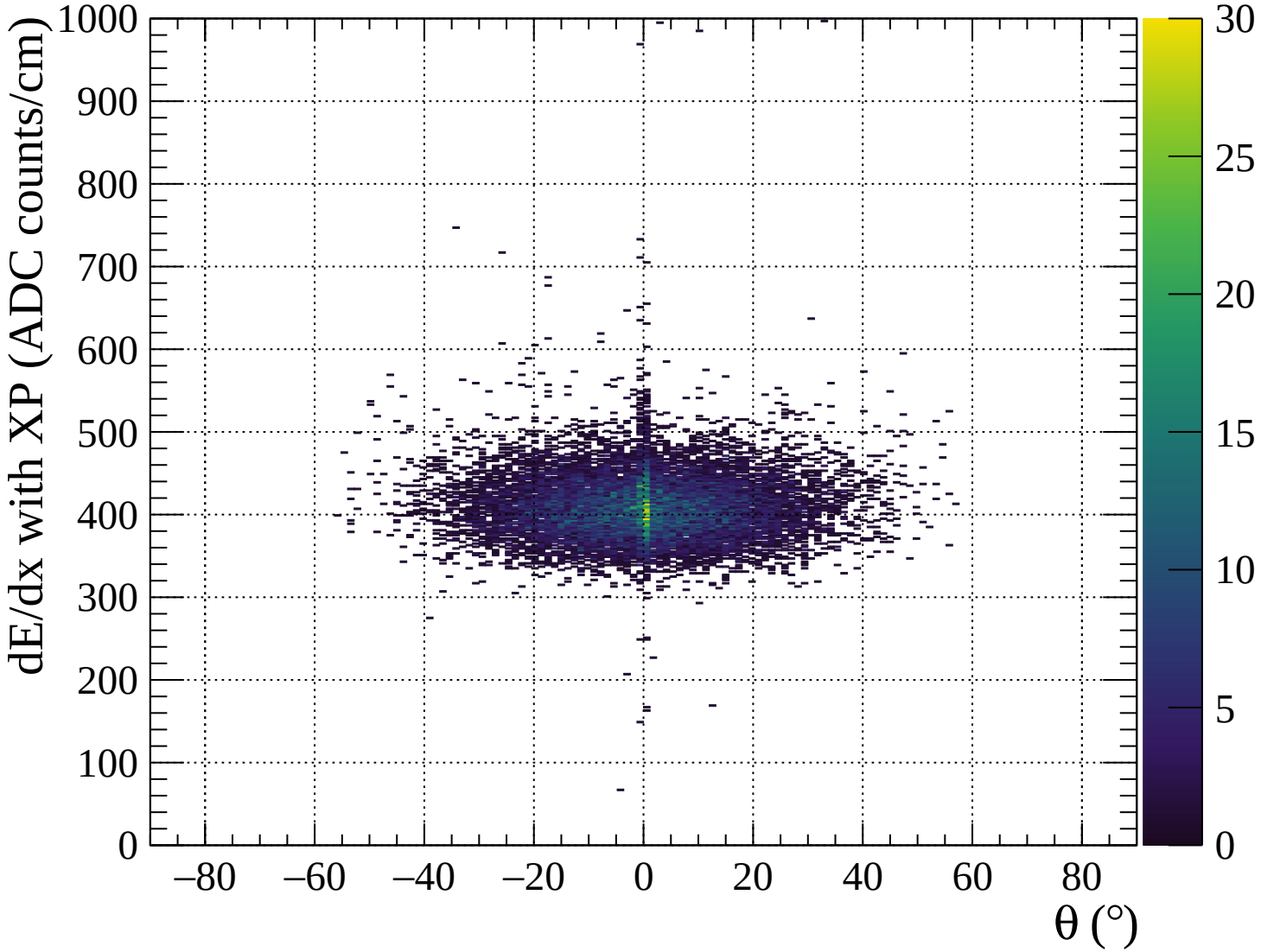


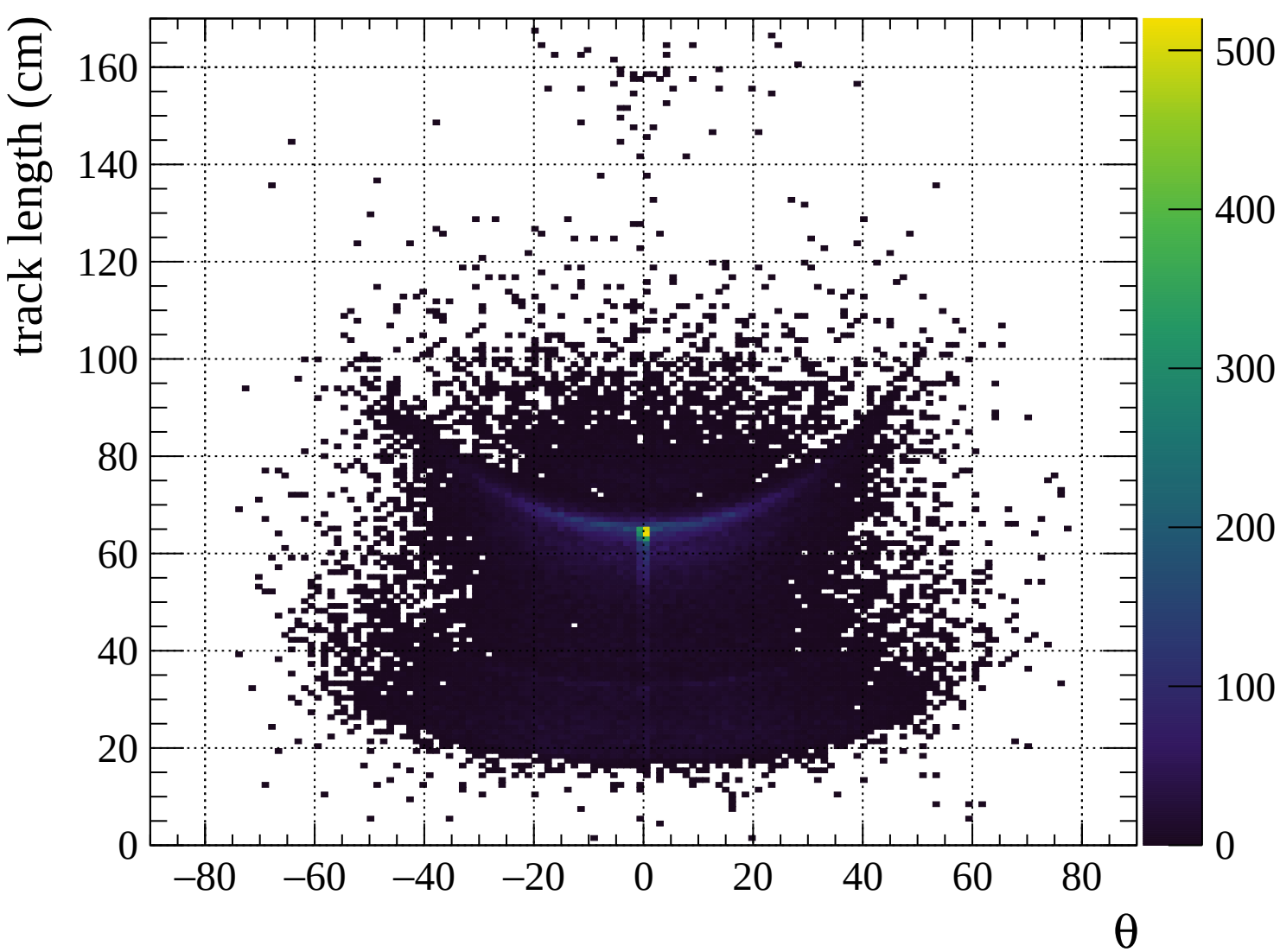


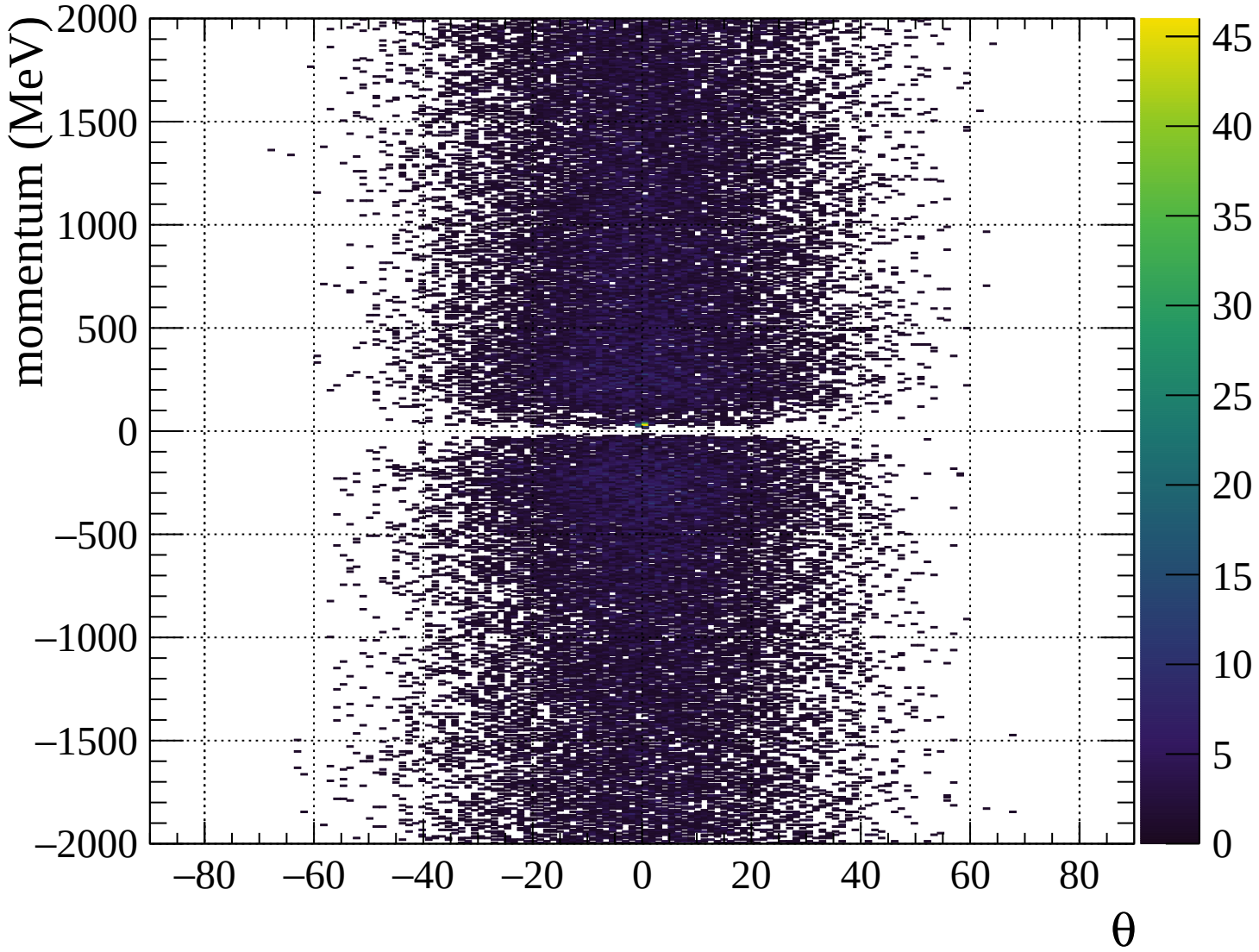


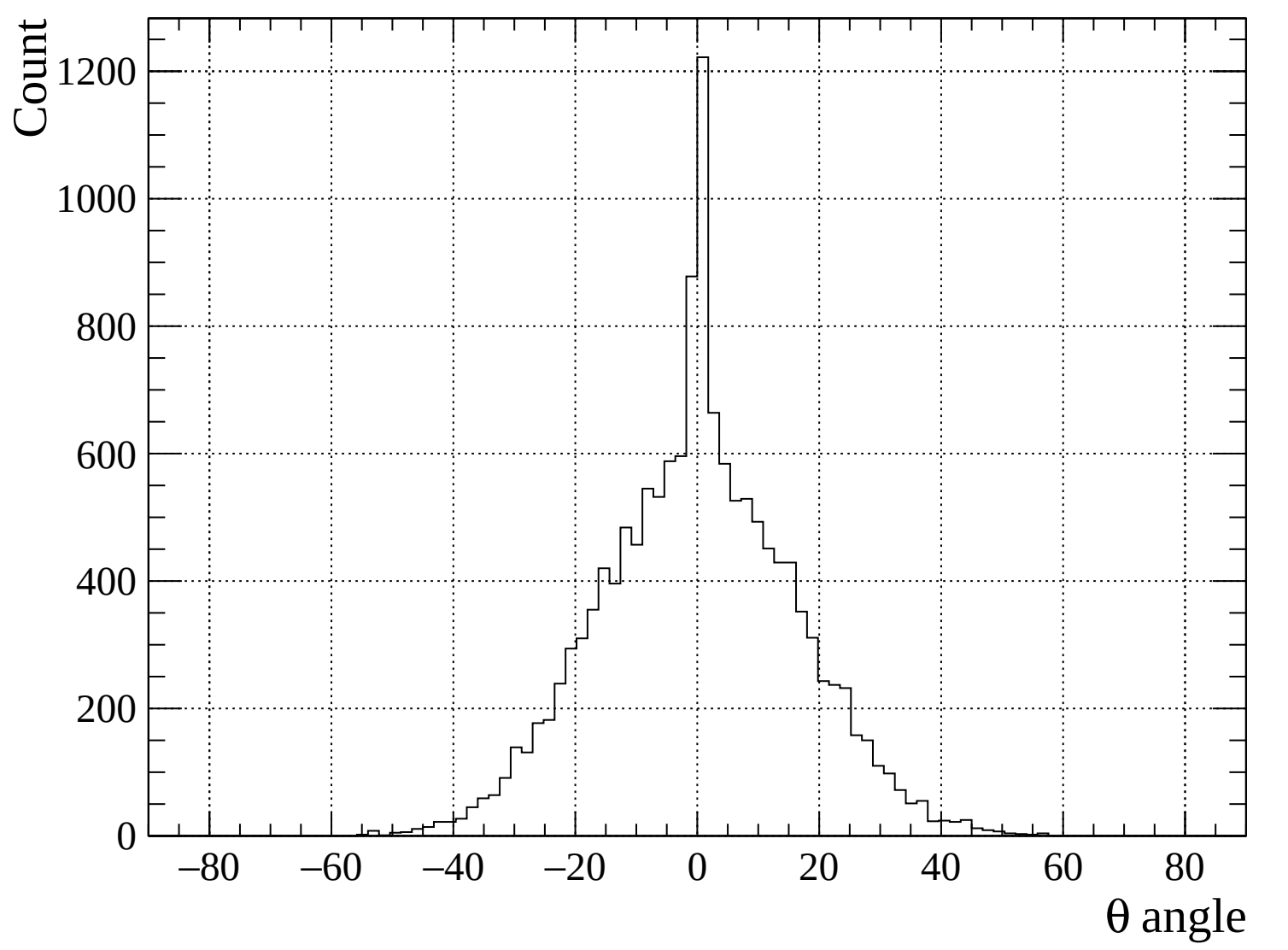


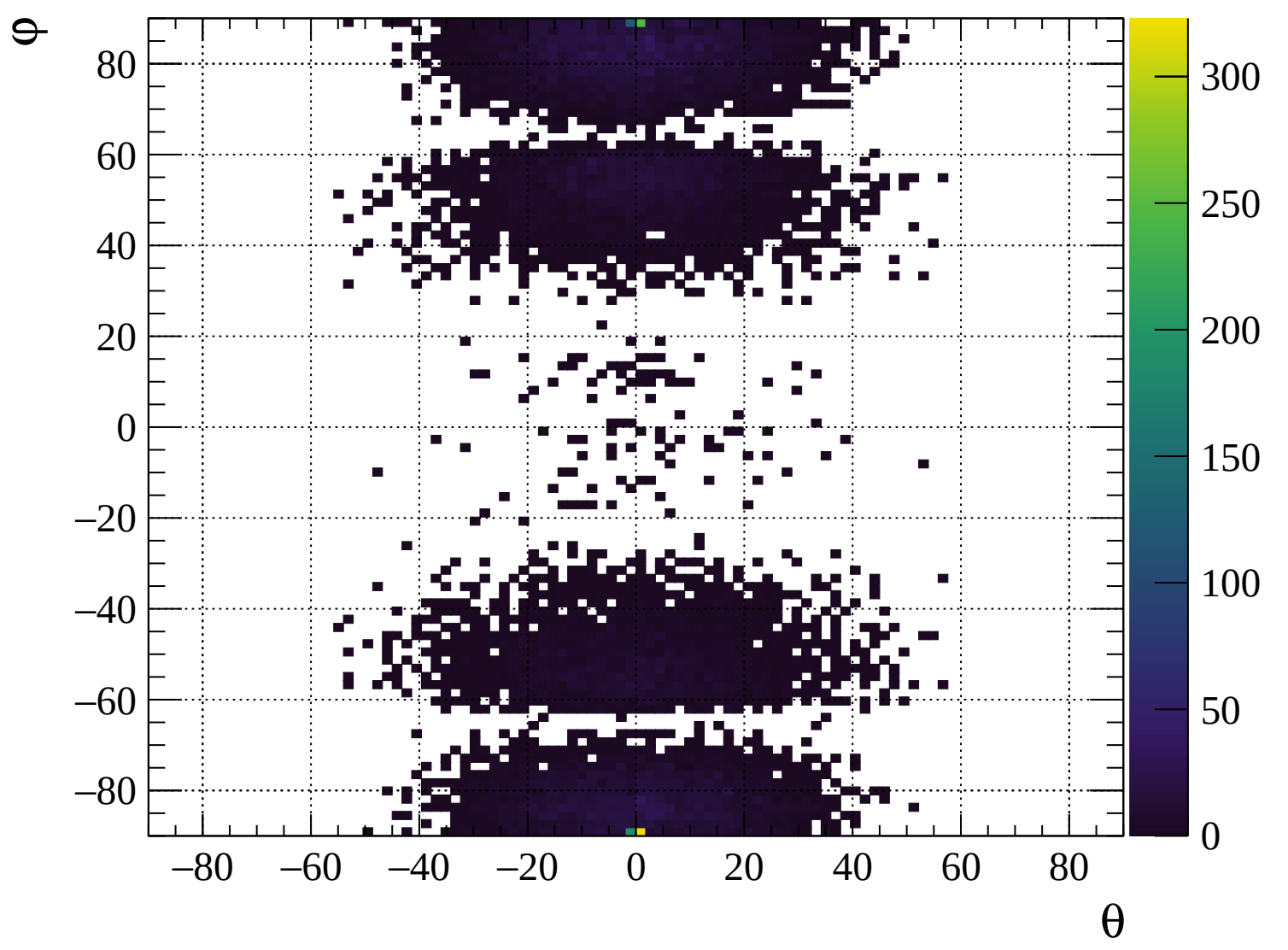


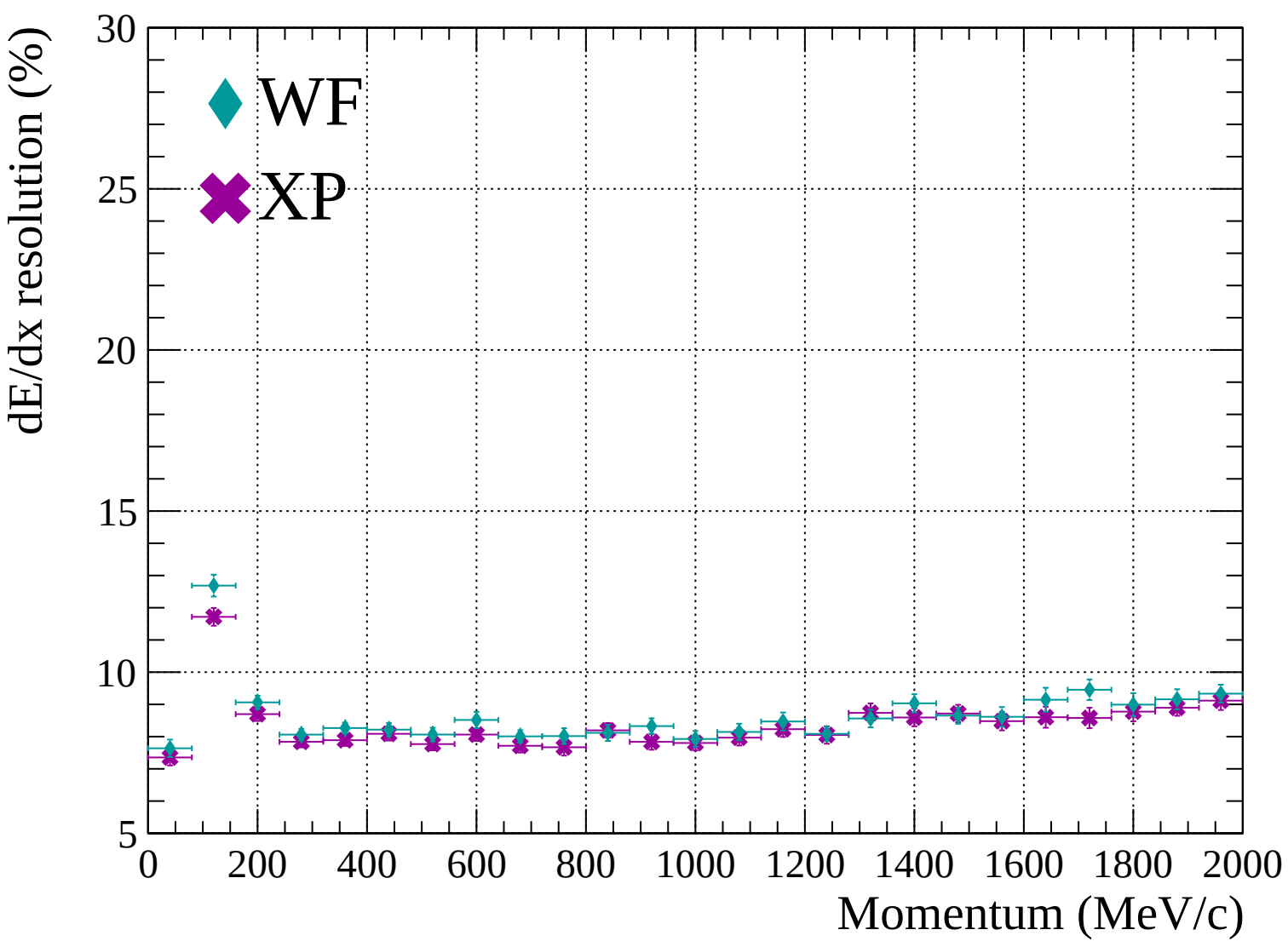


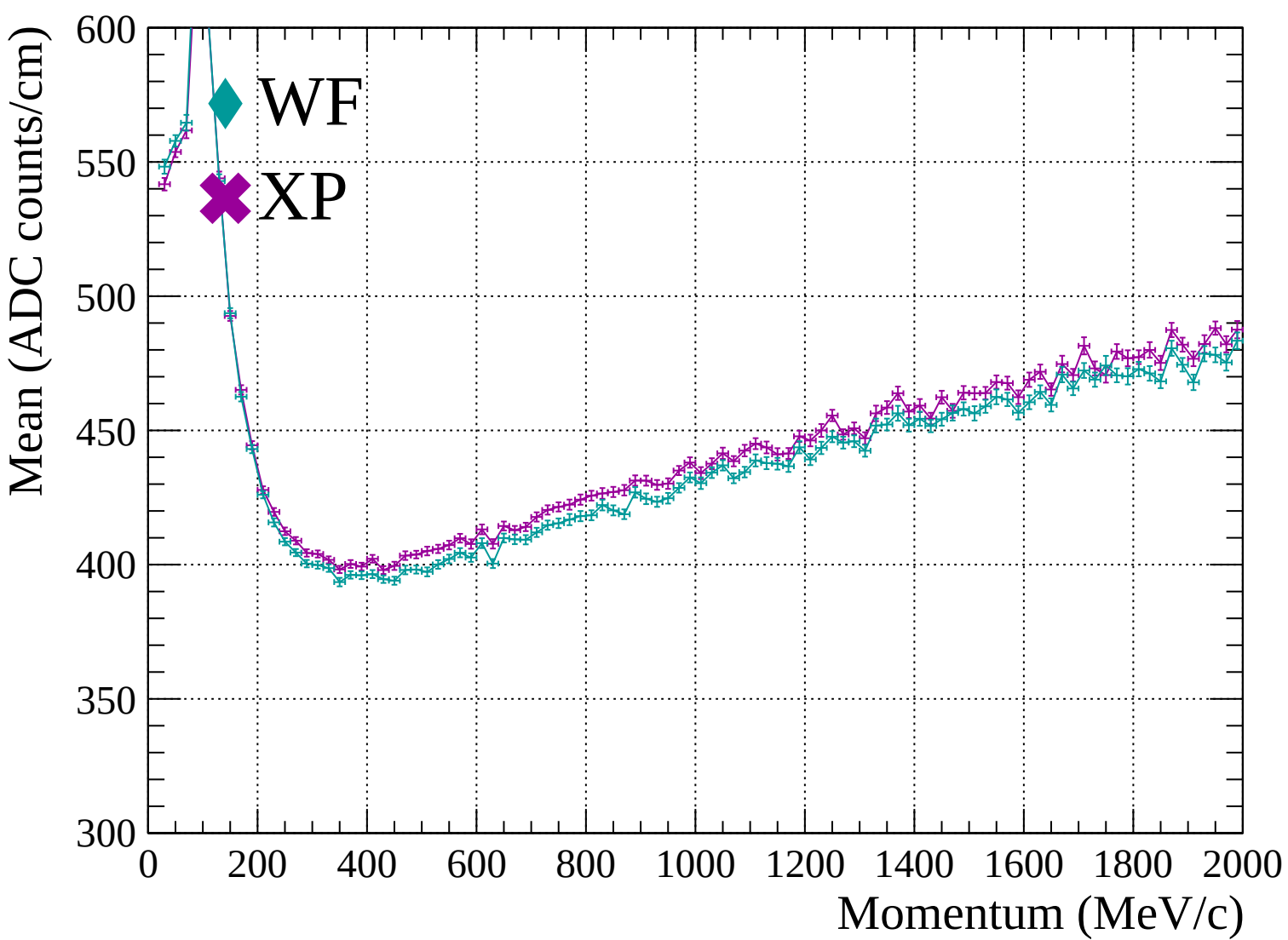


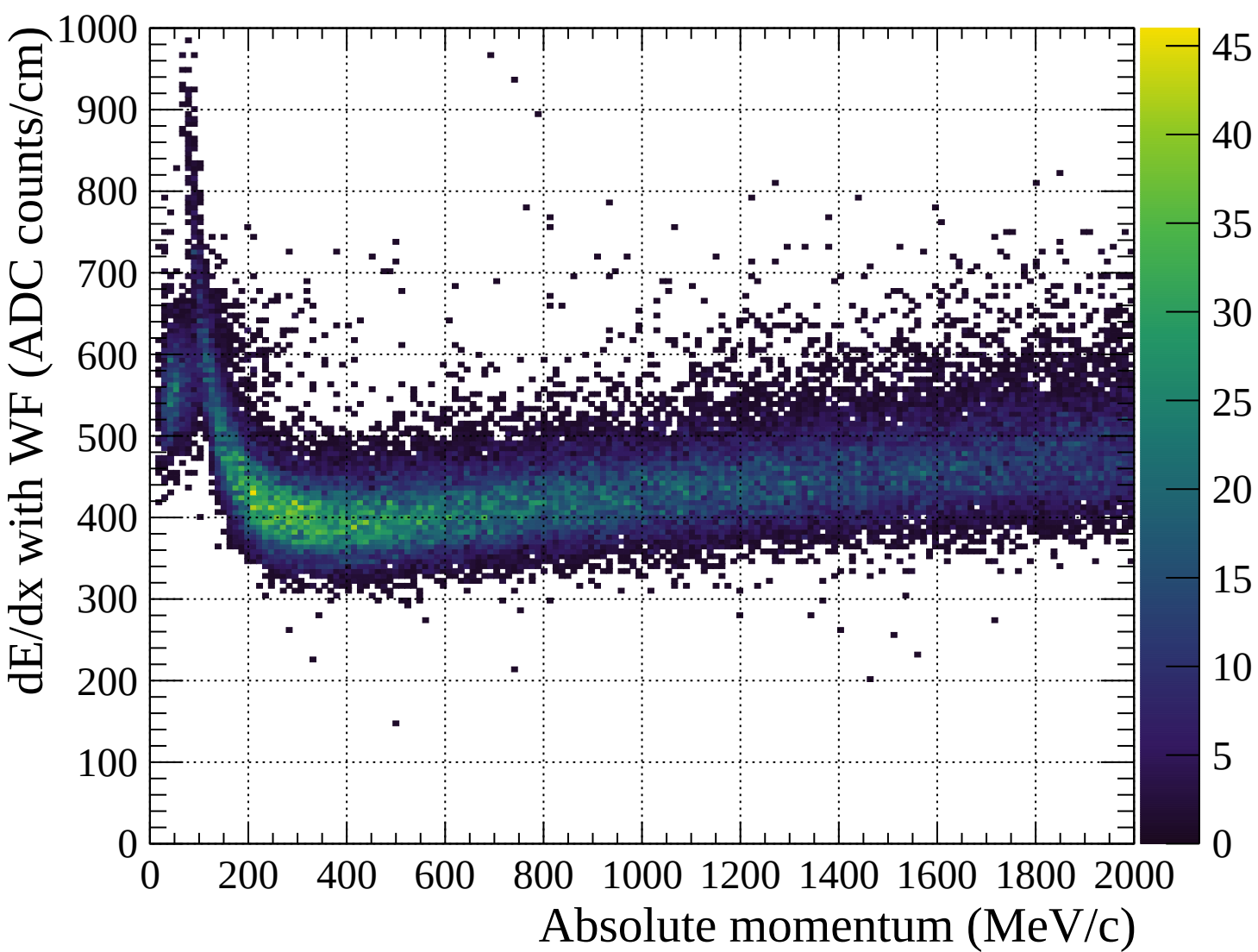


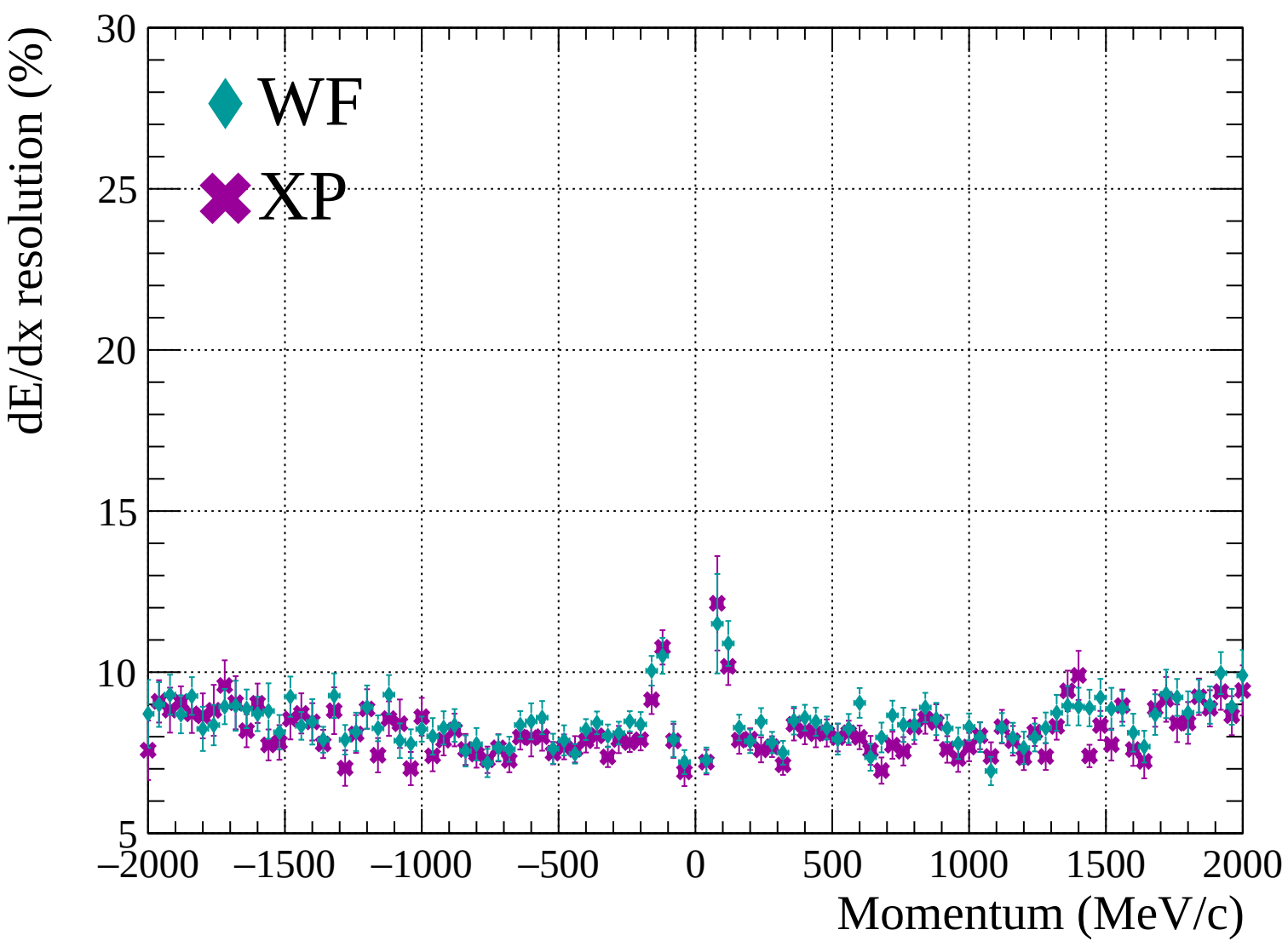


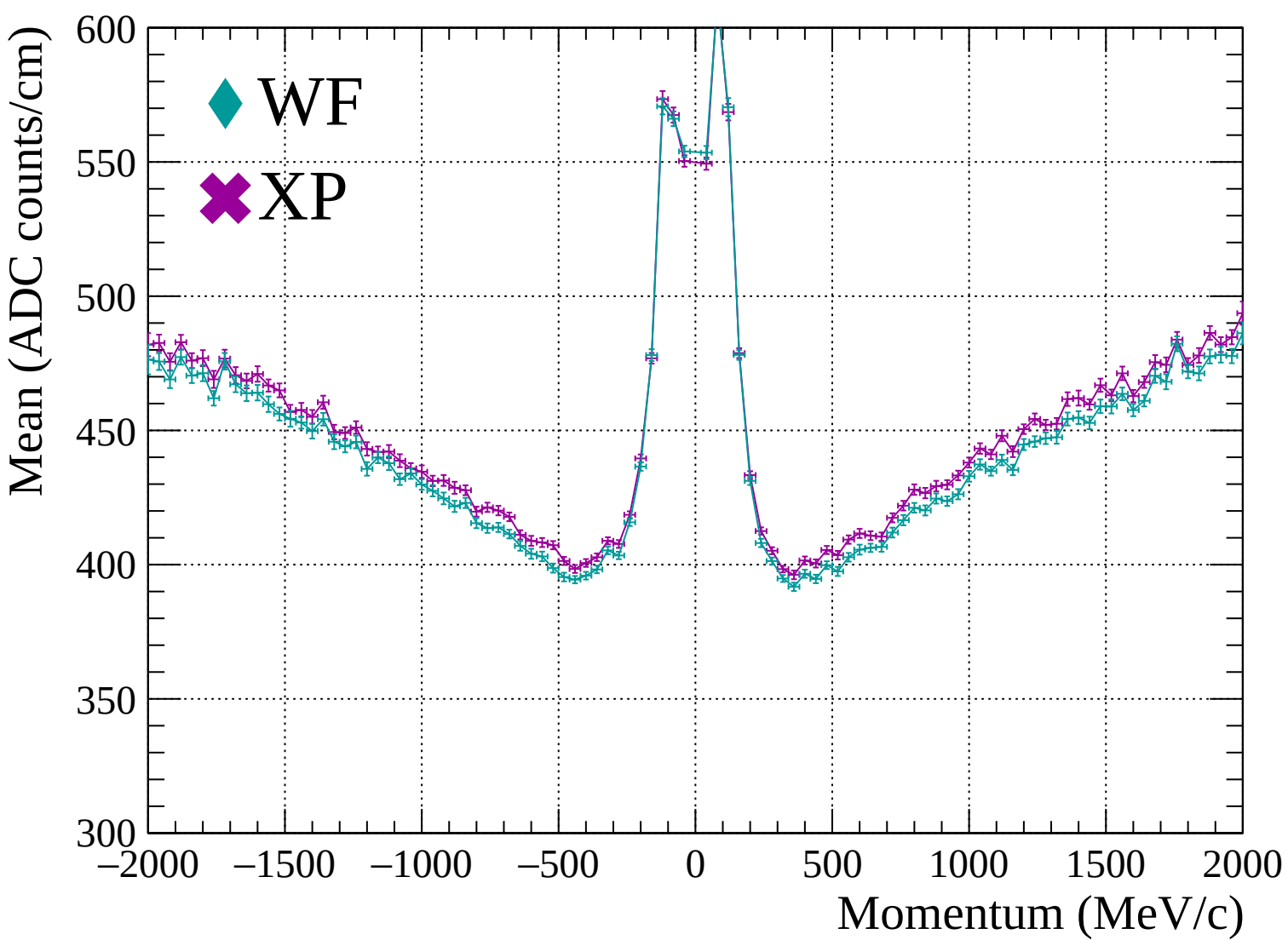


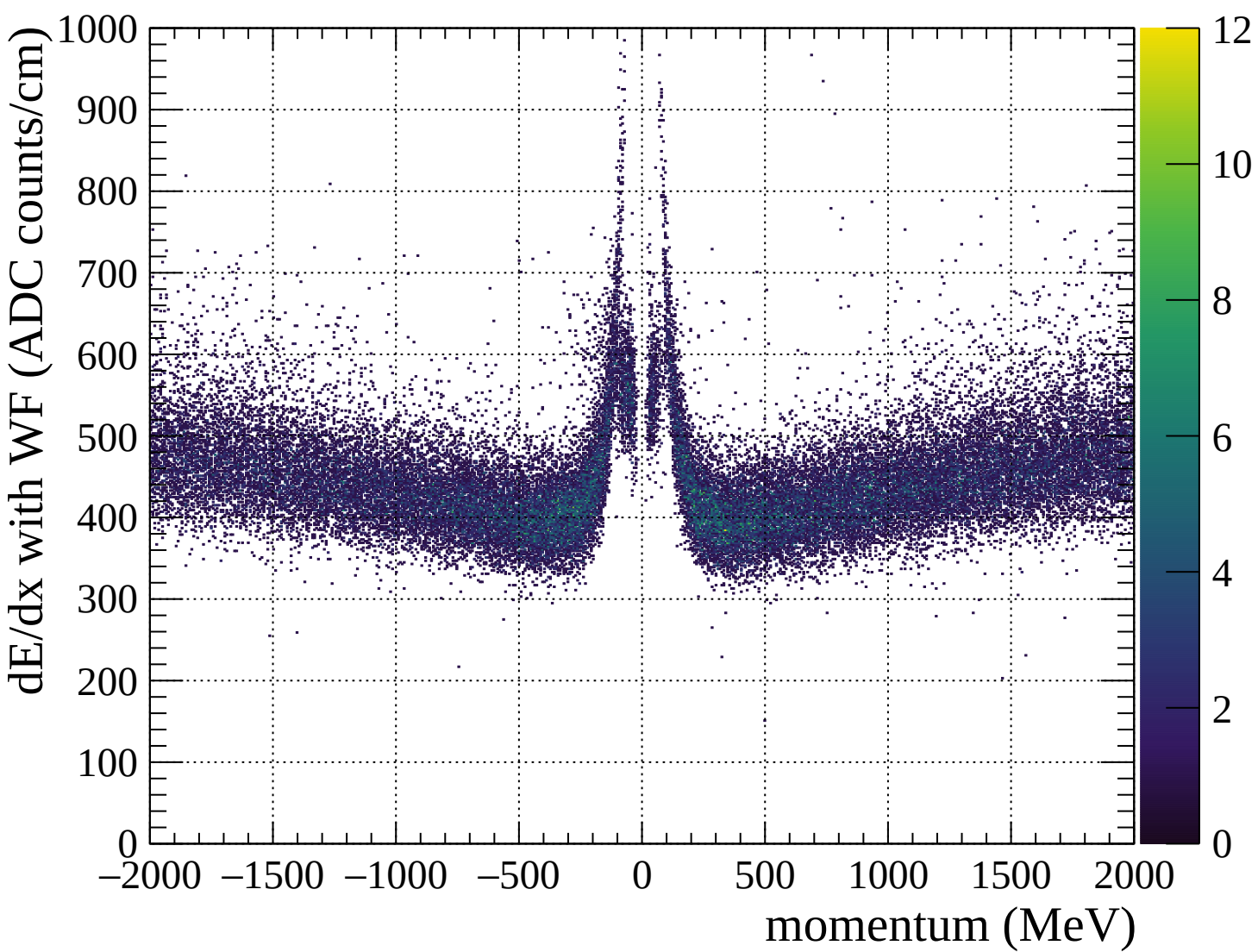


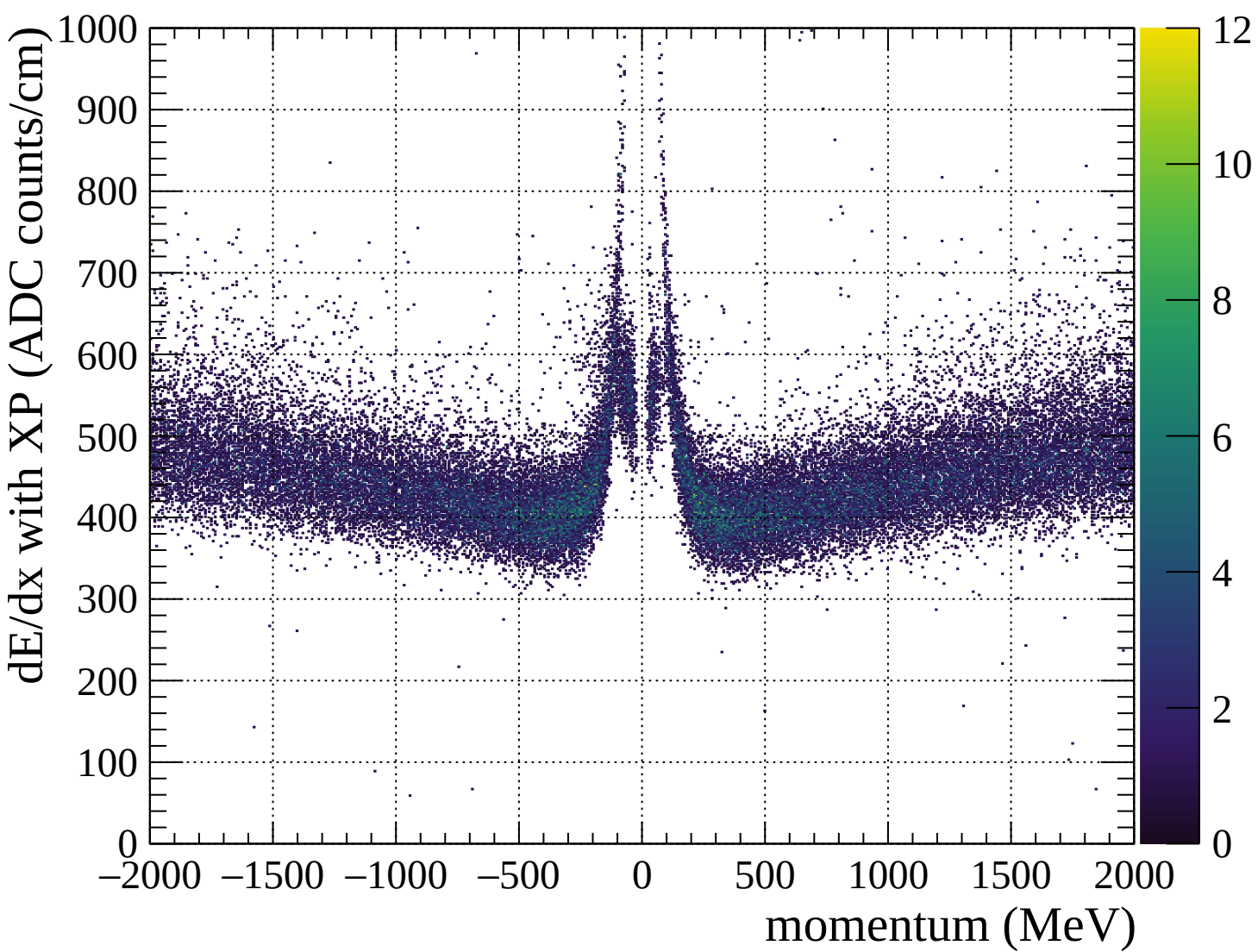


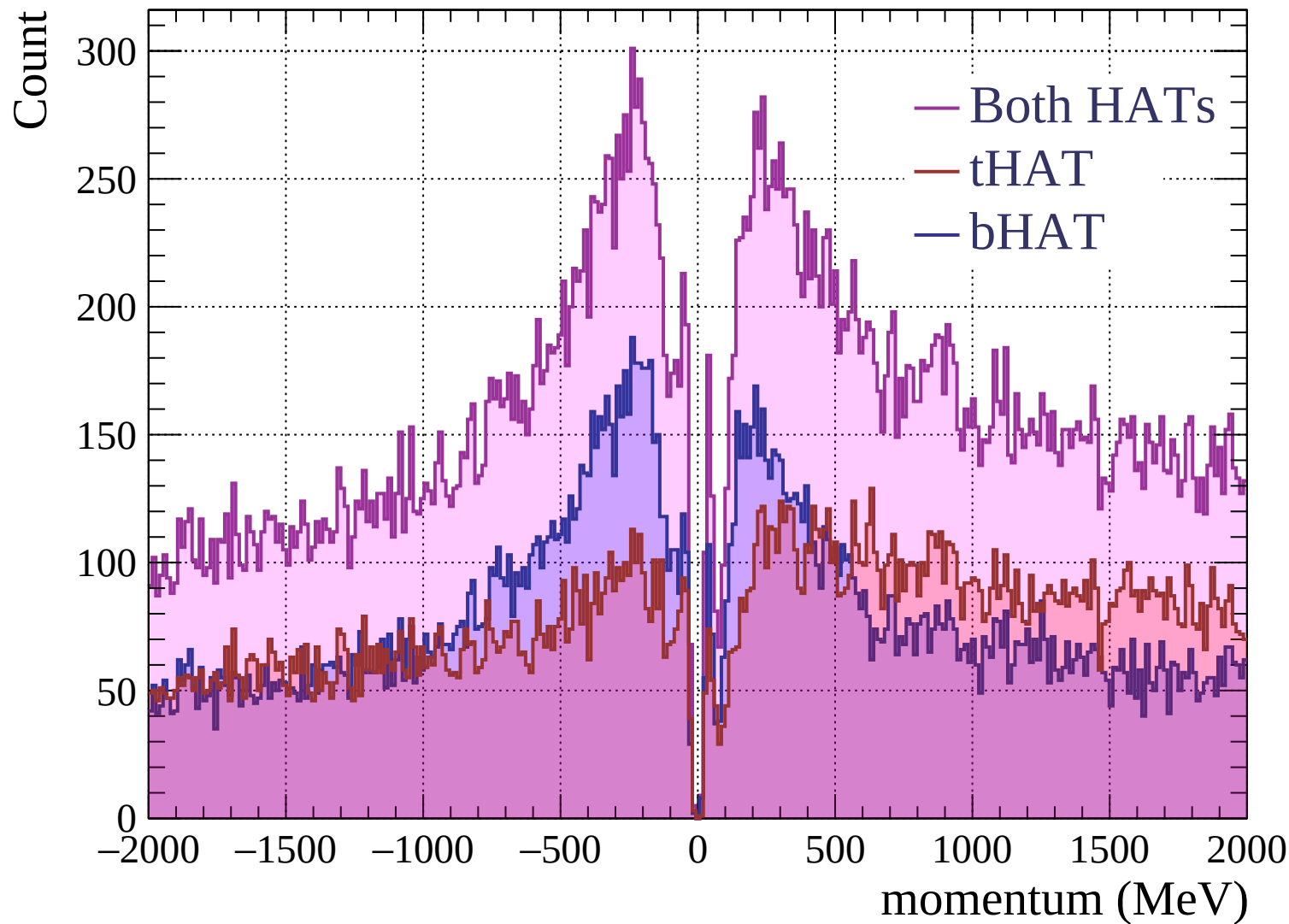


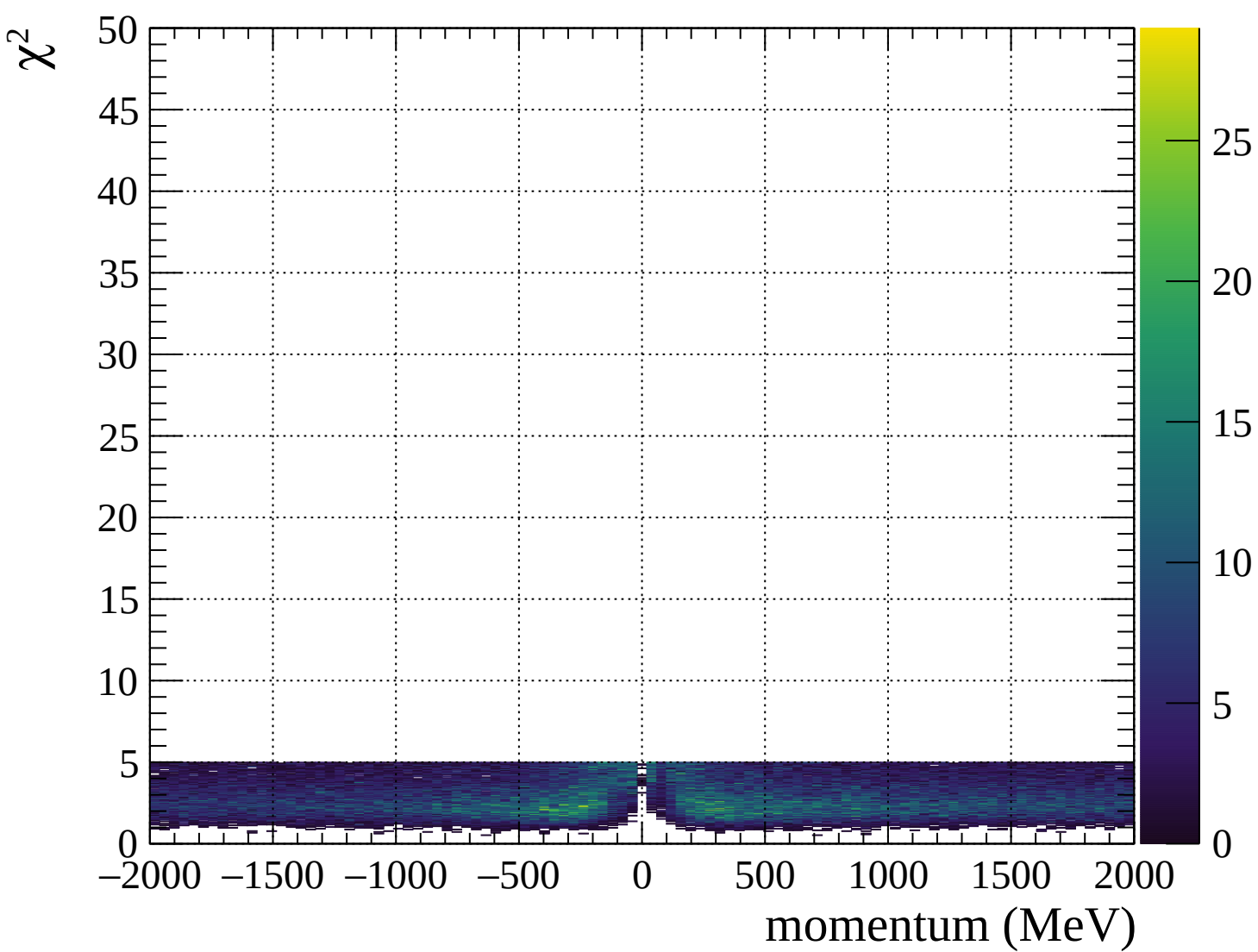




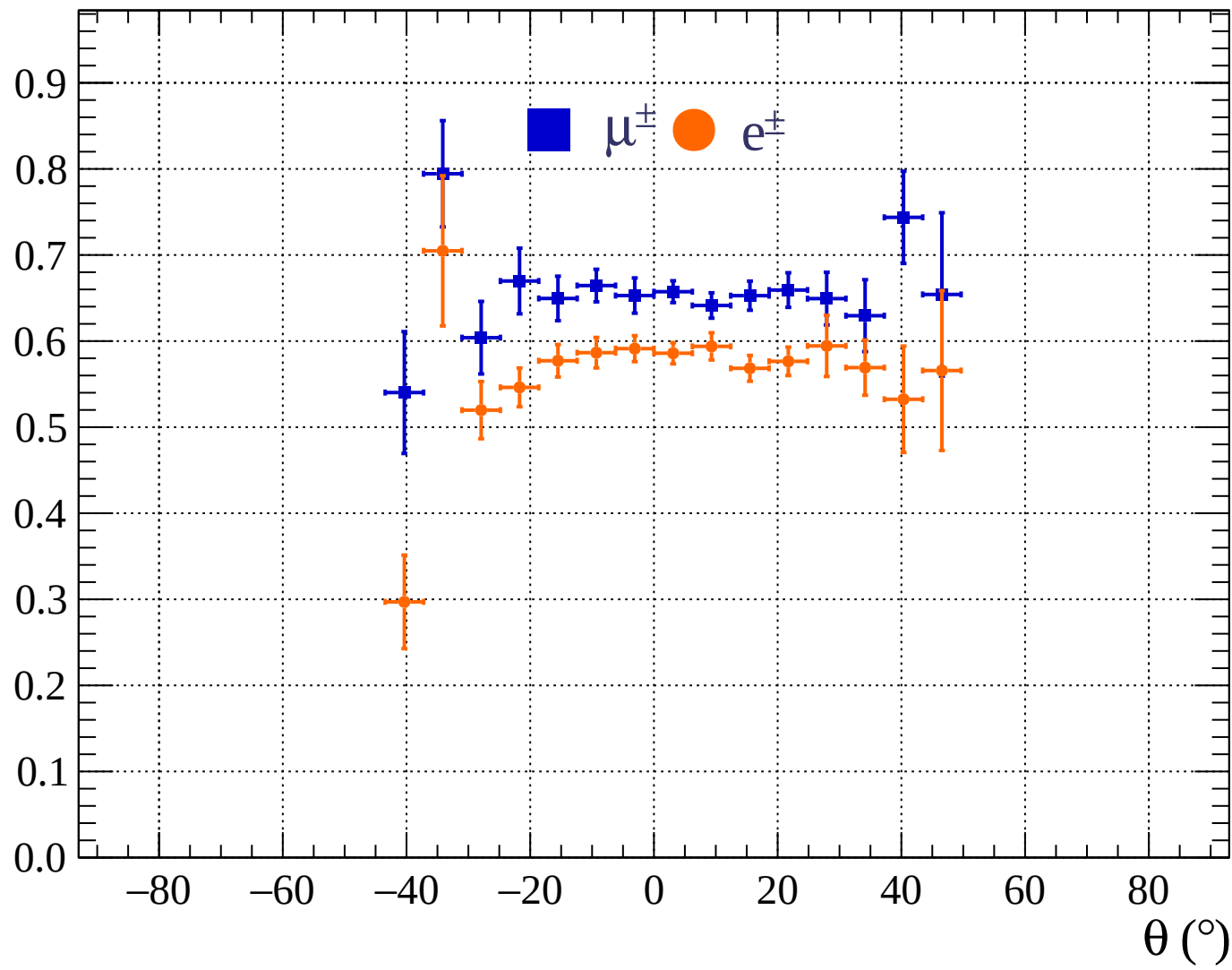




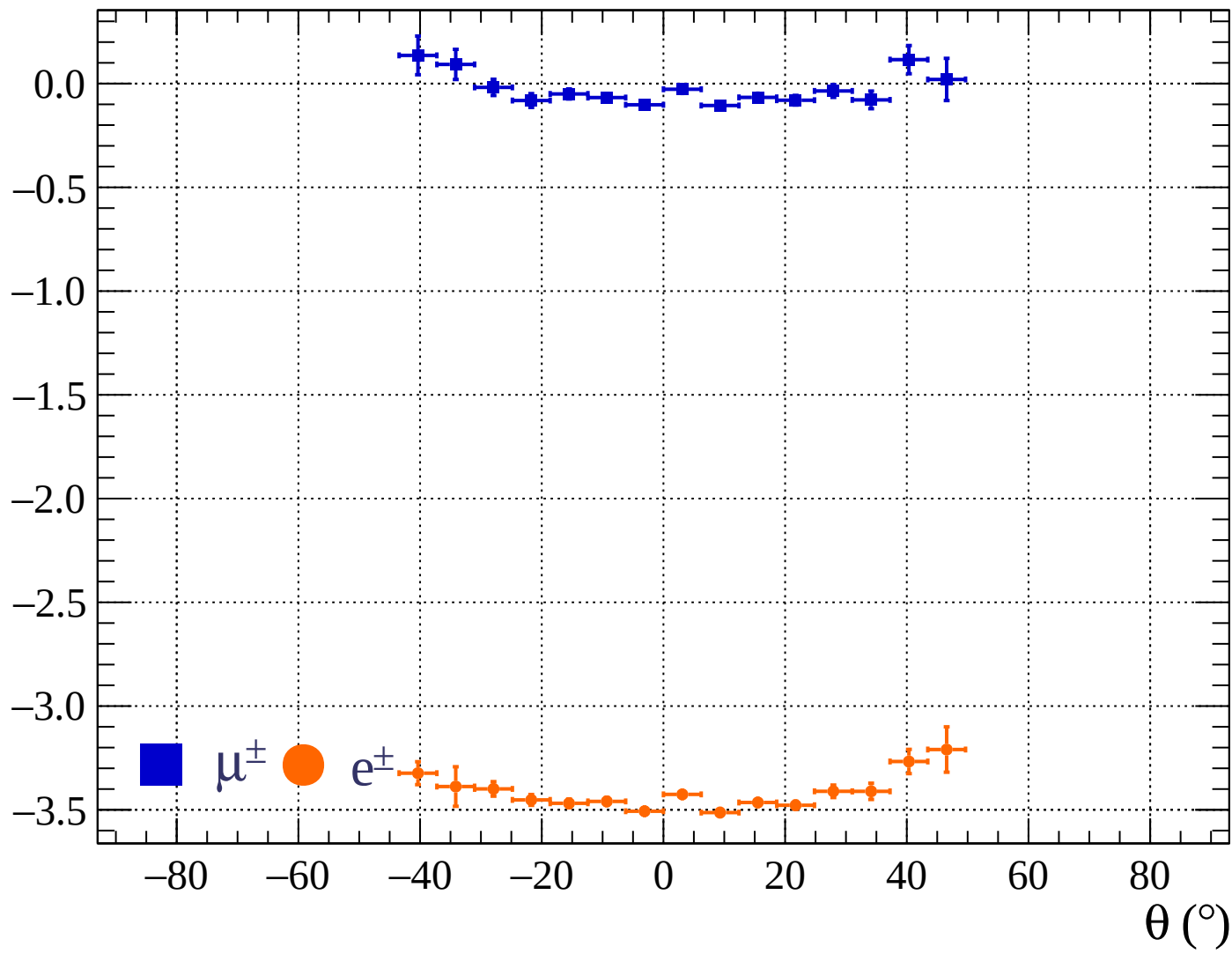




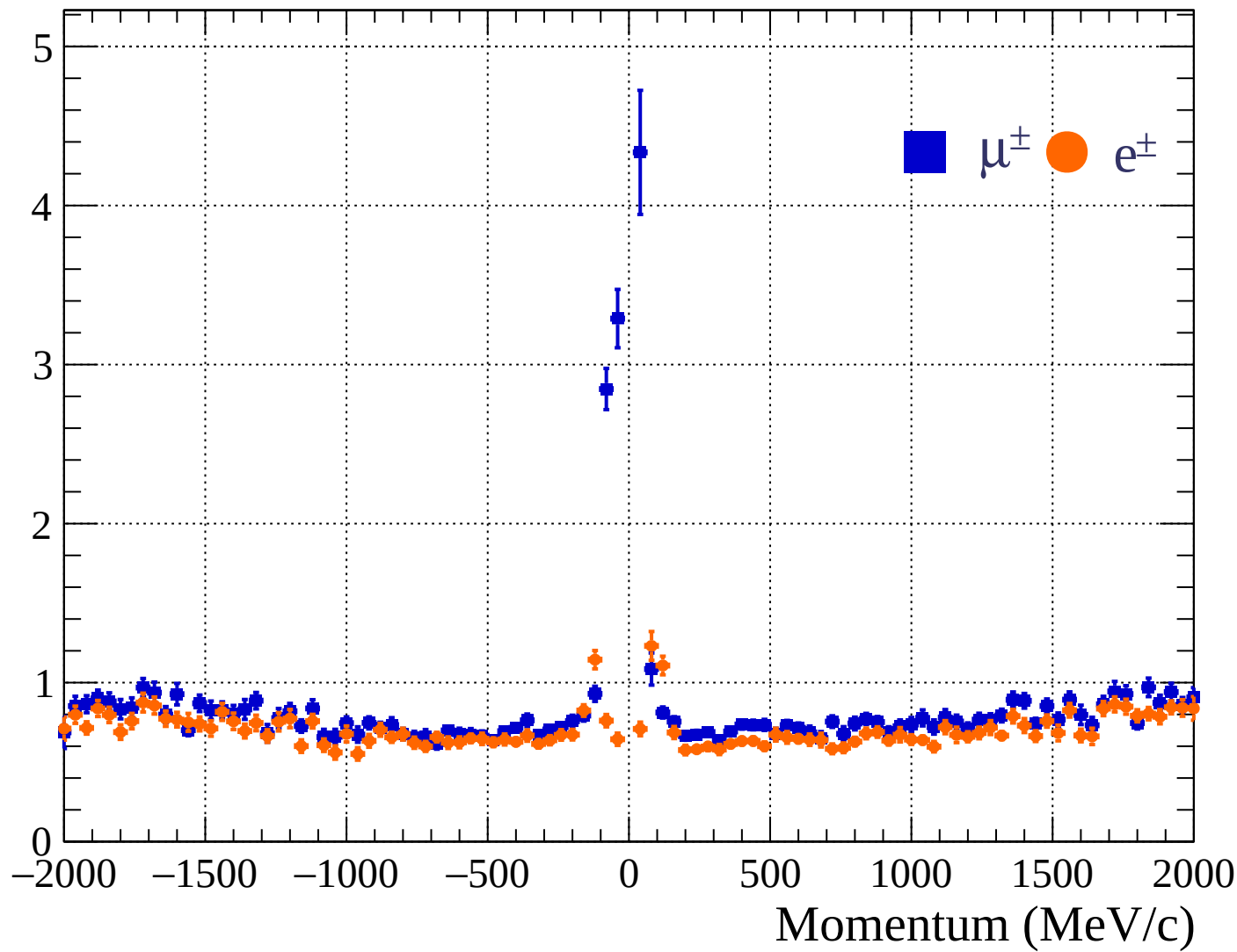
Pulls standard deviation



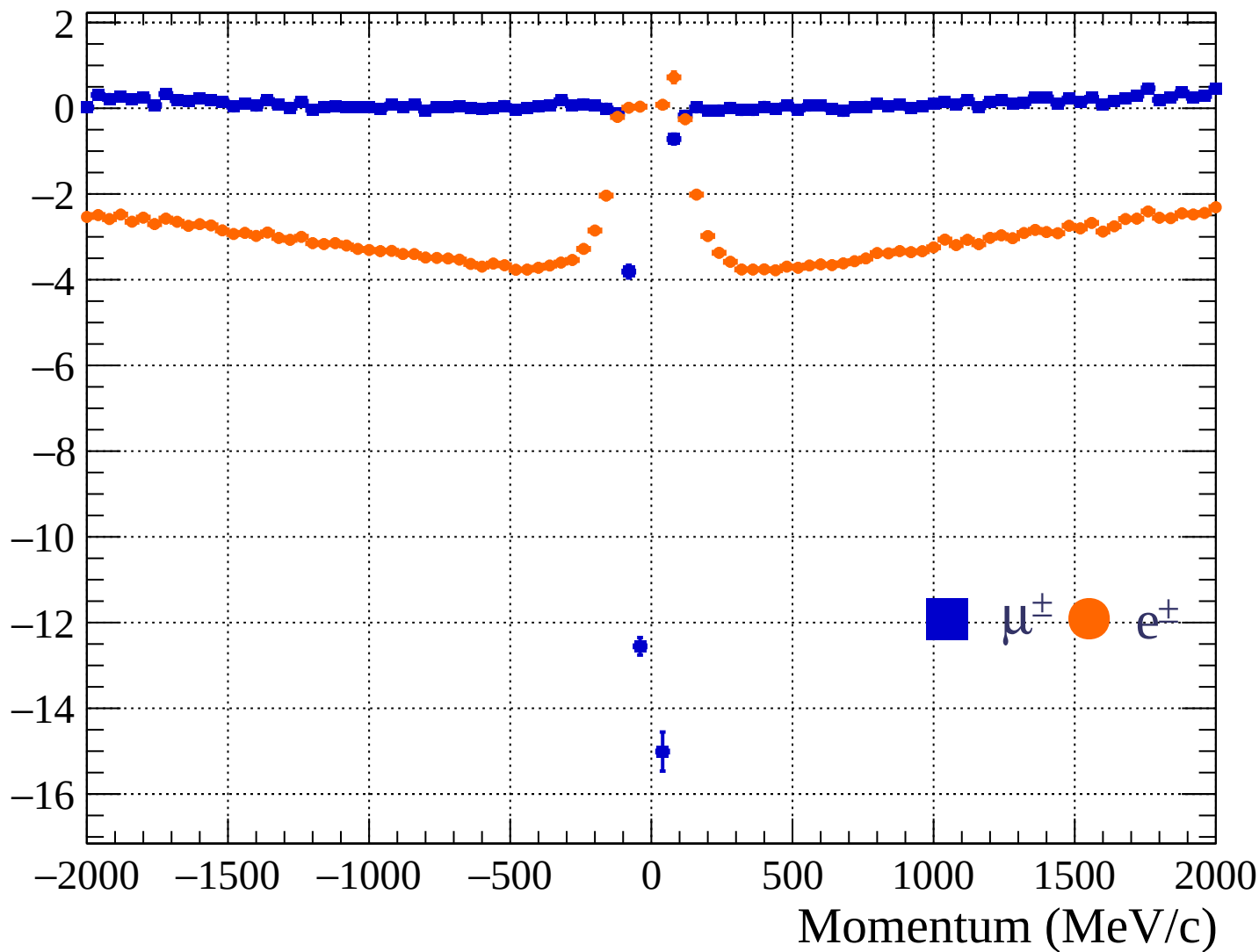
Pulls mean

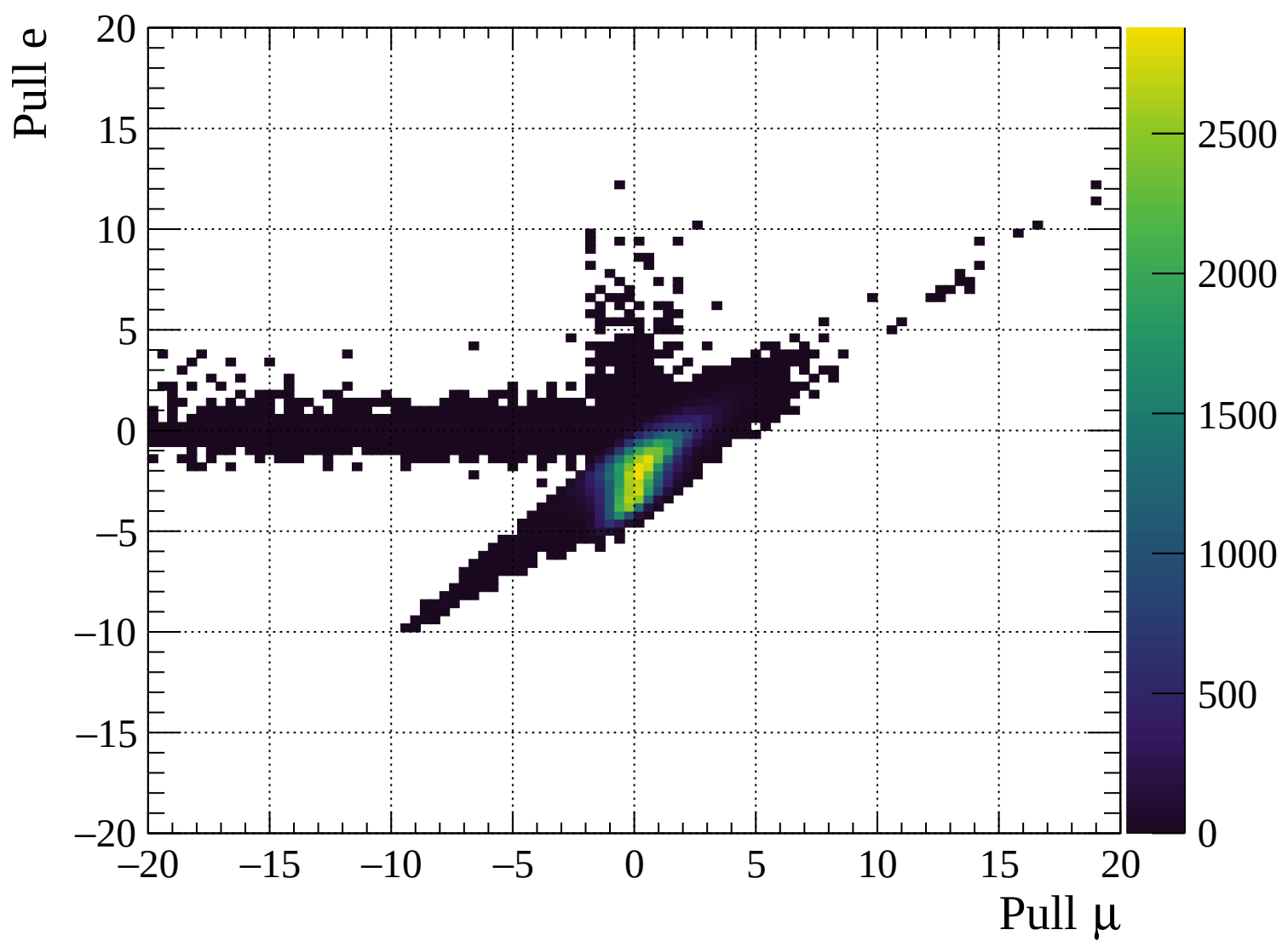


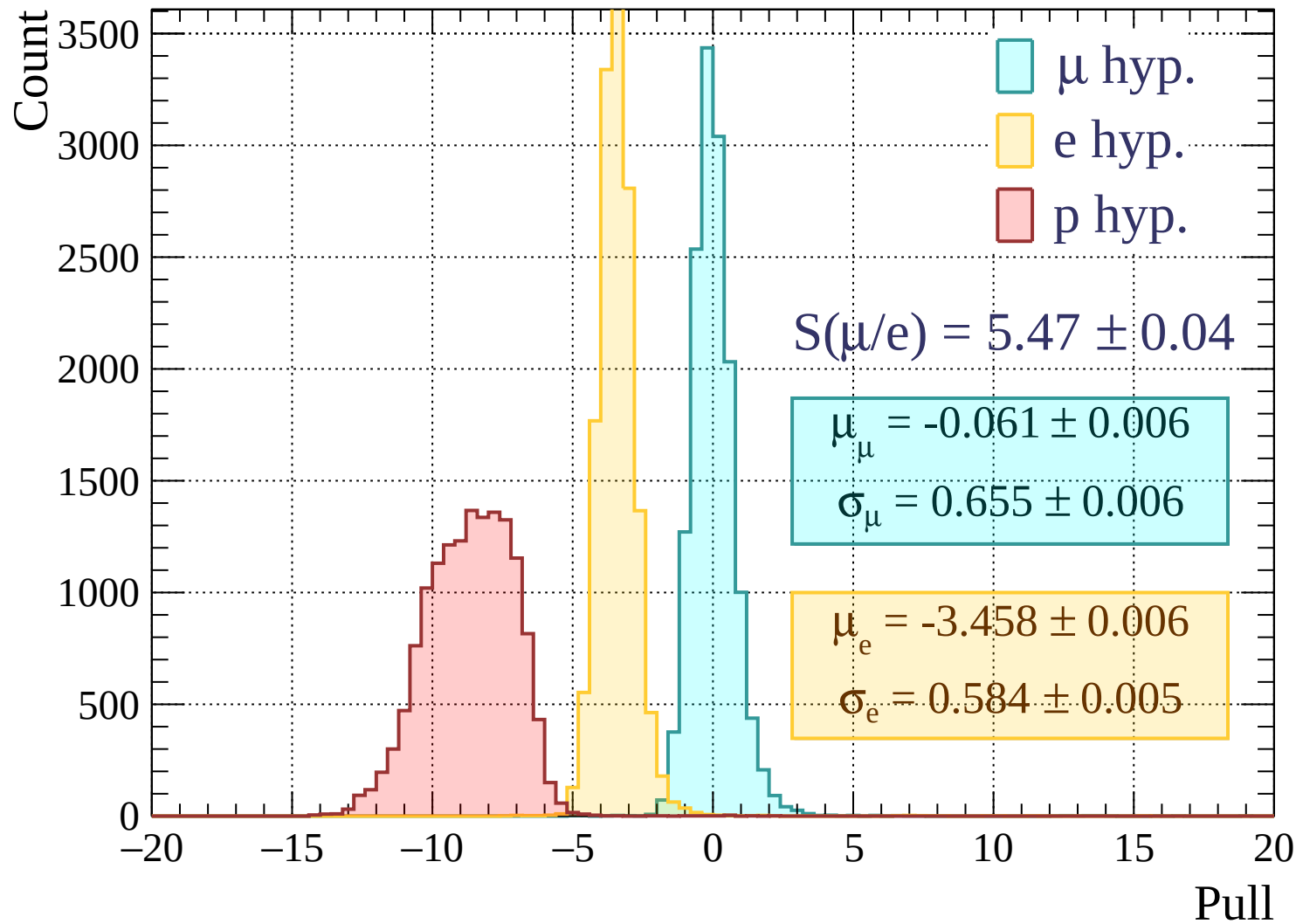
Pulls standard deviation

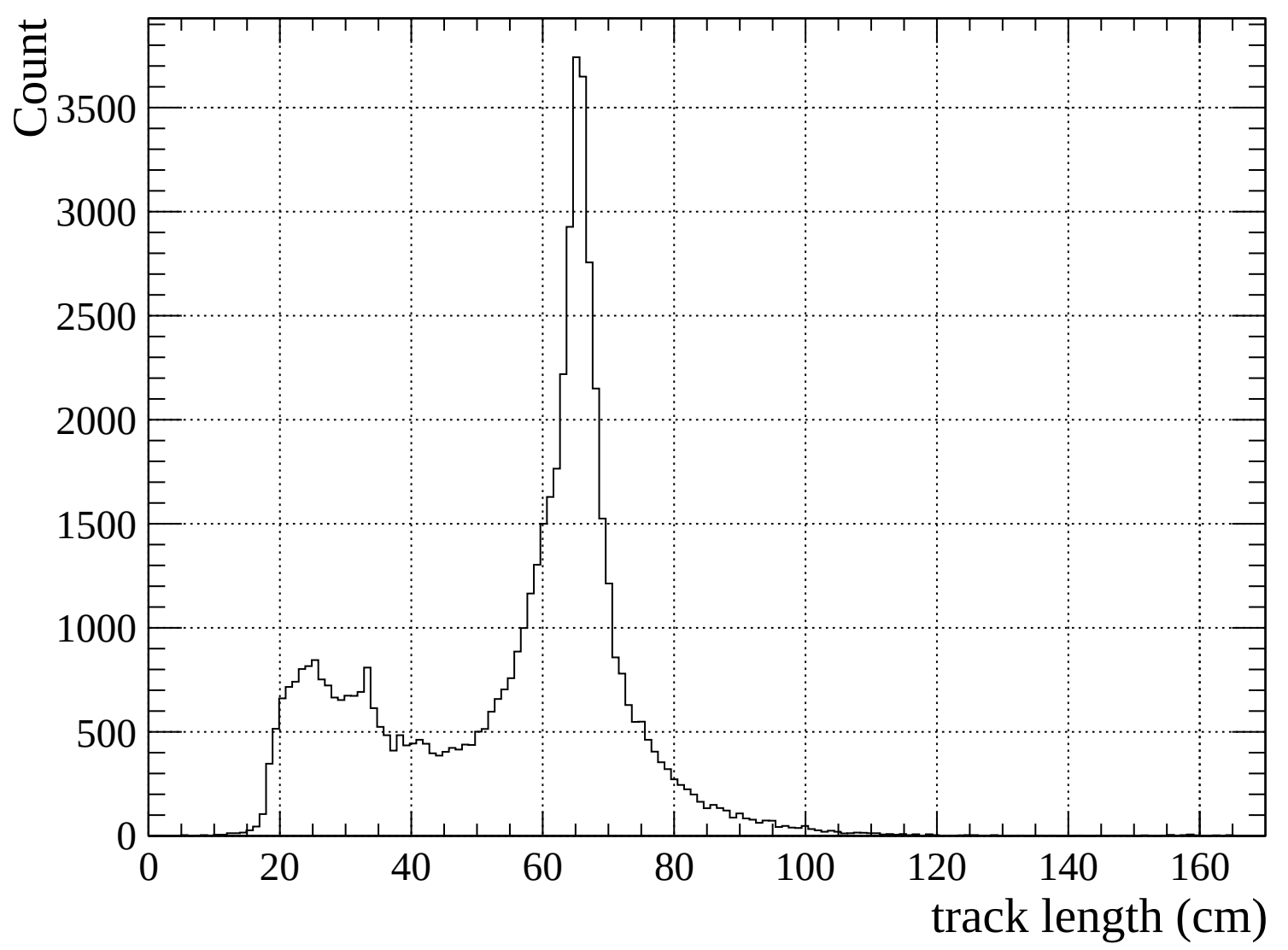


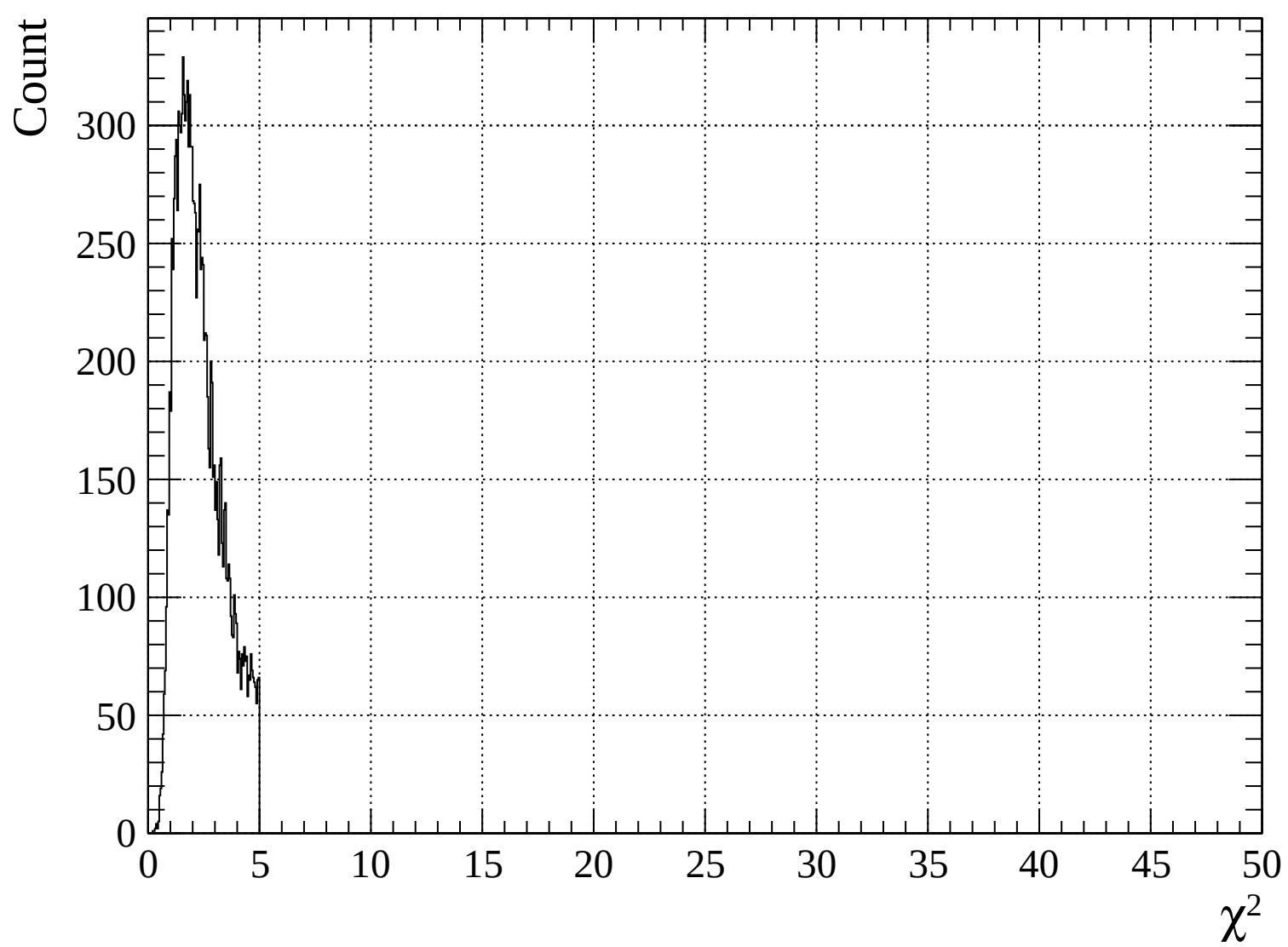
Pulls mean

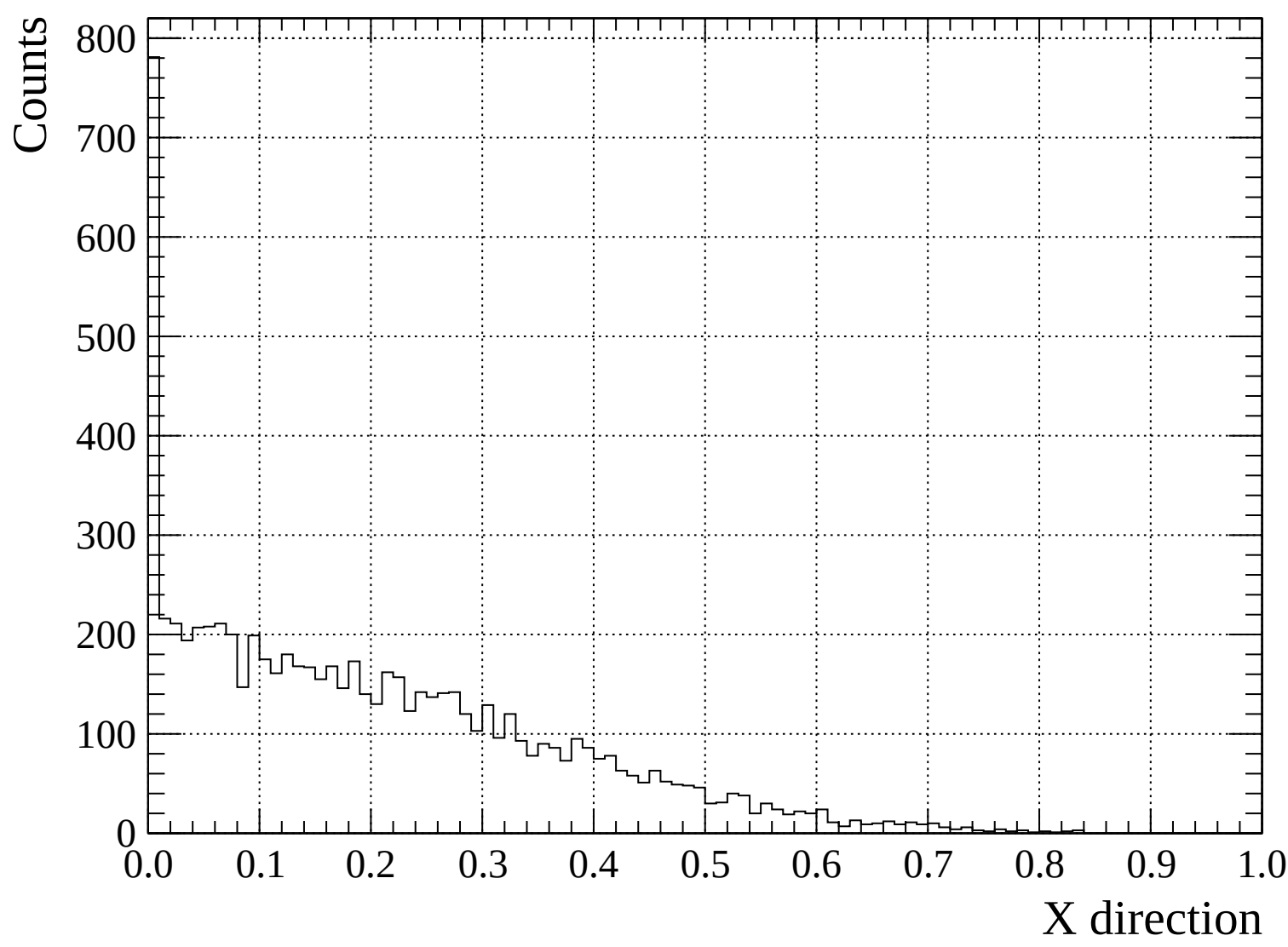


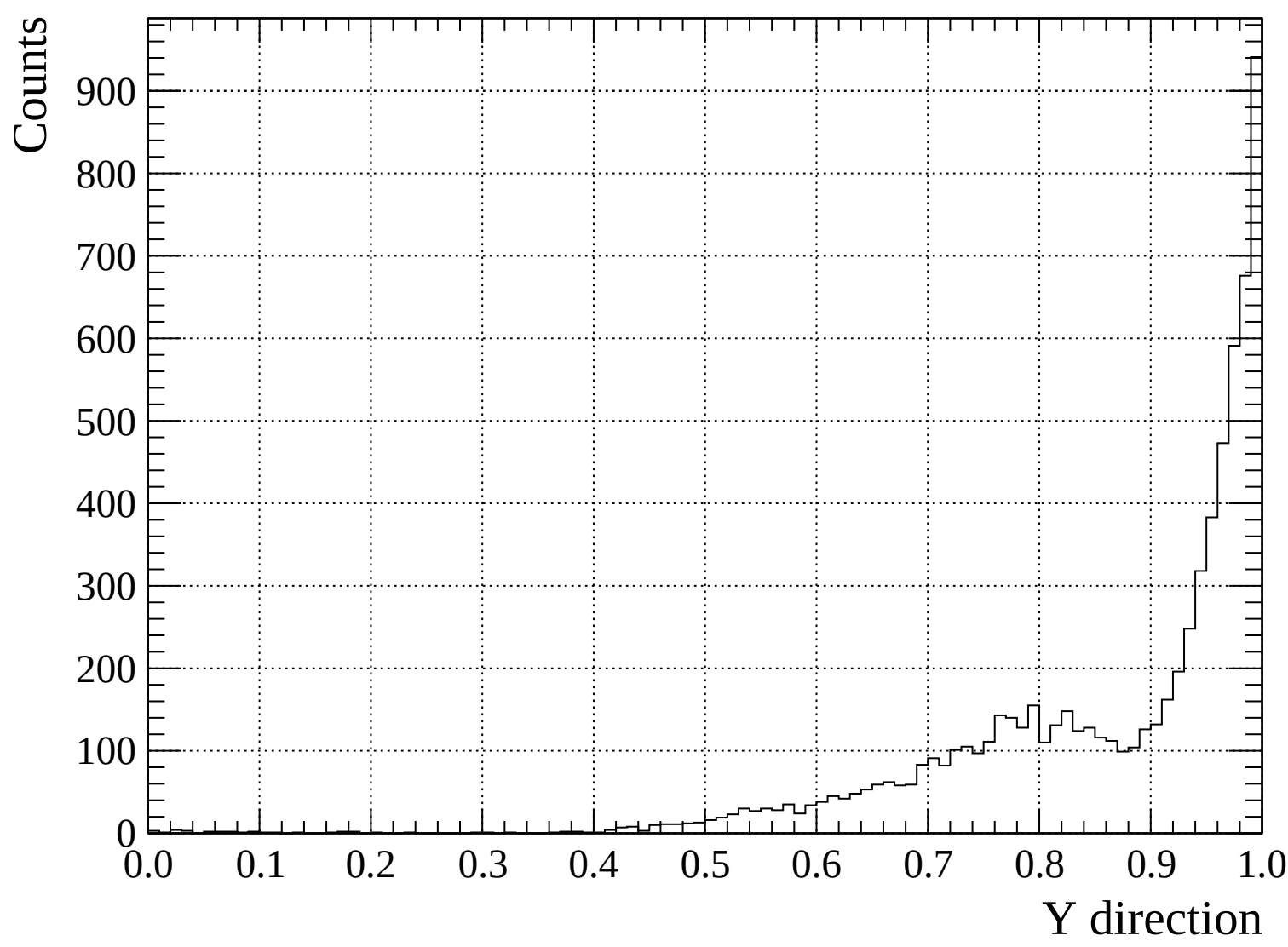


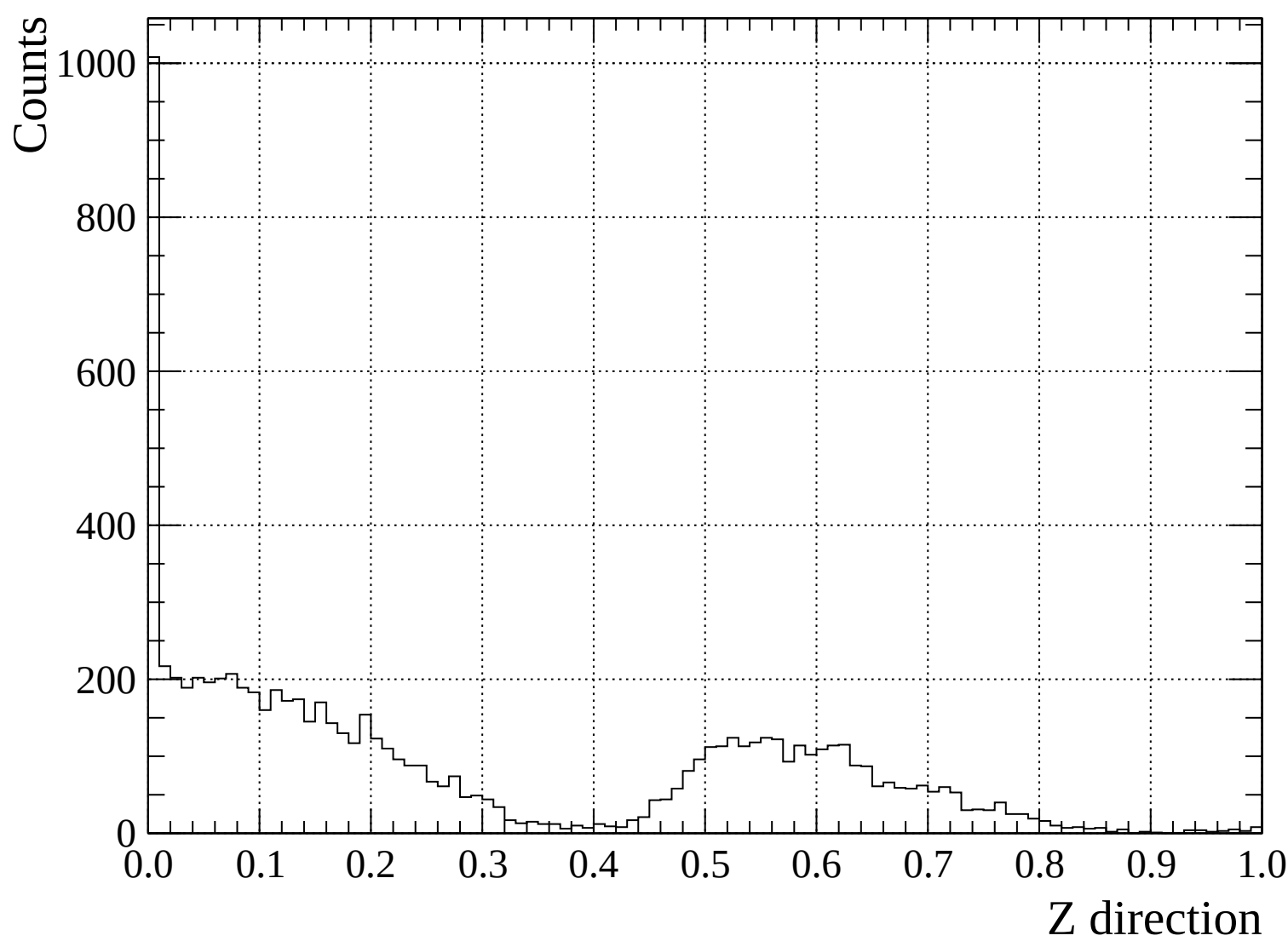


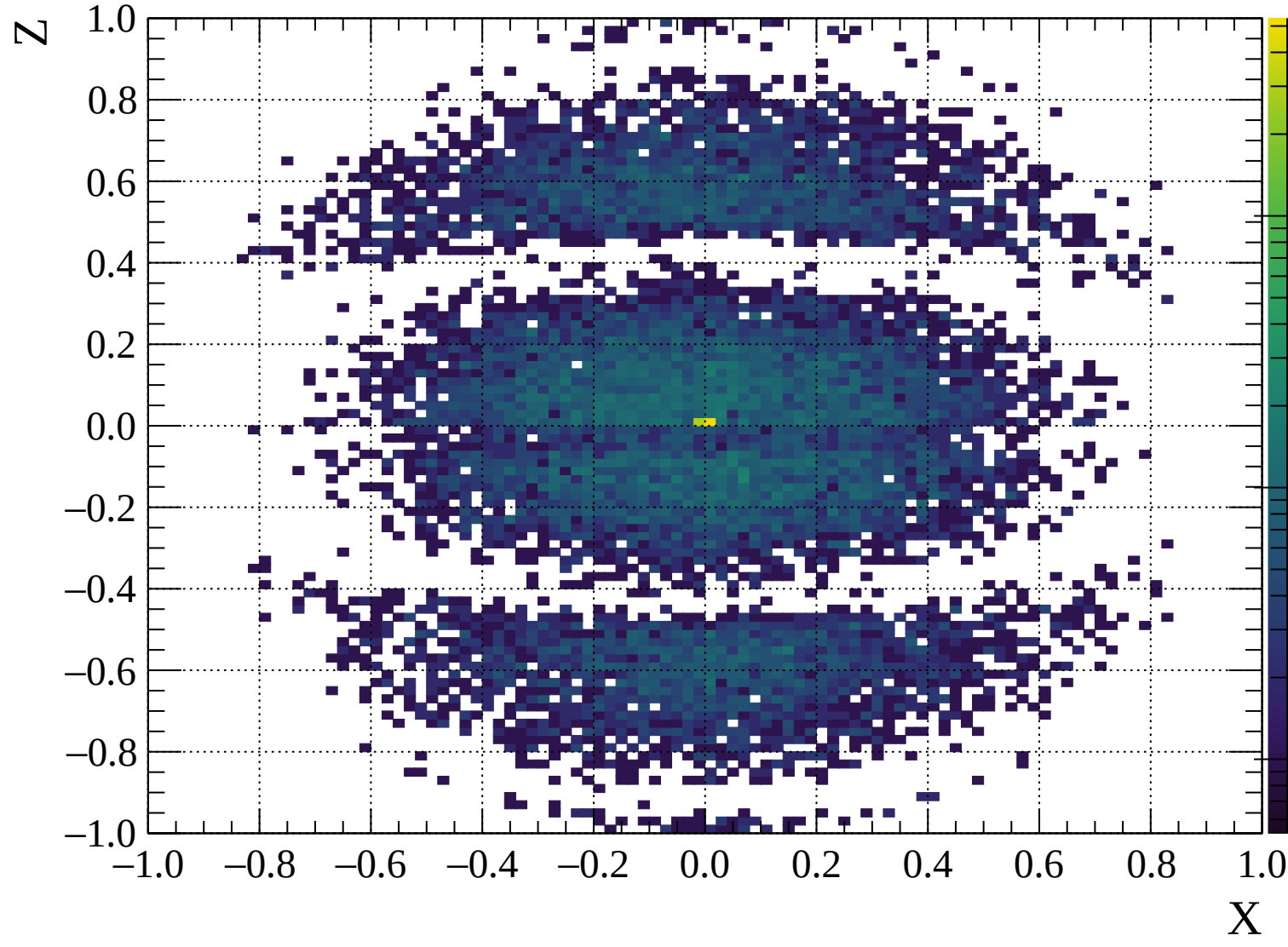


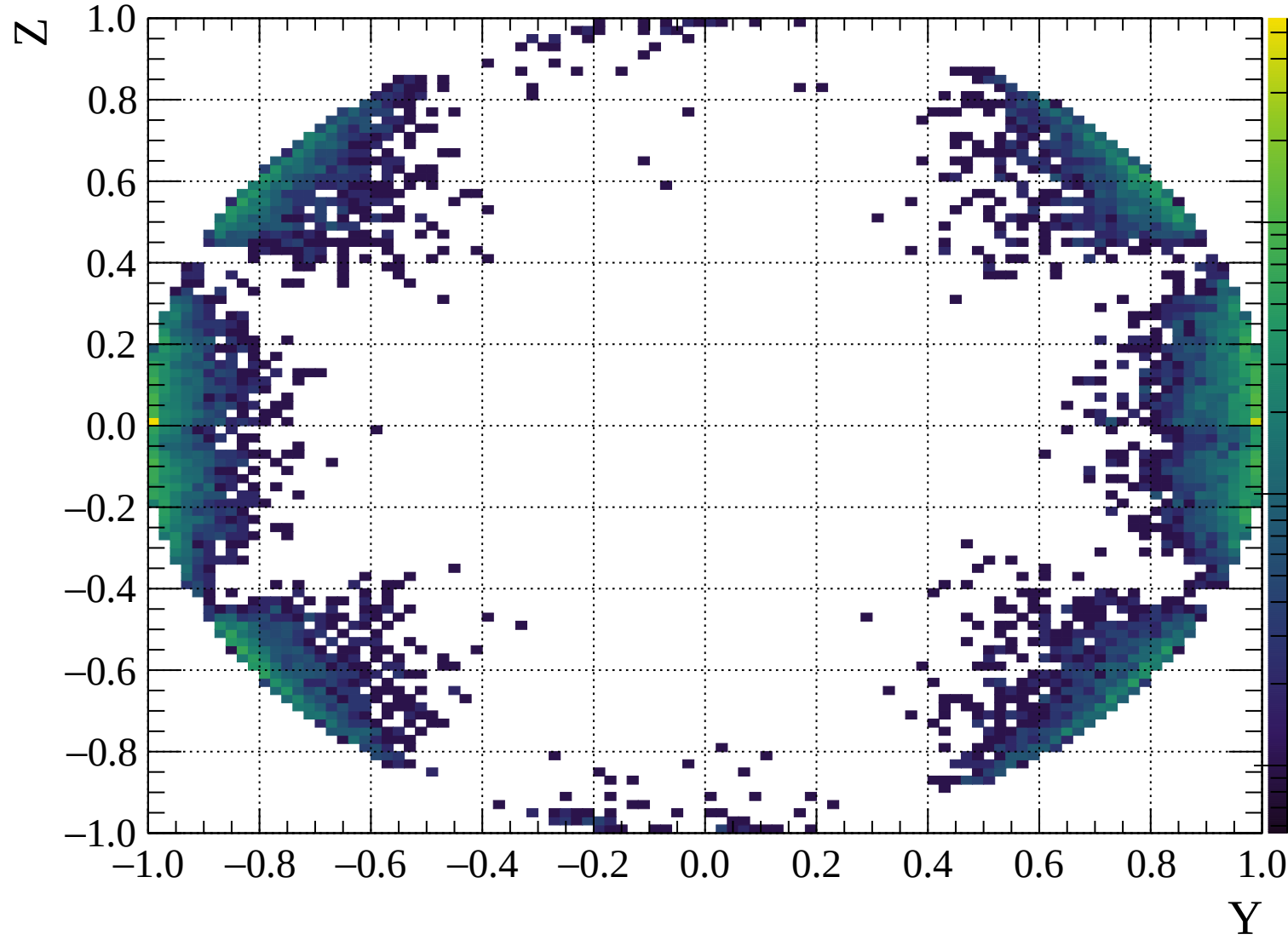


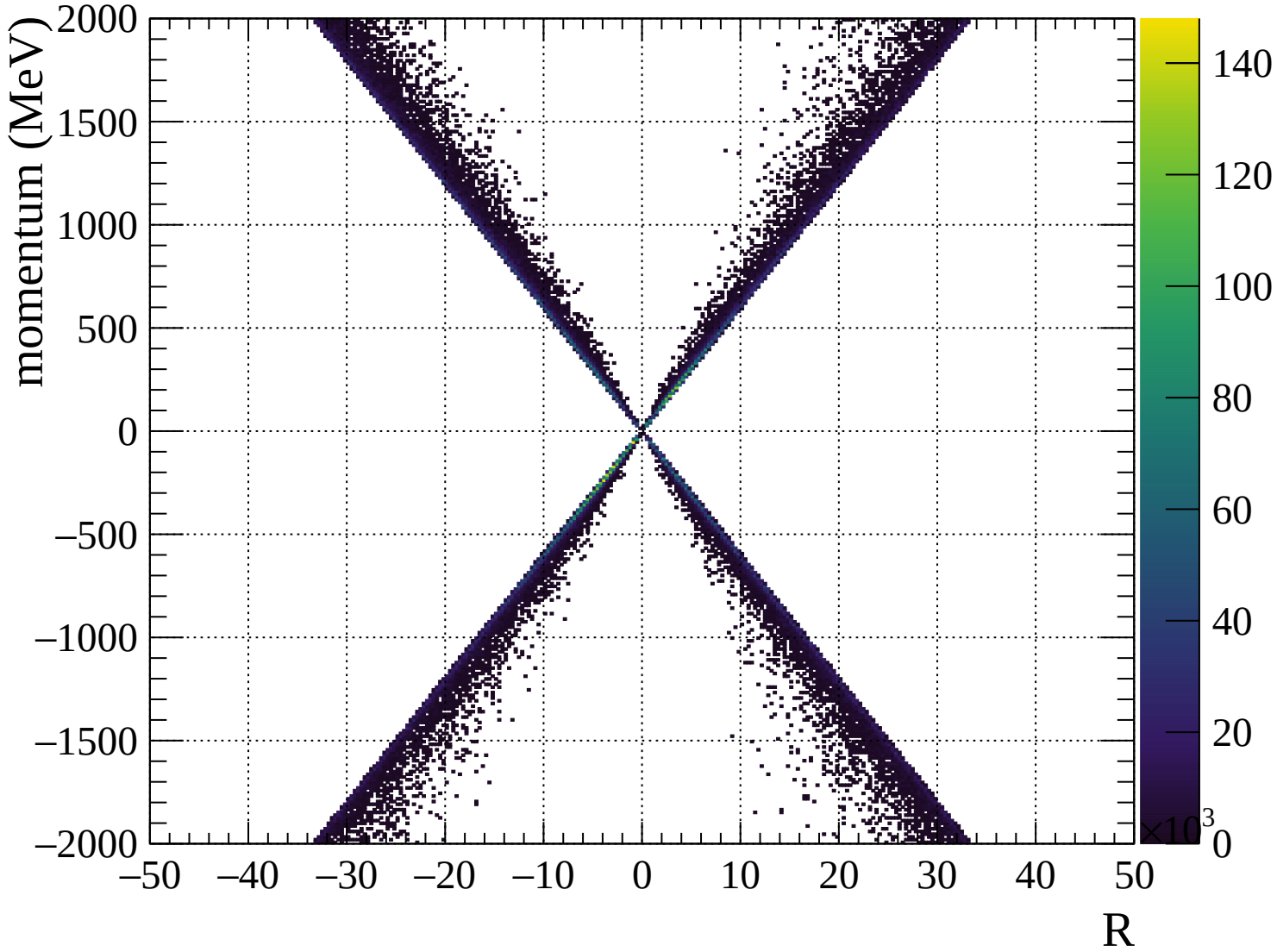


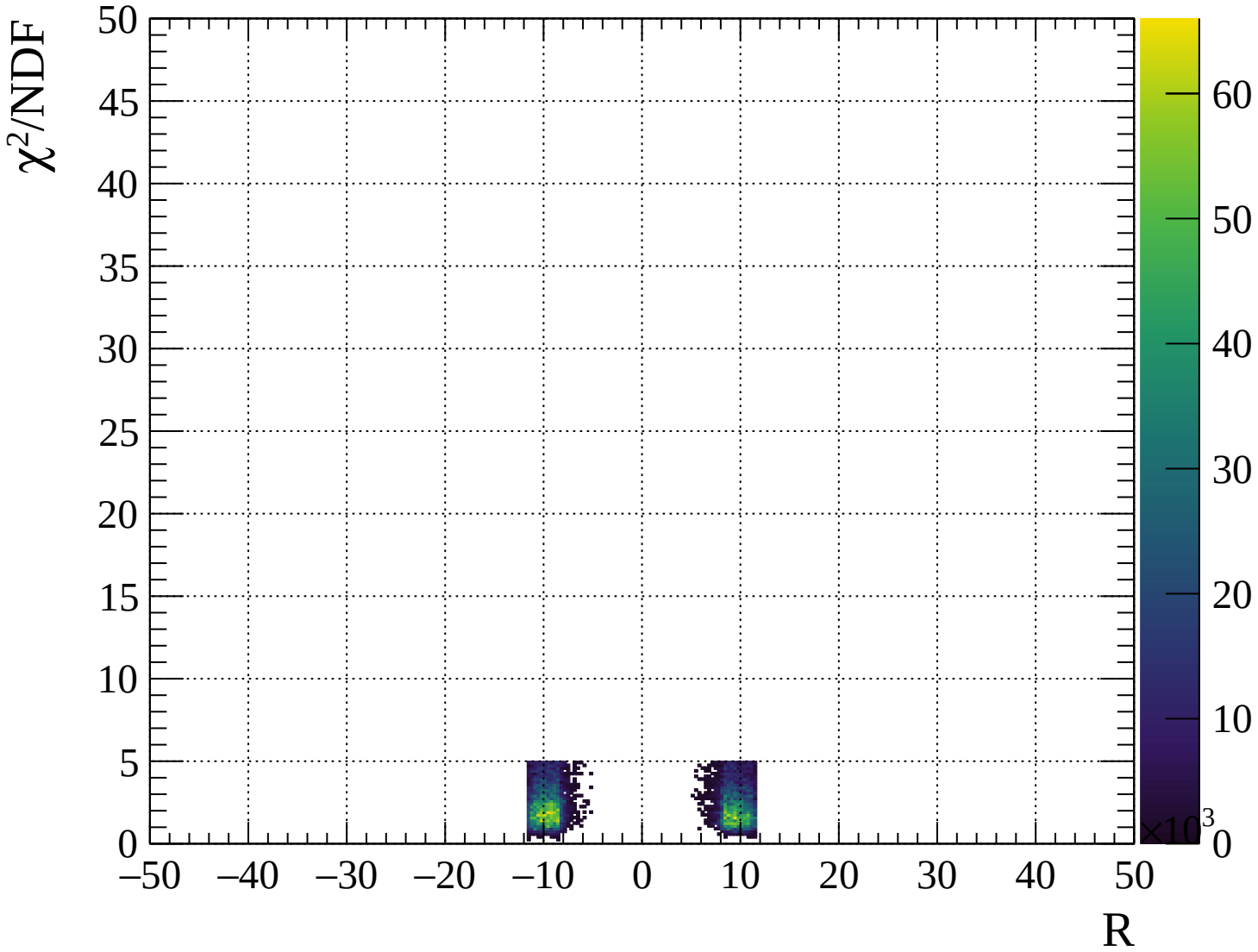




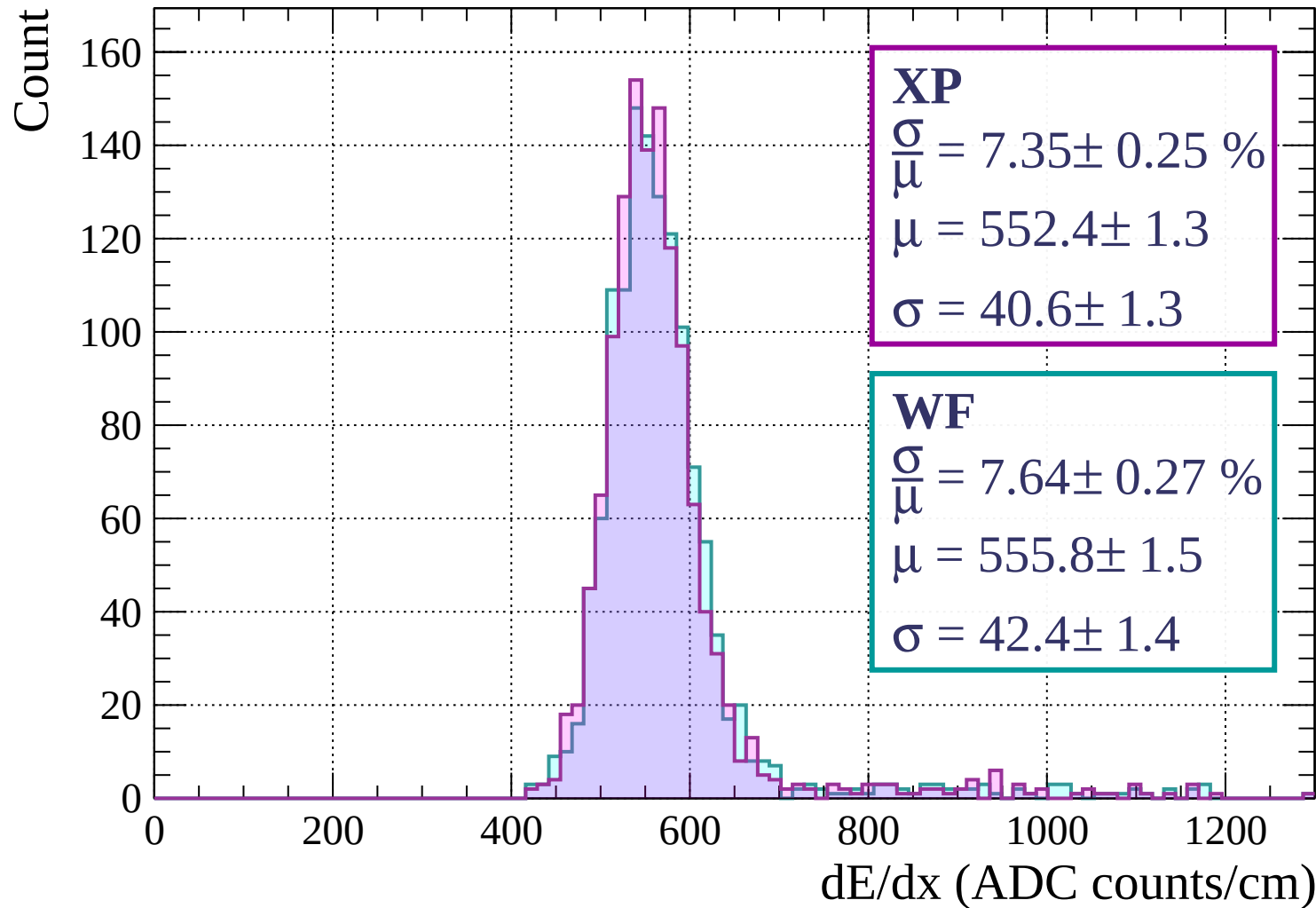




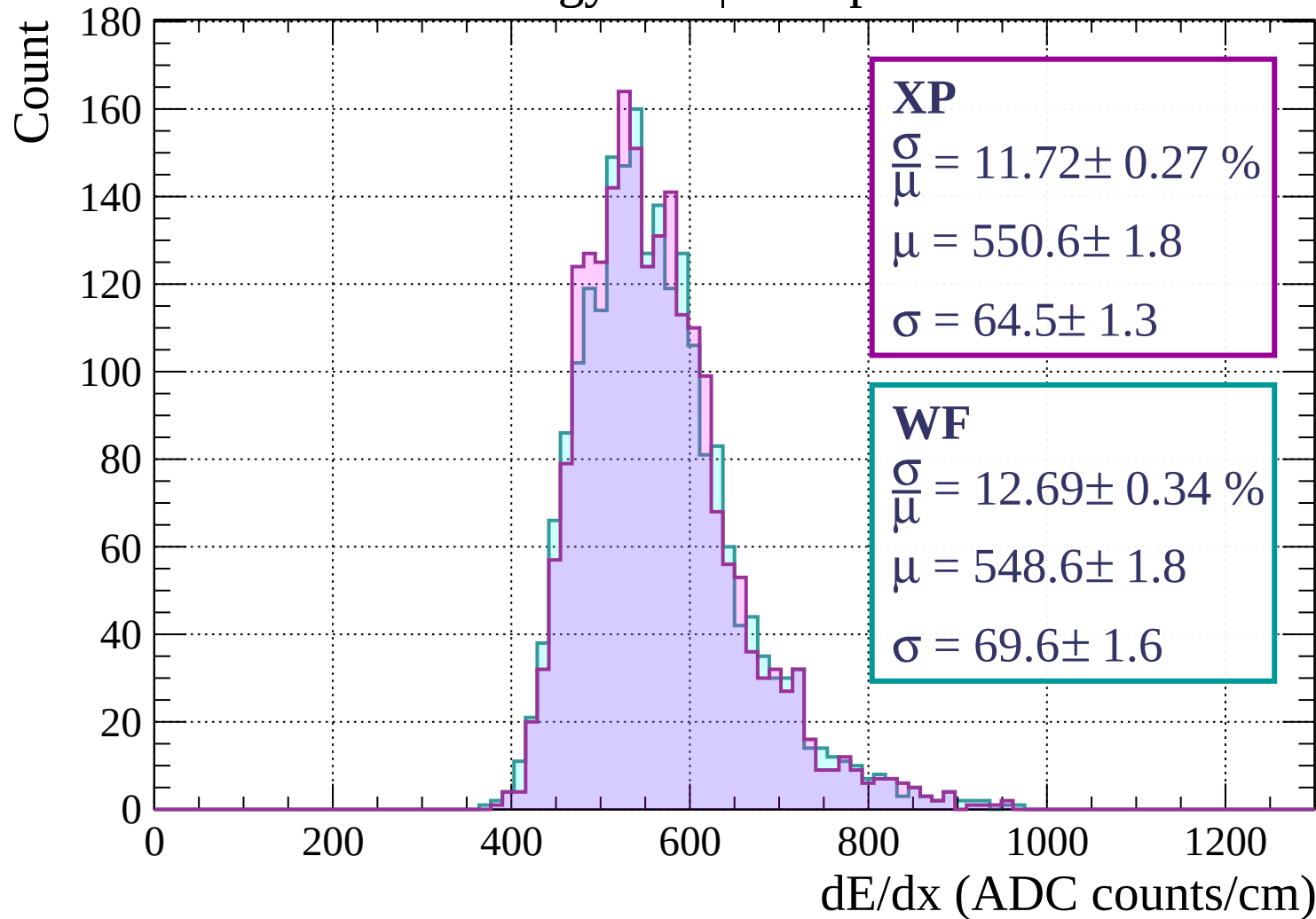




Energy loss | $0 < p < 80$

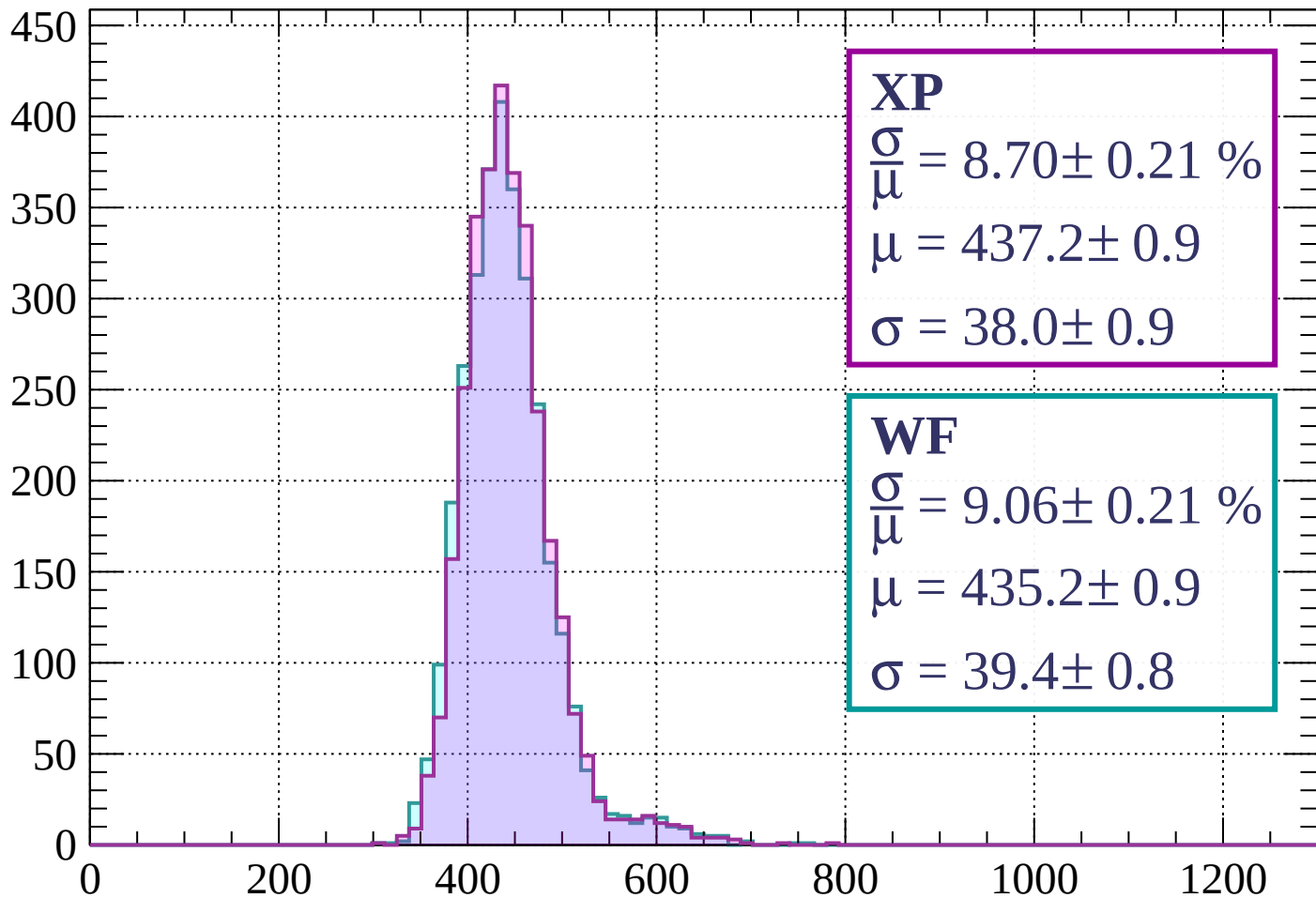


Energy loss | $80 < p < 160$



Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.21 \%$$

$$\mu = 437.2 \pm 0.9$$

$$\sigma = 38.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.21 \%$$

$$\mu = 435.2 \pm 0.9$$

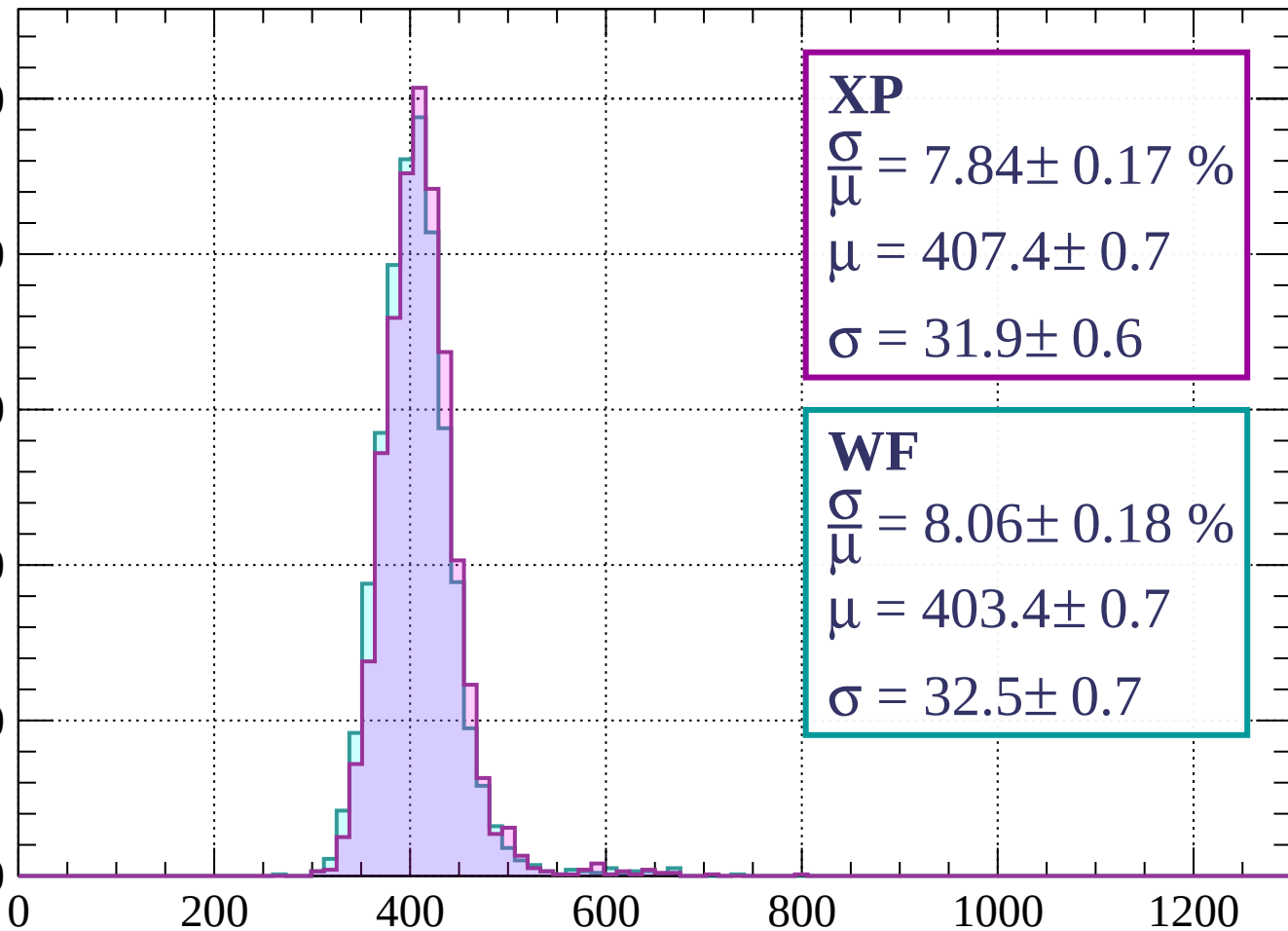
$$\sigma = 39.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.17 \%$$

$$\mu = 407.4 \pm 0.7$$

$$\sigma = 31.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.06 \pm 0.18 \%$$

$$\mu = 403.4 \pm 0.7$$

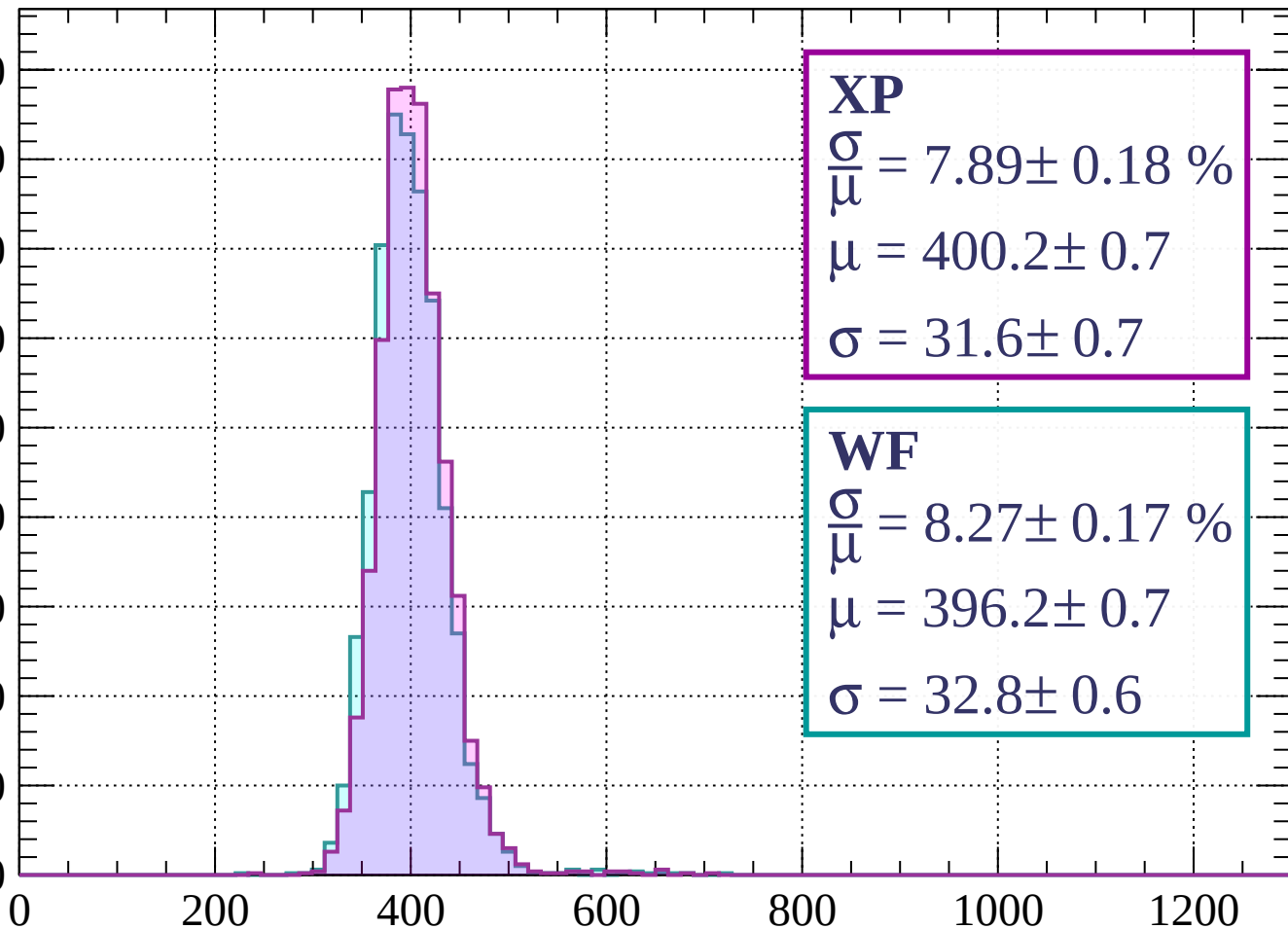
$$\sigma = 32.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.18 \%$$

$$\mu = 400.2 \pm 0.7$$

$$\sigma = 31.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.17 \%$$

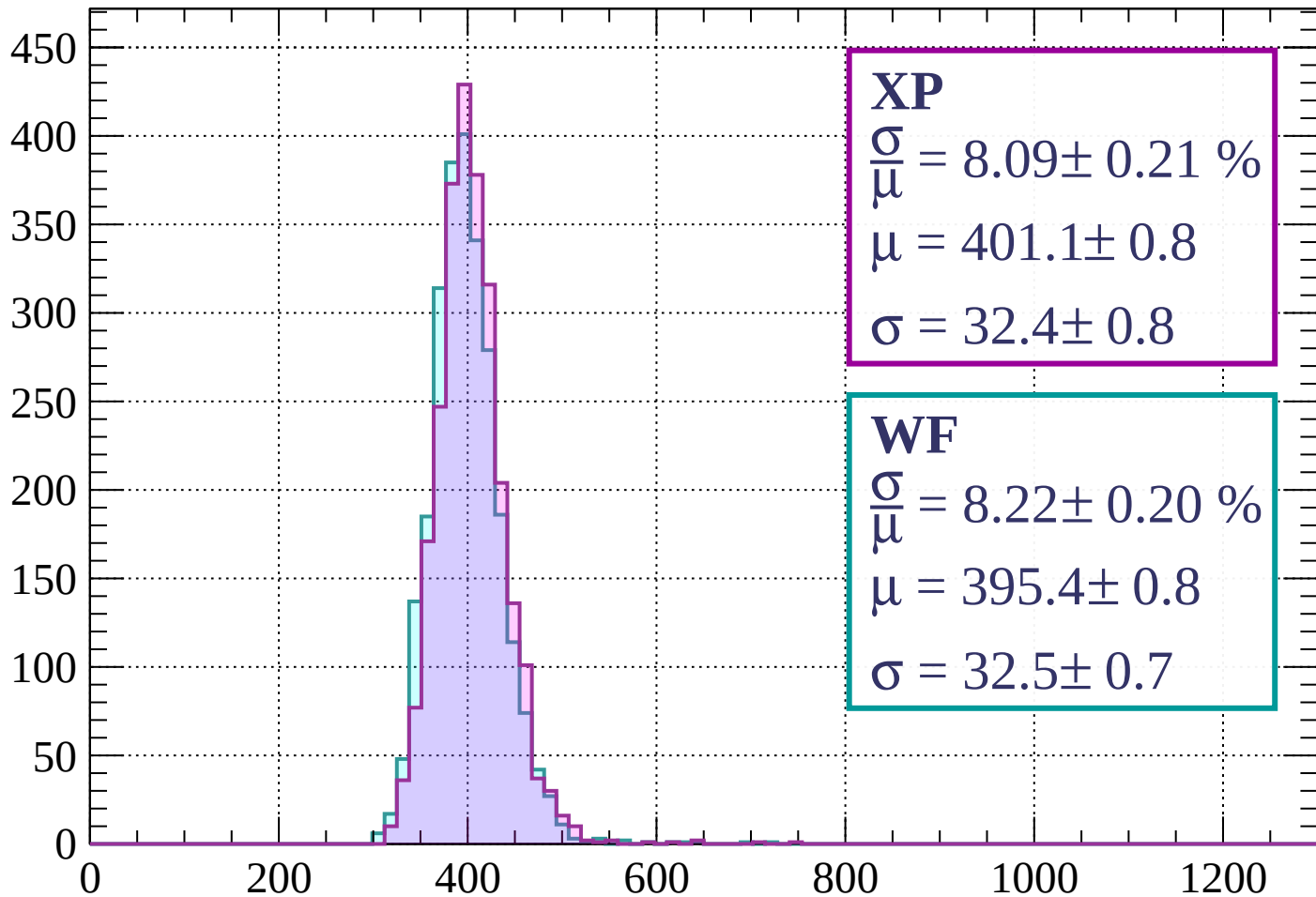
$$\mu = 396.2 \pm 0.7$$

$$\sigma = 32.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.21 \%$$

$$\mu = 401.1 \pm 0.8$$

$$\sigma = 32.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.20 \%$$

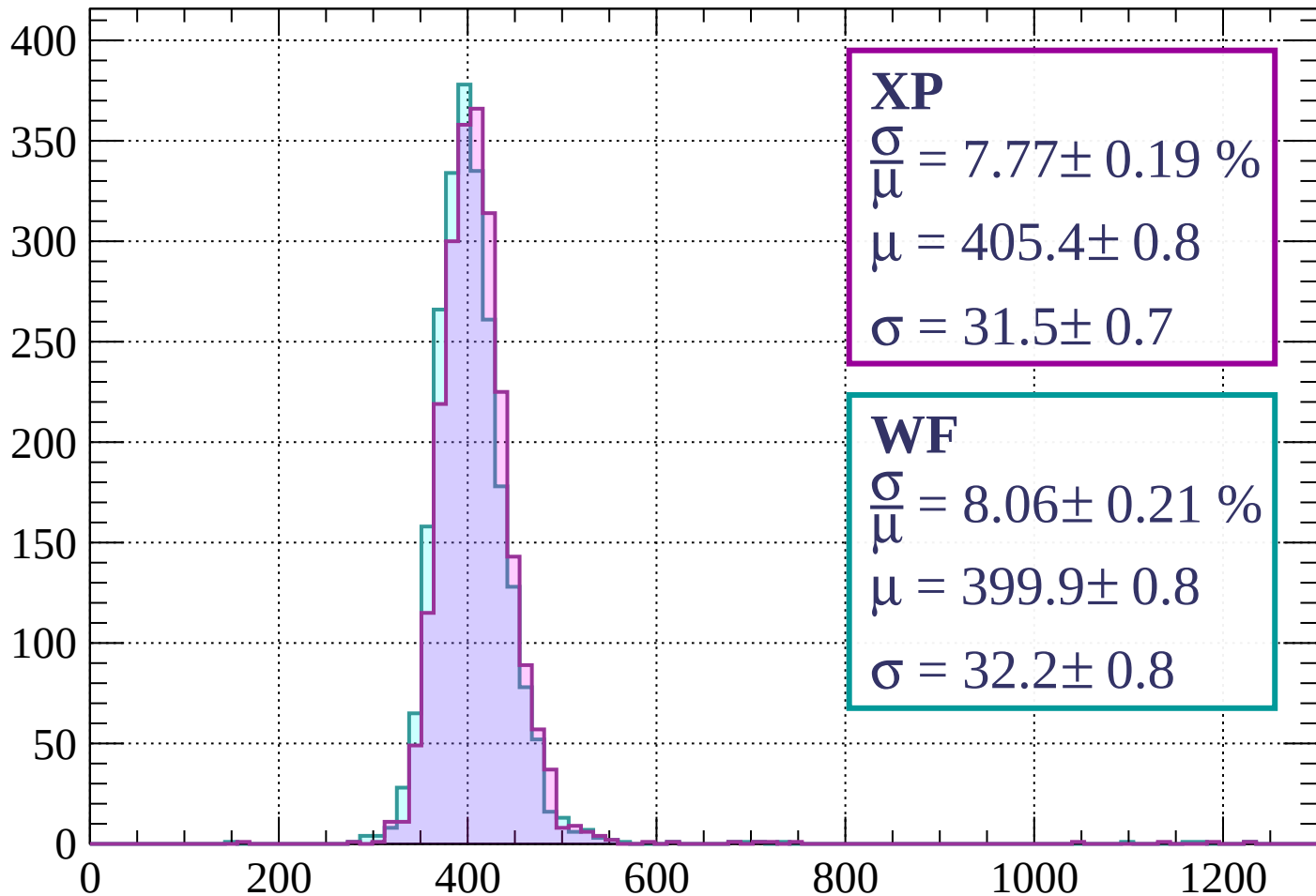
$$\mu = 395.4 \pm 0.8$$

$$\sigma = 32.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.77 \pm 0.19 \%$$

$$\mu = 405.4 \pm 0.8$$

$$\sigma = 31.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.06 \pm 0.21 \%$$

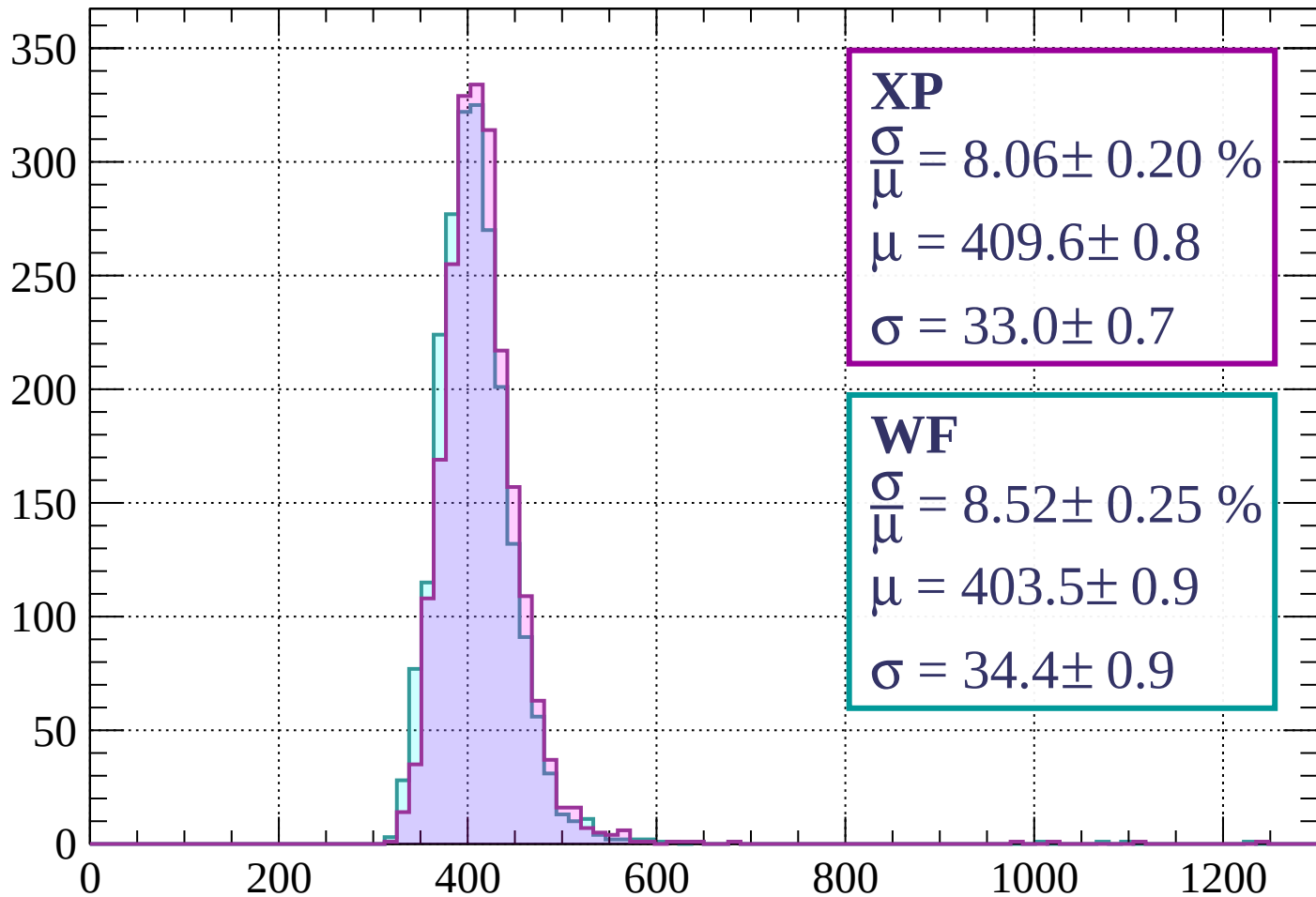
$$\mu = 399.9 \pm 0.8$$

$$\sigma = 32.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.20 \%$$

$$\mu = 409.6 \pm 0.8$$

$$\sigma = 33.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.52 \pm 0.25 \%$$

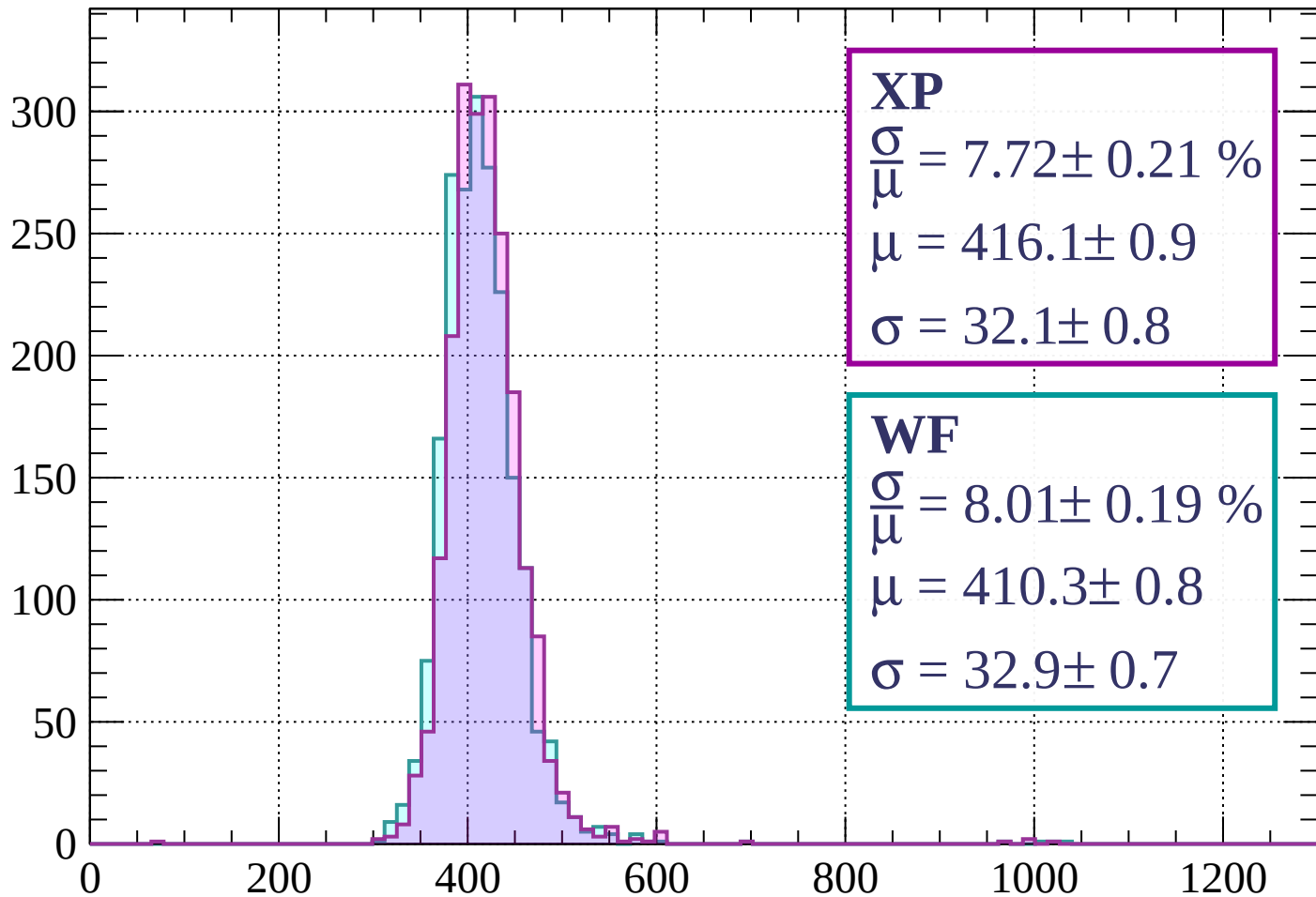
$$\mu = 403.5 \pm 0.9$$

$$\sigma = 34.4 \pm 0.9$$

dE/dx (ADC counts/cm)

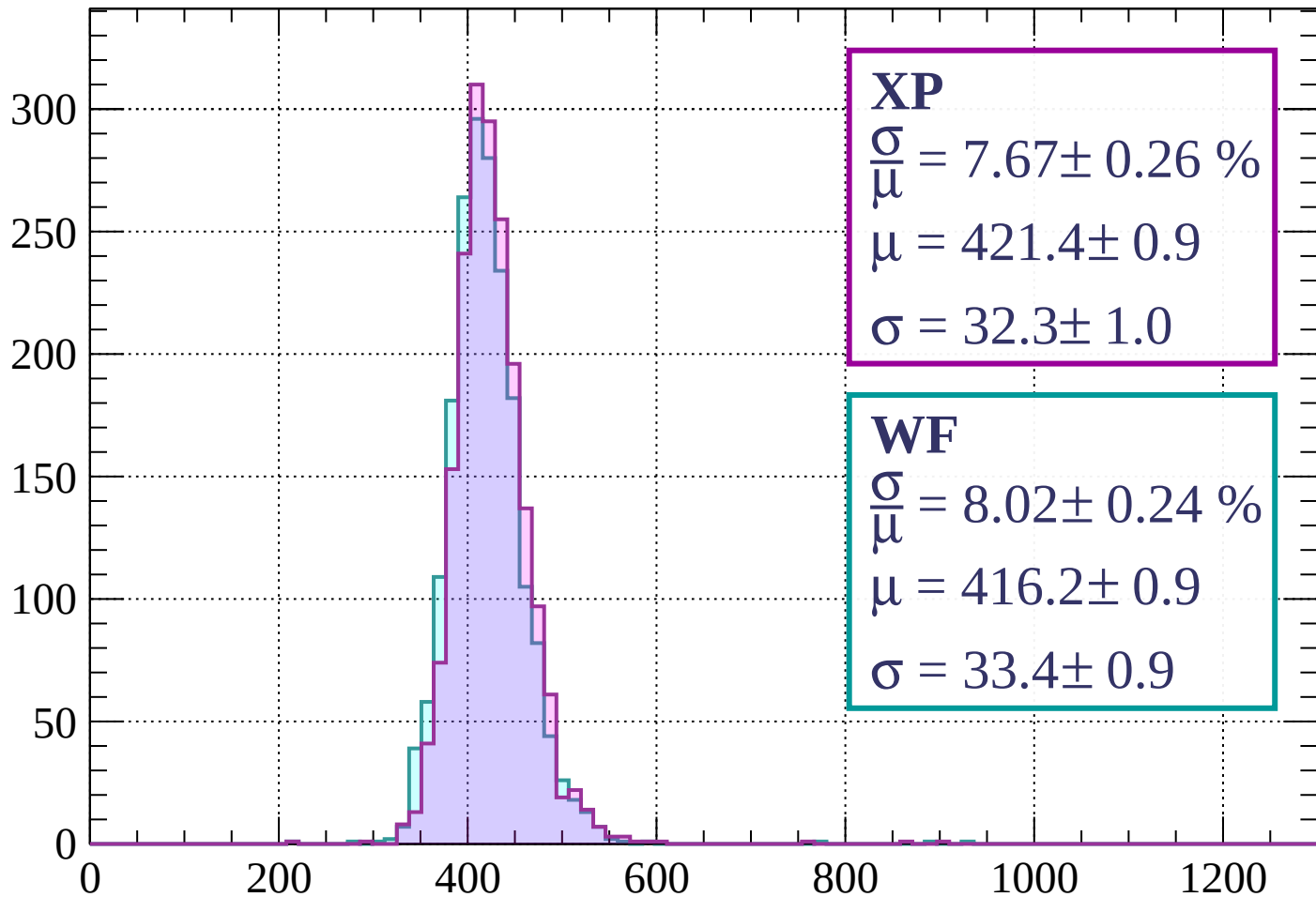
Energy loss | $640 < p < 720$

Count



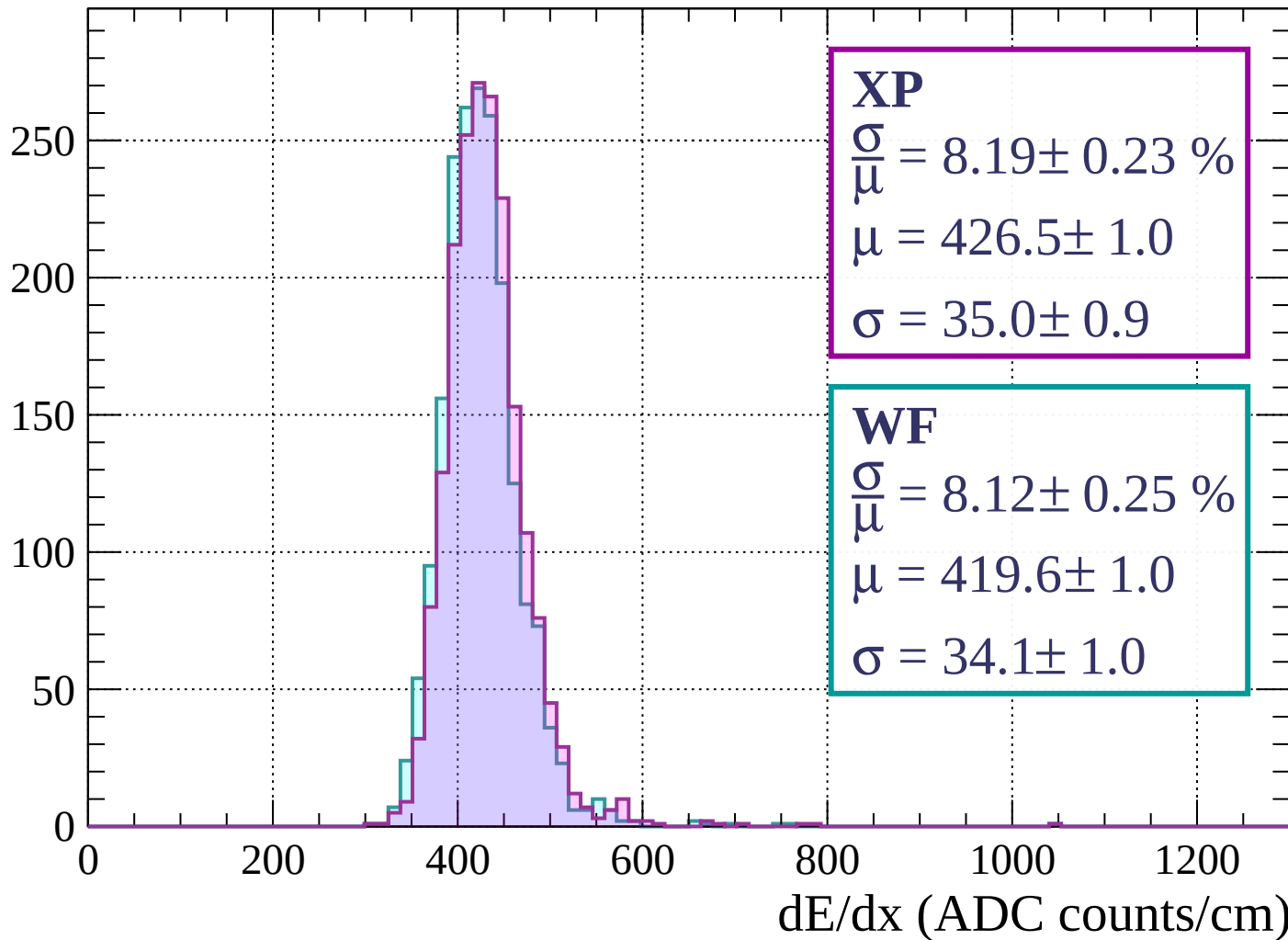
Energy loss | $720 < p < 800$

Count



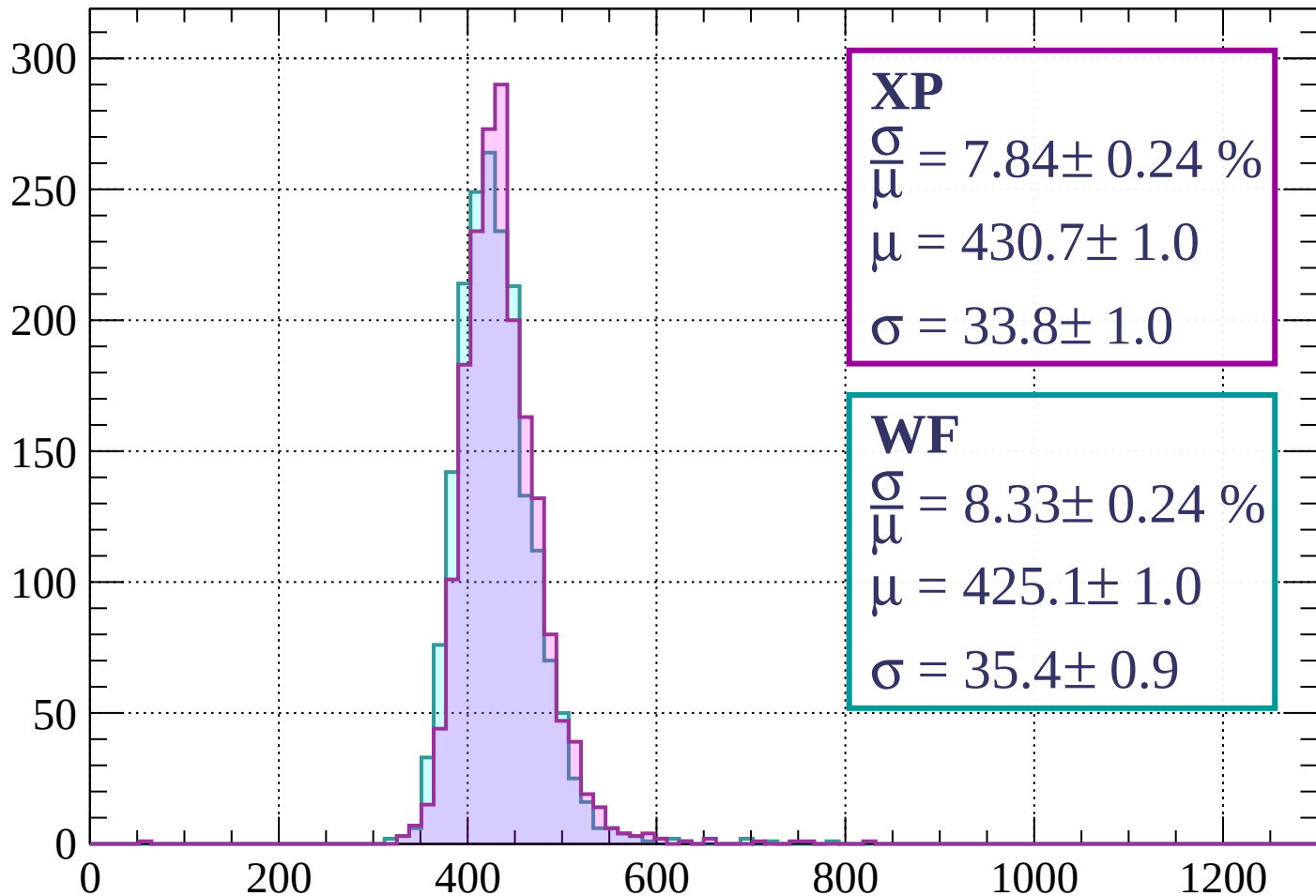
Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.24 \%$$

$$\mu = 430.7 \pm 1.0$$

$$\sigma = 33.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.33 \pm 0.24 \%$$

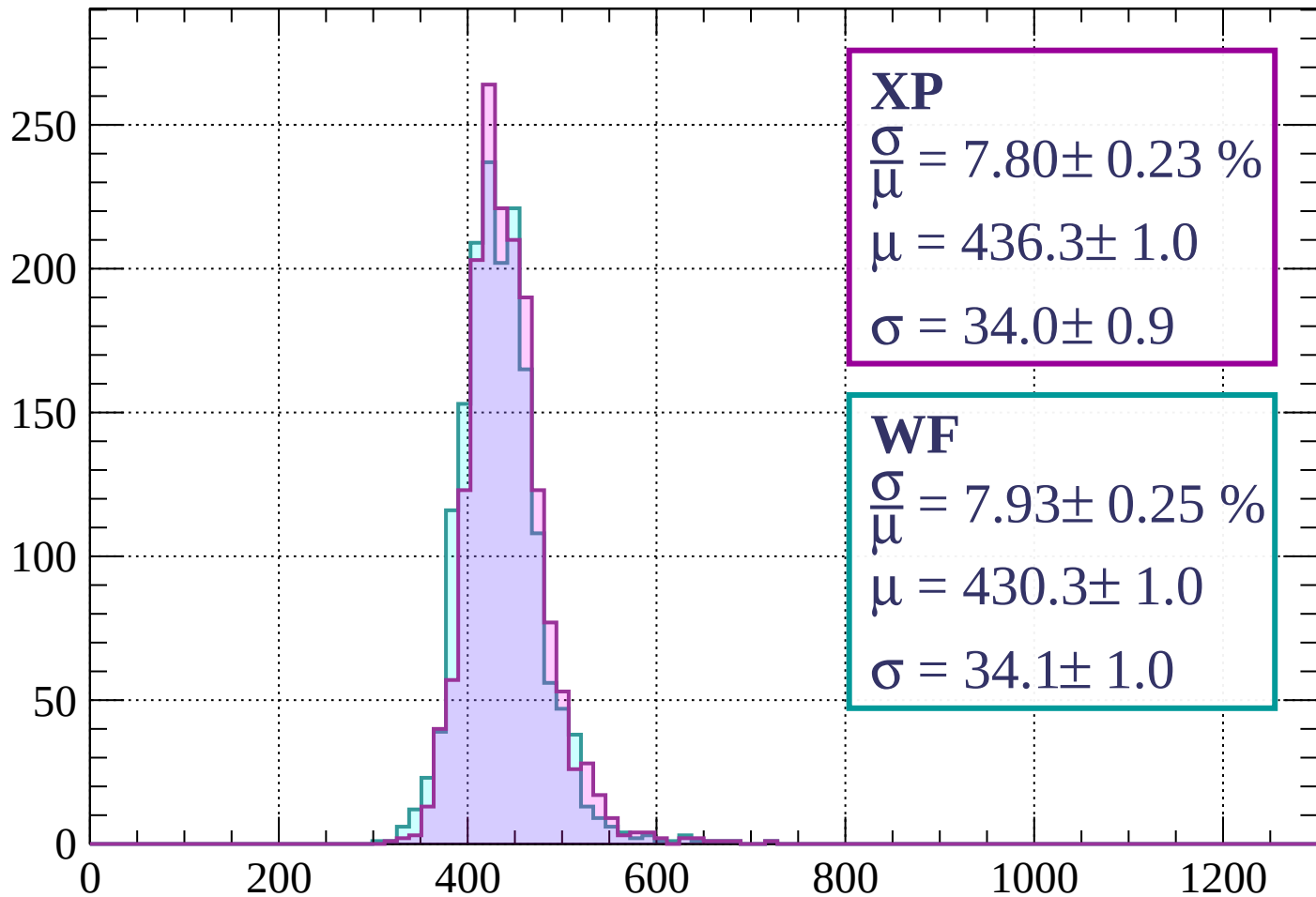
$$\mu = 425.1 \pm 1.0$$

$$\sigma = 35.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

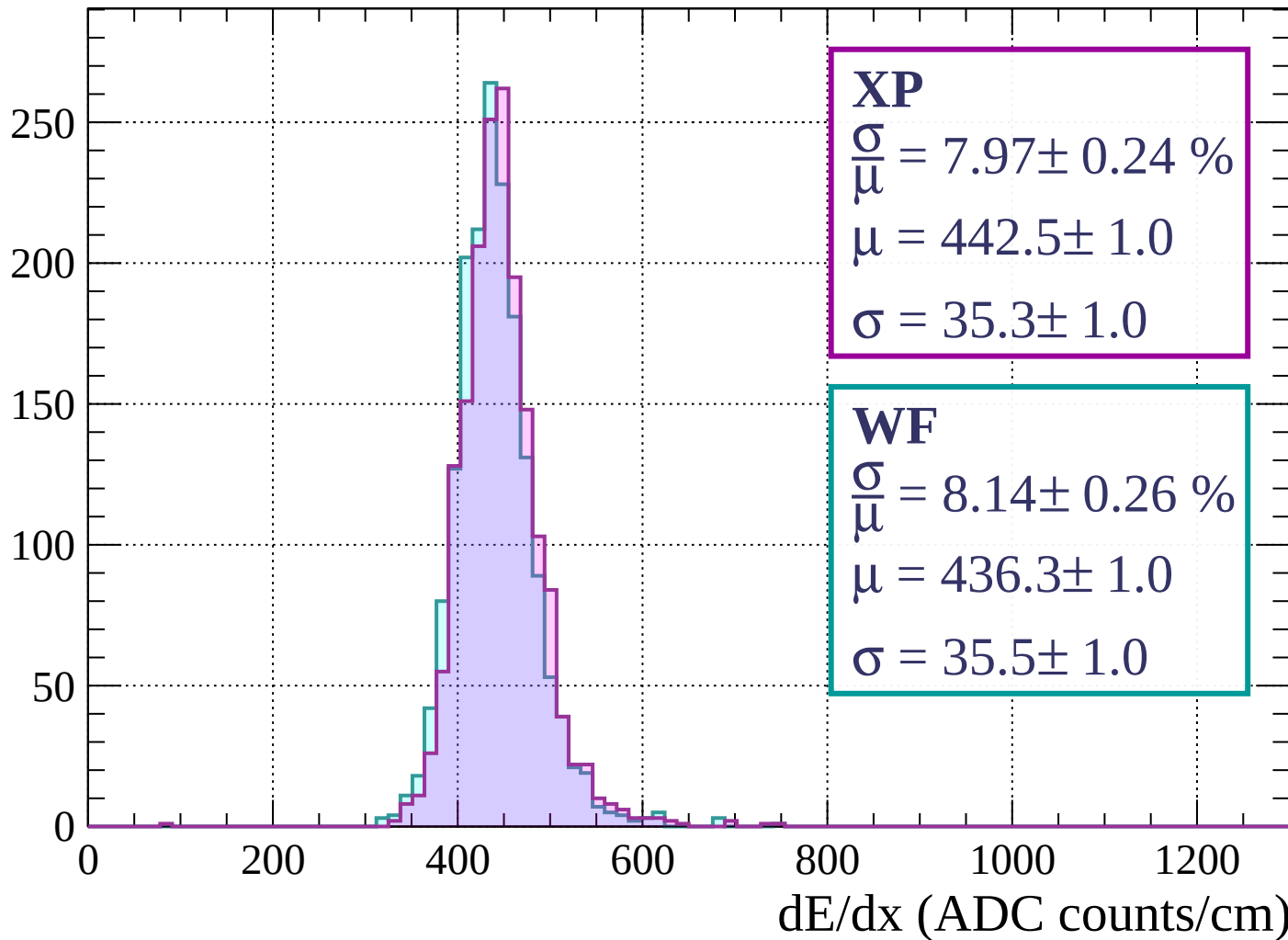
Count



dE/dx (ADC counts/cm)

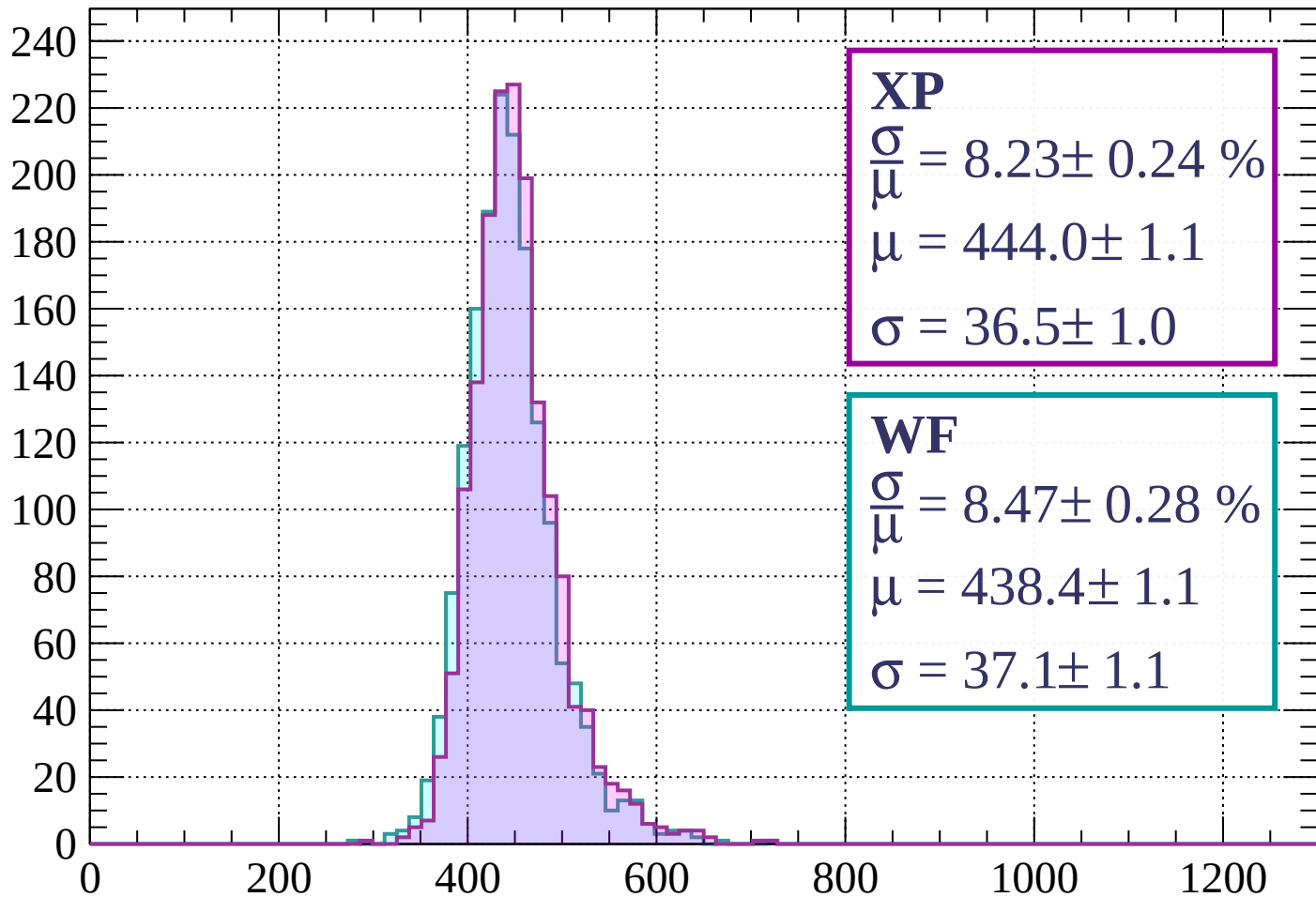
Energy loss | $1040 < p < 1120$

Count



Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.23 \pm 0.24 \%$$

$$\mu = 444.0 \pm 1.1$$

$$\sigma = 36.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.28 \%$$

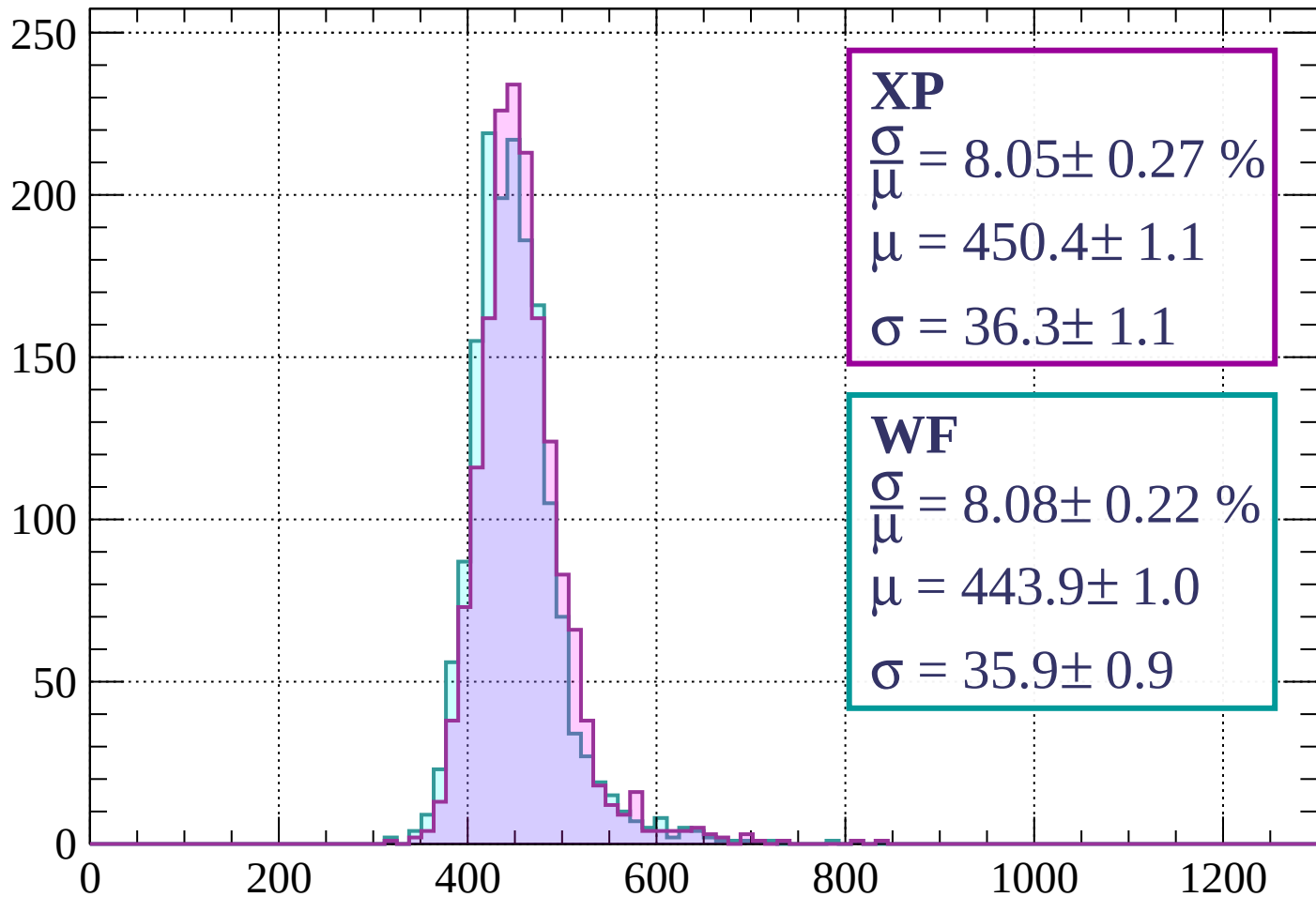
$$\mu = 438.4 \pm 1.1$$

$$\sigma = 37.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.27 \%$$

$$\mu = 450.4 \pm 1.1$$

$$\sigma = 36.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.22 \%$$

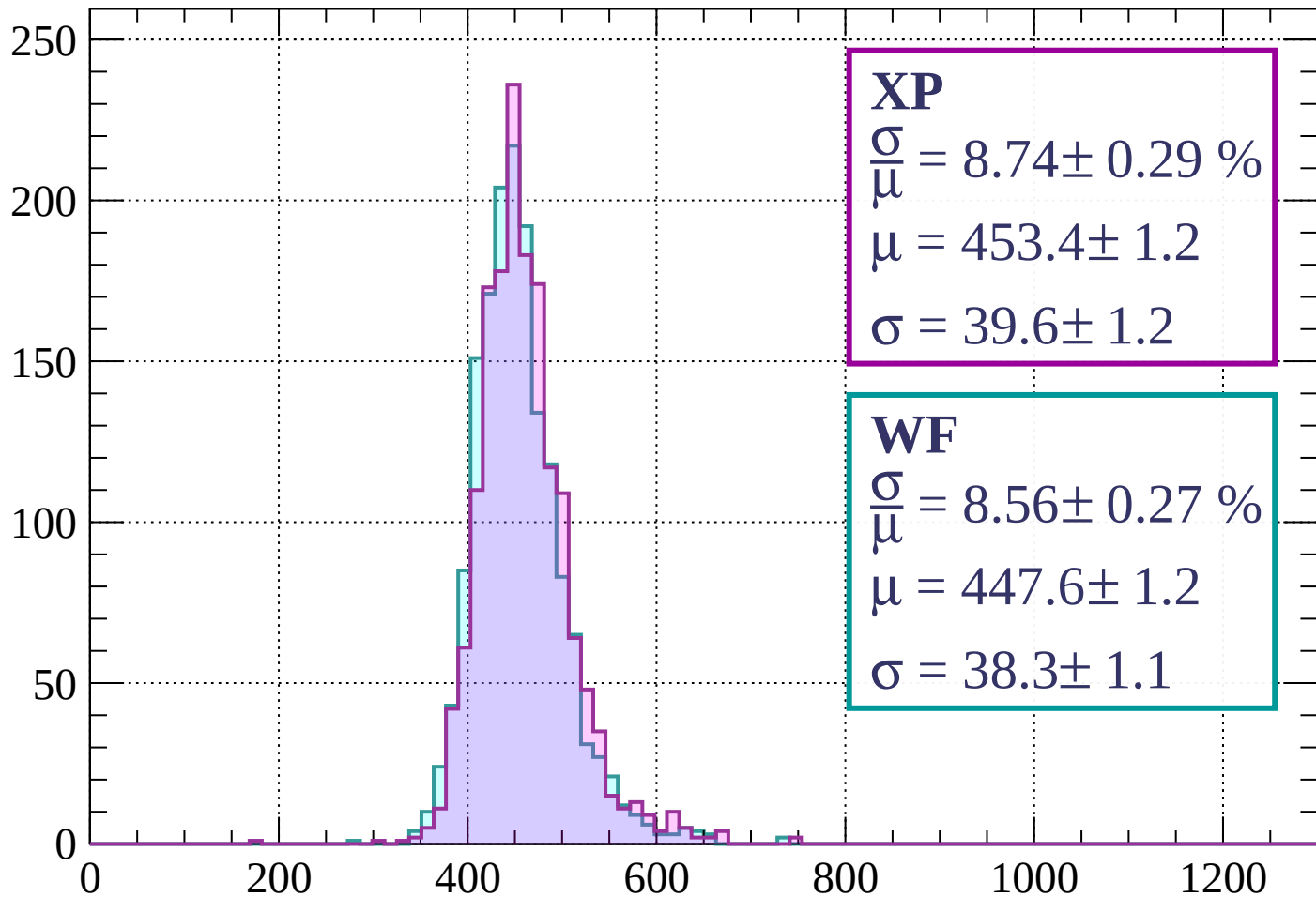
$$\mu = 443.9 \pm 1.0$$

$$\sigma = 35.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.29 \%$$

$$\mu = 453.4 \pm 1.2$$

$$\sigma = 39.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.56 \pm 0.27 \%$$

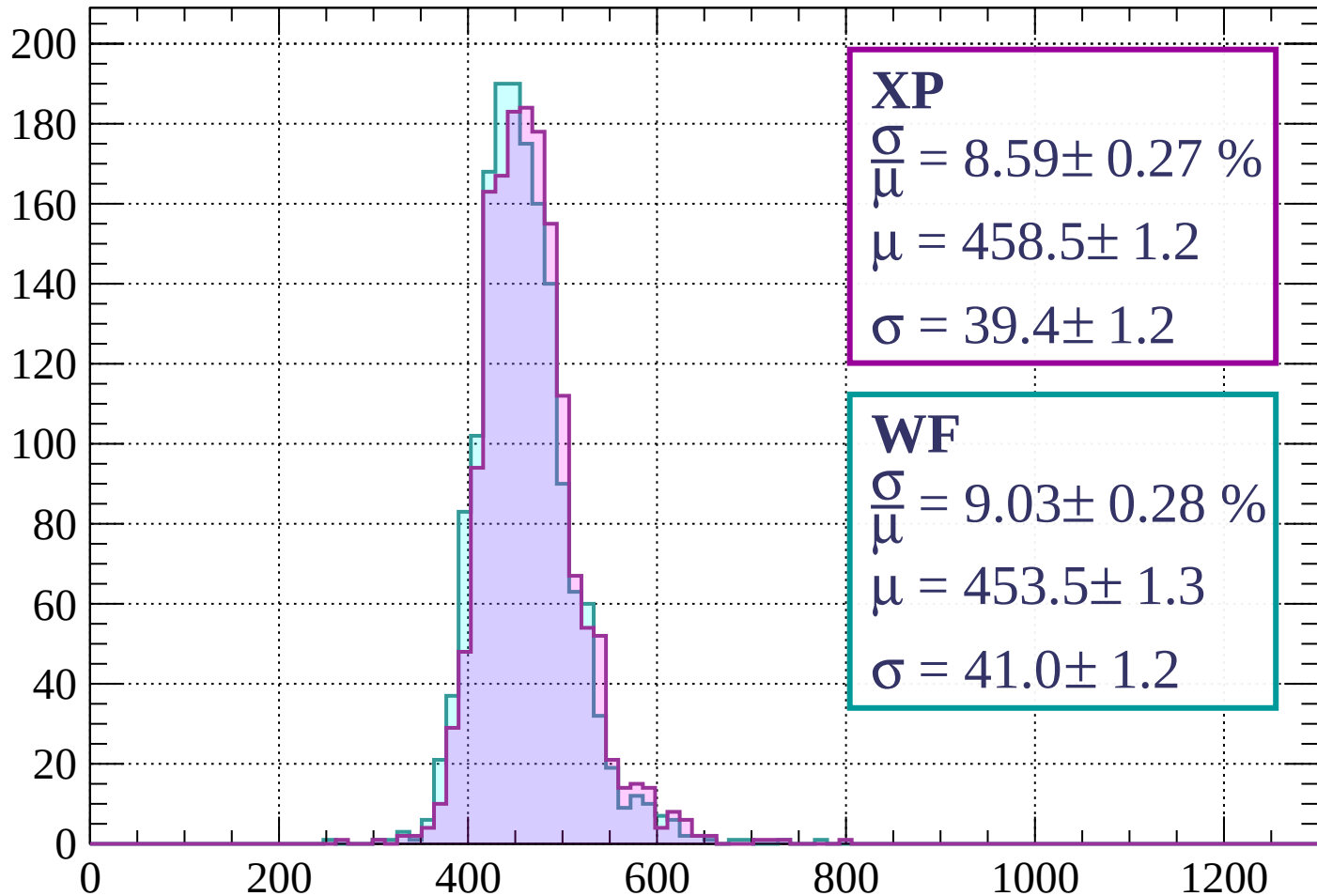
$$\mu = 447.6 \pm 1.2$$

$$\sigma = 38.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 8.59 \pm 0.27 \%$$

$$\mu = 458.5 \pm 1.2$$

$$\sigma = 39.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.28 \%$$

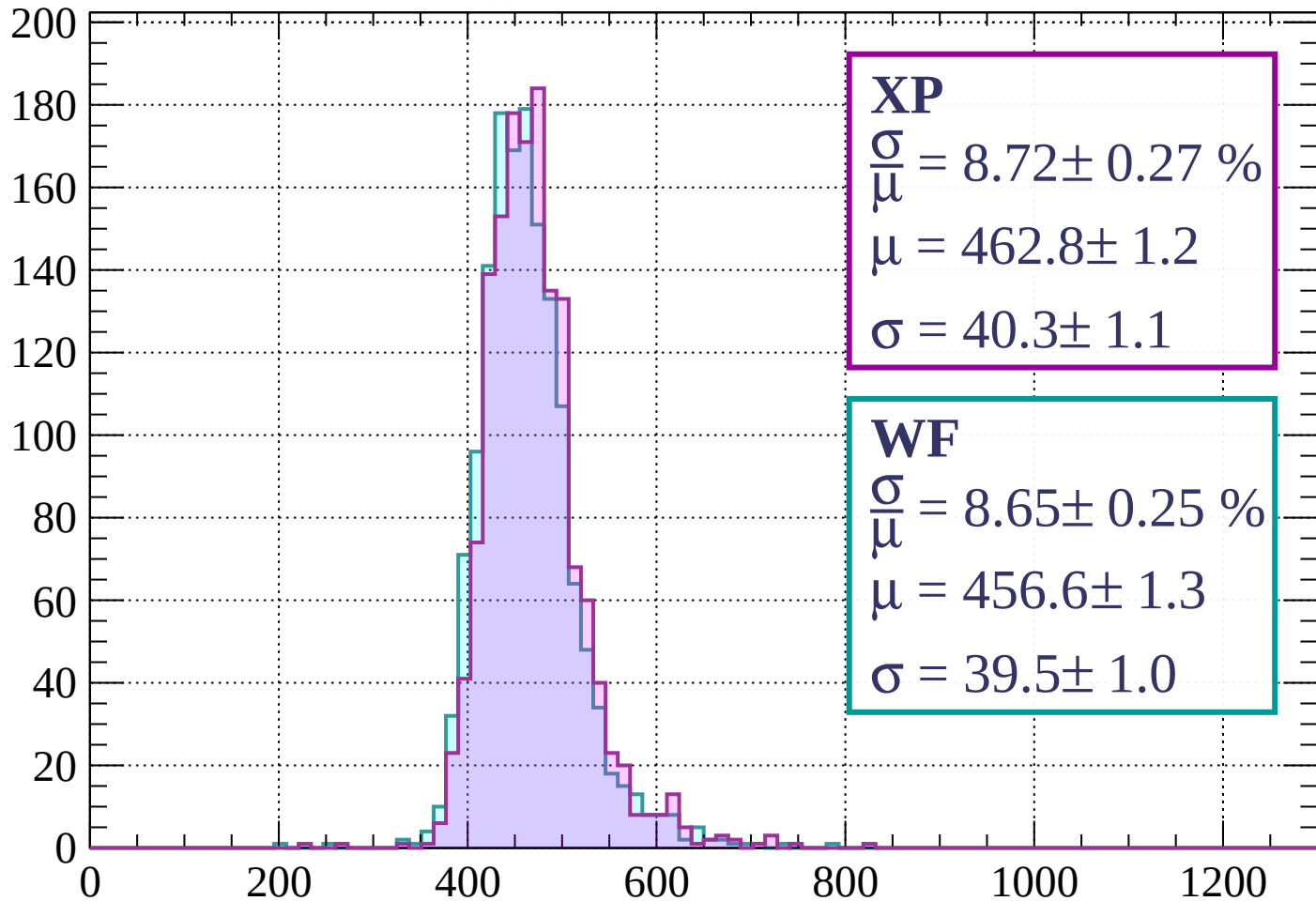
$$\mu = 453.5 \pm 1.3$$

$$\sigma = 41.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.27 \%$$

$$\mu = 462.8 \pm 1.2$$

$$\sigma = 40.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.65 \pm 0.25 \%$$

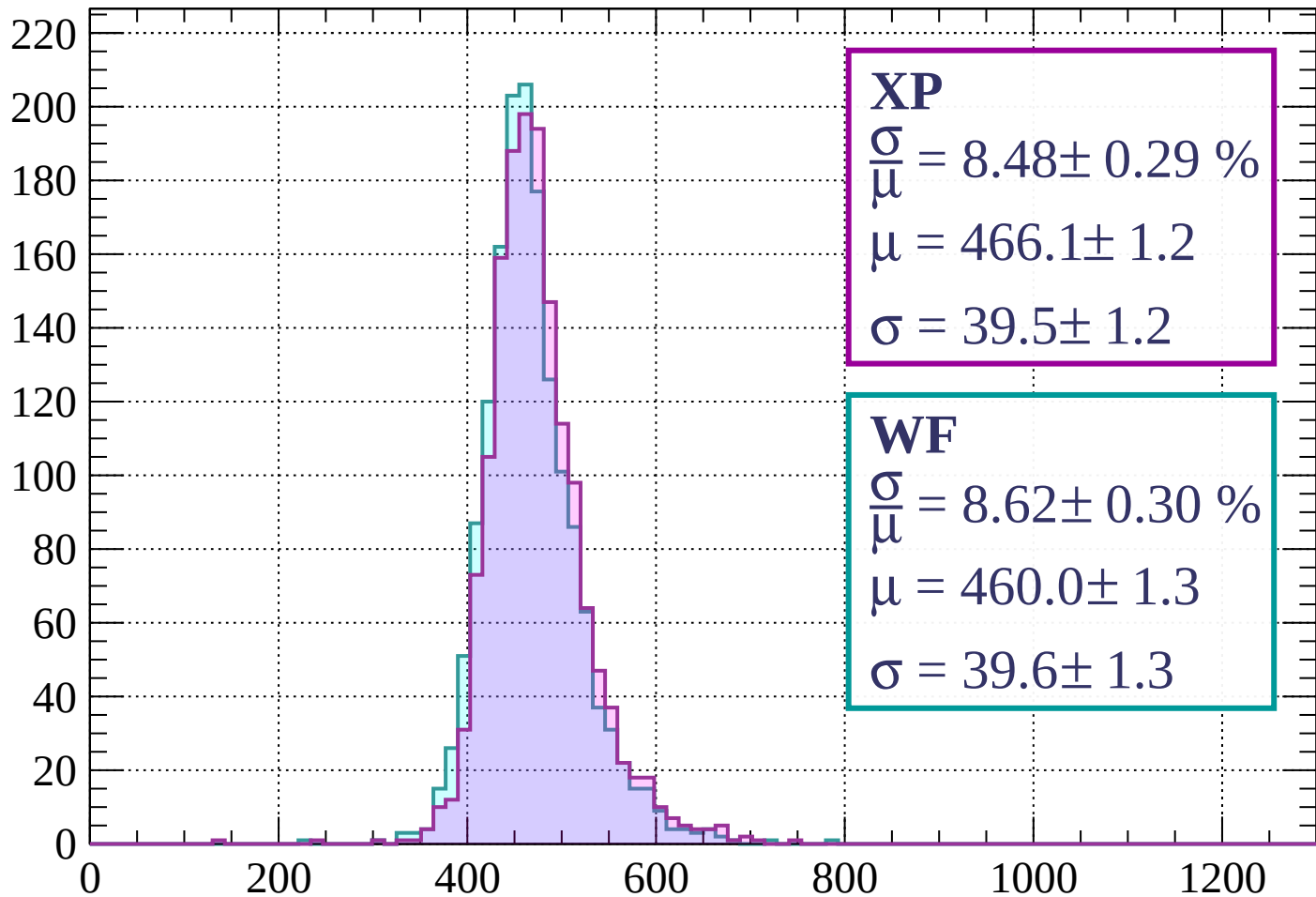
$$\mu = 456.6 \pm 1.3$$

$$\sigma = 39.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.48 \pm 0.29 \%$$

$$\mu = 466.1 \pm 1.2$$

$$\sigma = 39.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.30 \%$$

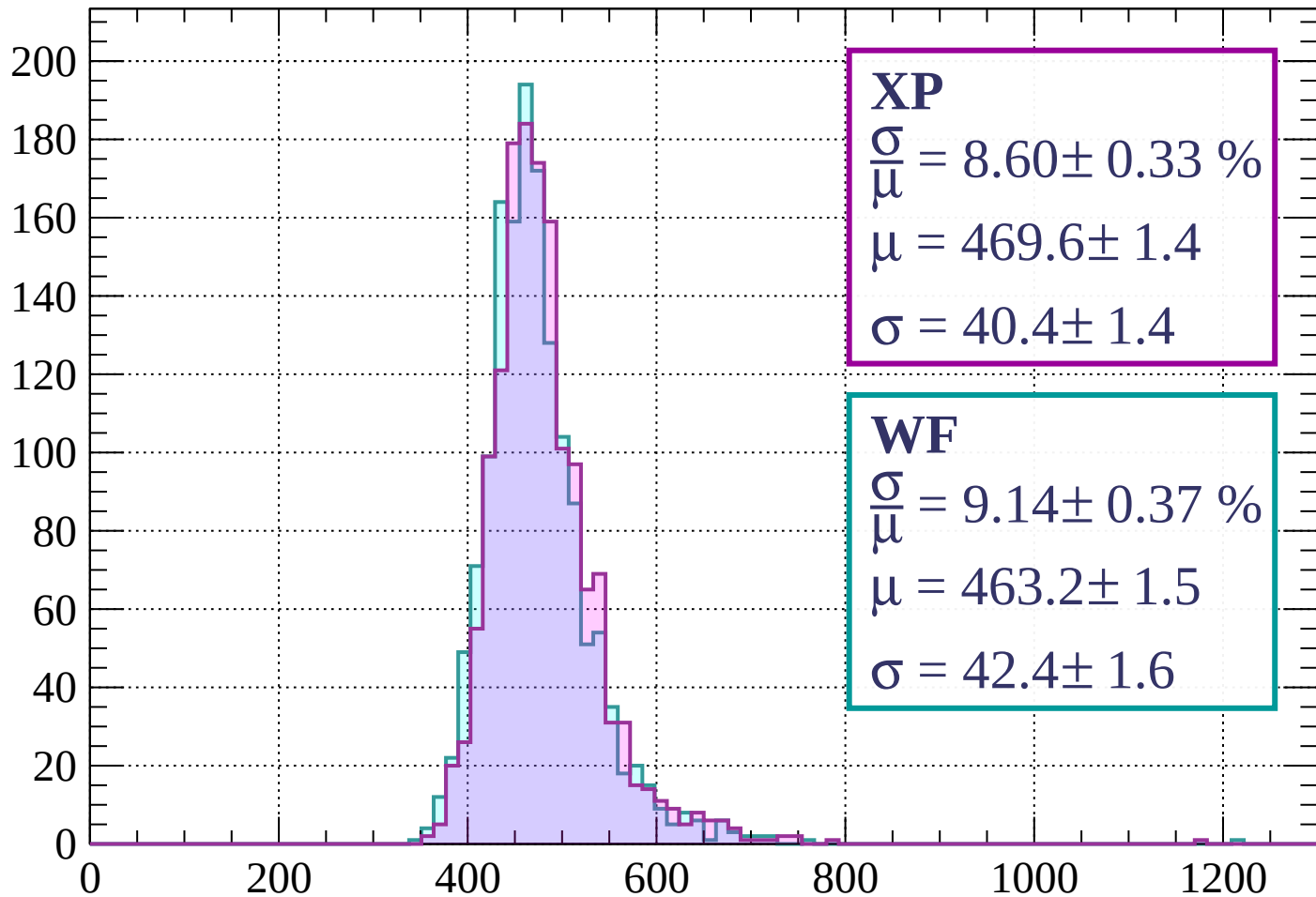
$$\mu = 460.0 \pm 1.3$$

$$\sigma = 39.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 8.60 \pm 0.33 \%$$

$$\mu = 469.6 \pm 1.4$$

$$\sigma = 40.4 \pm 1.4$$

WF

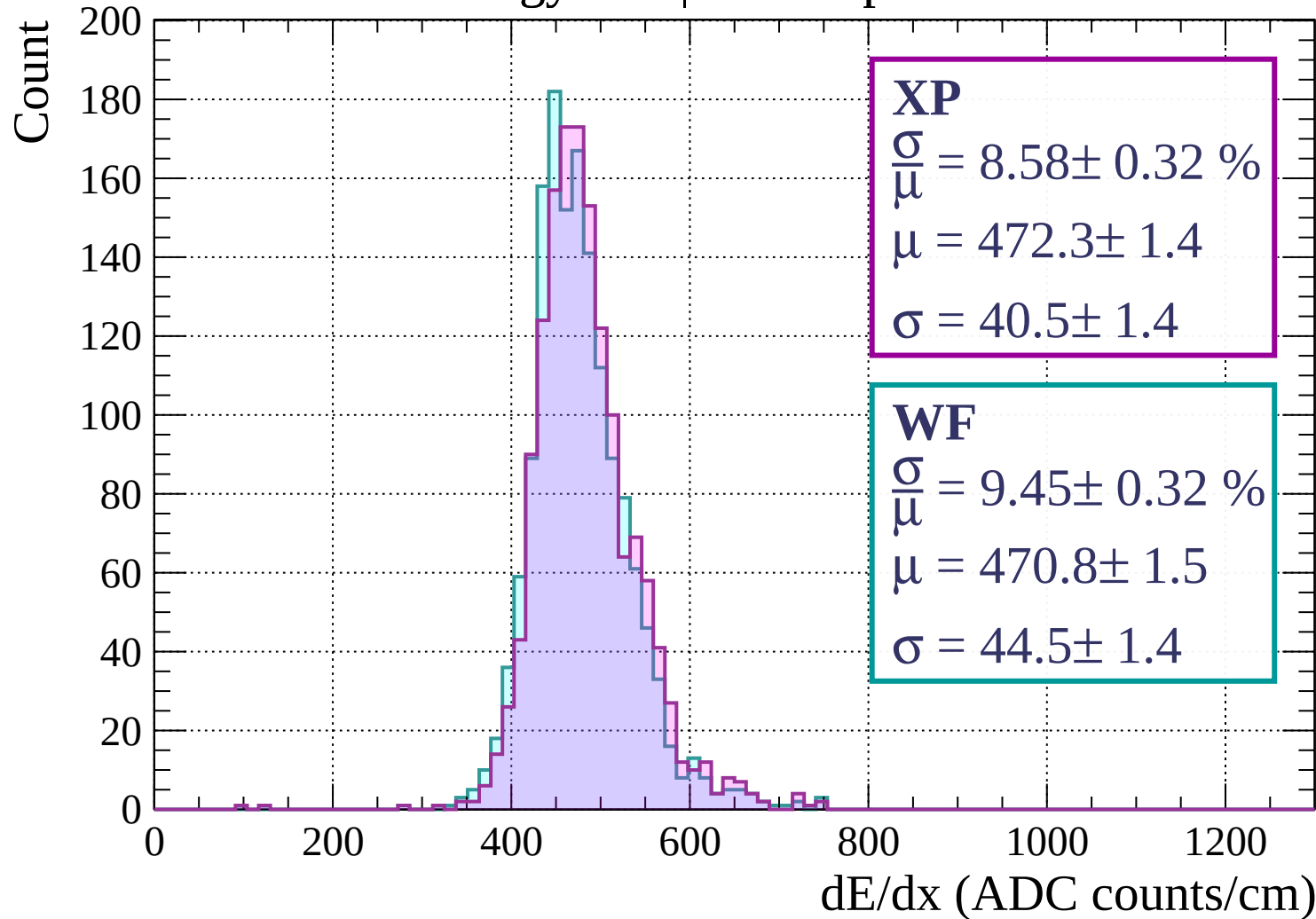
$$\frac{\sigma}{\mu} = 9.14 \pm 0.37 \%$$

$$\mu = 463.2 \pm 1.5$$

$$\sigma = 42.4 \pm 1.6$$

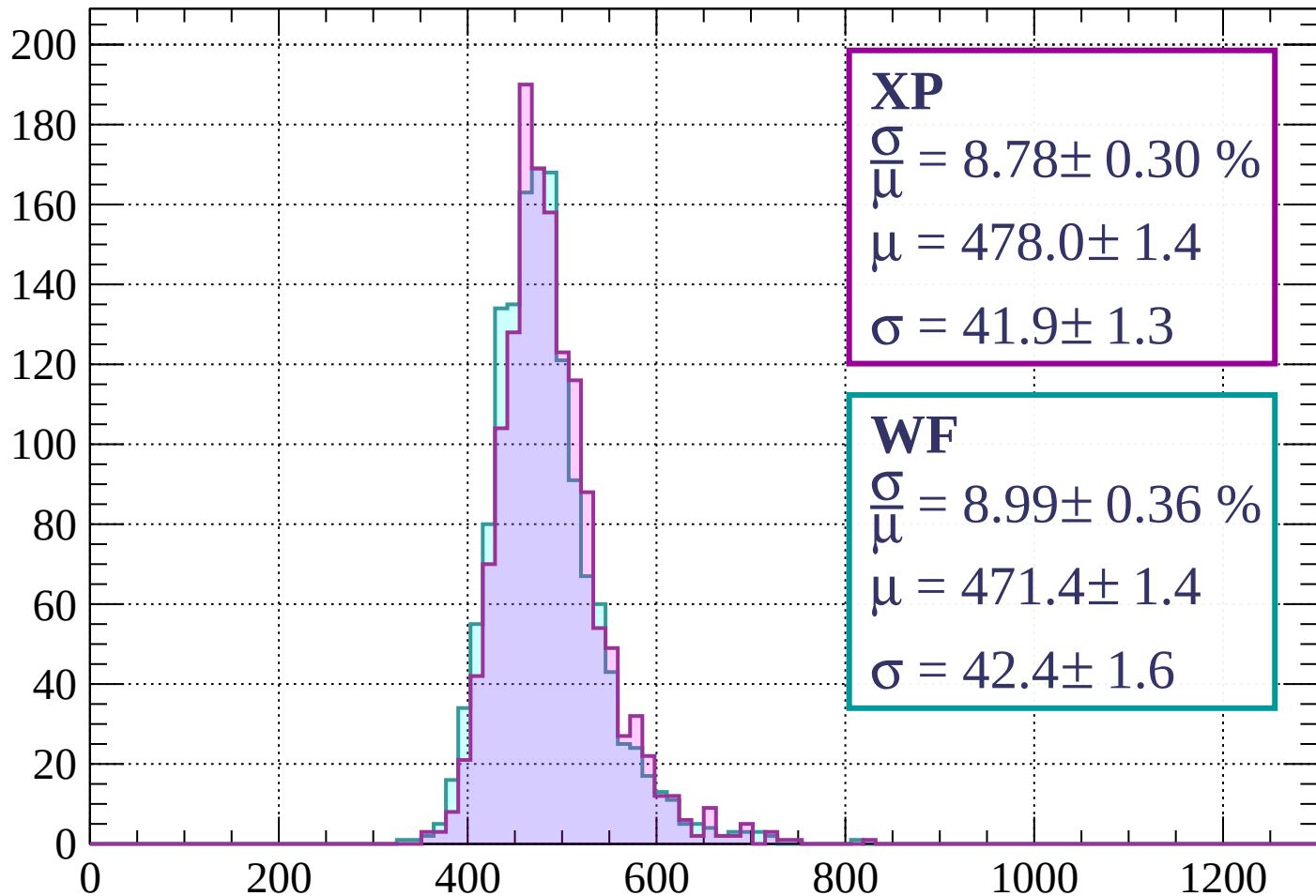
dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$



Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.30 \%$$

$$\mu = 478.0 \pm 1.4$$

$$\sigma = 41.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.36 \%$$

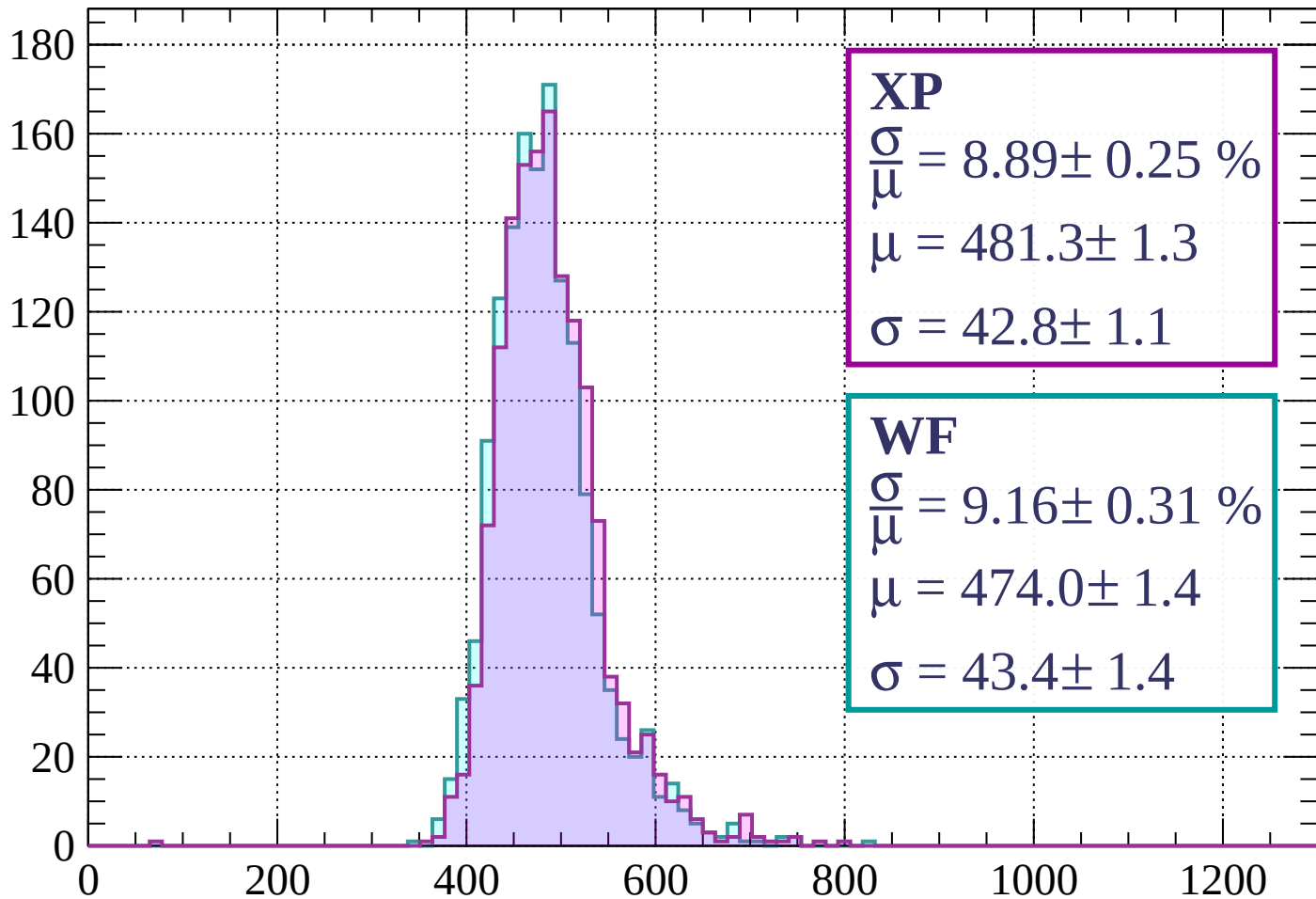
$$\mu = 471.4 \pm 1.4$$

$$\sigma = 42.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.25 \%$$

$$\mu = 481.3 \pm 1.3$$

$$\sigma = 42.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.16 \pm 0.31 \%$$

$$\mu = 474.0 \pm 1.4$$

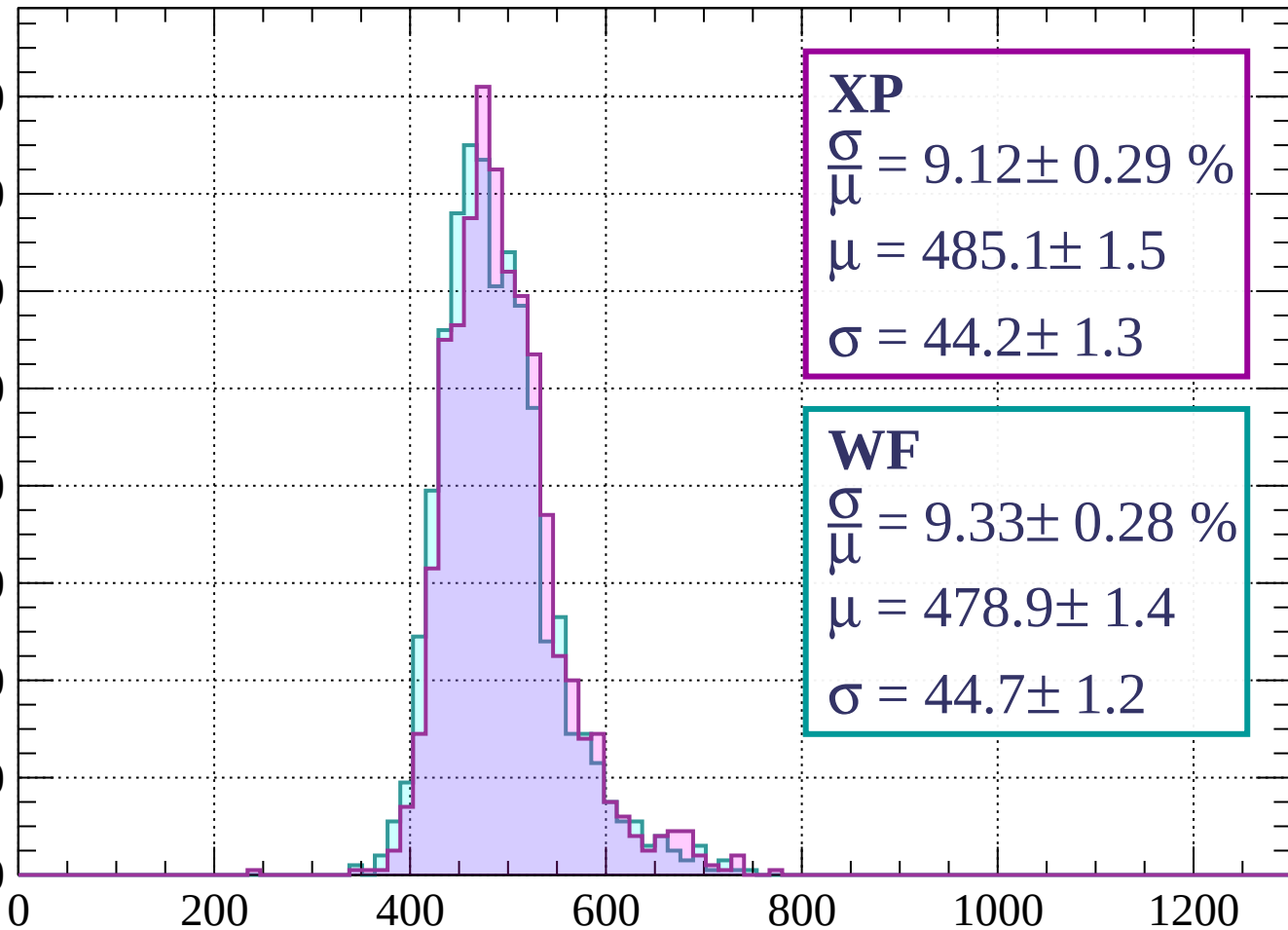
$$\sigma = 43.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.12 \pm 0.29 \%$$

$$\mu = 485.1 \pm 1.5$$

$$\sigma = 44.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.28 \%$$

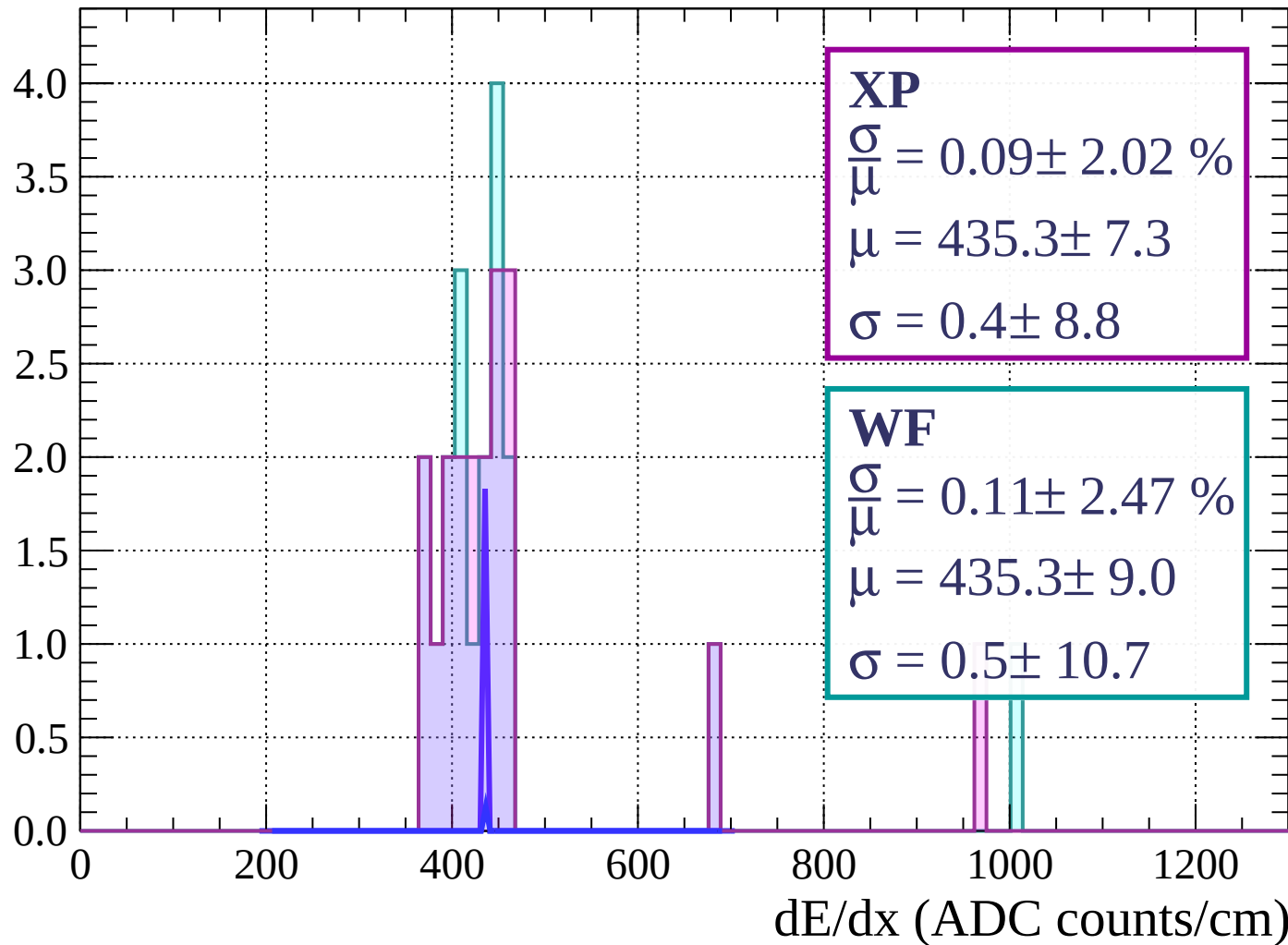
$$\mu = 478.9 \pm 1.4$$

$$\sigma = 44.7 \pm 1.2$$

dE/dx (ADC counts/cm)

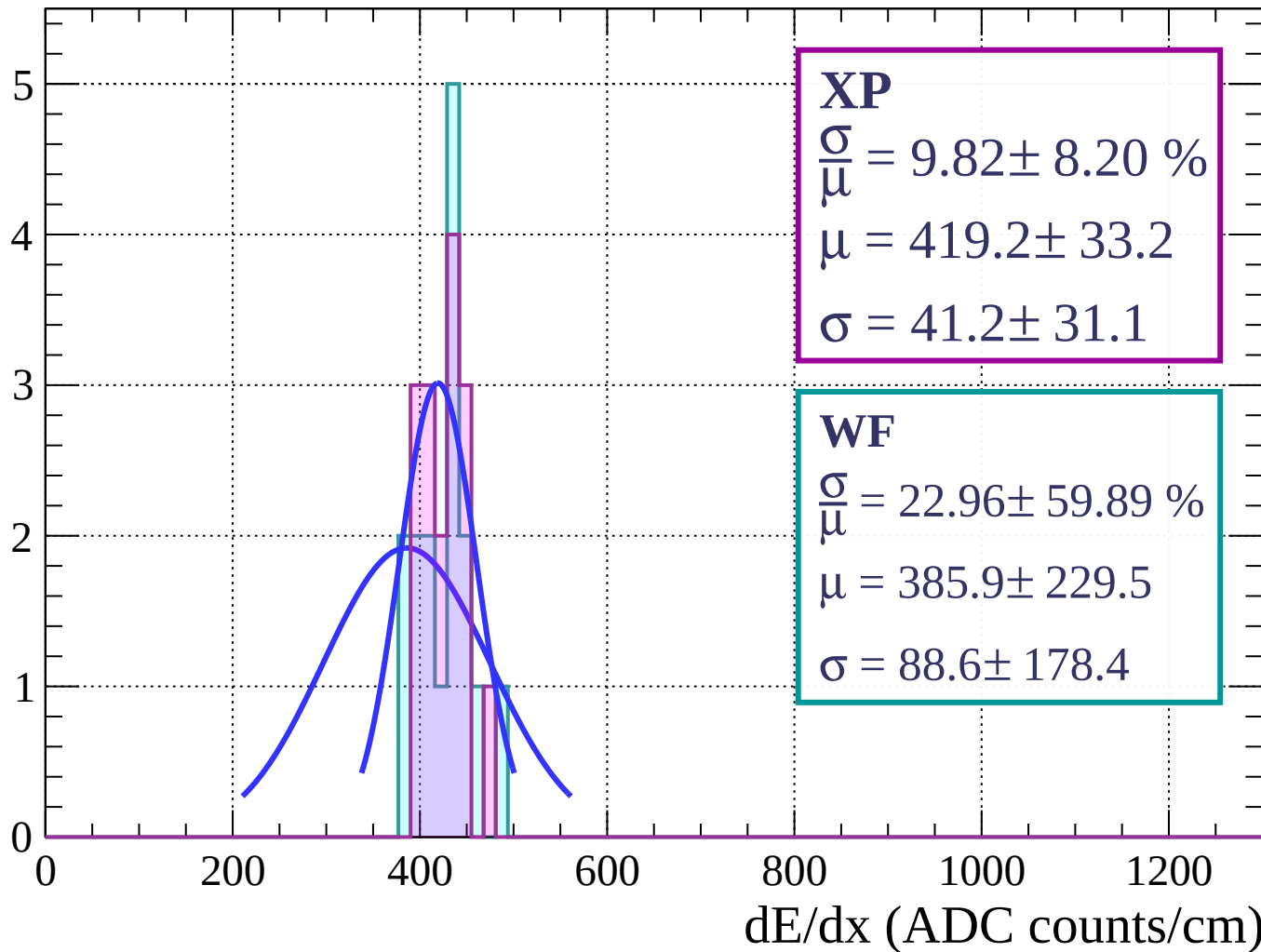
Energy loss | $0 < \phi < 3$

Count



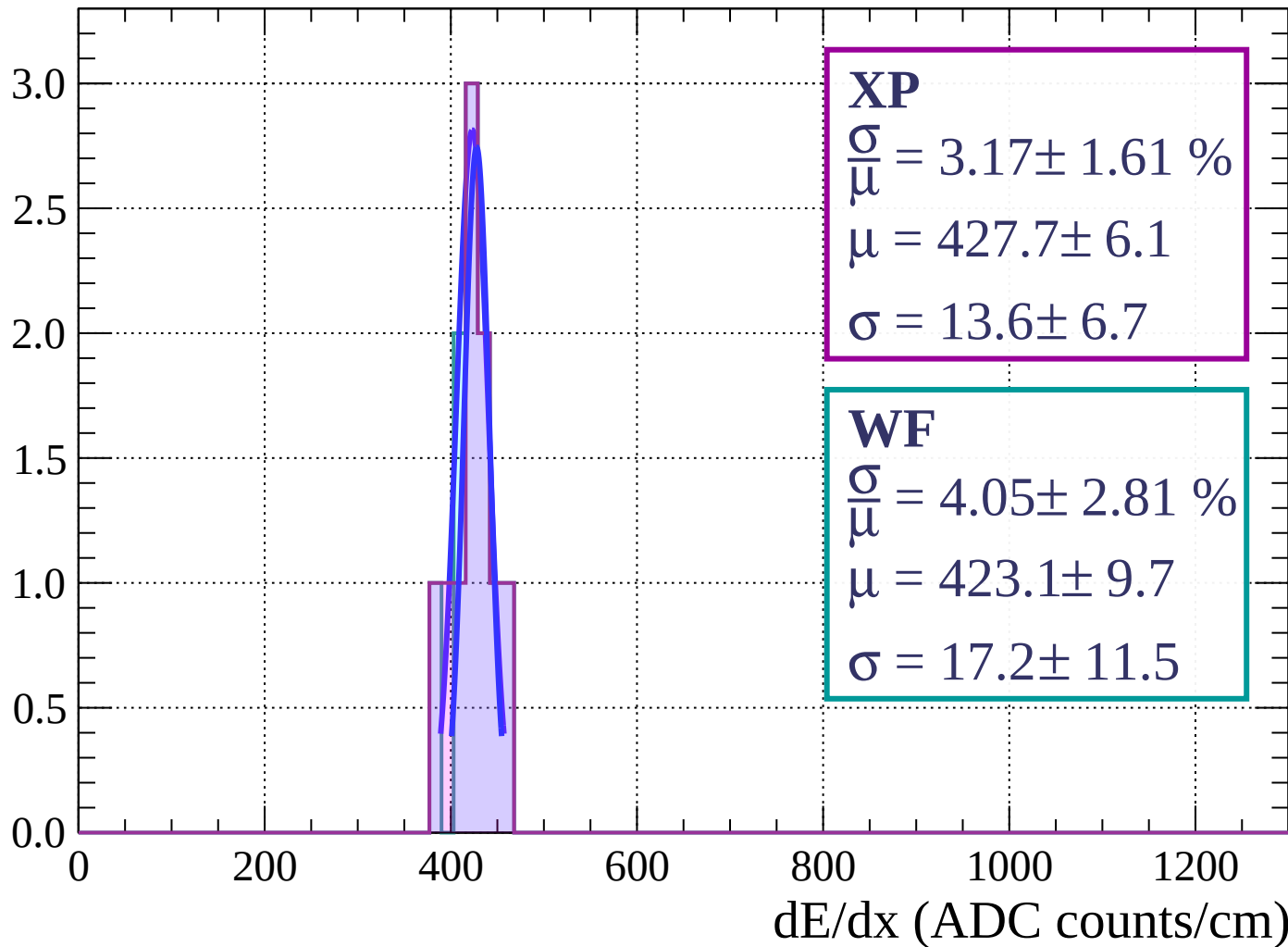
Energy loss | $3 < \phi < 6$

Count



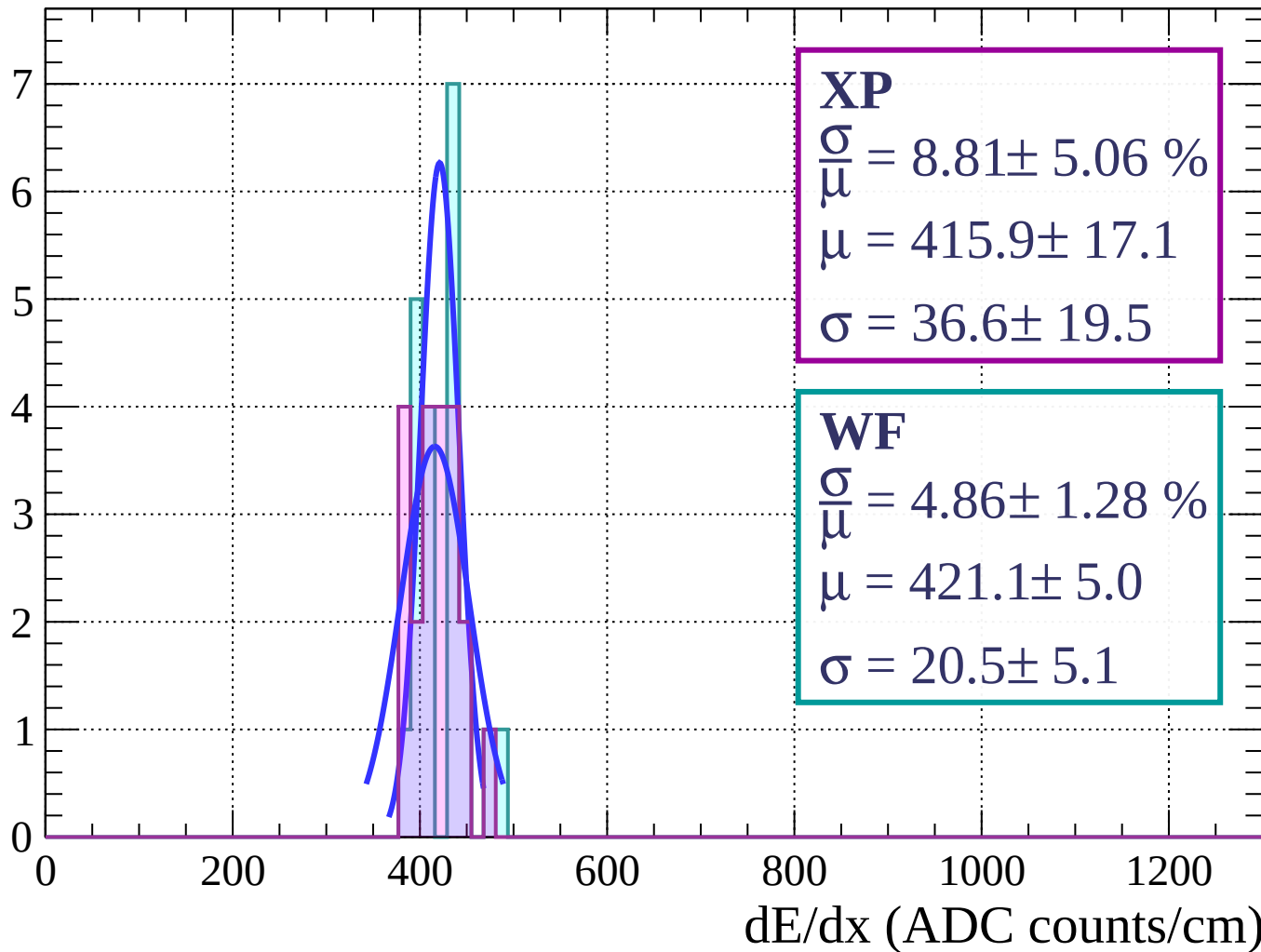
Energy loss | $6 < \phi < 9$

Count



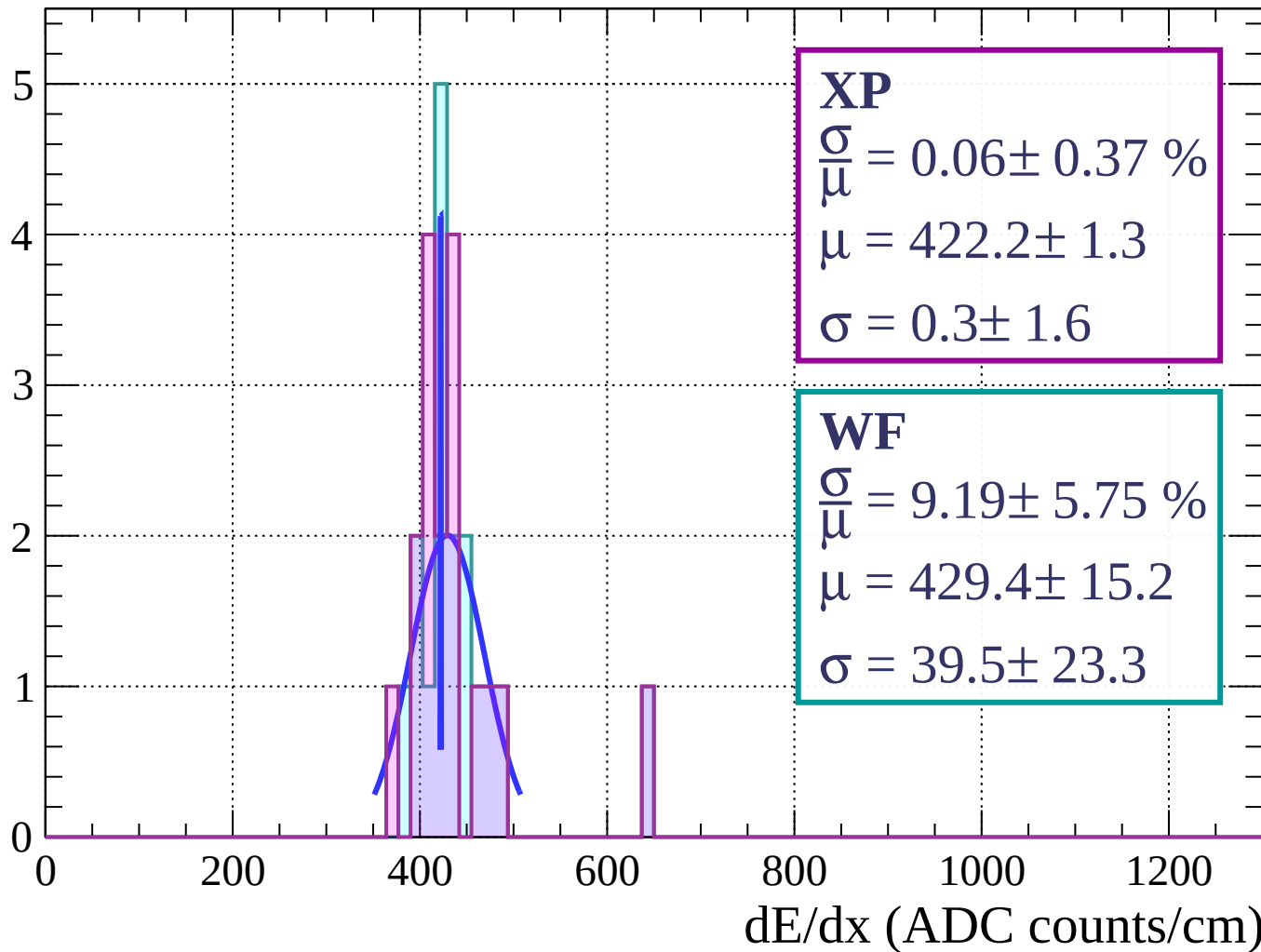
Energy loss | $9 < \phi < 12$

Count



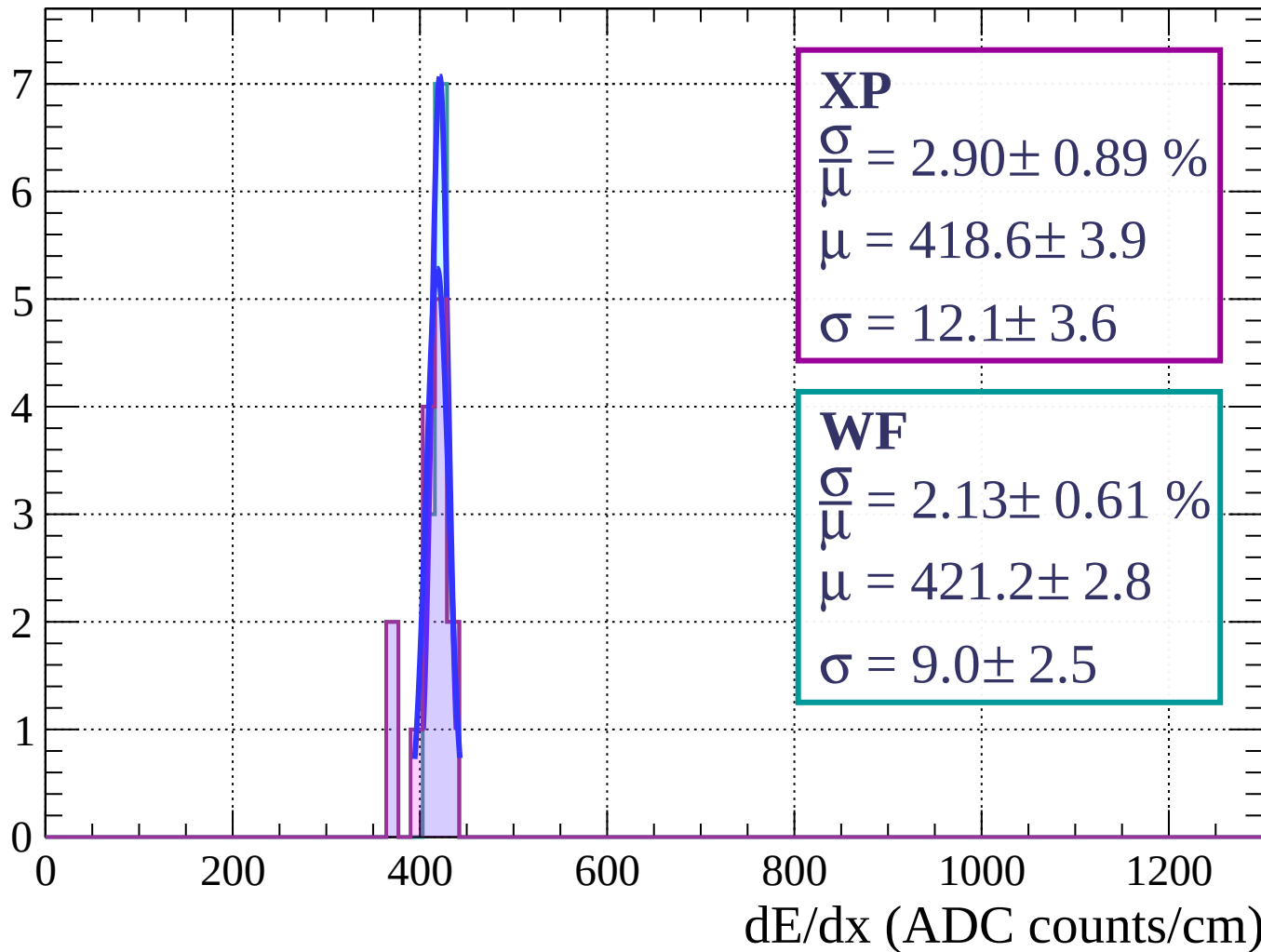
Energy loss | $12 < \phi < 15$

Count



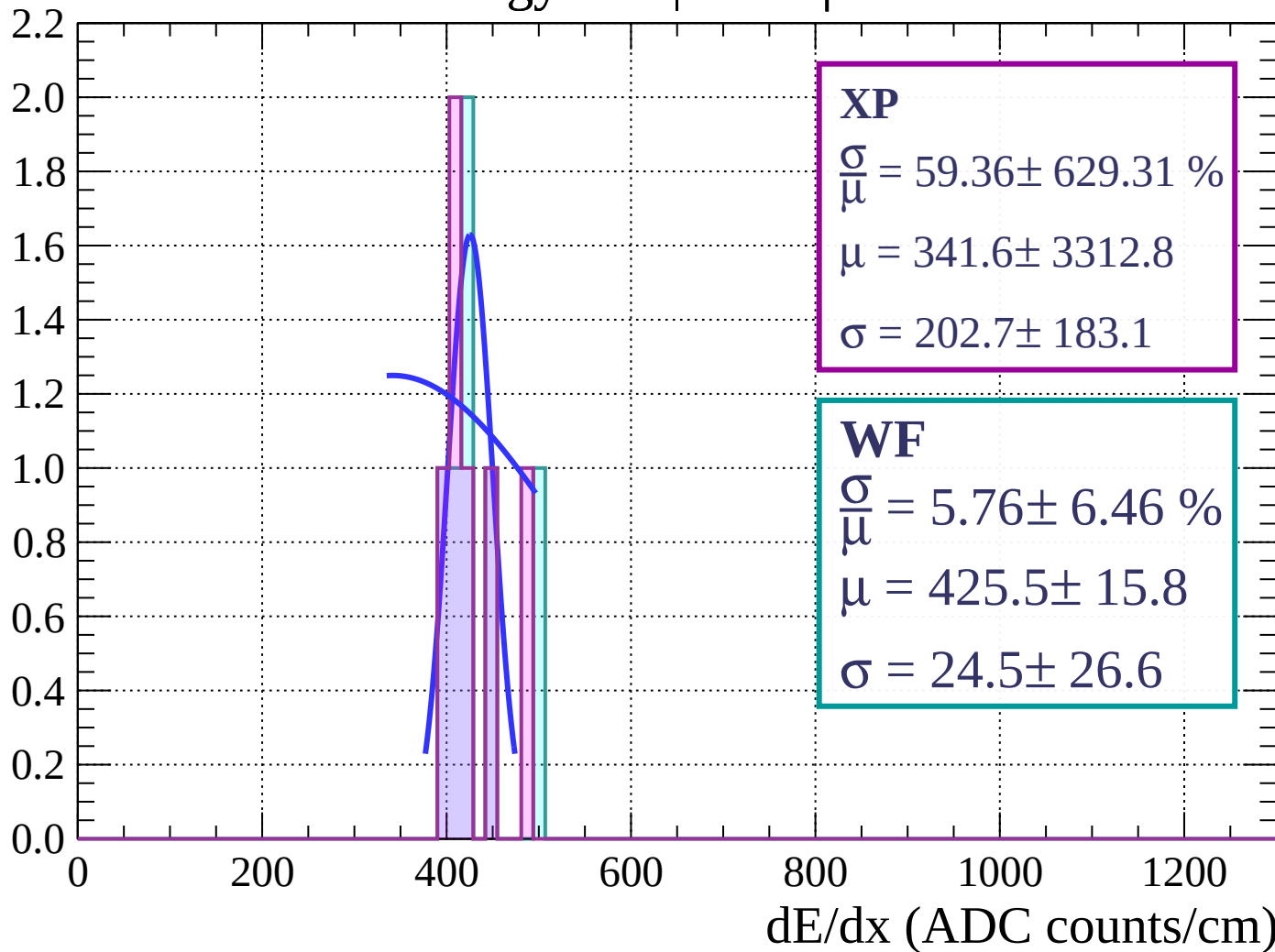
Energy loss | $15 < \phi < 18$

Count



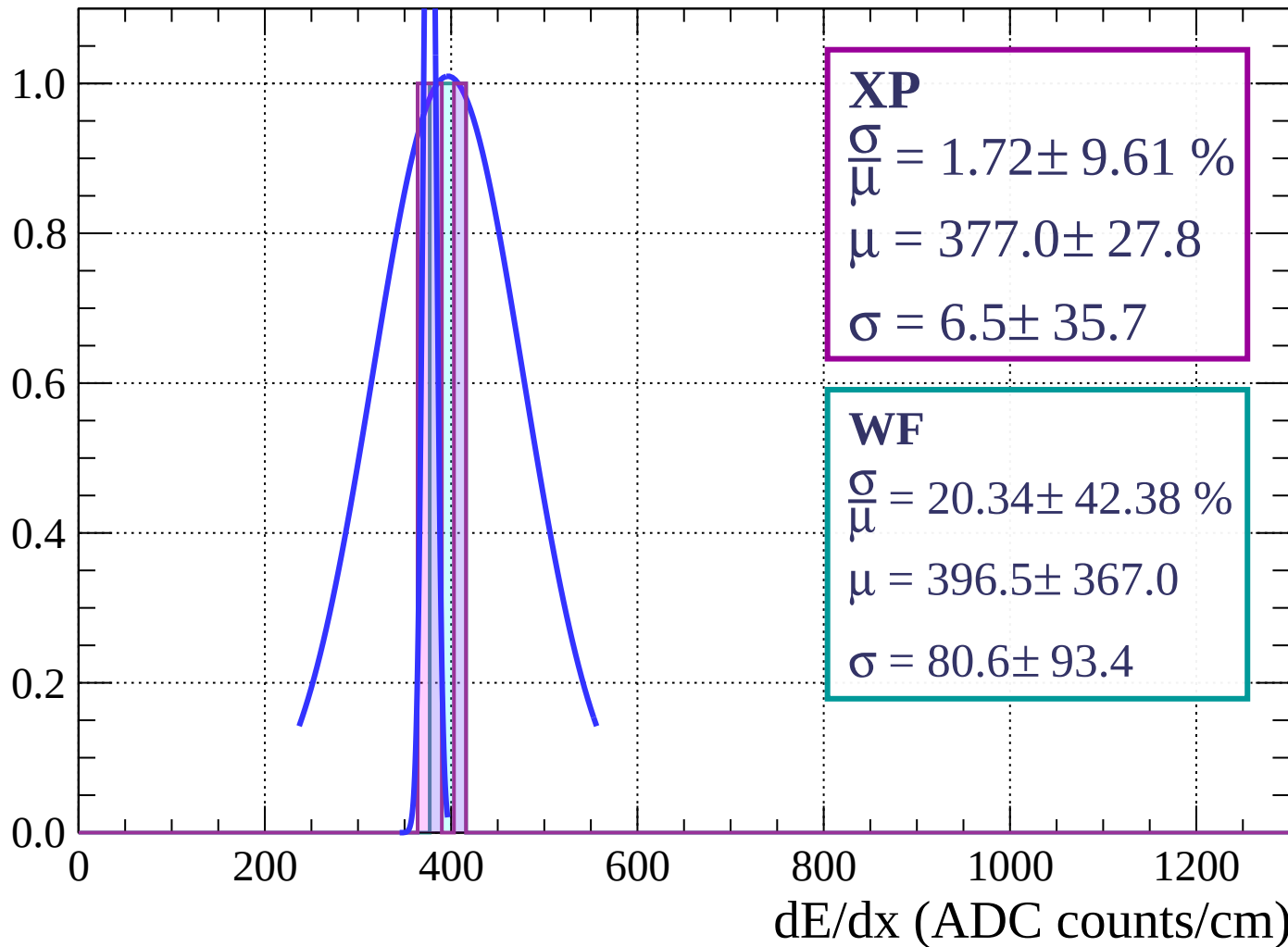
Energy loss | $18 < \phi < 21$

Count



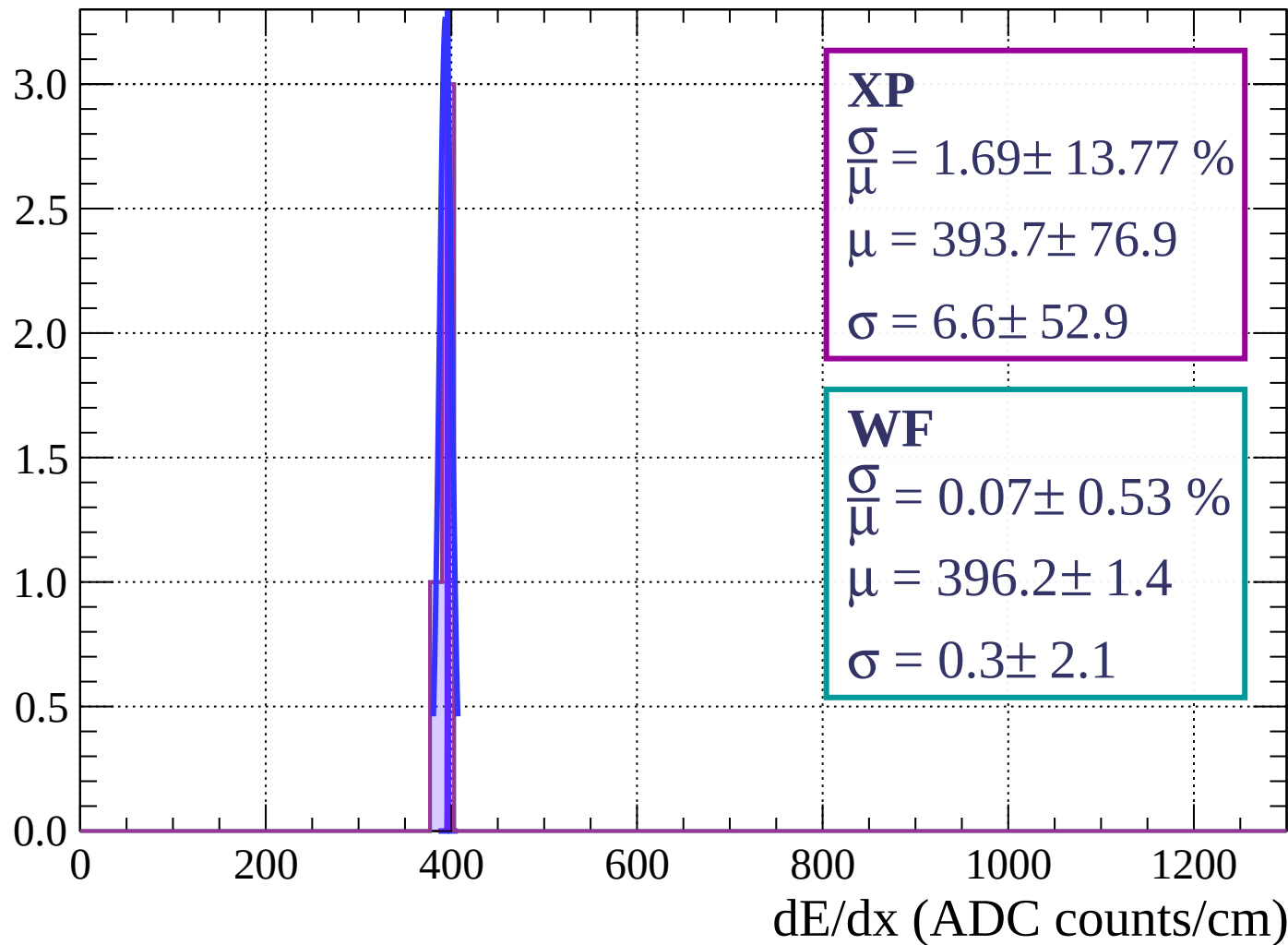
Energy loss | $21 < \phi < 24$

Count



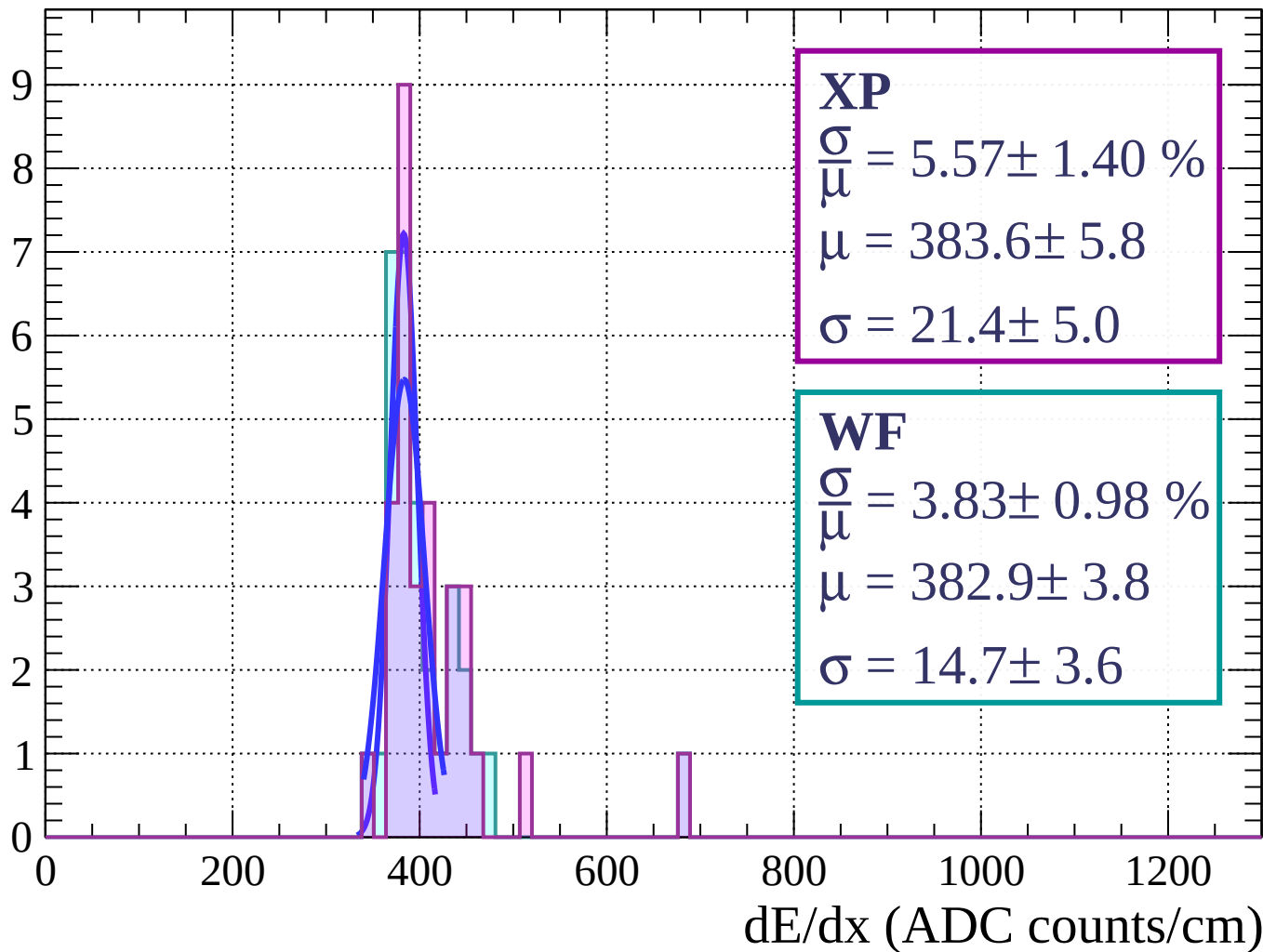
Energy loss | $24 < \phi < 27$

Count



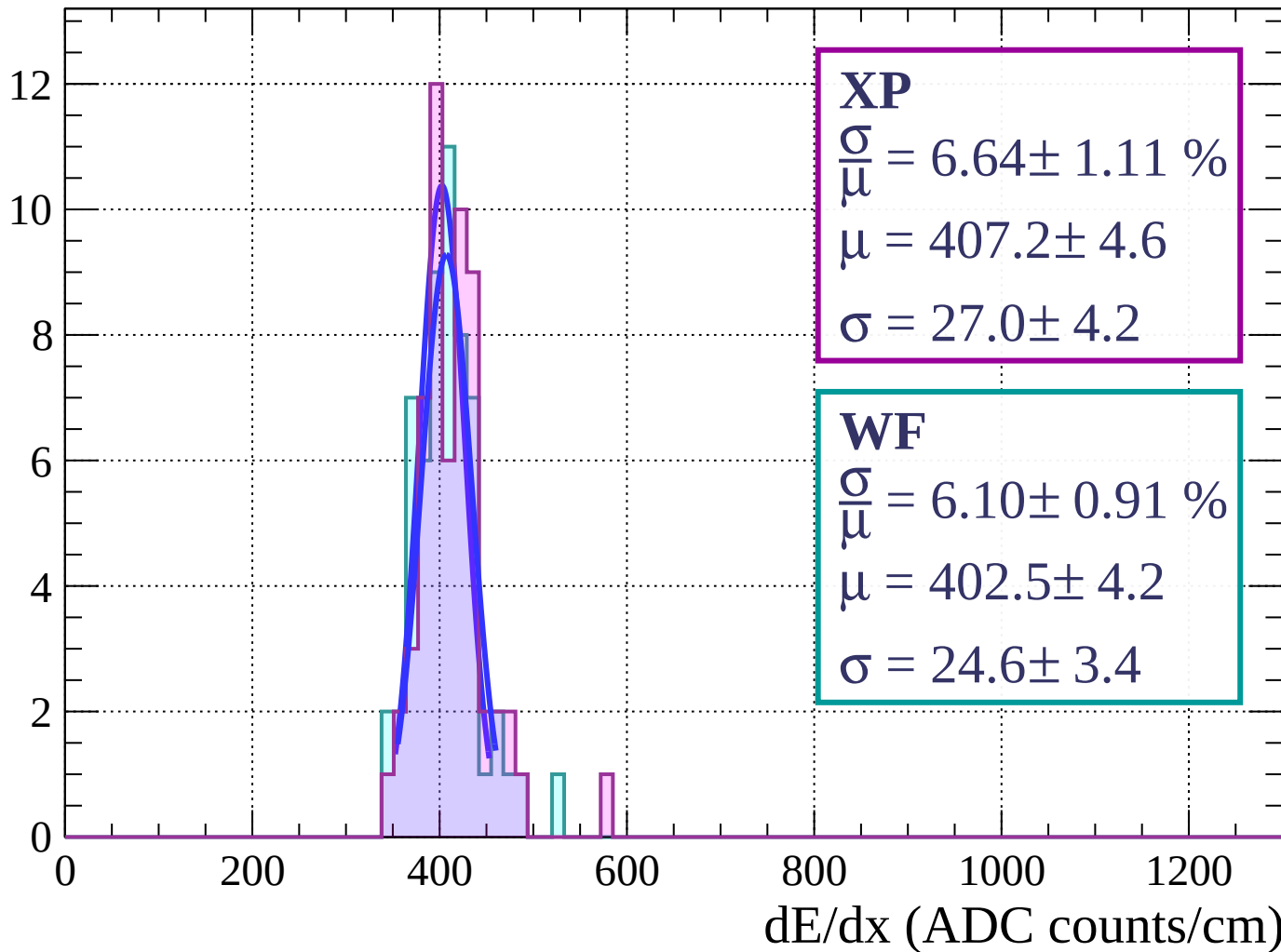
Energy loss | $27 < \phi < 30$

Count



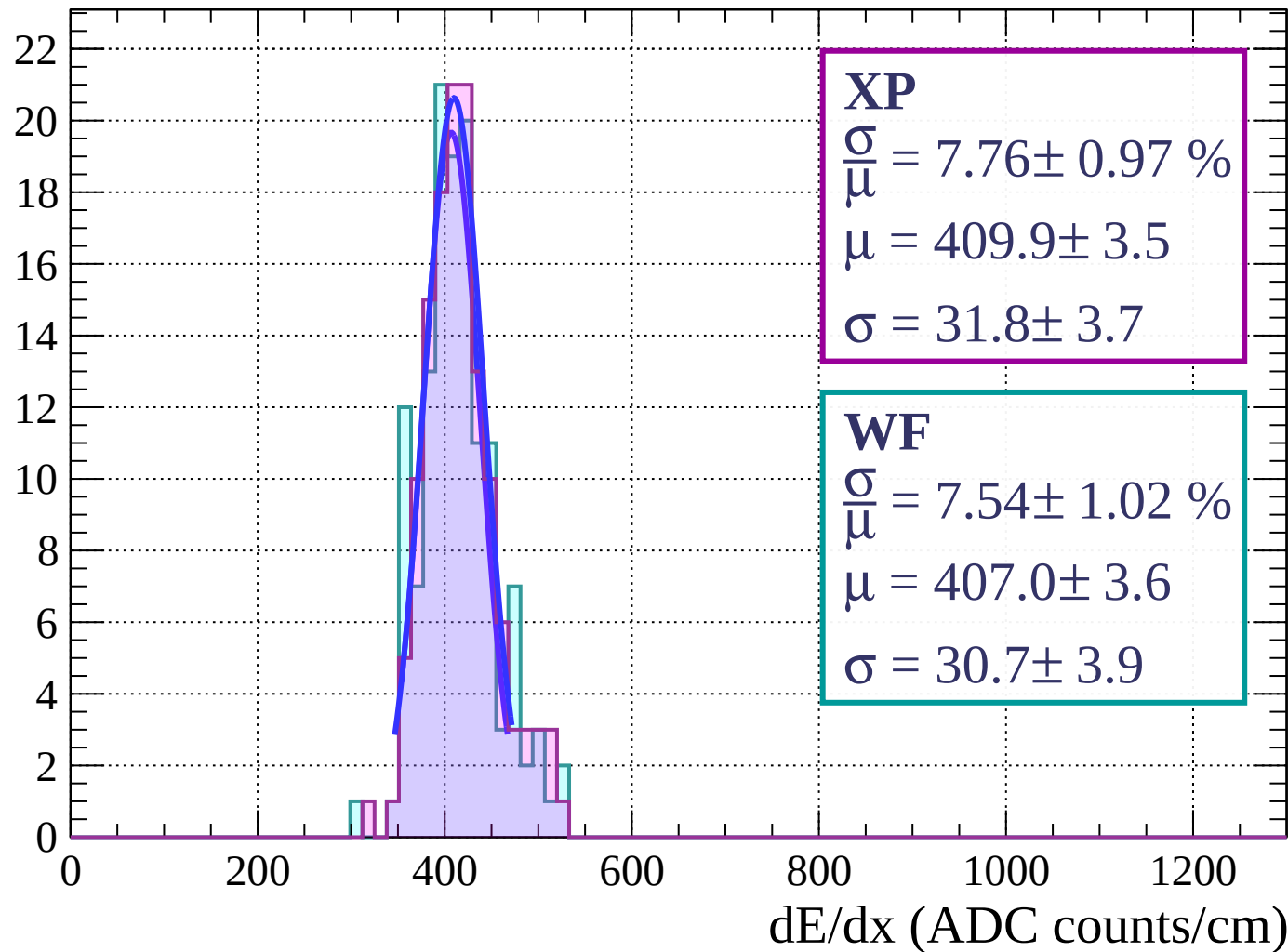
Energy loss | $30 < \phi < 33$

Count



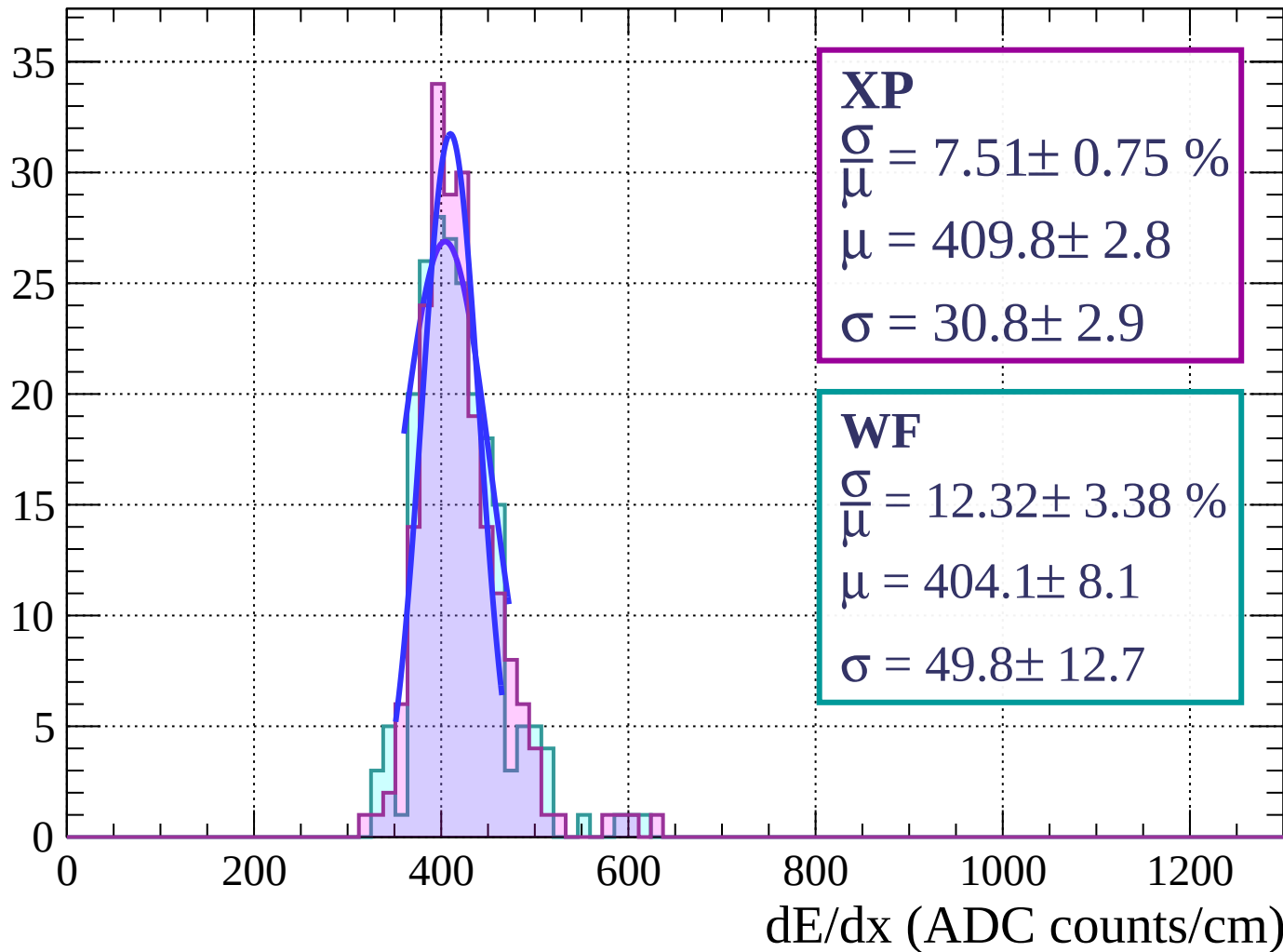
Energy loss | $33 < \phi < 36$

Count



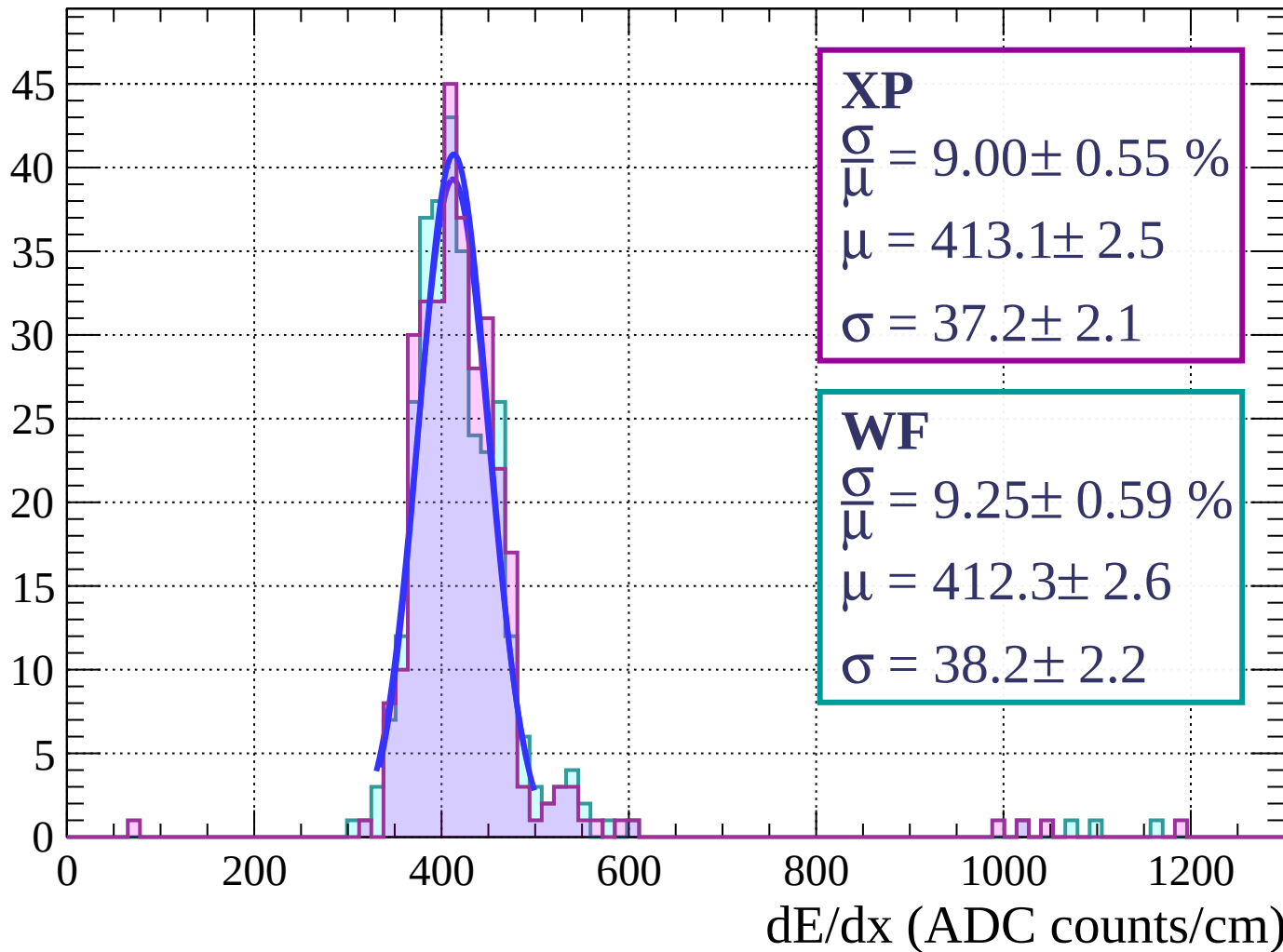
Energy loss | $36 < \phi < 39$

Count



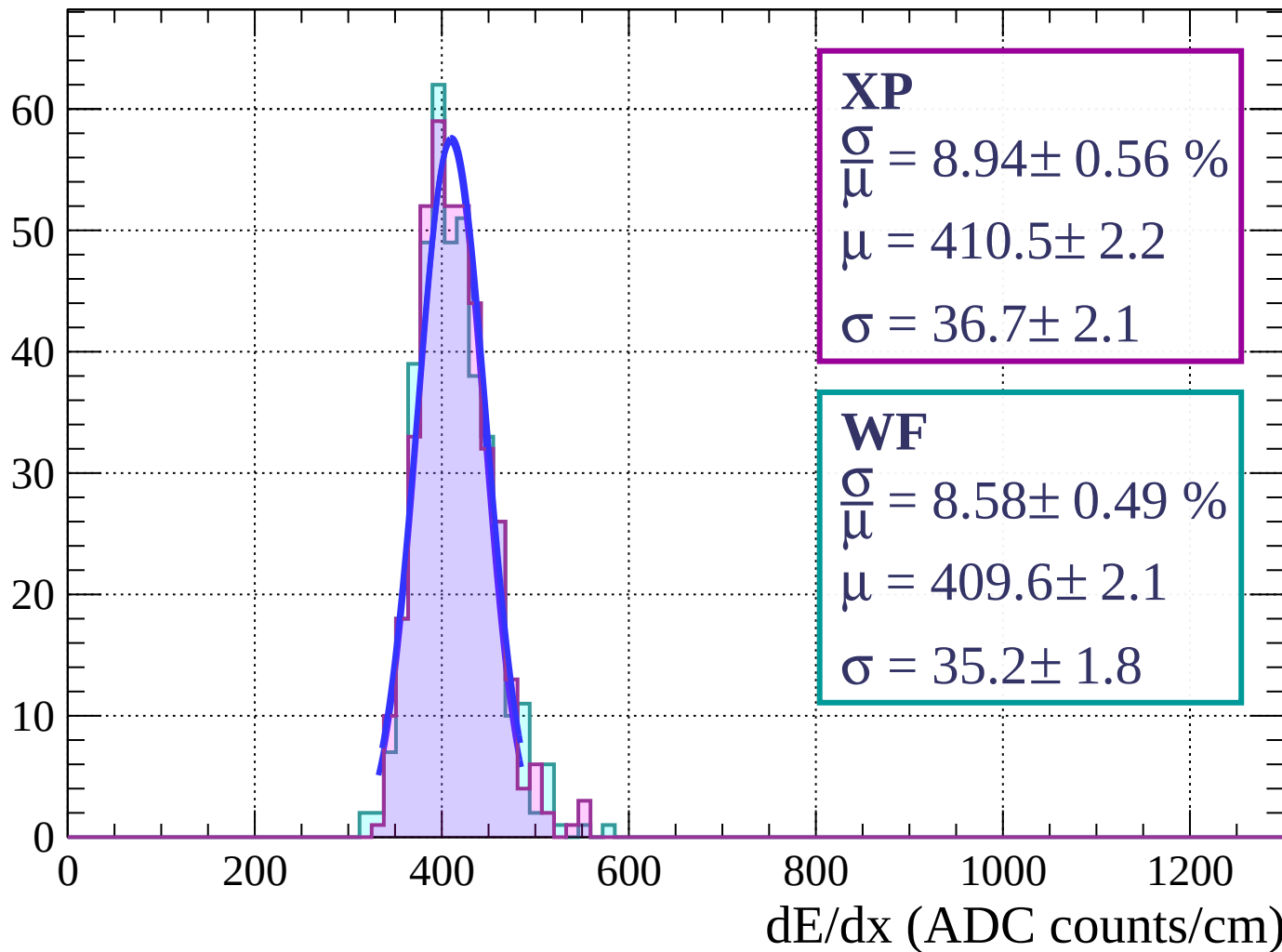
Energy loss | $39 < \phi < 42$

Count



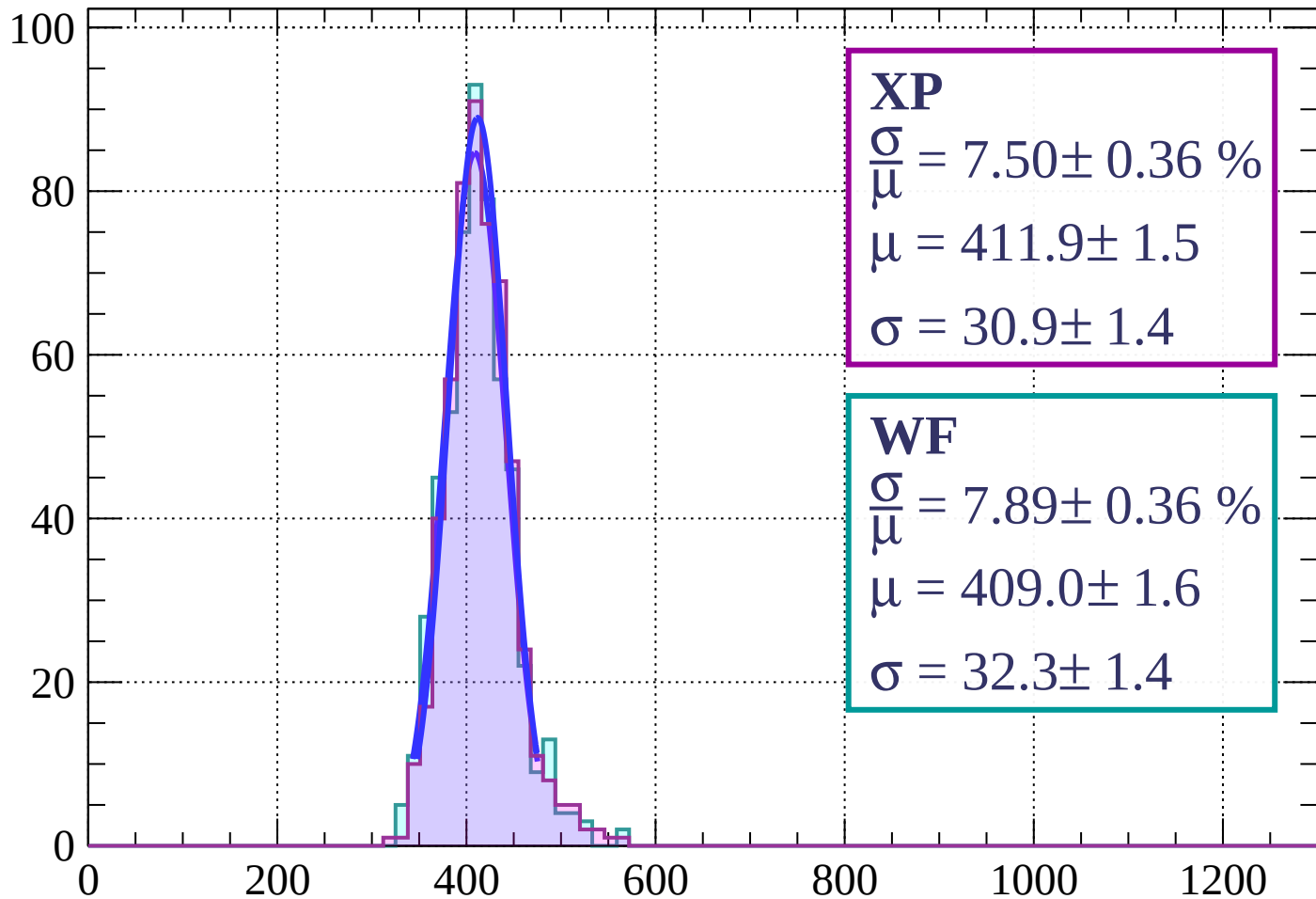
Energy loss | $42 < \phi < 45$

Count



Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 7.50 \pm 0.36 \%$$

$$\mu = 411.9 \pm 1.5$$

$$\sigma = 30.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.89 \pm 0.36 \%$$

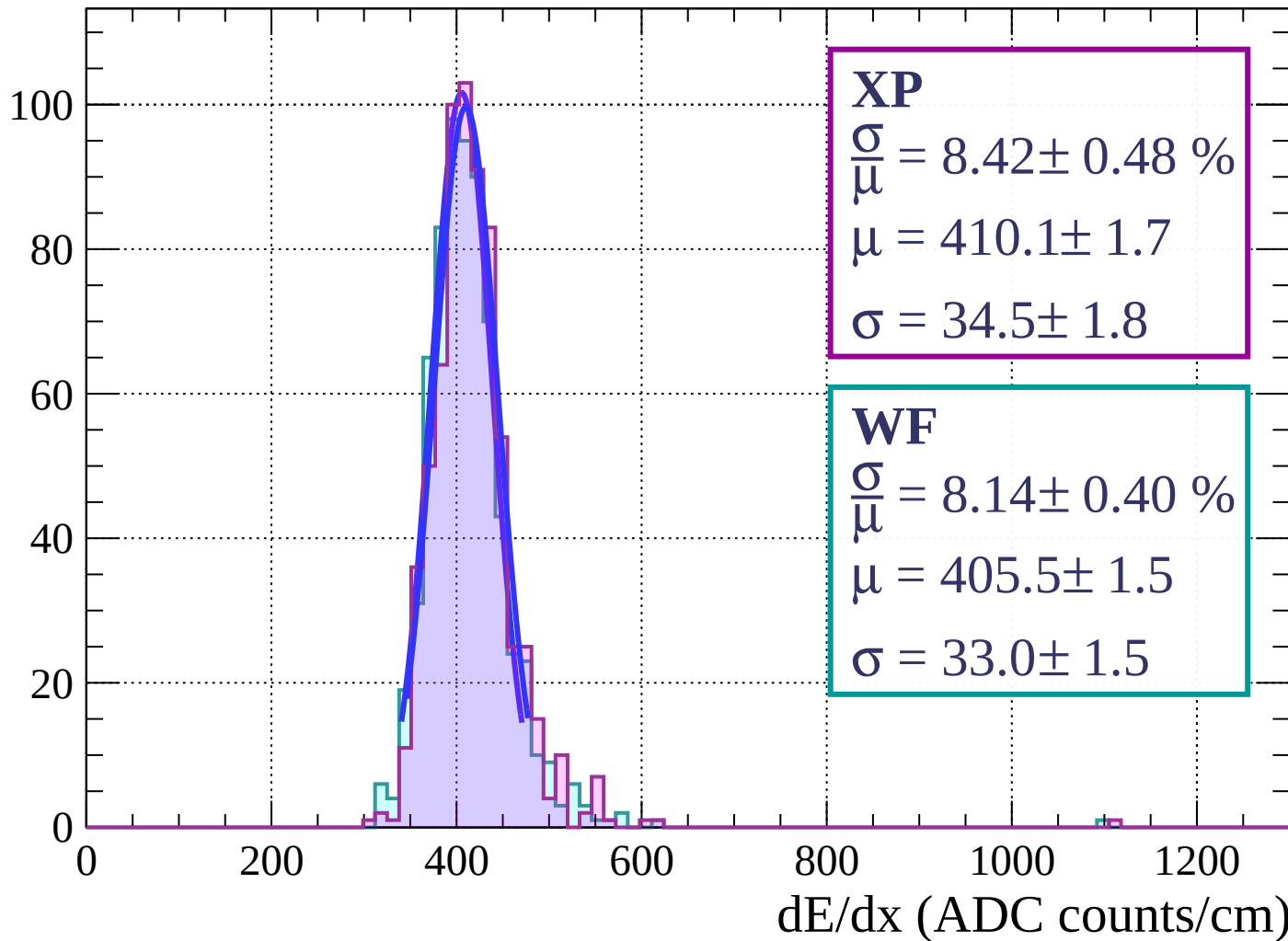
$$\mu = 409.0 \pm 1.6$$

$$\sigma = 32.3 \pm 1.4$$

dE/dx (ADC counts/cm)

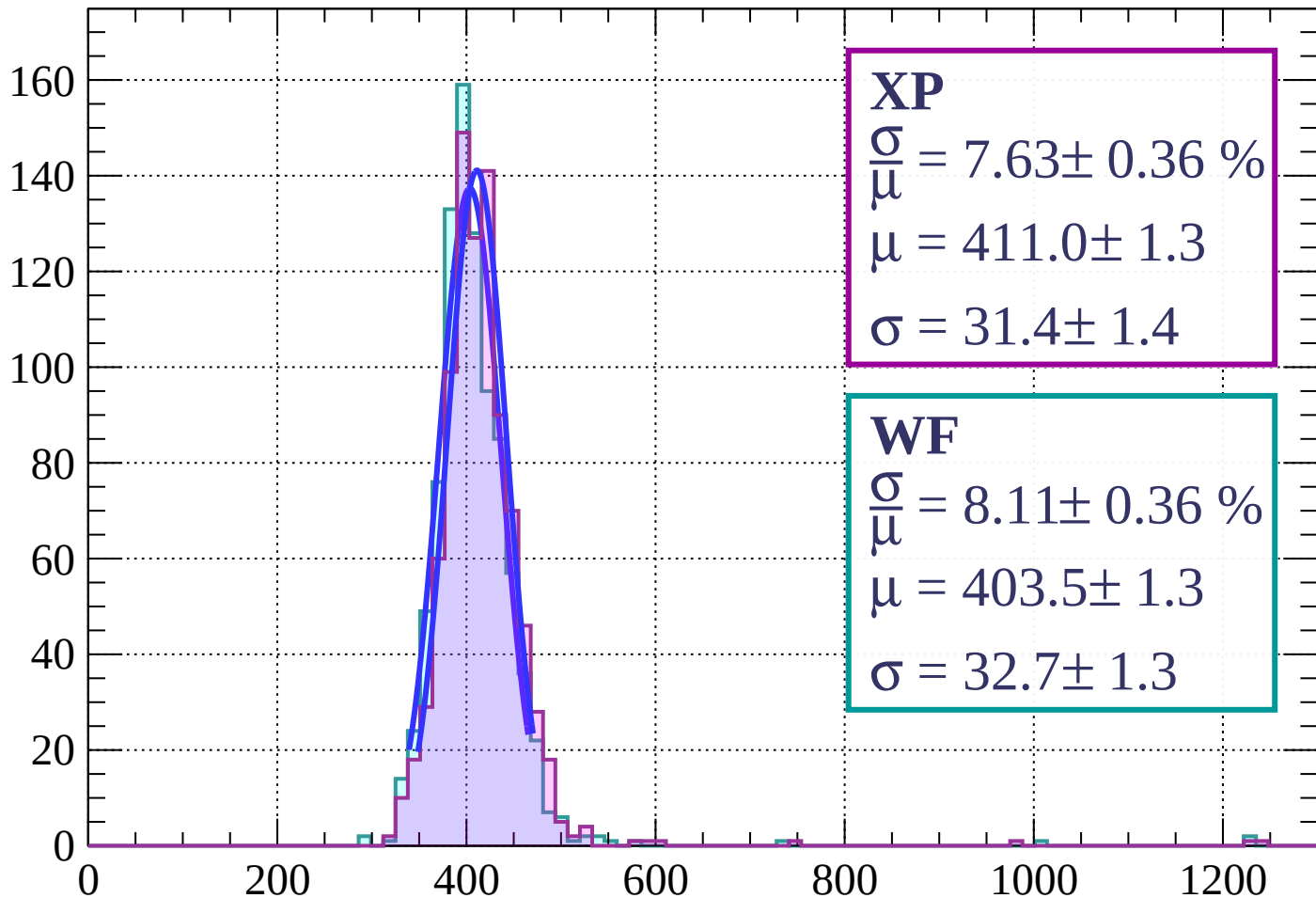
Energy loss | $48 < \phi < 51$

Count



Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 7.63 \pm 0.36 \%$$

$$\mu = 411.0 \pm 1.3$$

$$\sigma = 31.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.36 \%$$

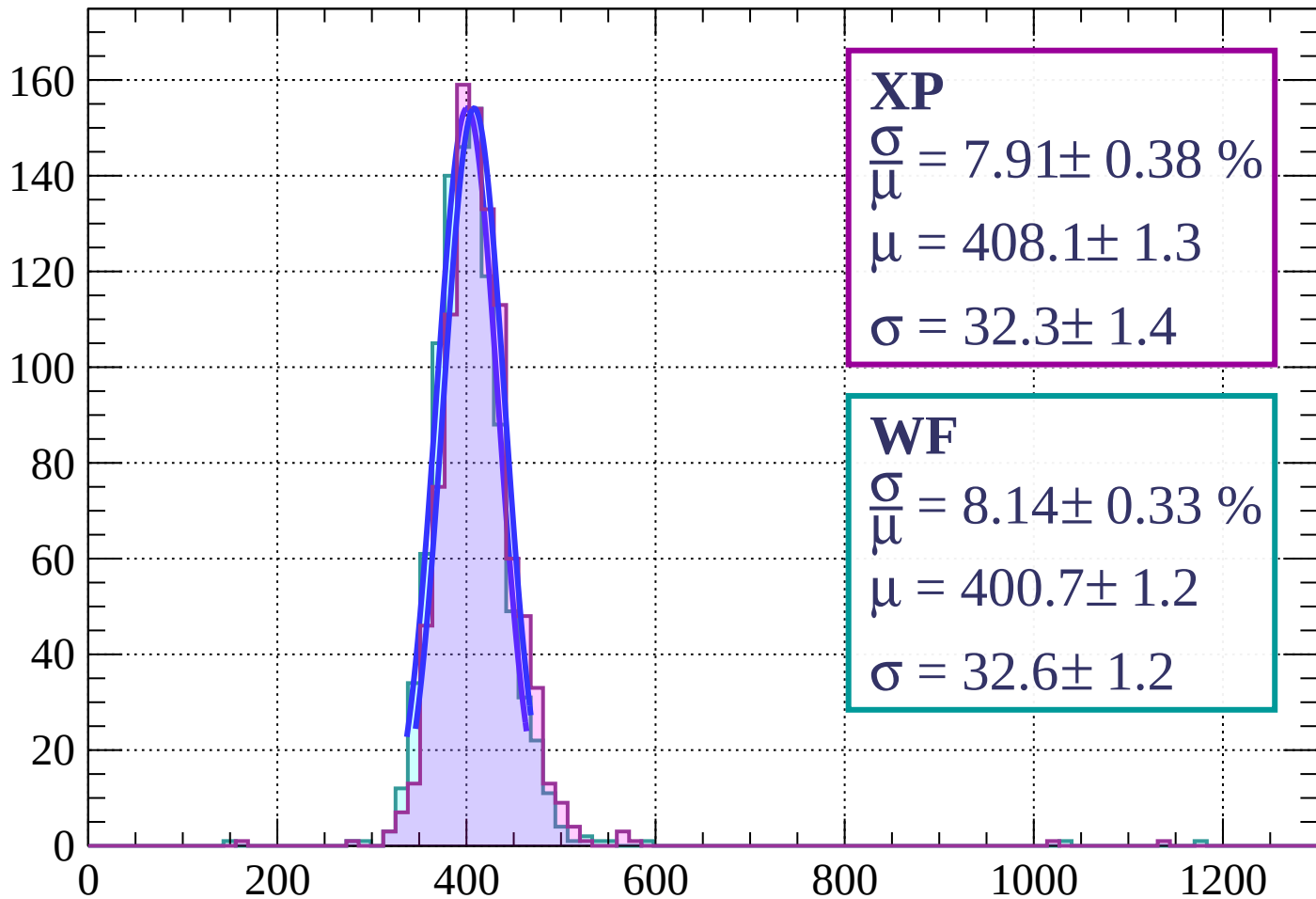
$$\mu = 403.5 \pm 1.3$$

$$\sigma = 32.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 7.91 \pm 0.38 \%$$

$$\mu = 408.1 \pm 1.3$$

$$\sigma = 32.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.14 \pm 0.33 \%$$

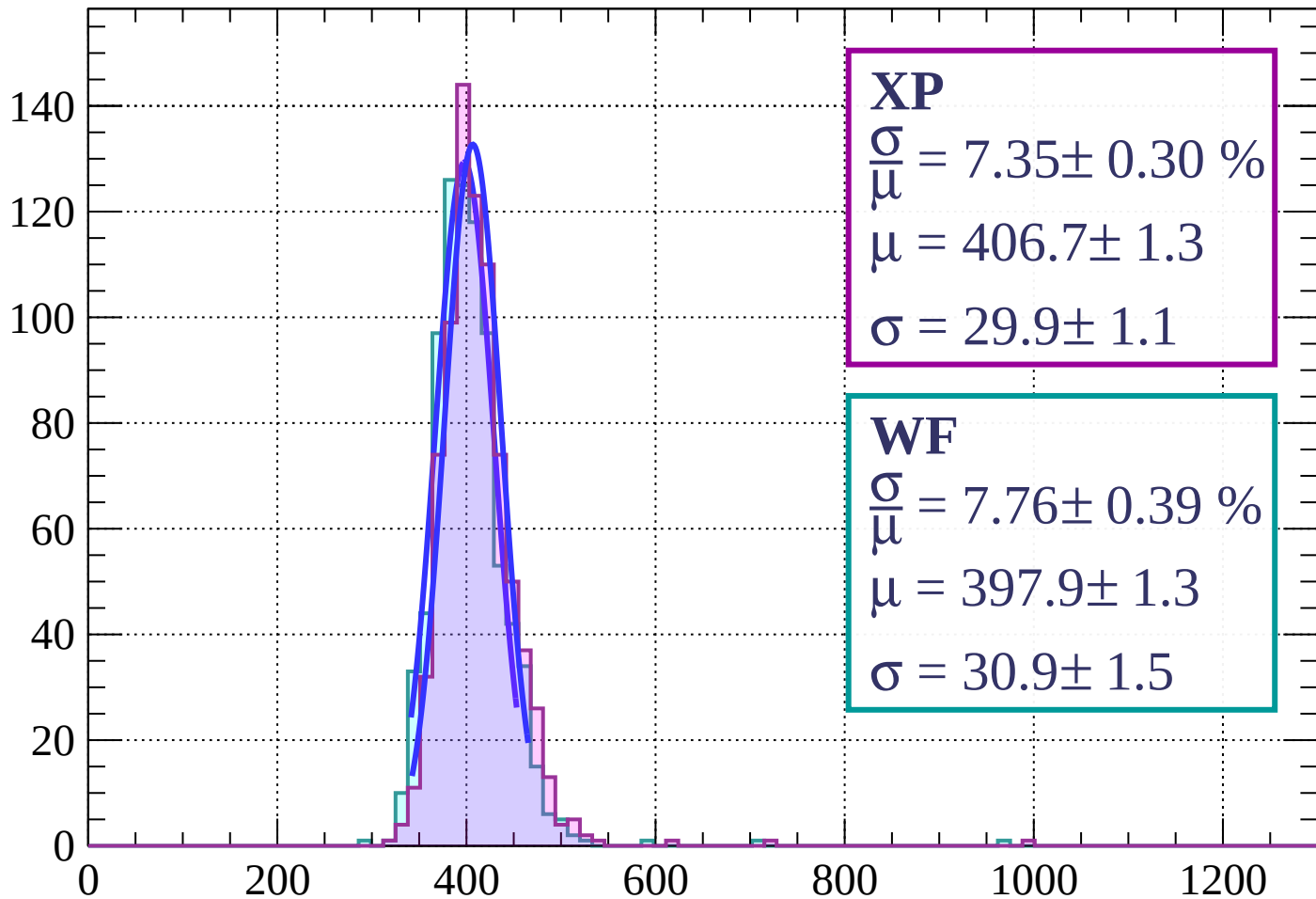
$$\mu = 400.7 \pm 1.2$$

$$\sigma = 32.6 \pm 1.2$$

dE/dx (ADC counts/cm)

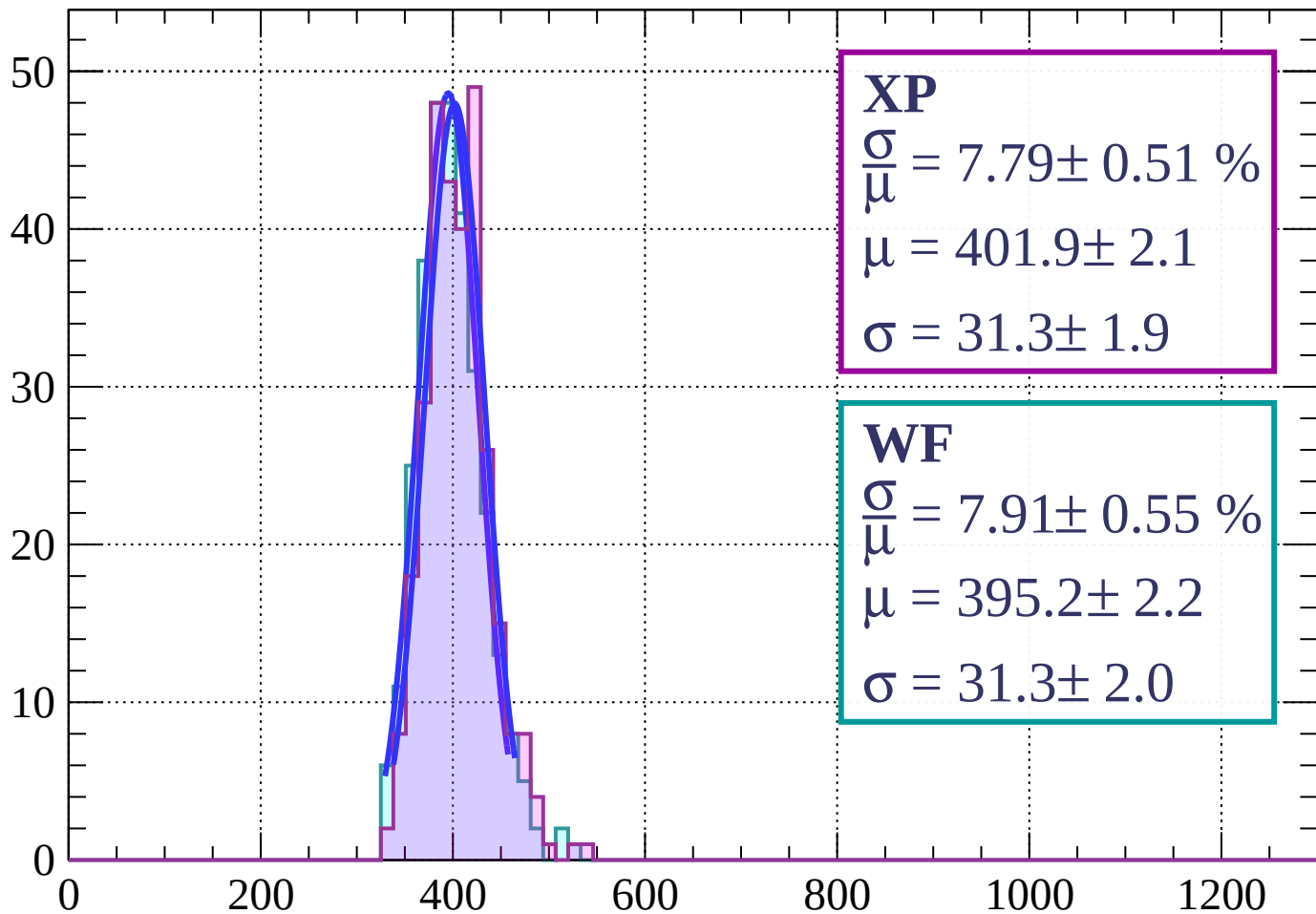
Energy loss | $57 < \phi < 60$

Count



Energy loss | $60 < \phi < 63$

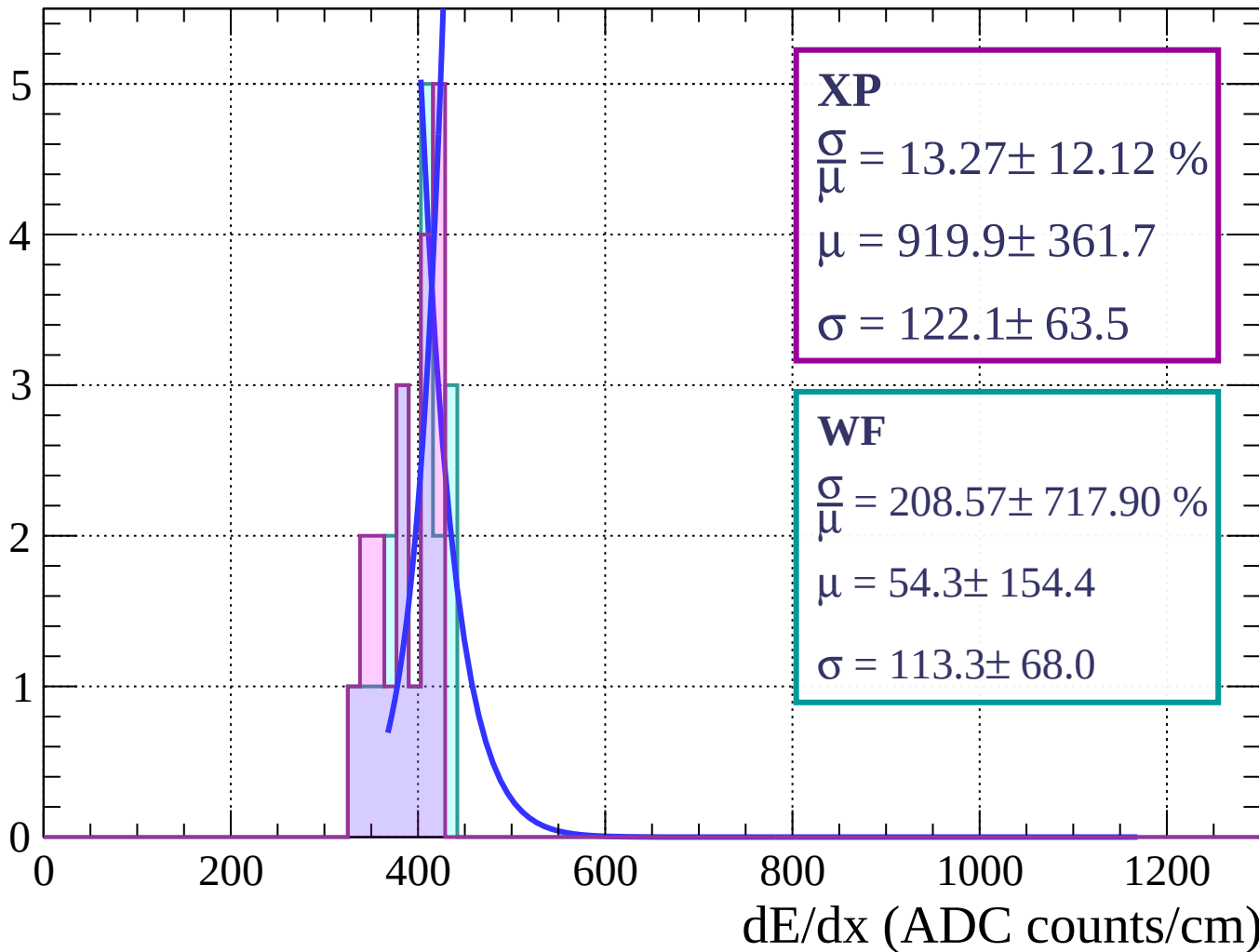
Count



dE/dx (ADC counts/cm)

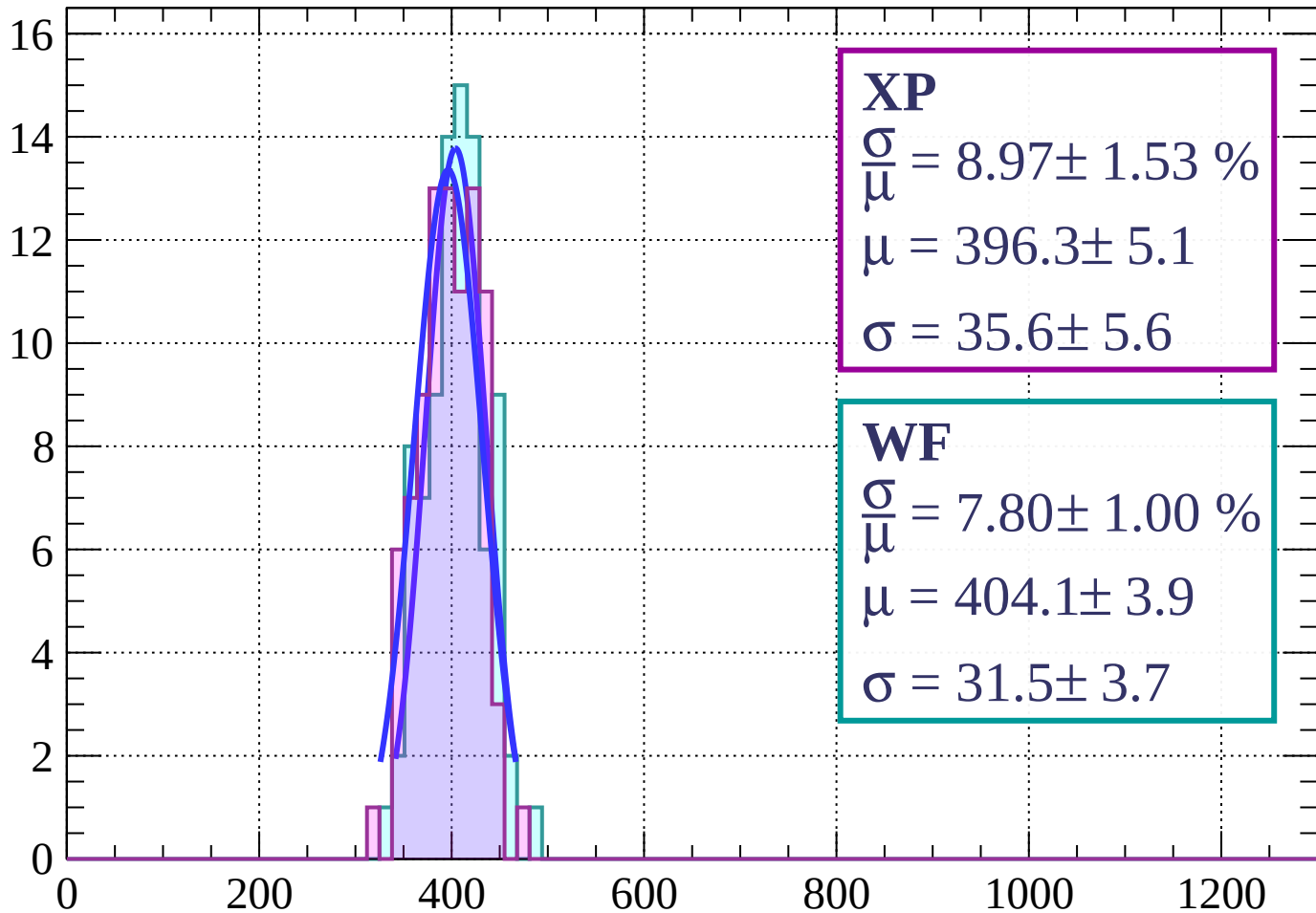
Energy loss | $63 < \phi < 66$

Count



Energy loss | $66 < \phi < 69$

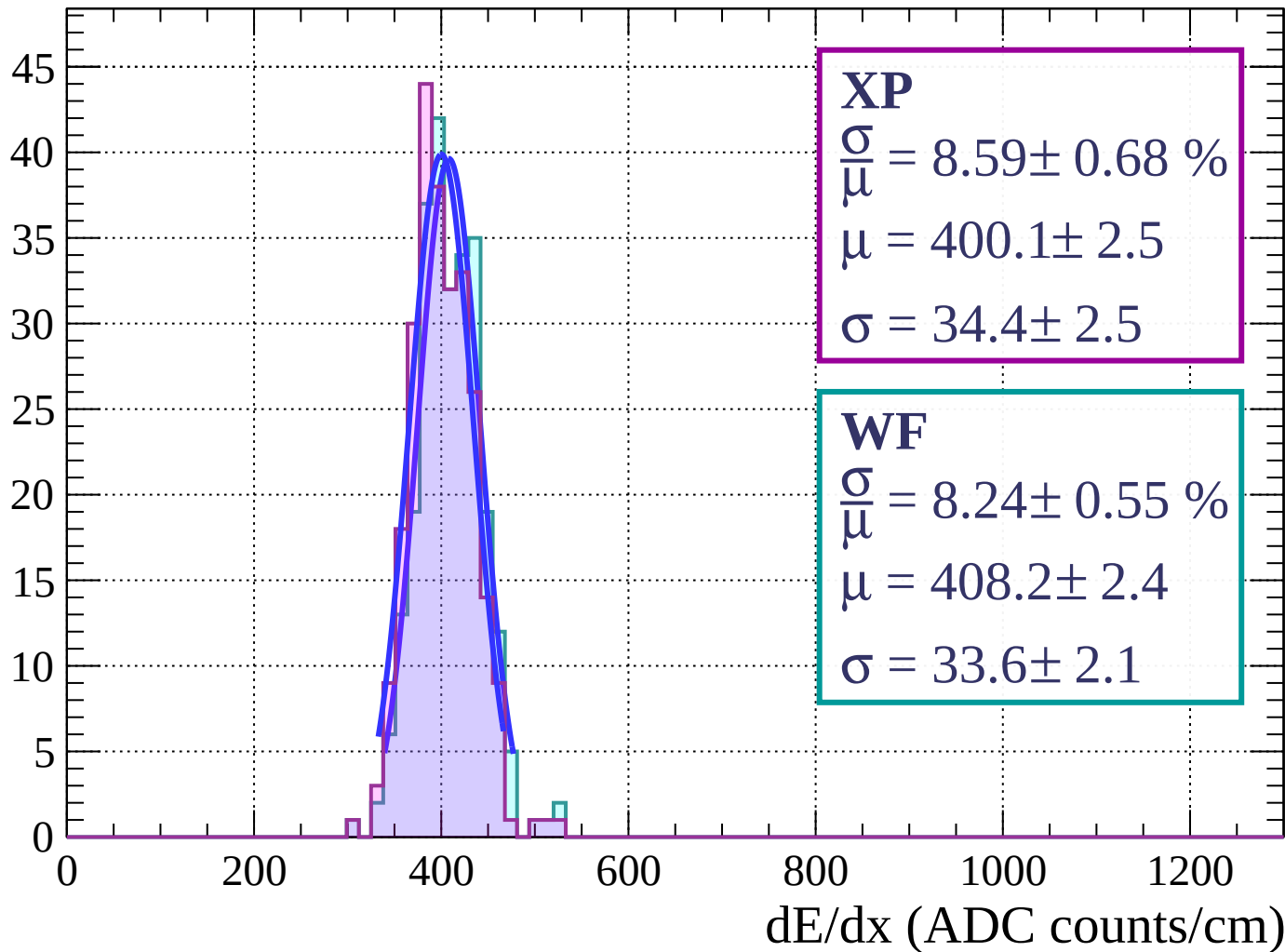
Count



dE/dx (ADC counts/cm)

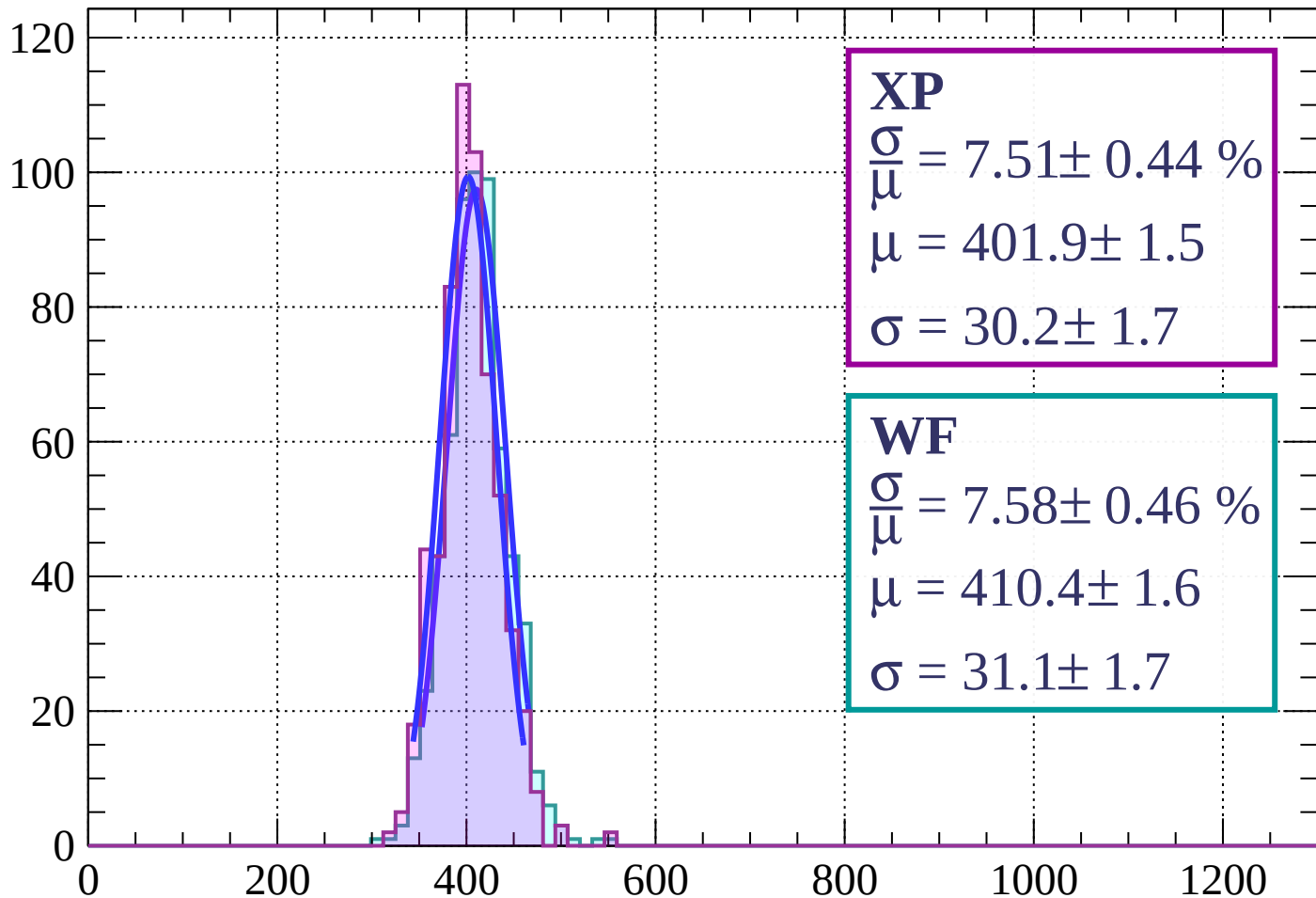
Energy loss | $69 < \phi < 72$

Count



Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.44 \%$$

$$\mu = 401.9 \pm 1.5$$

$$\sigma = 30.2 \pm 1.7$$

WF

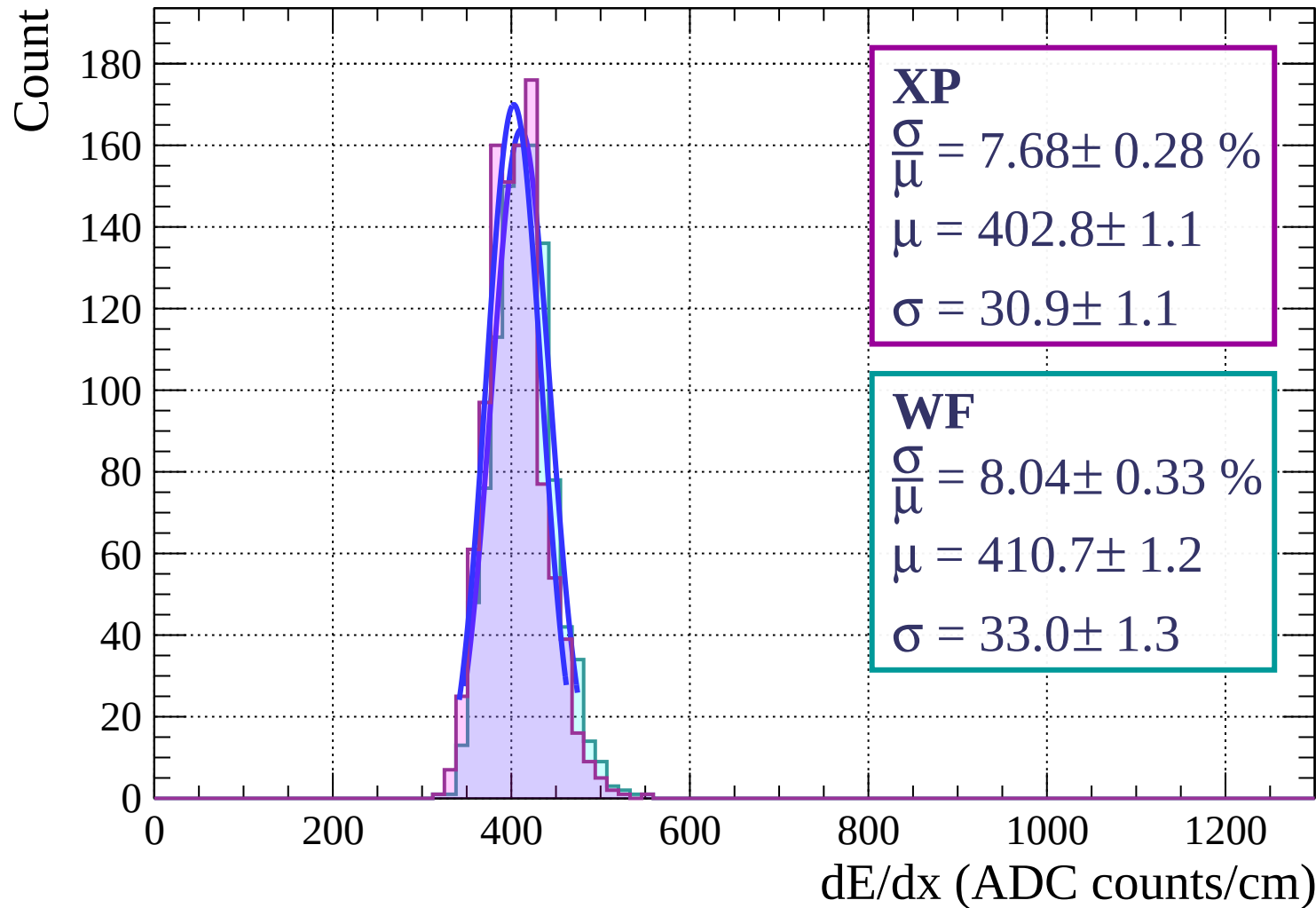
$$\frac{\sigma}{\mu} = 7.58 \pm 0.46 \%$$

$$\mu = 410.4 \pm 1.6$$

$$\sigma = 31.1 \pm 1.7$$

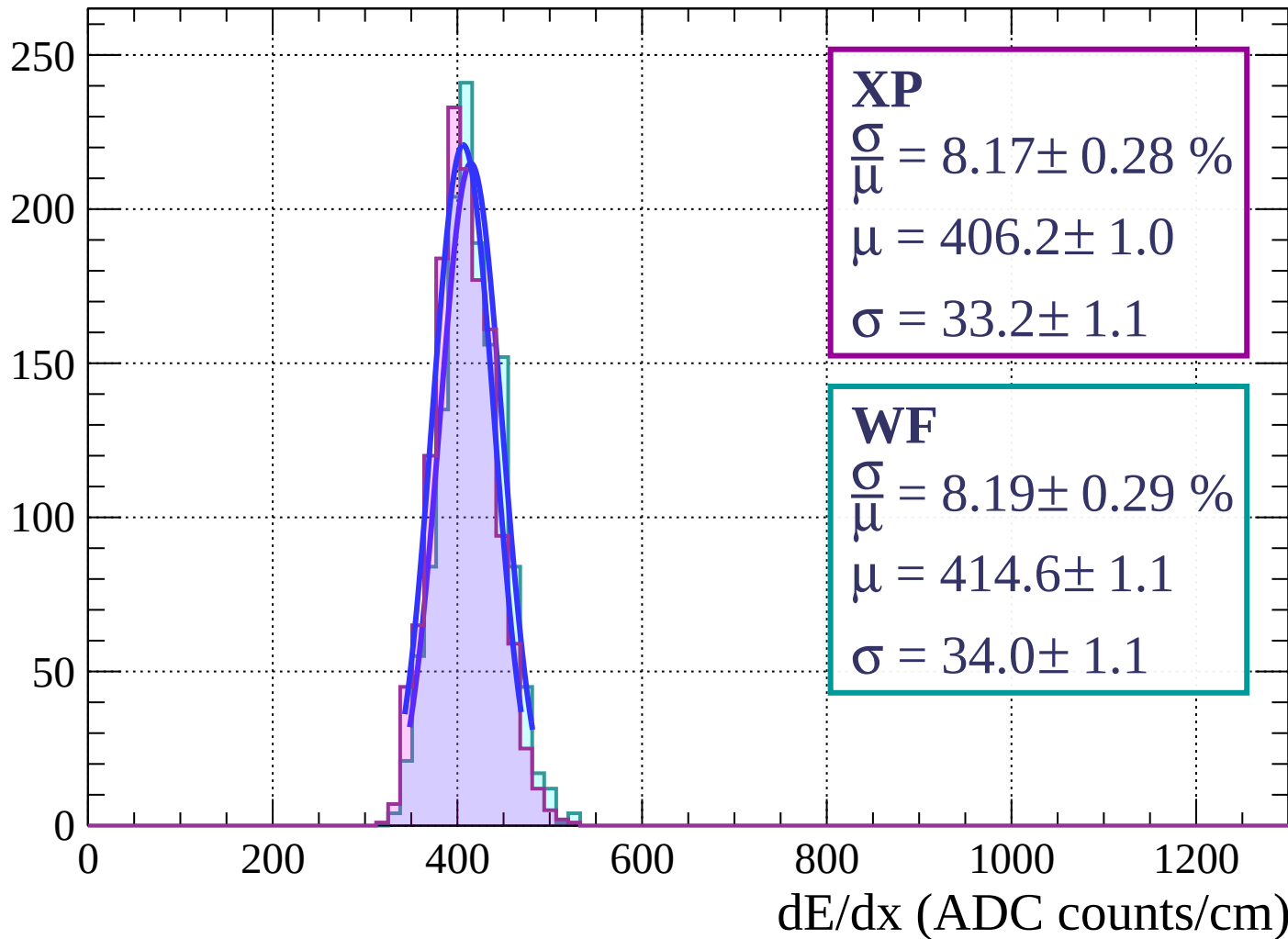
dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

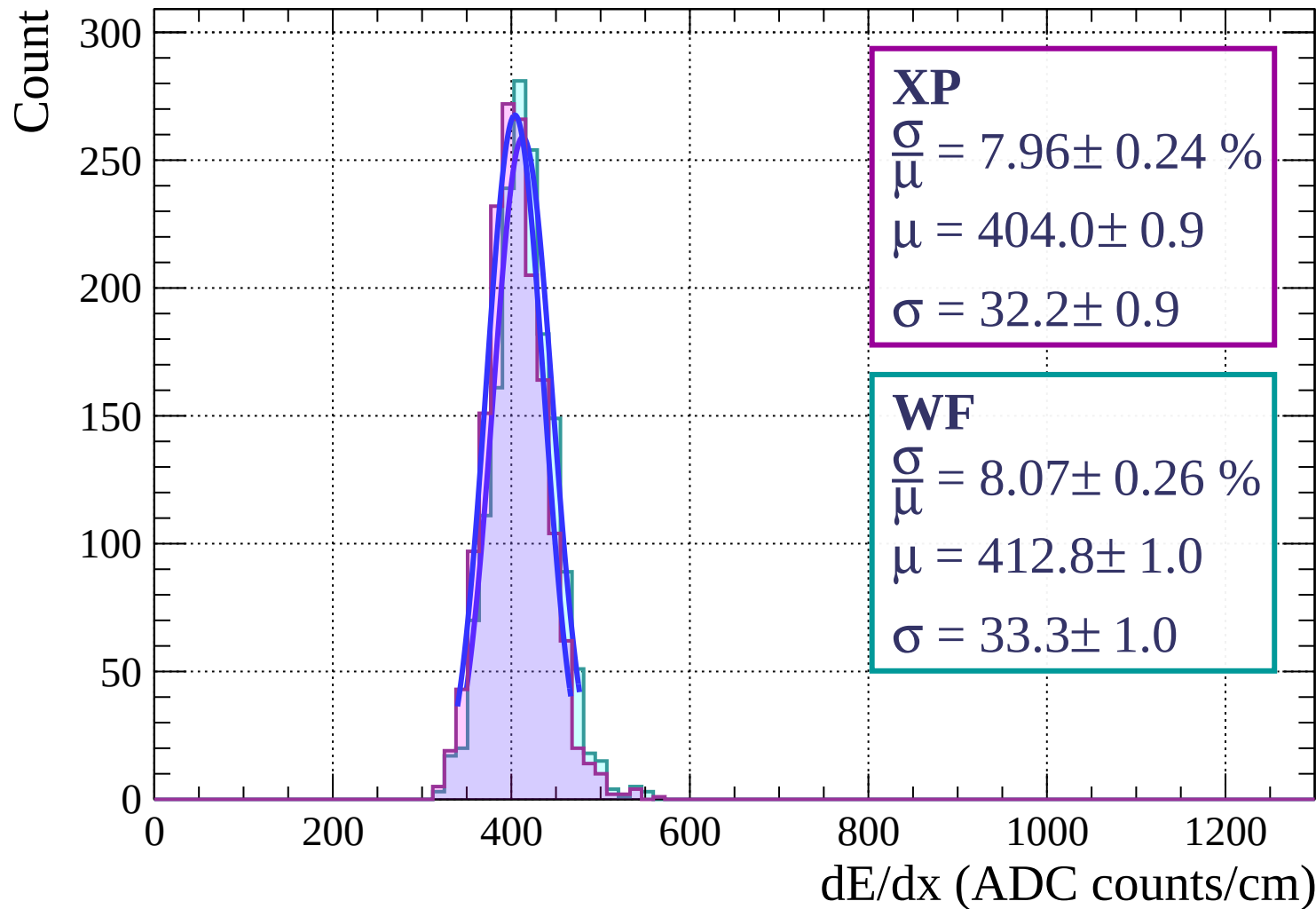


Energy loss | $78 < \phi < 81$

Count

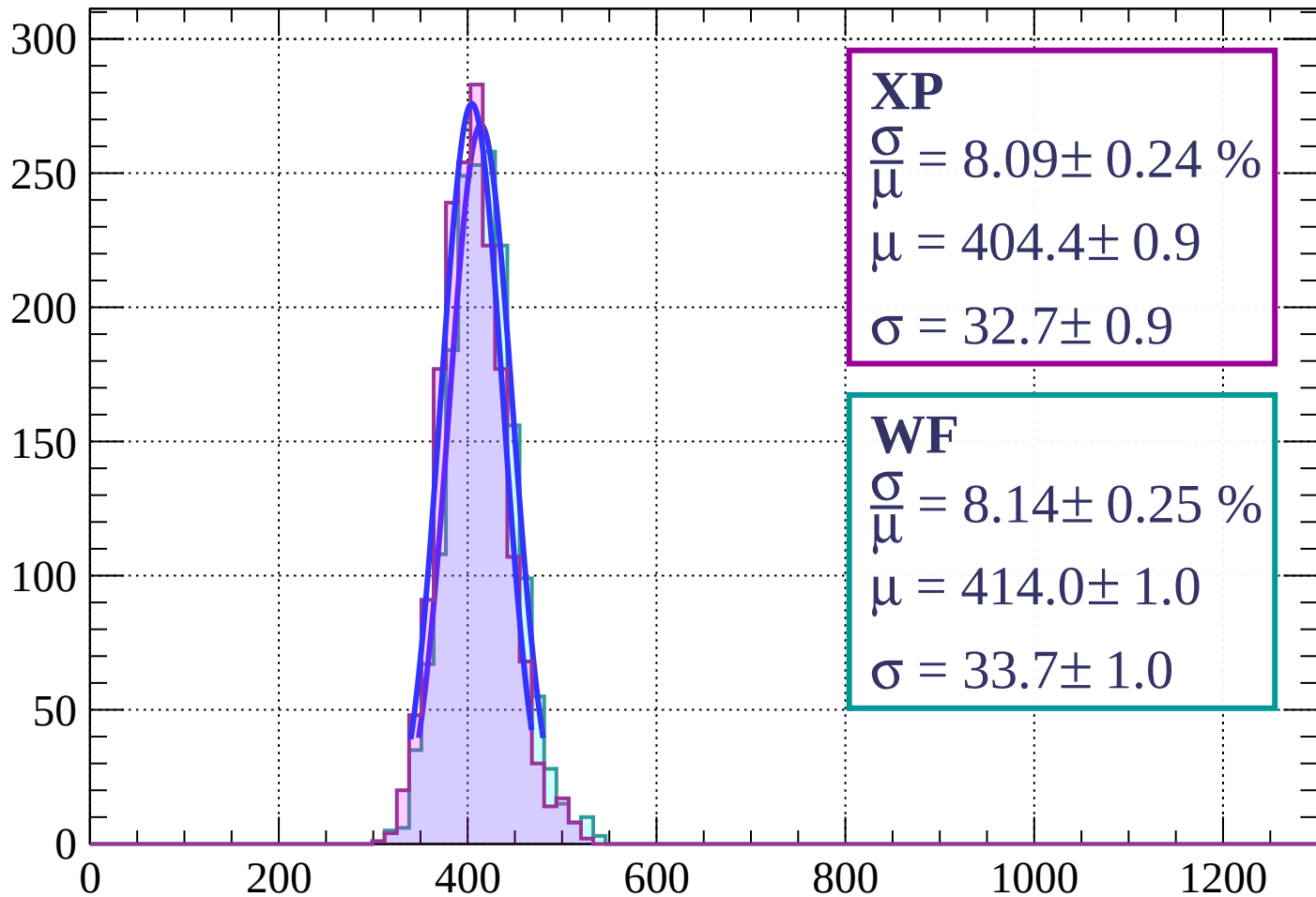


Energy loss | $81 < \phi < 84$



Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.24 \%$$

$$\mu = 404.4 \pm 0.9$$

$$\sigma = 32.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.14 \pm 0.25 \%$$

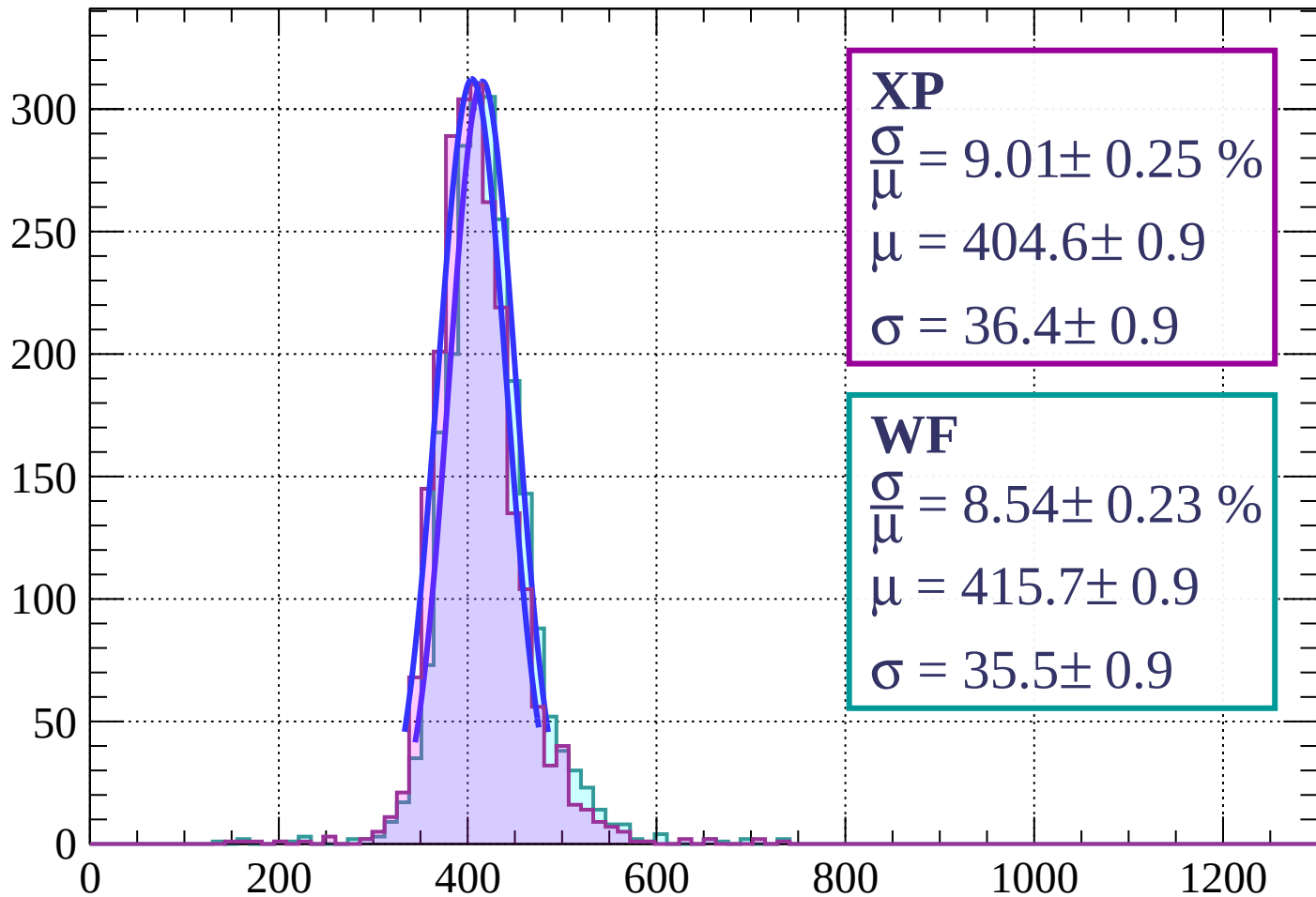
$$\mu = 414.0 \pm 1.0$$

$$\sigma = 33.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $87 < \phi < 90$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.25 \%$$

$$\mu = 404.6 \pm 0.9$$

$$\sigma = 36.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.54 \pm 0.23 \%$$

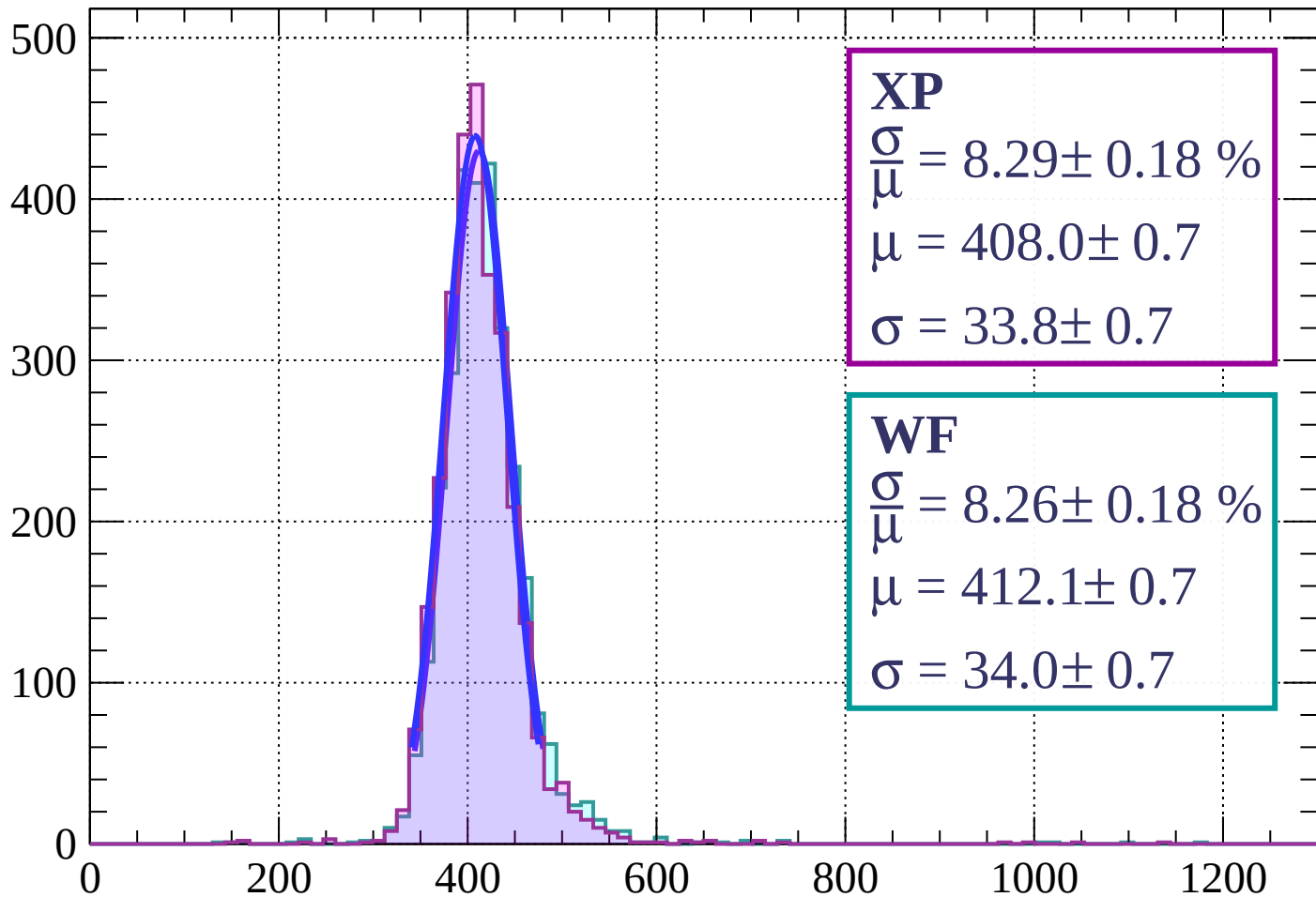
$$\mu = 415.7 \pm 0.9$$

$$\sigma = 35.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.18 \%$$

$$\mu = 408.0 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

WF

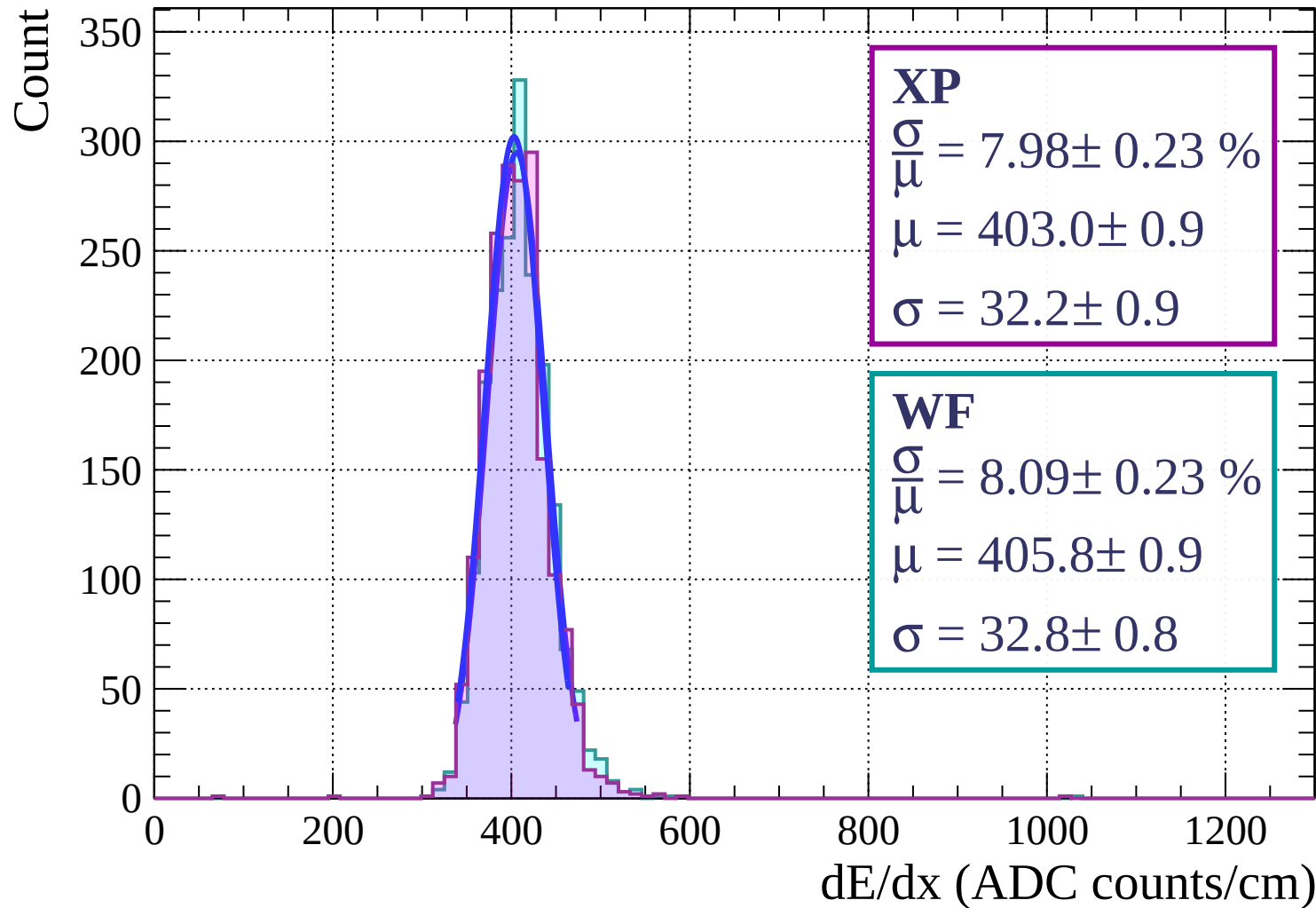
$$\frac{\sigma}{\mu} = 8.26 \pm 0.18 \%$$

$$\mu = 412.1 \pm 0.7$$

$$\sigma = 34.0 \pm 0.7$$

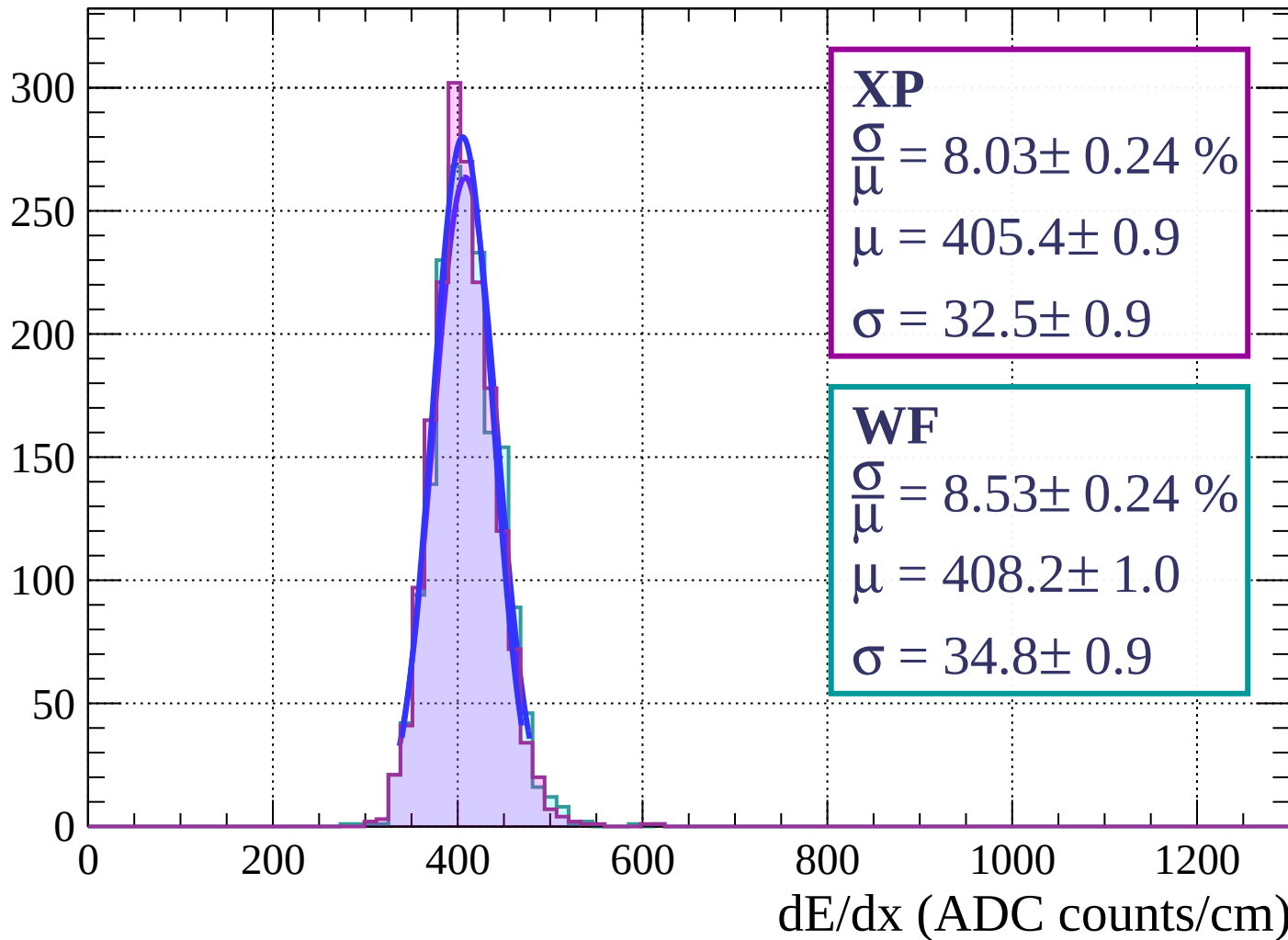
dE/dx (ADC counts/cm)

Energy loss | $3 < \theta < 6$



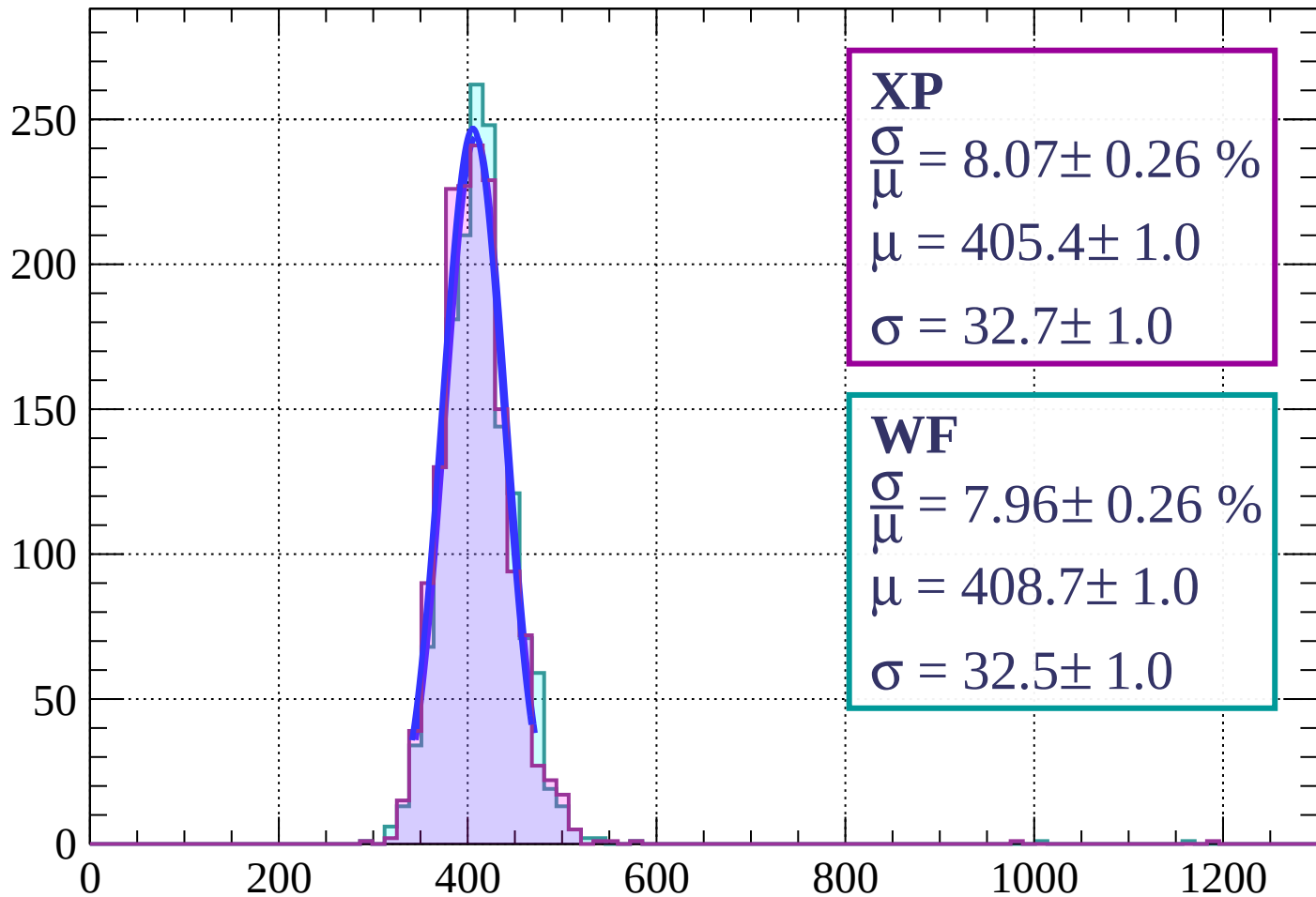
Energy loss | $6 < \theta < 9$

Count



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.26 \%$$

$$\mu = 405.4 \pm 1.0$$

$$\sigma = 32.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.96 \pm 0.26 \%$$

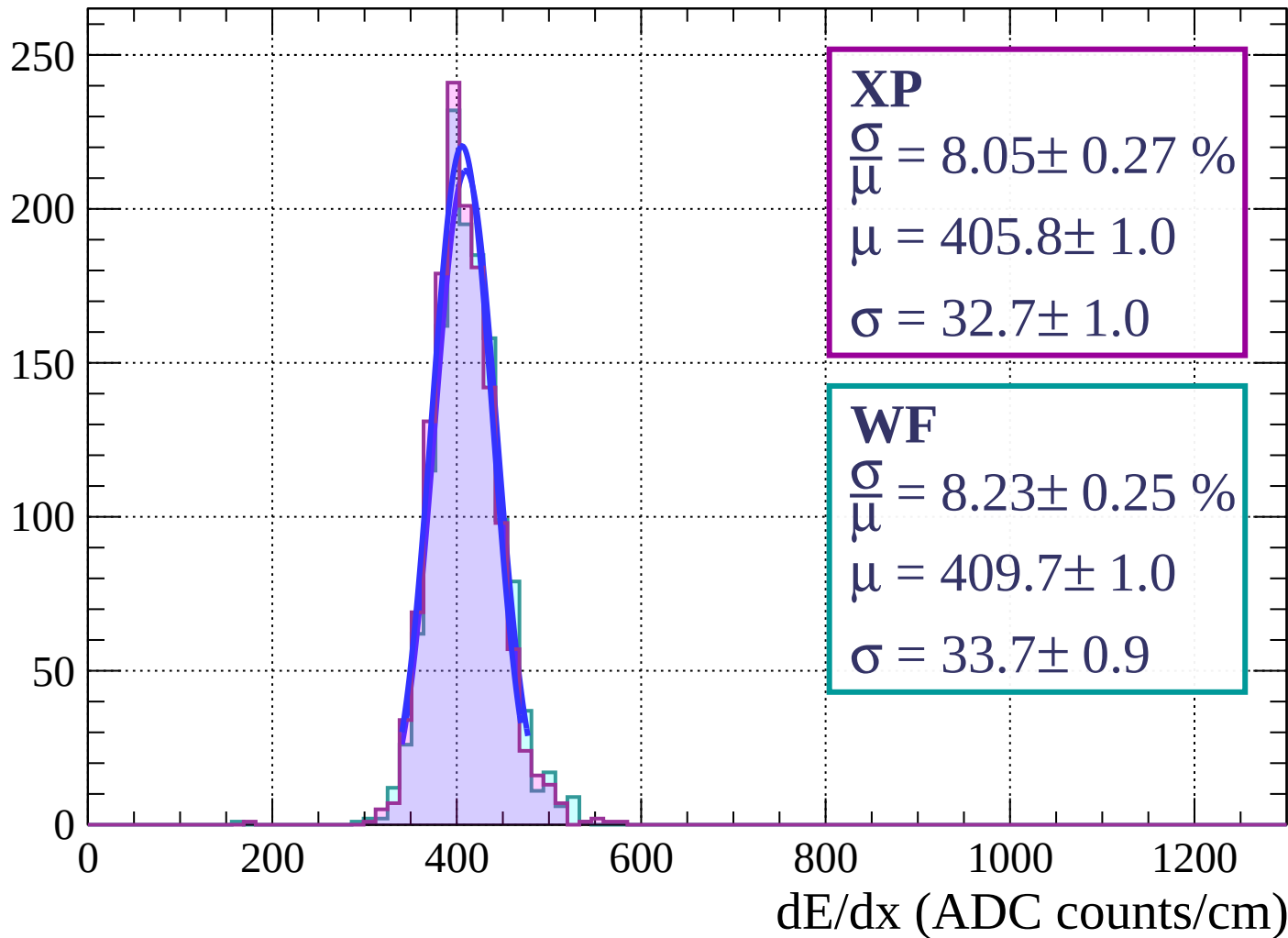
$$\mu = 408.7 \pm 1.0$$

$$\sigma = 32.5 \pm 1.0$$

dE/dx (ADC counts/cm)

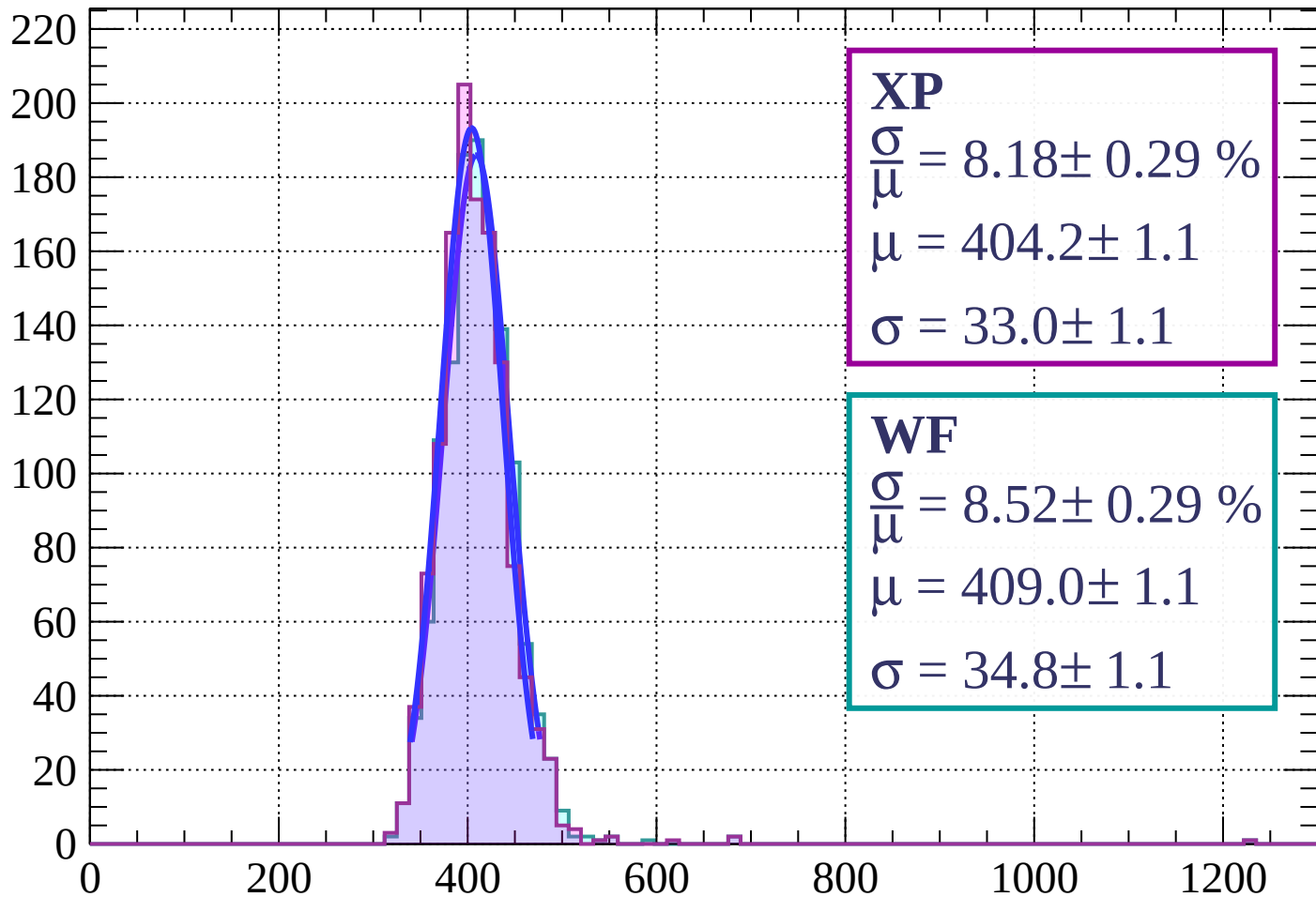
Energy loss | $12 < \theta < 15$

Count



Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 8.18 \pm 0.29 \%$$

$$\mu = 404.2 \pm 1.1$$

$$\sigma = 33.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.52 \pm 0.29 \%$$

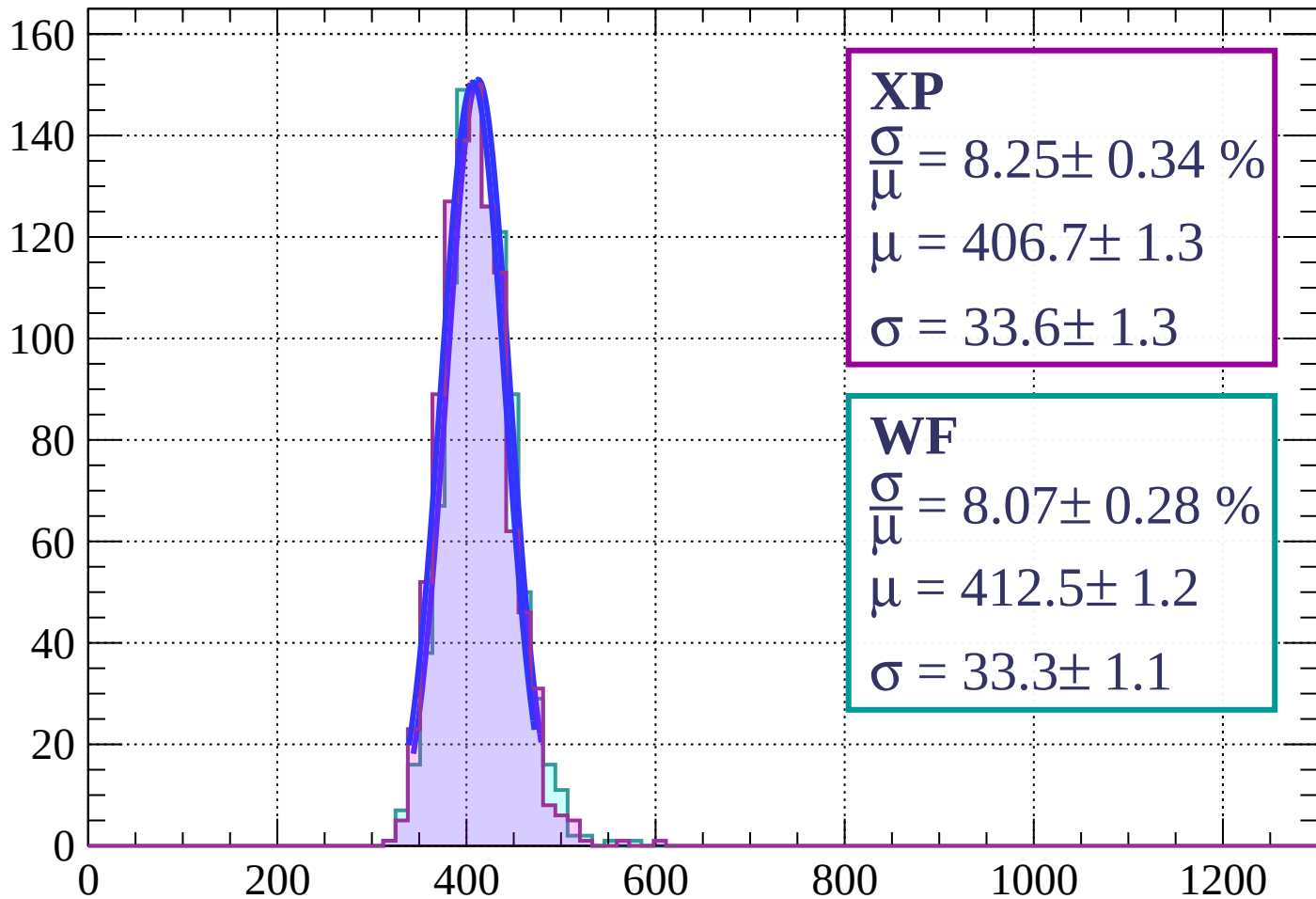
$$\mu = 409.0 \pm 1.1$$

$$\sigma = 34.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 8.25 \pm 0.34 \%$$

$$\mu = 406.7 \pm 1.3$$

$$\sigma = 33.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.07 \pm 0.28 \%$$

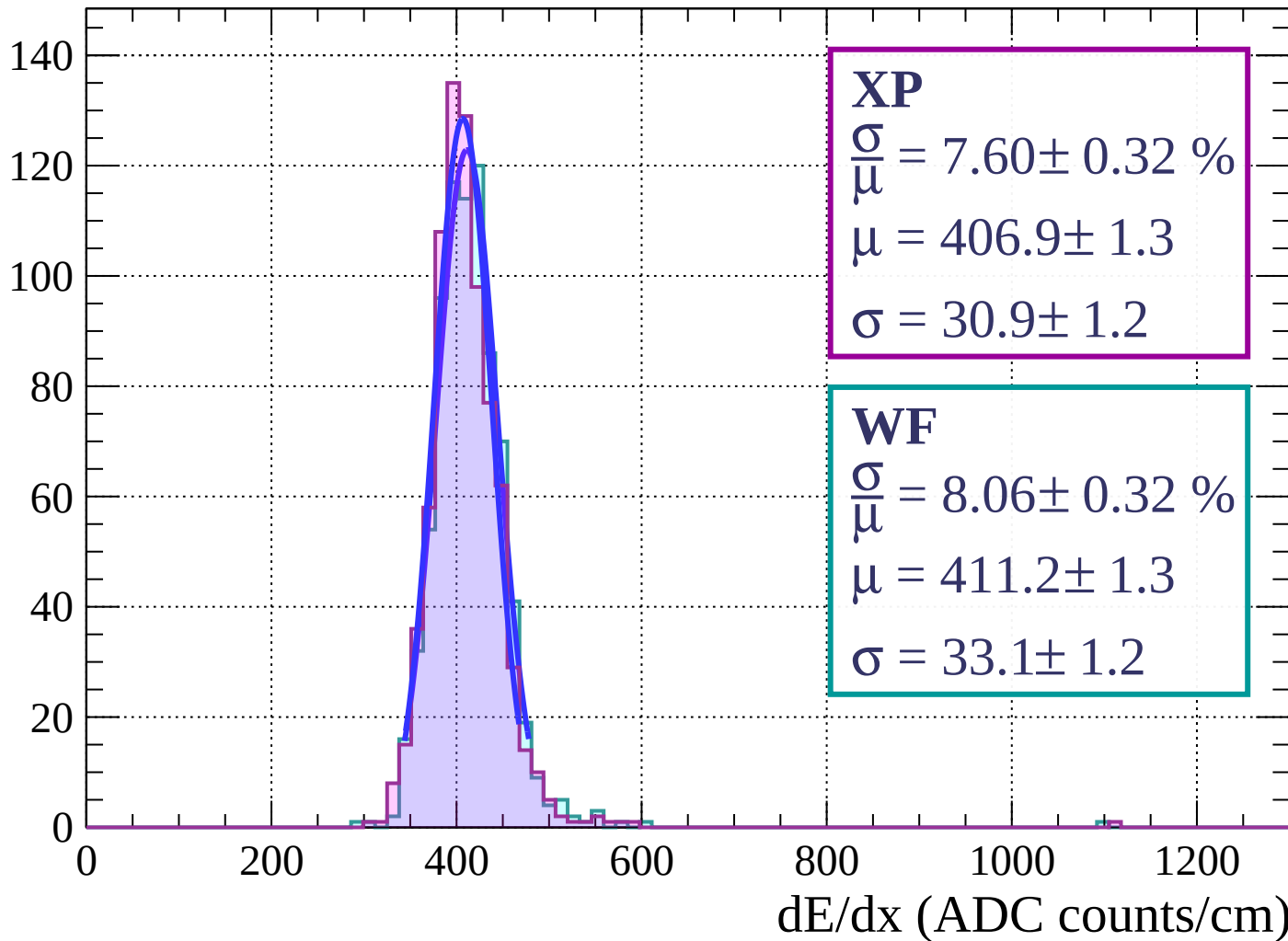
$$\mu = 412.5 \pm 1.2$$

$$\sigma = 33.3 \pm 1.1$$

dE/dx (ADC counts/cm)

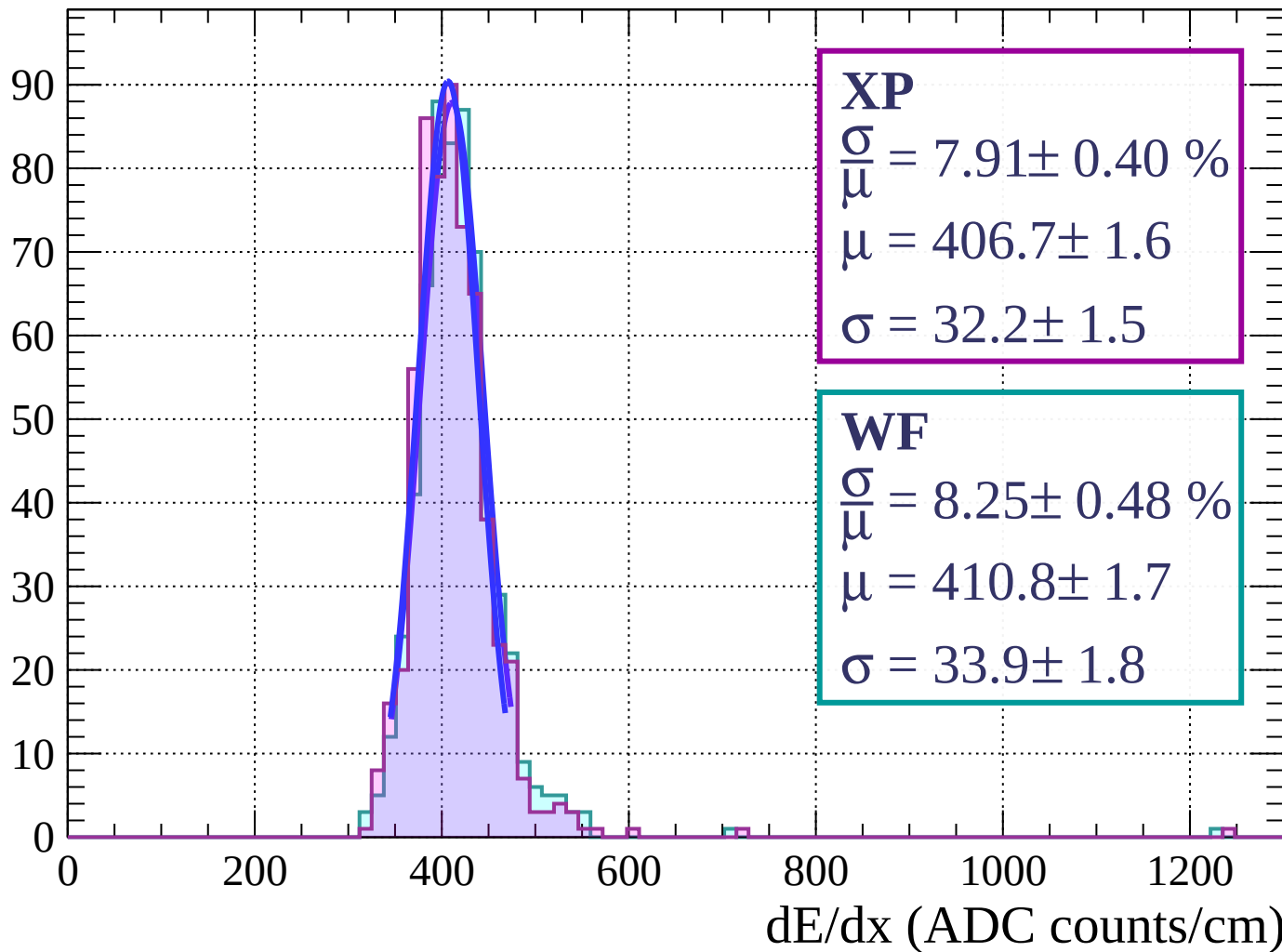
Energy loss | $21 < \theta < 24$

Count



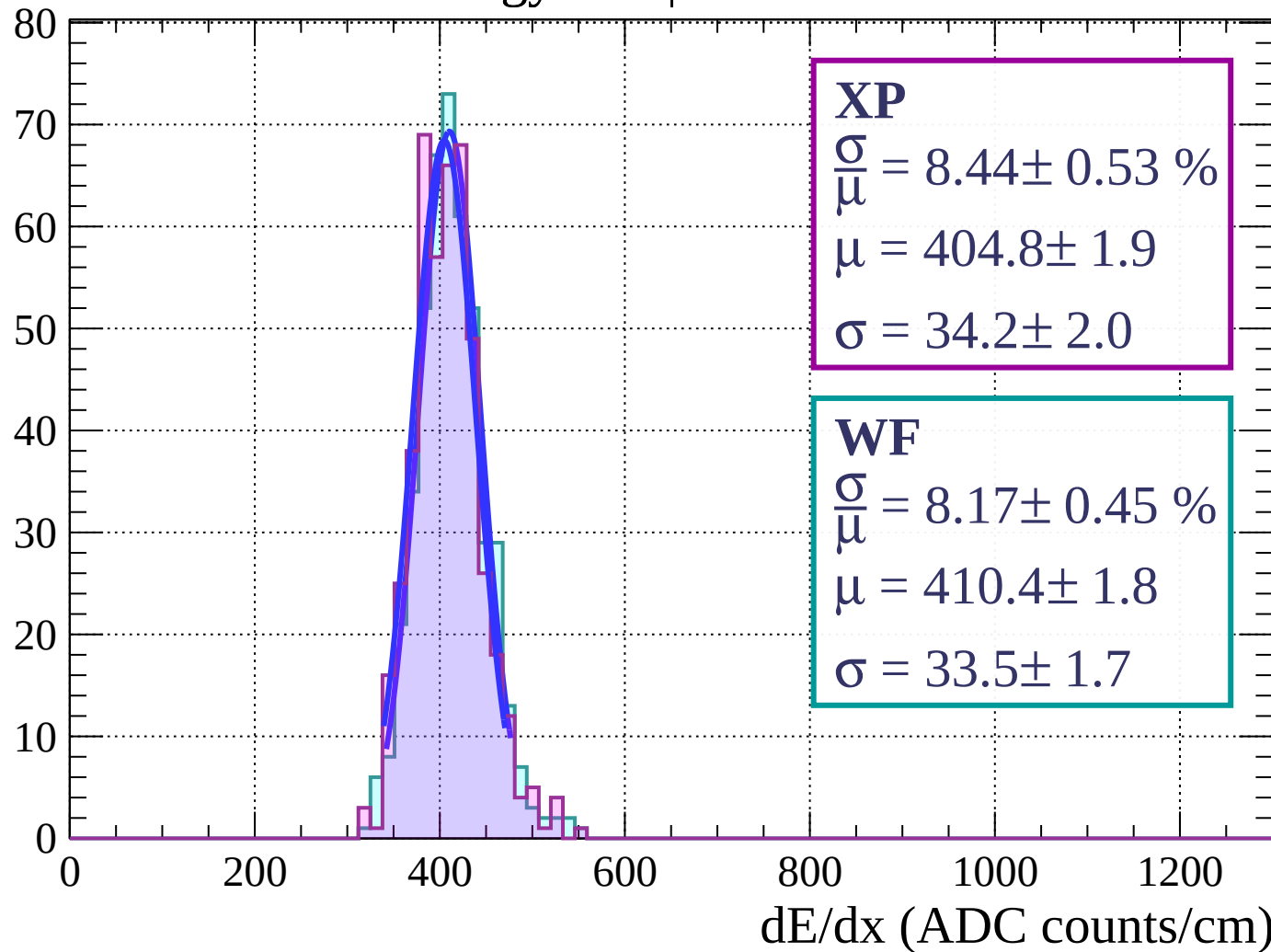
Energy loss | $24 < \theta < 27$

Count



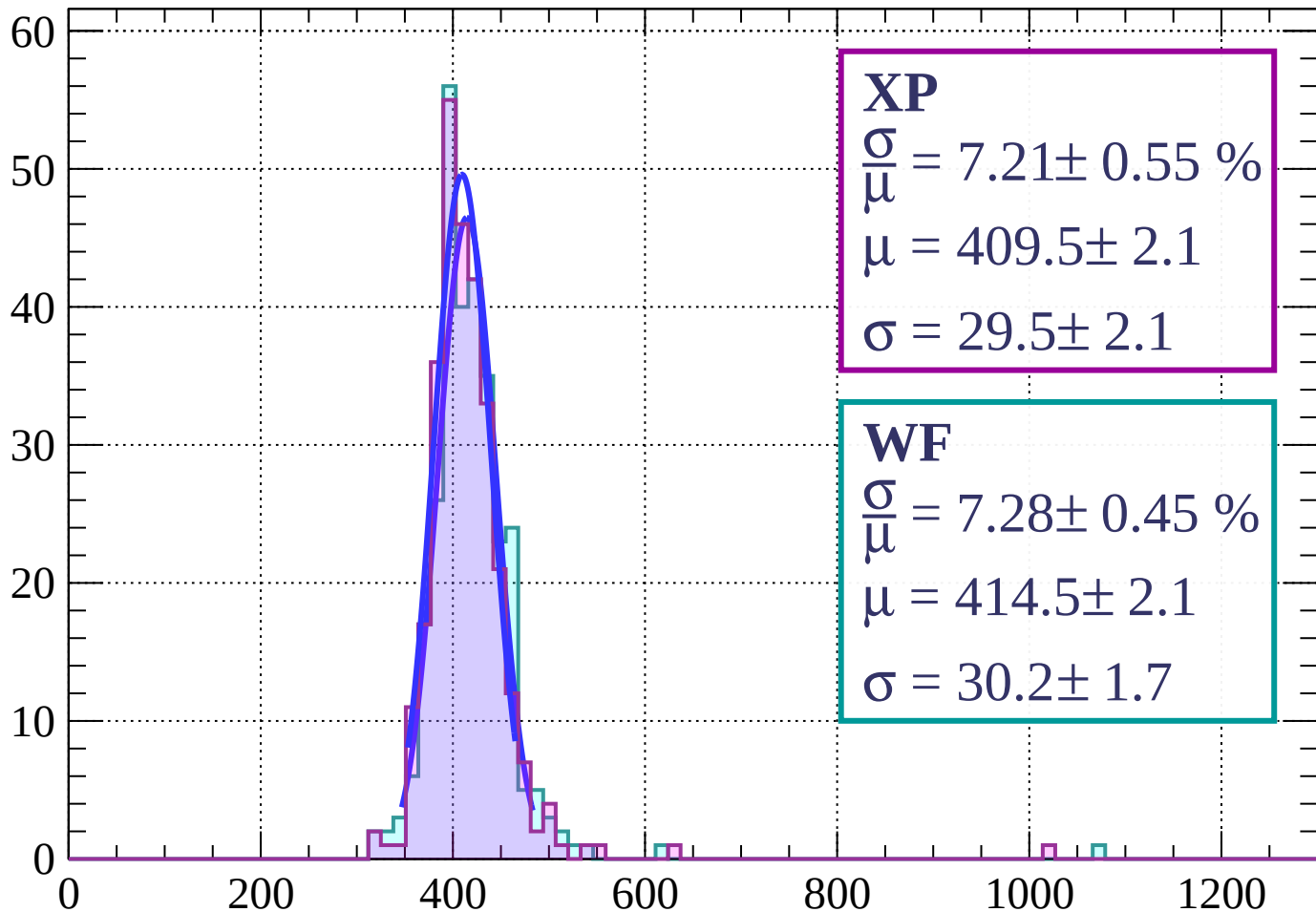
Energy loss | $27 < \theta < 30$

Count



Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.55 \%$$

$$\mu = 409.5 \pm 2.1$$

$$\sigma = 29.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.28 \pm 0.45 \%$$

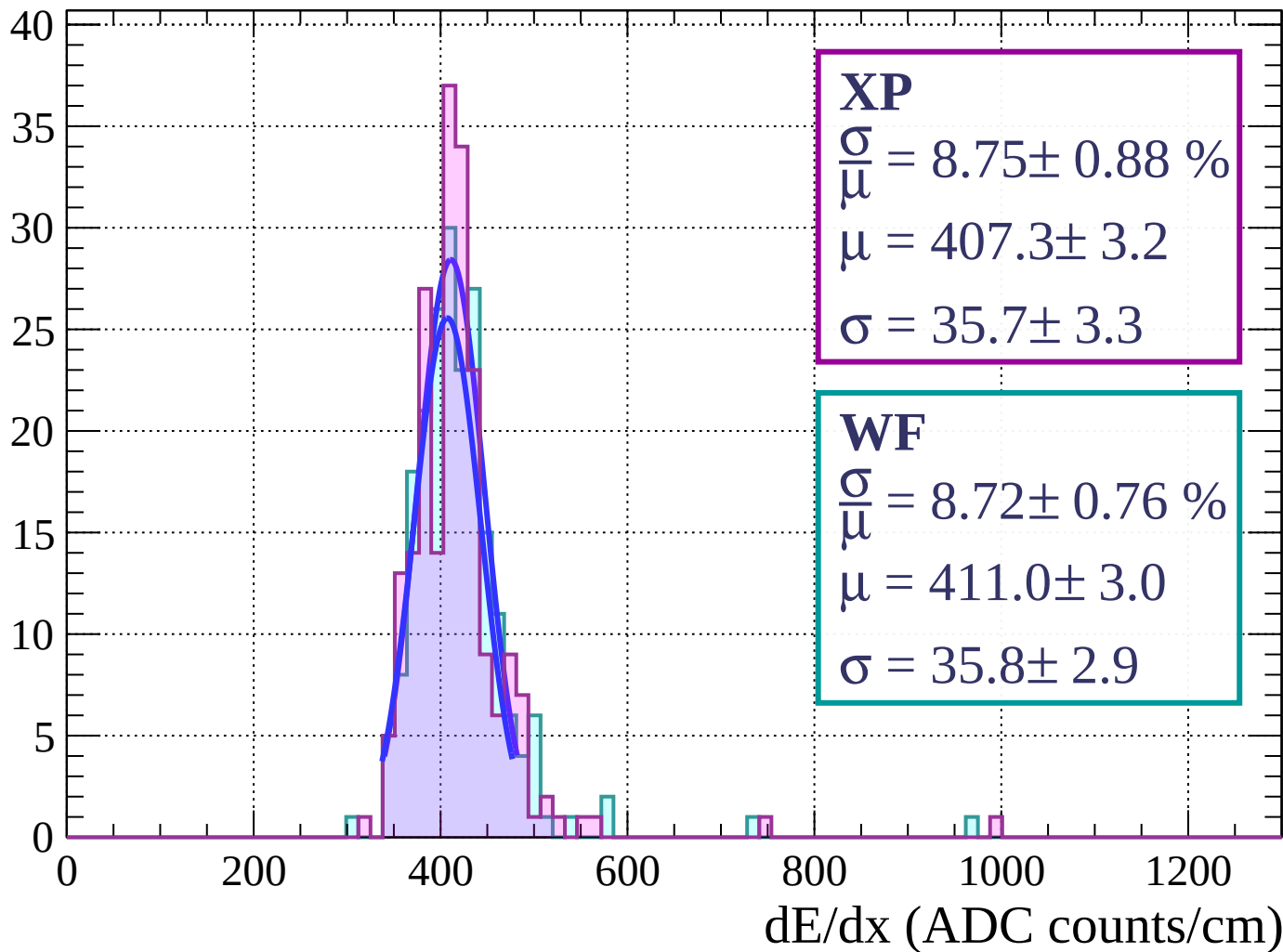
$$\mu = 414.5 \pm 2.1$$

$$\sigma = 30.2 \pm 1.7$$

dE/dx (ADC counts/cm)

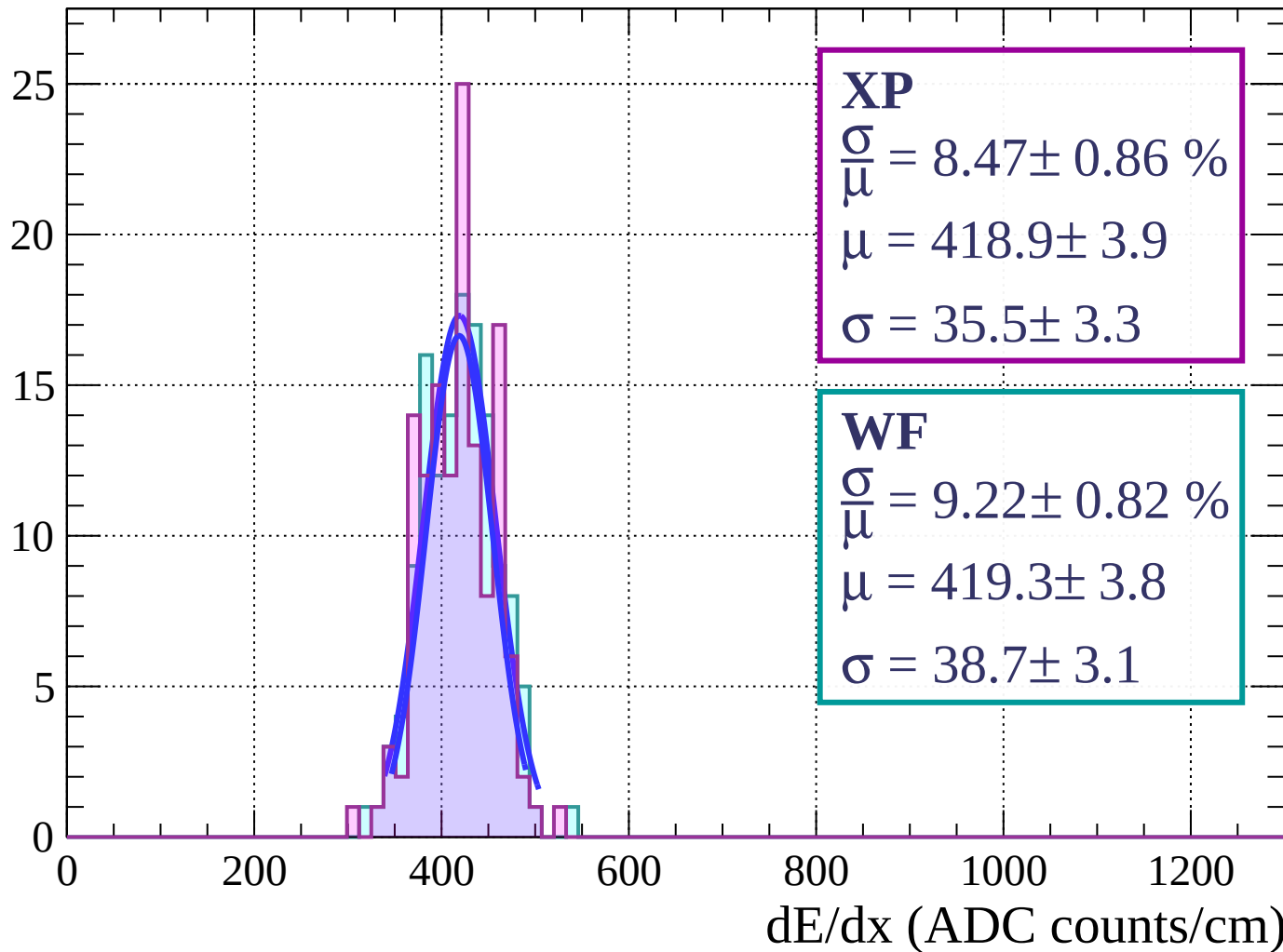
Energy loss | $33 < \theta < 36$

Count



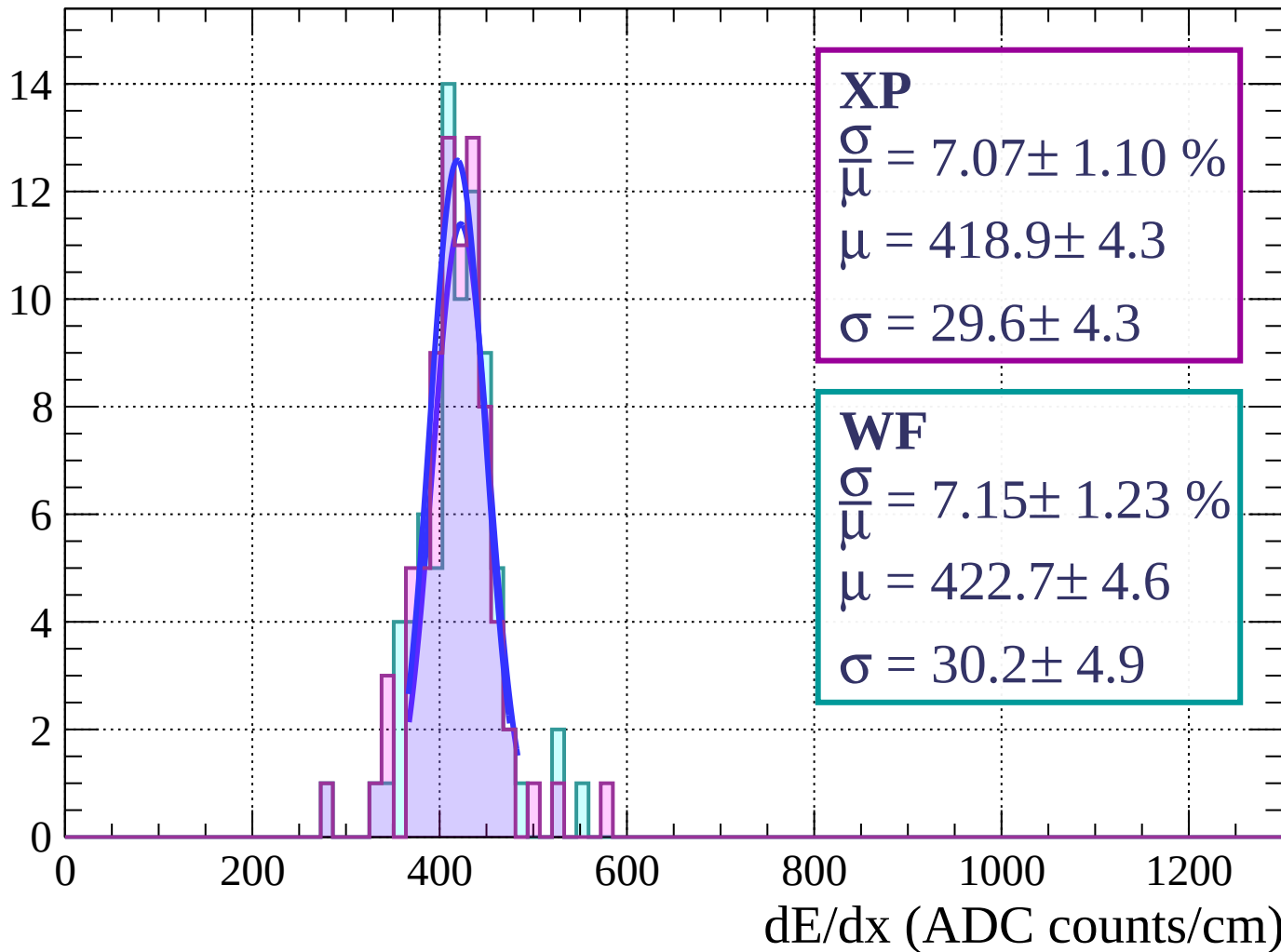
Energy loss | $36 < \theta < 39$

Count



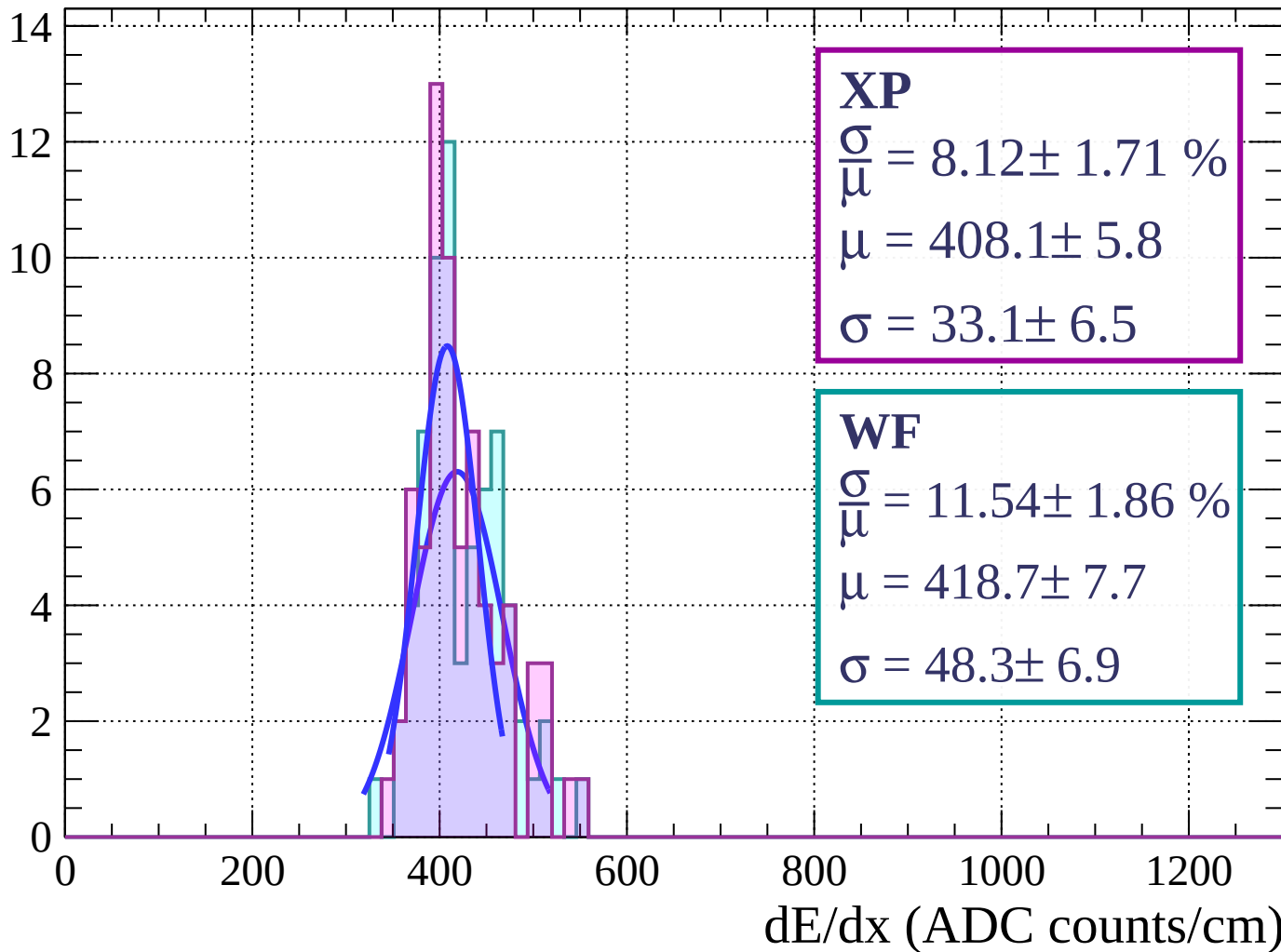
Energy loss | $39 < \theta < 42$

Count



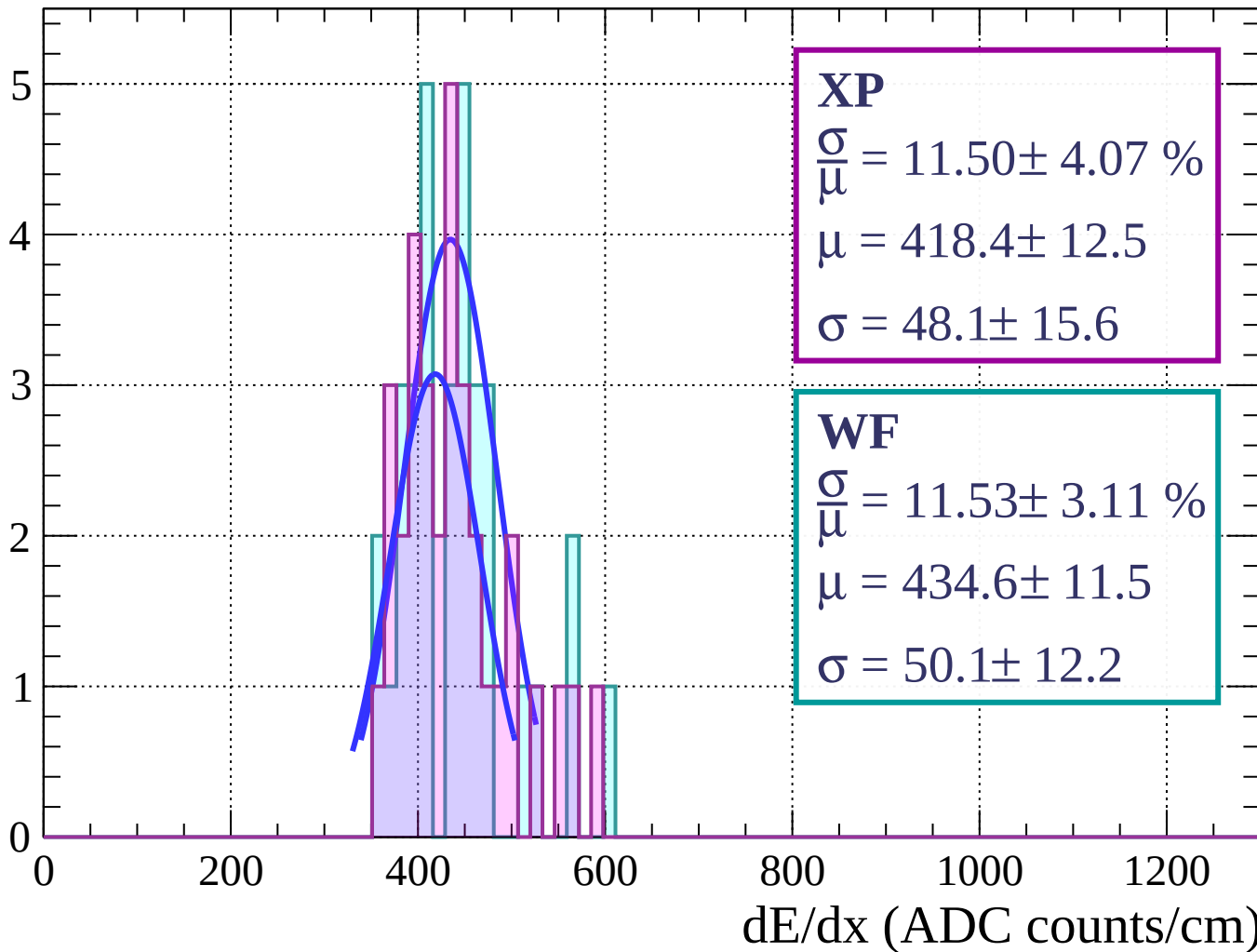
Energy loss | $42 < \theta < 45$

Count



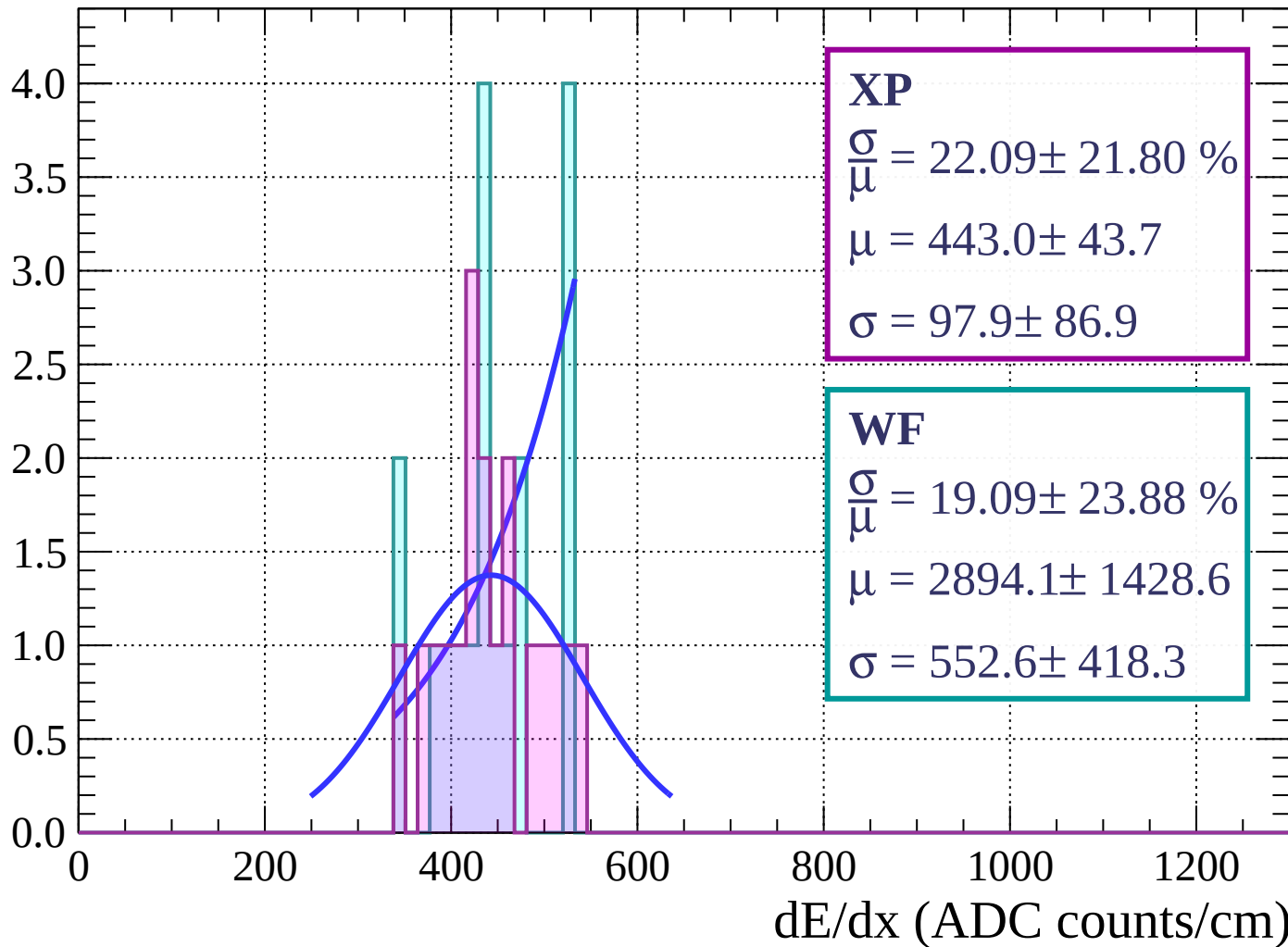
Energy loss | $45 < \theta < 48$

Count



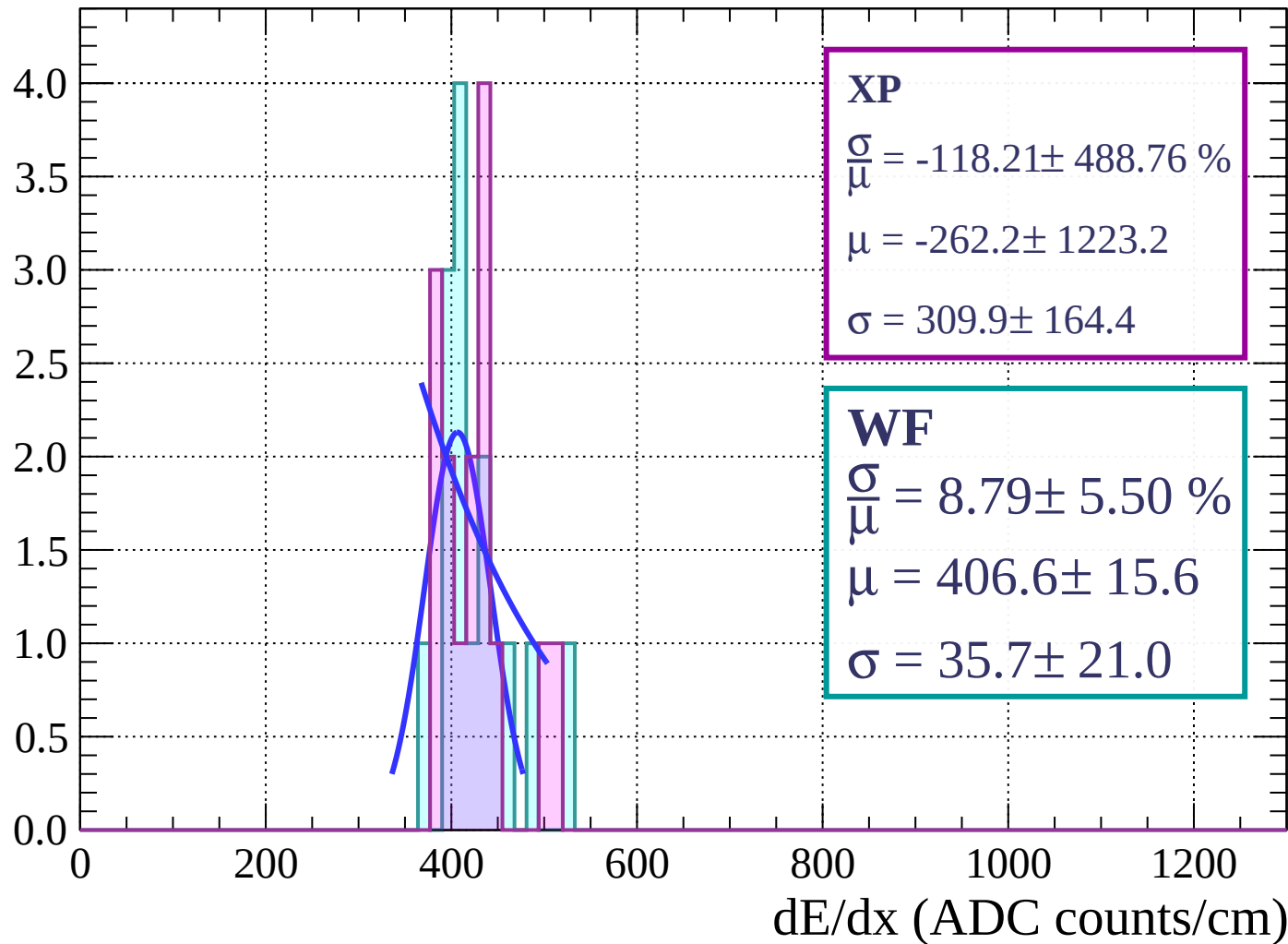
Energy loss | $48 < \theta < 51$

Count



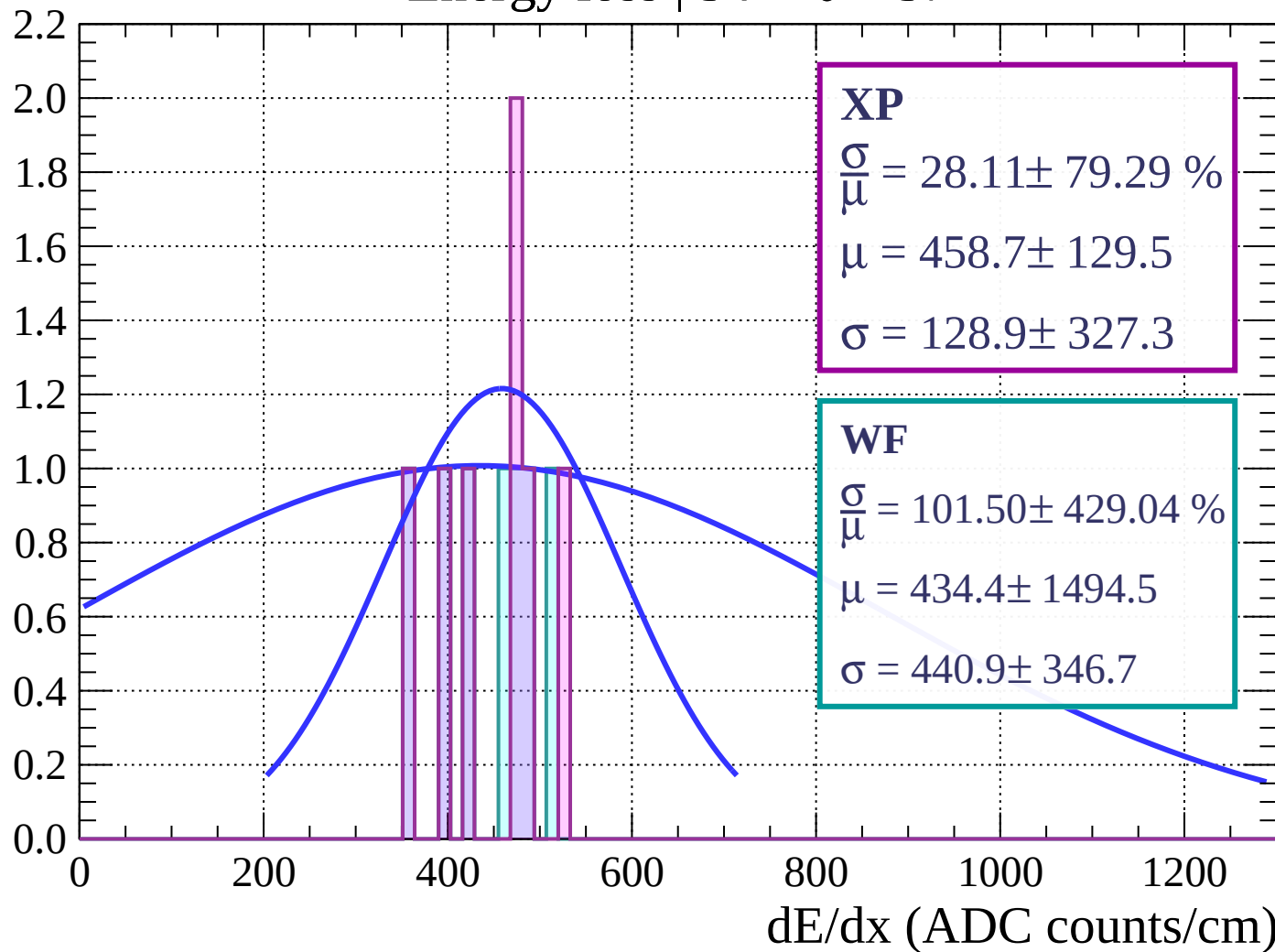
Energy loss | $51 < \theta < 54$

Count



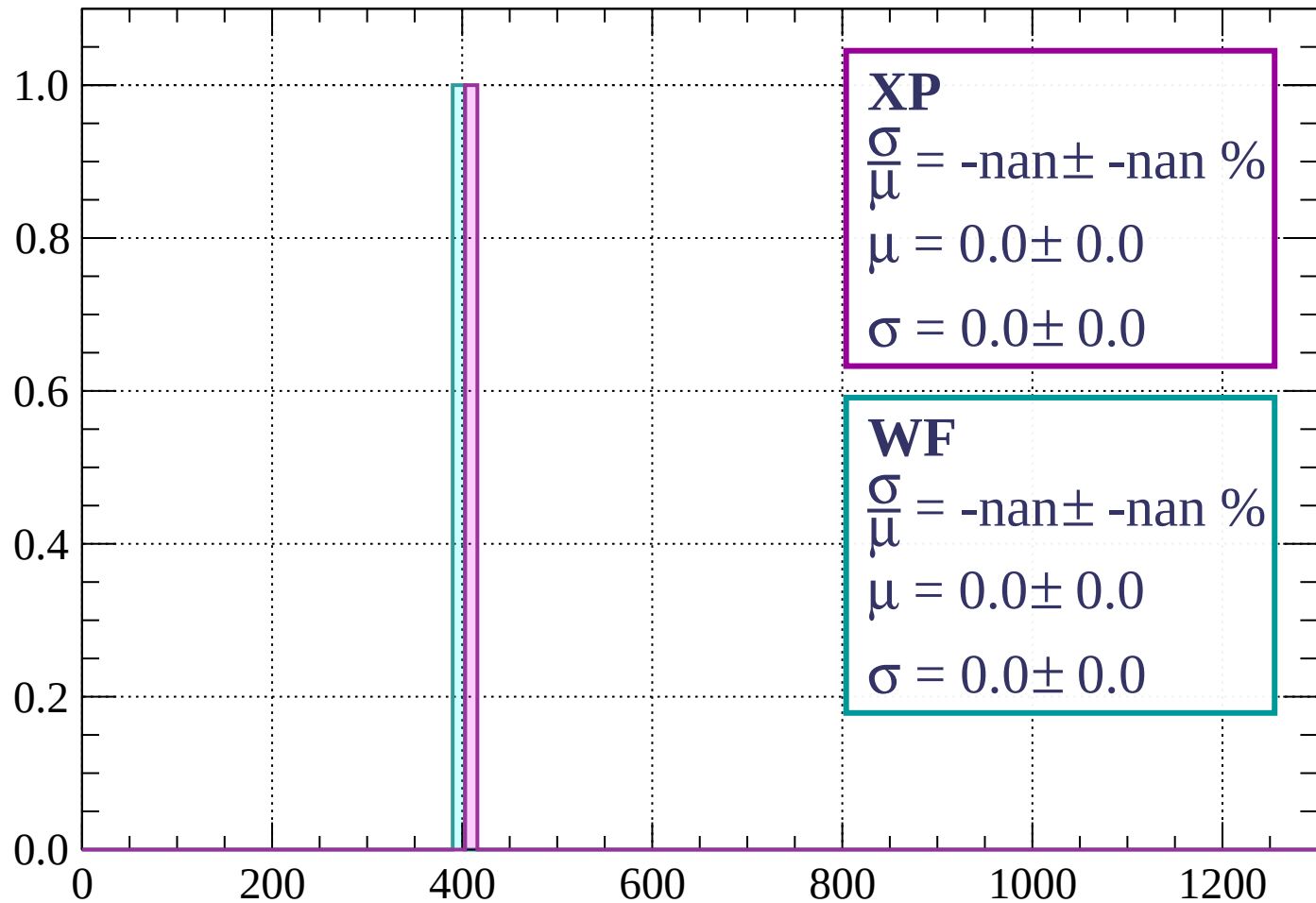
Energy loss | $54 < \theta < 57$

Count



Energy loss | $57 < \theta < 60$

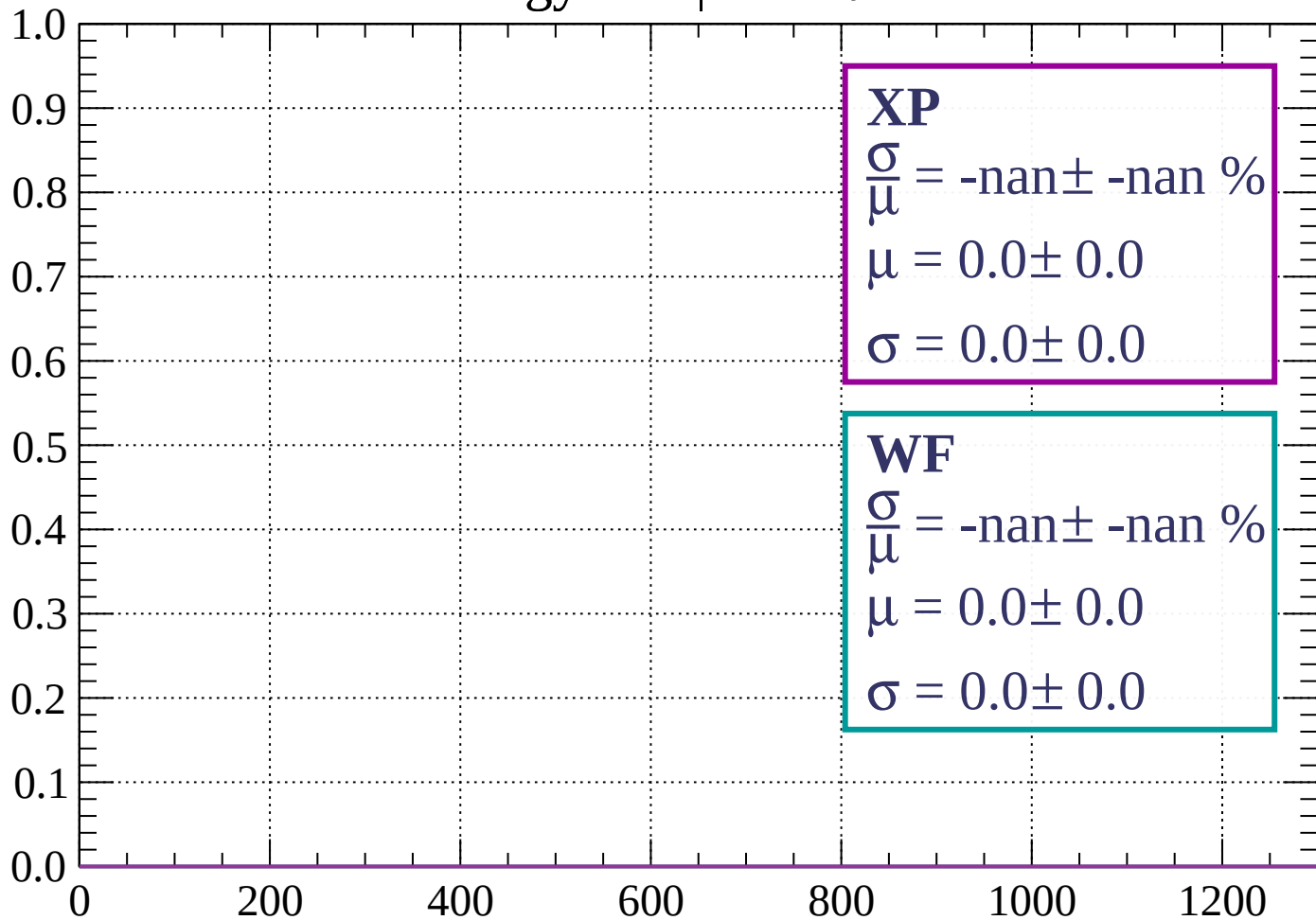
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

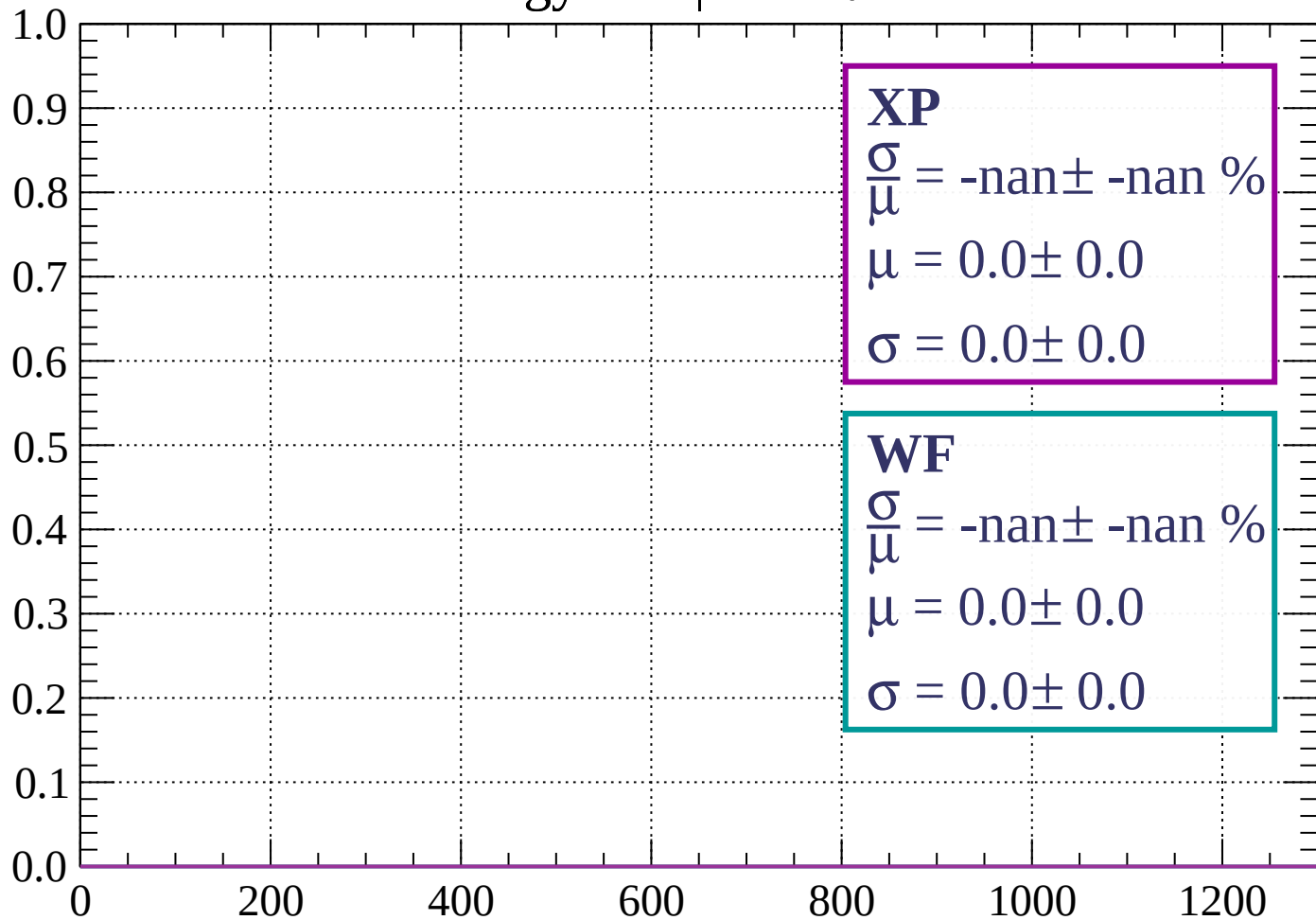
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

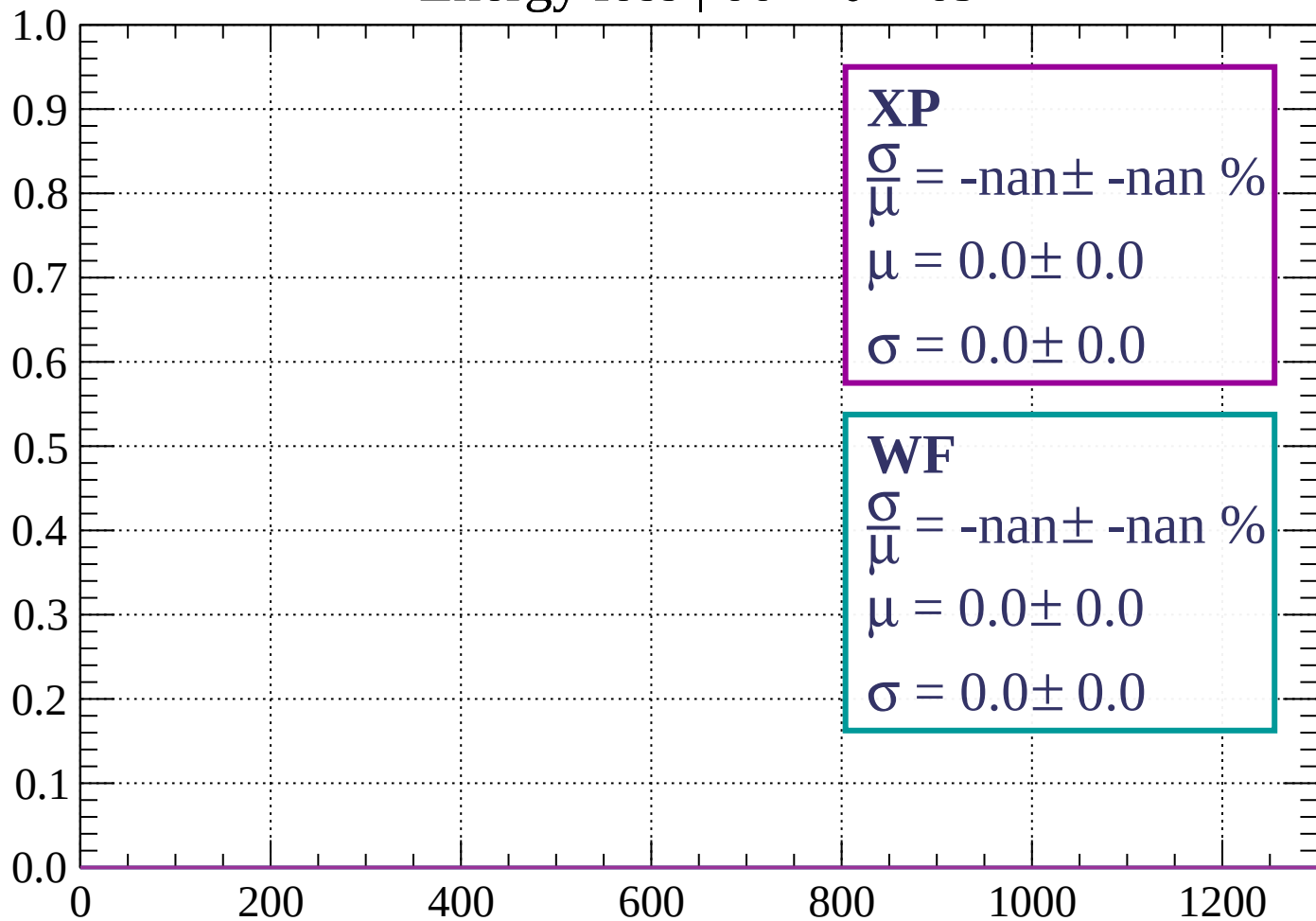
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

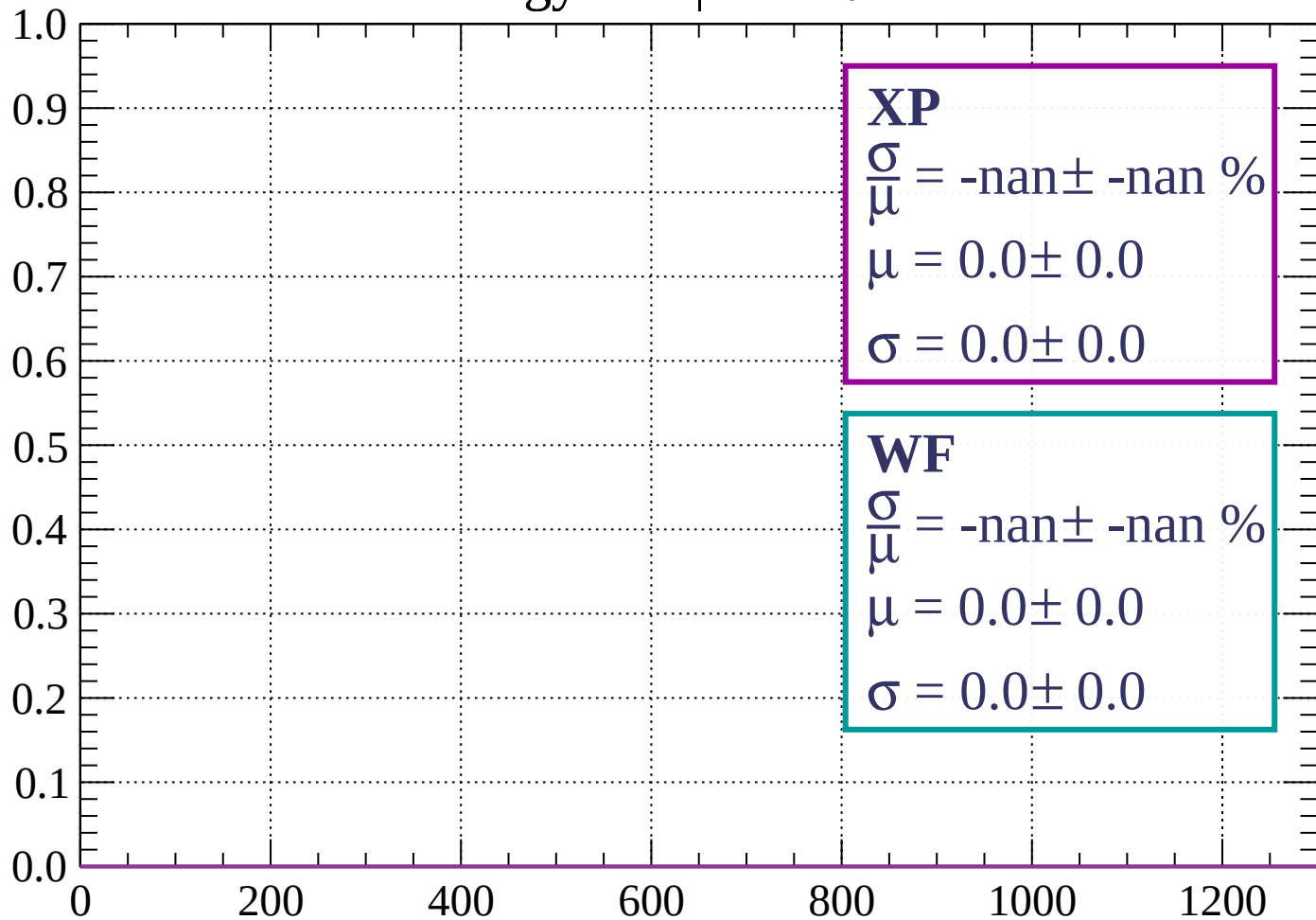
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

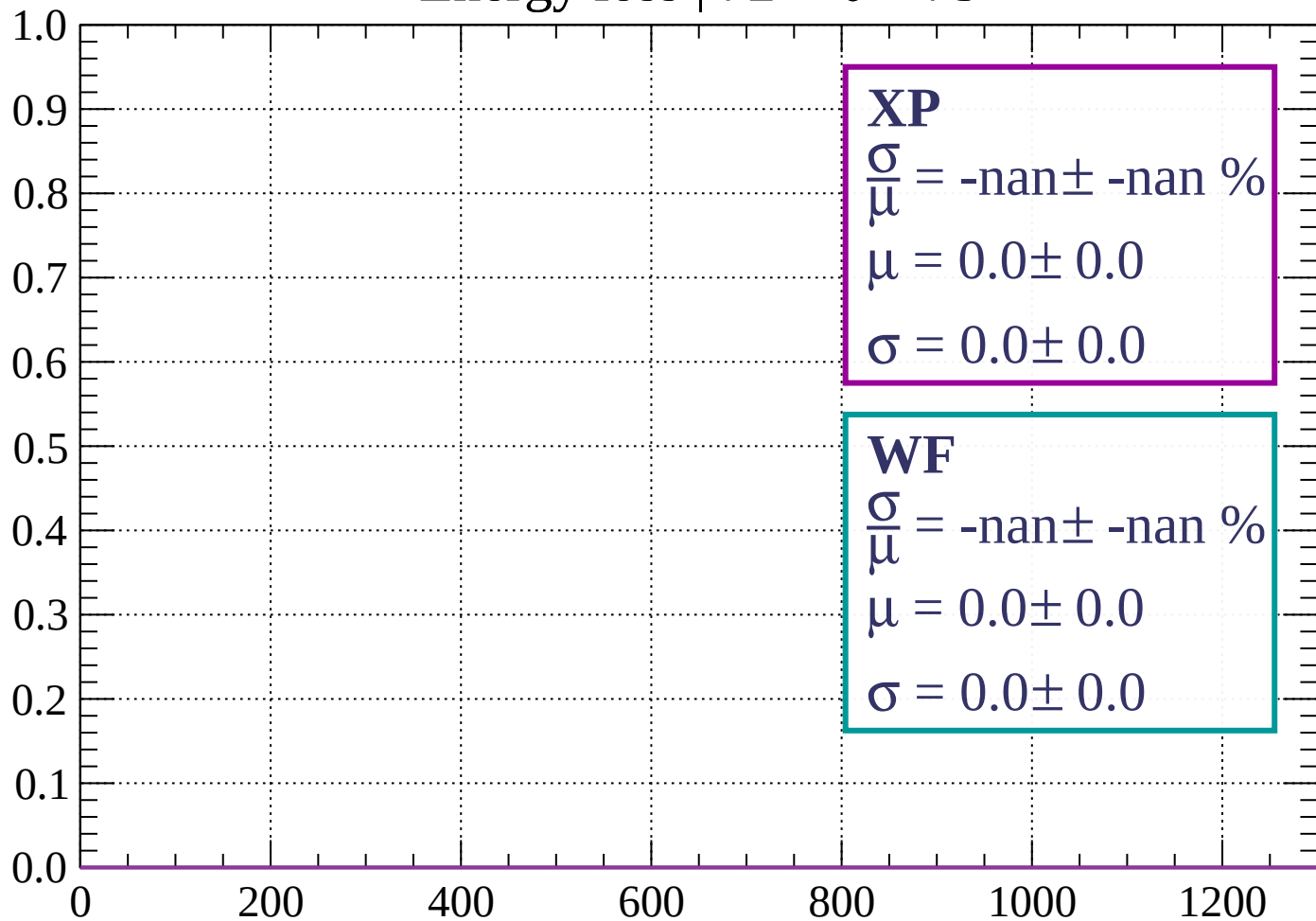
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

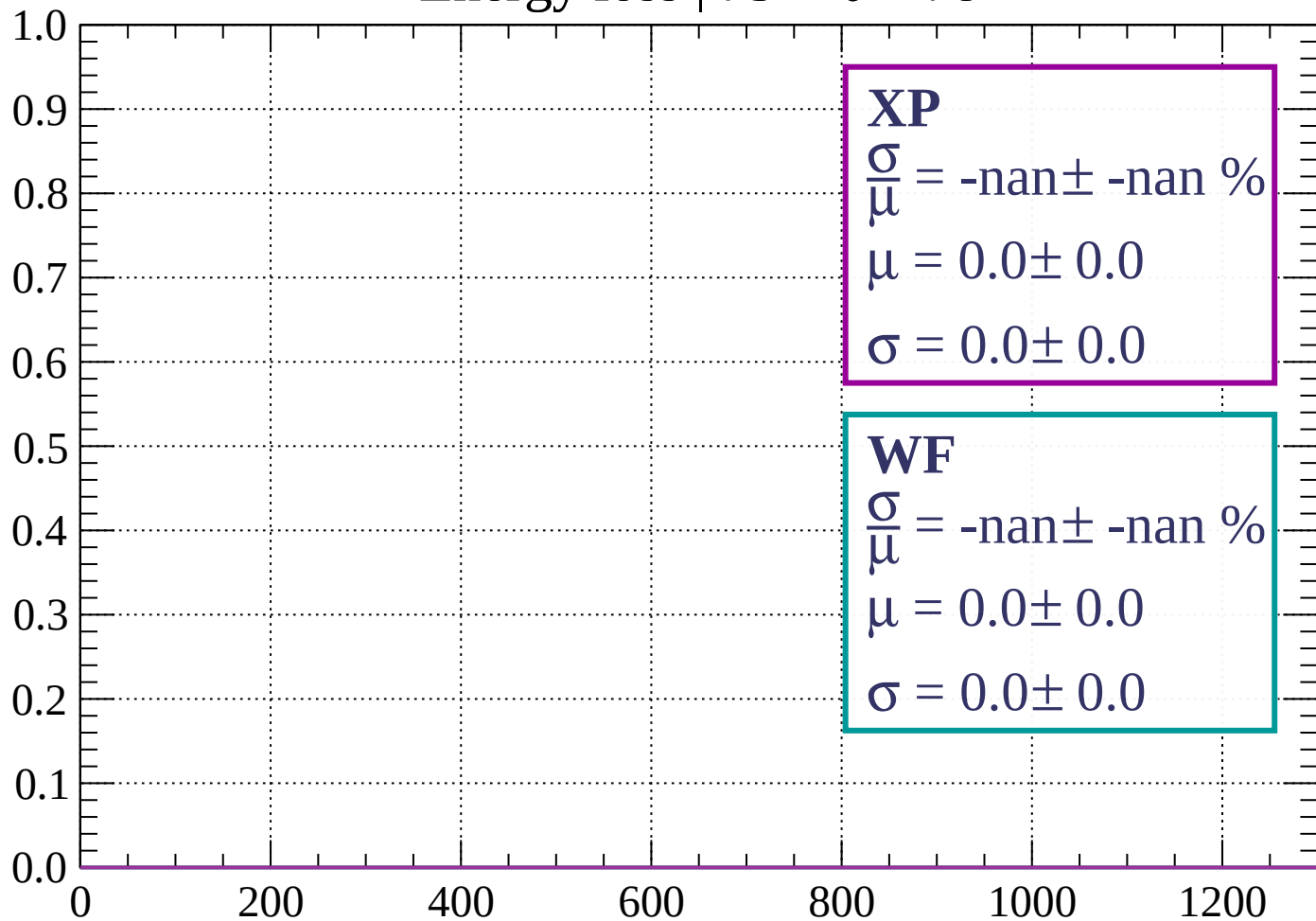
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

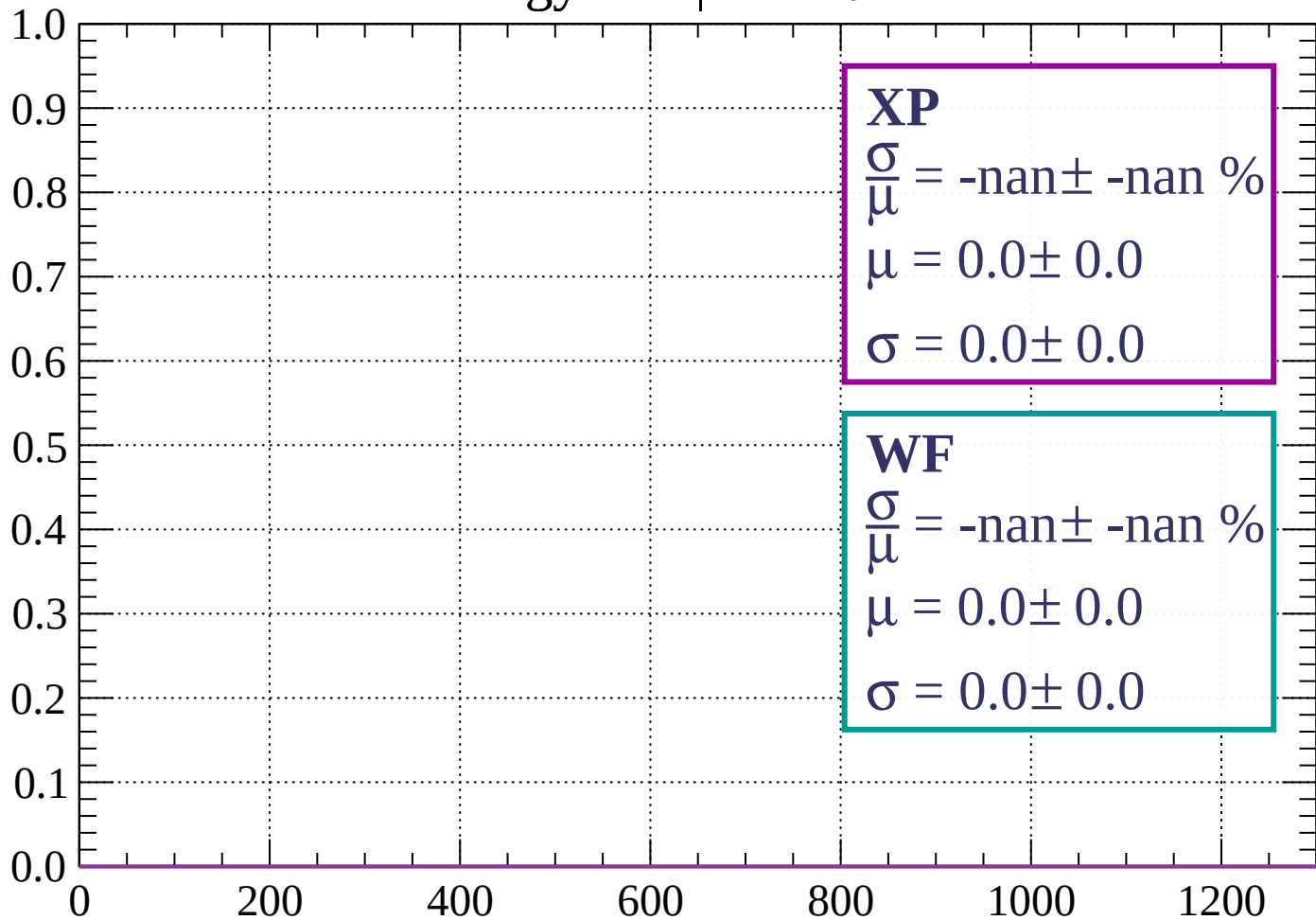
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

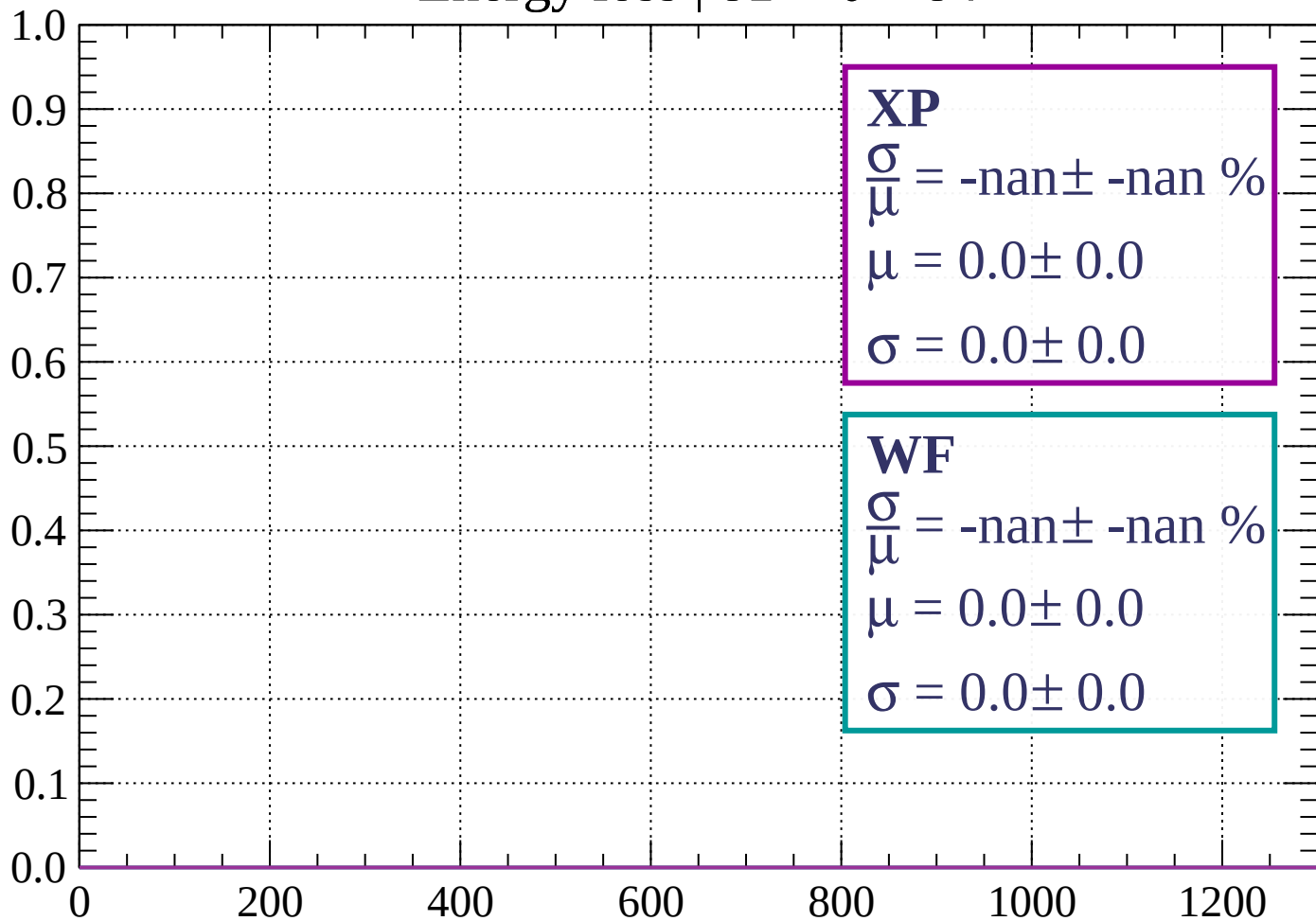
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

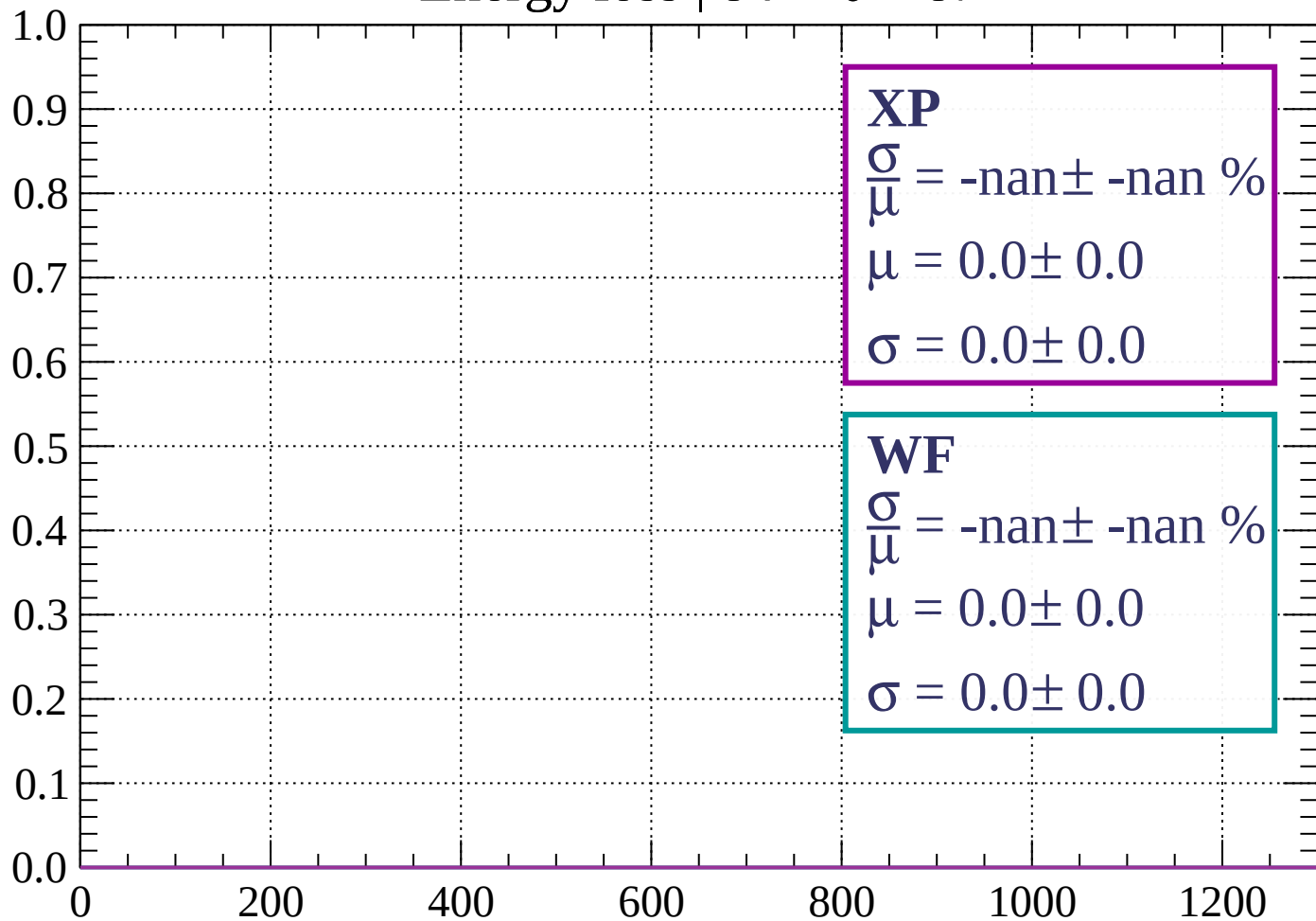
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

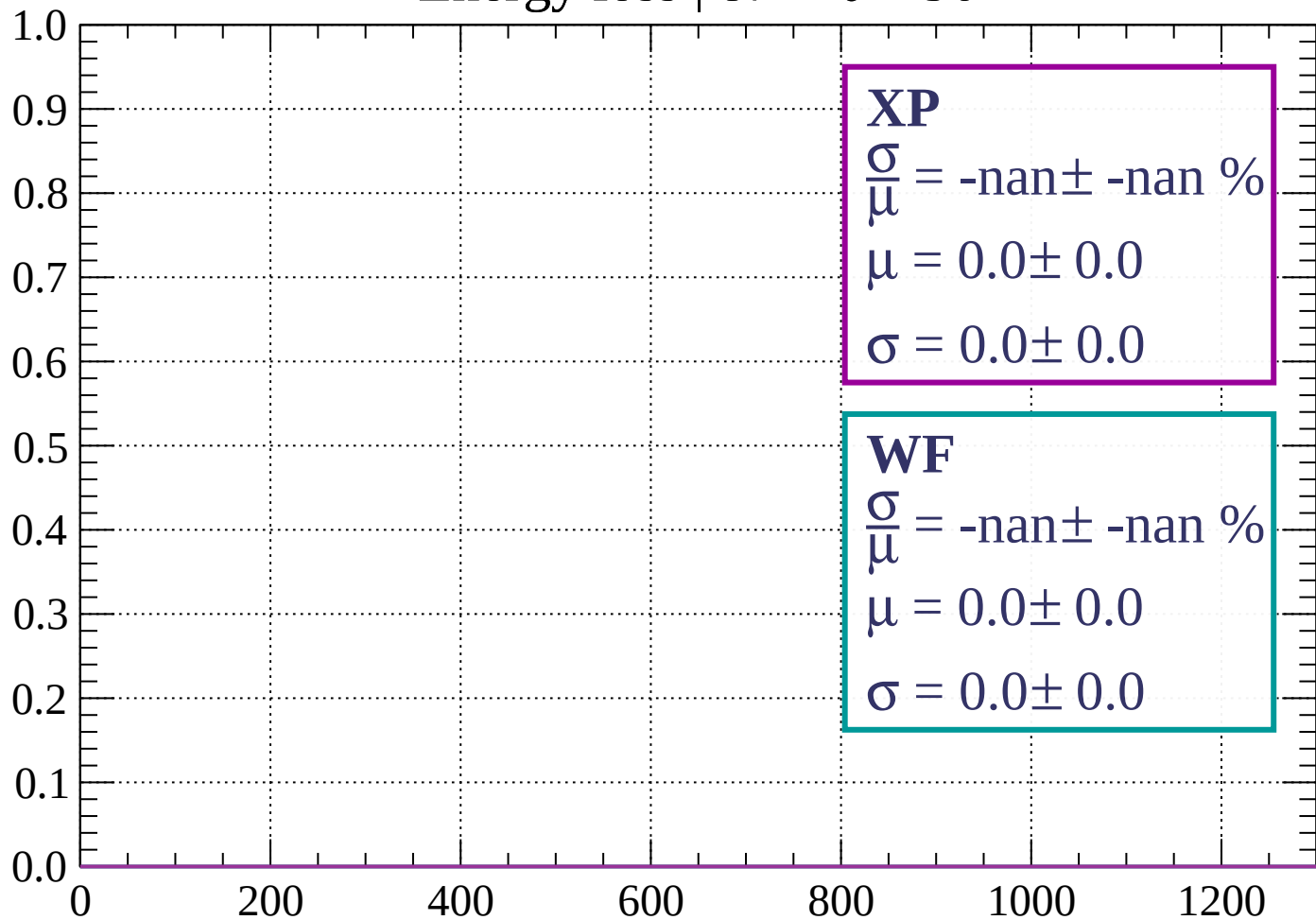
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



XP

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

WF

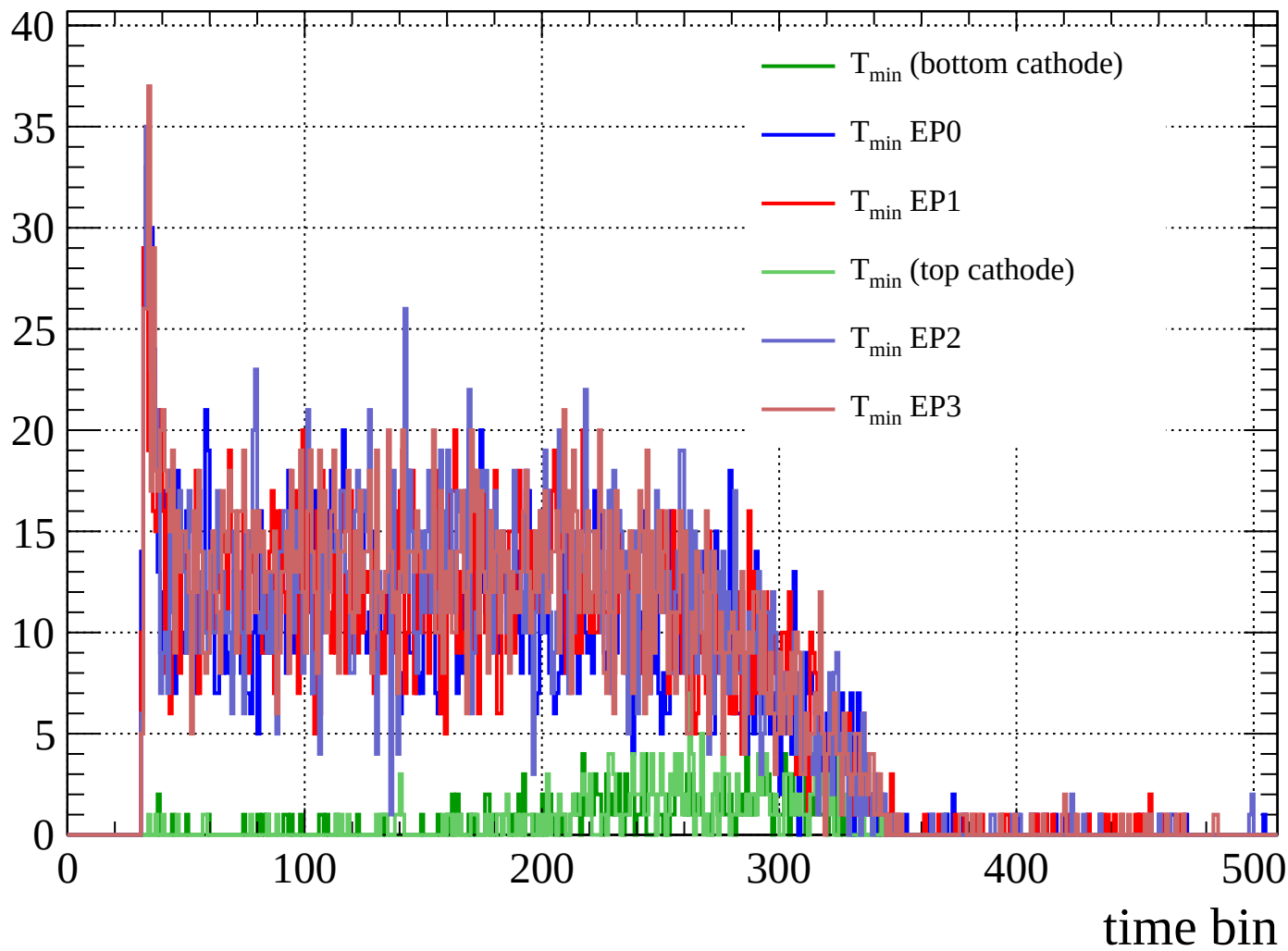
$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

dE/dx (ADC counts/cm)

Count



Count

